

 motorola a lenovo company	Material or Methods Specification	No. 12014512001
Title: Factory Test Specification for LTE / 3GSM / CDMA / TD-SCDMA Handsets		
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ANDROID

Vertex

GSM/UMTS/CDMA/LTE/TD-SCDMA

Factory 12M Test Specification

MSM8953 Platform

MSM8953, Adreno 506, PM8953 + PMI8952

Models: Vertex (Addison)

12M-014512001

Issue 03

June 22, 2016



Material or Methods Specification

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Revision History

Issue	ECR #	Description of Release Change(s)	Originator	Date
01		Initial Pre-release of MSM8953 Vertex Platform 12M Issue 01	Joe Pacella	February 19, 2016
02		The following TCNs and changes are included: Update XO Cal to remove collect samples part of procedure as not required Update to indicate 2 sections for Camera LED Calibration Test, one Internal and one External like Vector TCN-MMI-5459-SXXXB-(MSM8953 Vertex - remove Universal Headset Detect IC Test) TCN-MMI-5460-SXXXB-(MSM8953 Vertex - update RF Connector Detection Test) TCN-MMI-5456-SXXXB-(MSM8953 Vertex - Add FRONT HOUSING COLOR UTAG write test) TCN-MMI-5471-SXXXB-(MSM8953 Vertex - Remove TOUCH_SCREEN I2C Lines Test from Vertex 12m) Update User Reset Test per latest Vector TCN-5325. TCN-MMI-5486-SXXXB-(MSM8953 Vertex - Add Batt ID Resistance Test) TCN-MMI-5492-SXXXB-(MSM8953 Vertex - update WLAN RSSI Test) TCN-MMI-5488-SXXXB-(MSM8953 Vertex - update Display Tests) TCN-MMI-5487-SXXXB-(MSM8953 Vertex - add USB-C Powered Cable and Accessory Test and USB Reverse Plug In Test) TCN-MMI-5476-SXXXB-(MSM8953 Vertex - Turn On) TCN-MMI-5498-SXXXB-(MSM8953 Vertex - update limits for Battery Thermistor Test) TCN-MMI-5514-SXXXB-(MSM8953 Vertex - update Turn On)	Joe Pacella Joe Pacella Joe Pacella Joe Pacella Joe Pacella Dave Broschka Dave Broschka Dave Broschka Dave Broschka Dave Broschka Dave Broschka Joe Pacella Dave Broschka Joe Pacella	March 2, 2016 March 7, 2016 March 7, 2016 March 7, 2016 March 9, 2016 March 15, 2016 March 24, 2016 March 25, 2016 March 25, 2016 March 29, 2016 March 29, 2016 April 4, 2016 April 6, 2016 April 11, 2016
03		The following TCNs and changes are included: TCN-MMI-5515-SXXXB-(MSM8953 Vertex - Add UTAG RF ID read test) TCN-MMI-5545-SXXXB-(MSM8953 Vertex - Remove earpiece mode test) TCN-MMI-5533-SXXXB-(MSM8953 Vertex - Magnetometer Test) TCN-MMI-5531-SXXXB-(MSM8953 Vertex - Light Sensor Test) TCN-MMI-5556-SXXXB-(MSM8953 Vertex - Add AOV Audio Interrupt Line Test) TCN-MMI-5532-SXXXB-(MSM8953 Vertex - Hall Effect Sensor Test) TCN-MMI-5559-SXXXB-(MSM8953 Vertex - Update upper limit for battery thermistor temp check at CFC) TCN-MMI-5570-SXXXB-(MSM8953 Vertex - Update Blemish Test Limits) TCN-MMI-5529-SXXXB-(MSM8953 Vertex - Update USB PMIC Line Check Test) TCN-MMI-5536-SXXXB-(MSM8953 Vertex - update Sensor Hub Wake Line Test) TCN-MMI-5565-SXXXB-(MSM8953 Vertex - Add Camera VSYNC to Sensor Hub Line Test) TCN-MMI-5581-SXXXB-(MSM8953 Vertex - Change XO cal WCDMA Band from 1 to 5) TCN-MMI-5583-SXXXB-(MSM8953 Vertex - Add FPS Image Quality Test no contact condition) TCN-MMI-5499-SXXXB-(MSM8953 Vertex - Audio Calibration) TCN-MMI-5591-SXXXB-(MSM8953 Vertex - Update common test limits section - Audio Calibration Limits) TCN-MMI-5588-SXXXB-(MSM8953 Vertex - Update common test limits	Joe Pacella Joe Pacella Joe Pacella Joe Pacella Dave Broschka Joe Pacella Joe Pacella Bill Schmitt Dave Broschka Dave Broschka Joe Pacella Joe Pacella Dave Broschka Joe Pacella Dave Broschka Dave Broschka	April 15, 2016 April 25, 2016 April 25, 2016 April 25, 2016 May 5, 2016 May 5, 2016 May 6, 2016 May 11, 2016 May 13, 2016 May 13, 2016 May 13, 2016 May 13, 2016 May 16, 2016 May 18, 2016 May 18, 2016 May 18, 2016



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		section - Camera Tests Limits)		
		TCN-MMI-5586-SXXXB-(MSM8953 Vertex - Update LTE B41 non-signalling channels for PRC and aXGP)	Joe Pacella	May 20, 2016
		TCN-MMI-5590-SXXXB-(MSM8953 Vertex - Update common test limits section - Light Sources and Display Limits)	Dave Broschka	May 20, 2016
		TCN-MMI-5595-SXXXB-(MSM8953 Vertex - Update vibrator limits)	Bill Schmitt	May 20, 2016
		TCN-MMI-5530-SXXXB-(MSM8953 Vertex - Proximity Sensor test)	Joe Pacella	May 1, 2016
		TCN-MMI-5574-SXXXB-(MSM8953 Vertex - update limit for Display Vendor ID Check Test)	Joe Pacella	May 11, 2016
		TCN-MMI-5523-SXXXB-(MSM8953 Vertex - update Laser Focus Test)	Joe Pacella	May 20, 2016
		TCN-MMI-5598-SXXXB-(MSM8953 - Update Vertex front led cal limit)	Joe Pacella	May 20, 2016
		TCN-MMI-5477-SXXXB-(MSM8953 Vertex - Touchscreen Test)	Joe Pacella	May 23, 2016
		TCN-MMI-5605-SXXXB-(MSM8953 Vertex - Update B41 Full band channels)	Joe Pacella	May 24, 2016
		TCN-MMI-5589-SXXXB-(MSM8953 Vertex - Update common test limits section - WLAN BT RSSI GPS limits)	Dave Broschka	May 24, 2016
		TCN-MMI-5613-SXXXB-(MSM8953 Vertex - Update Led Cal Procedures)	Dave Broschka	May 26, 2016
		TCN-MMI-5619-SXXXB-(MSM8953 Vertex - Update Upper Limit for Battery Off Current in Board Test)	Joe Pacella	May 31, 2016
		TCN-MMI-5550-SXXXB-(MSM8953 Vertex - Update PA ON NO FLASH Test)	Dave Broschka	May 31, 2016
		TCN-MMI-5621-SXXXB-(MSM8953 Vertex - Update Led Calibration Limits)	Bill Schmitt	June 1, 2016
		TCN-MMI-5627-SXXXB-(MSM8953 Vertex - Update Led Cal Exp_Idx Values)	Dave Broschka	June 6, 2016
		TCN-MMI-5629-SXXXB-(MSM8953 Vertex - Update Front Led Calibration due to color of housing)	Joe Pacella	June 7, 2016
		TCN-MMI-5567-SXXXB-(MSM8953 Vertex - Add AMPS support for factory testing)	Dave Broschka	June 8, 2016
		TCN-MMI-5576-SXXXB-(MSM8953 Vertex - Update Charge percent method for battery verification at build up level)	Bill Schmitt	June 9, 2016
		TCN-MMI-5628-SXXXB-(MSM8953 Vertex - update limits for HW and Revision Identifier Test)	Joe Pacella	June 10, 2016
		TCN-MMI-5587-SXXXB-(MSM8953 Vertex - Update common test limits section - on off current)	Dave Broschka	June, 10 2016
		TCN-MMI-5634-SXXXB-(MSM8953 Vertex - Update antenna test test channels)	Joe Pacella	June 13, 2016
		TCN-MMI-5632-SXXXB-(MSM8953 Vertex - Update WLAN Current Drain)	Bill Schmitt	June 13, 2016
		TCN-MMI-5636-CXXXB-(MSM8953 Vertex - Update Auto Focus Test Limits)	Joe Pacella	June 14, 2016
		TCN-MMI-5637-CXXXB-(MSM8953 Vertex - Update UTAG Write and Read Front housing color limits)	Joe Pacella	June 16, 2016
		TCN-MMI-5640-SXXXB-(MSM8953 Vertex - Update suspend current limits)	Joe Pacella	June 17, 2016
		TCN-MMI-5612-SXXXB-(MSM8953 Vertex - Ultrasonic Test)	Joe Pacella	June 20, 2016
		TCN-MMI-5646-SXXXB-(MSM8953 Vertex - Update Internal Battery Test)	Dave Broschka	June 22, 2016
		TCN-MMI-5649-SXXXB-(MSM8953 Vertex - Update Auto Focus Test Limits)	Dave Broschka	June 22, 2016

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APPROVALS

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1.0 Introduction

1.1 Scope

This document describes the test procedures required to validate the parts and manufacturing processes used to build 4G LTE / WCDMA / GSM Android Leadership Category products. The test procedures contained herein apply specifically to the product models defined in section 1.4.

Engineers and technicians may use this document as a basis for the design of manual or automated test systems. Each section describes the product setup, test equipment setup, test commands, and testing procedure required to validate proper product operation. Test limits are listed in the tables found in section 2.0. It is assumed that the reader is familiar with the "P2K Test Command" protocol [1] and general test command usage. It is also assumed that the systems and/or equipment used for testing are calibrated to an ANSI or IEEE traceable standard of measurement.

1.2 Applicable Documents and Specifications

[1] Common Test Command Interface Specification; Latest Version

<https://drive.google.com/a/motorola.com/#folders/0BxpO5vaJFvVSS2RSZS1nQklKRVU>

[2] 3GPP TS 51.010; Latest Version

http://www.3gpp.org/ftp/Specs/archive/51_series/

[3] 3GPP UE Transmission and Reception (FDD) (Release 5); 3G TS 25.101; Latest Version

ftp://ftp.3gpp.org/Specs/archive/25_series/25.101

[4] Factory Tracking Info 12M; Latest Version

<https://drive.google.com/a/motorola.com/#folders/0BxpO5vaJFvVSNzNNNGRnT2EtLWM>

[5] 3GPP TS 34.121; Latest Version

http://www.3gpp.org/ftp/Specs/archive/34_series/

[6] 3GPP TS 36.101; Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) radio transmission and reception; Latest Version

http://www.3gpp.org/ftp/specs/archive/36_series/36.101/

[7] 3GPP TS 36.508; Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Common test environments for User Equipment (UE) conformance testing; Latest Version

http://www.3gpp.org/ftp/specs/archive/36_series/36.508/

[8] 3GPP TS 36.521; Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) conformance specification; Radio transmission and reception; Part 1: Conformance testing; Latest Version

http://www.3gpp.org/ftp/specs/archive/36_series/36.521-1/

[9] CFC 3G Recipe Specifications (71M88454W); CFC CDMA Recipe Specifications (71M88458W); CFC GSM Recipe Specifications (71M88457W)

1.3 Terms and Acronyms

Acronyms / Terms	Definition / Description
3G	Third Generation Wireless
3GPP	Third Generation Partnership Project
4G	Fourth Generation Wireless
A/D	Analog to Digital Converter
AGC	Automatic Gain Control
AOC	Automatic Output Control
AP	Application Processor
B+	Supply Voltage
BATT	Battery Voltage
BER	Bit Error Rate
CIT	Customer Interface Test
CM	Configuration Management
DAC	Digital to Analog Converter
ETSI	European Telecommunications Standards Institute
EVM	Error Vector Magnitude
FAC	Factory Code
GHz	Gigahertz
GND	Electrical Ground
GSM	Global System for Mobile Communications
HAB	High Assurance Boot
HSDPA	High Speed Downlink Packet Access
HW	Hardware
IMEI	International Mobile Equipment Identifier
LFFS	Language Format File System
LTE	Long Term Evolution
LUHN	Check digit for the IMEI
MA	Model Architecture
MMC	Multi-Media Card
MHz	Megahertz
N/A	Not Applicable
NVM	Non-Volatile Memory
OCNS	Orthogonal Channel Noise Source
PA	Power Amplifier
PCS	Personal Communications System
PN9	Pseudo-random Number 9 (bit sequence)
RF	Radio Frequency Signal
RSCP	Receive Signal Code Power
RSSI	Receive Signal Strength Indicator
RX	Receive
SW	Software
TAC	Type Approval Code
TBD	To Be Determined
TX	Transmit
UARFCN	UTRA Absolute Radio Frequency Channel Number
UTRA	Universal Terrestrial Radio Access
UE	User Equipment (module)
WCDMA	Wideband Code Division Multiple Access

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1.4 Model Descriptions

1.4.1 Vertex

Vertex is one of Motorola's Smartphones using the latest Android OS and RF technologies.

Bands/Modes:

Vertex Verizon: WCDMA 850/900/1900/2100, CDMA BC0/BC1; GSM 850/900/1800/1900; LTE Band 02/03/04/05/07/13; LTE CAT 4, HSDPA DC 42.2 Mbps (Category 24), HSUPA 5.76 Mbps, EDGE Class 12, WiFi 802.11b/g/n (2.4GHz), Bluetooth Class 4.0 LE Version 4.1 LE+EDR, aGPS (assisted)/ AGPS (autonomous)/Galileo/GLONASS/sGPS (simultaneous)/ Standalone GPS

Vertex ROW: WCDMA 850/900/1700/1900/2100, GSM 850/900/1800/1900, LTE Band 01/02/03/04/05/07/08/12/17/19/20/28/38/40/41, LTE CAT 4, HSDPA DC 42.2 Mbps (Category 24), HSUPA 5.76 Mbps, EDGE Class 12, WiFi 802.11b/g/n (2.4GHz), Bluetooth Class 4.0 LE Version 4.1 LE+EDR, aGPS (assisted)/ AGPS (autonomous)/Galileo/GLONASS/sGPS (simultaneous)/ Standalone GPS

Vertex China: WCDMA 850/900/1900/2100, CDMA BC0, GSM 850/900/1800/1900, LTE Band 01/03/07/26/38/39/40/41, TD-SCDMA B34/29, LTE CAT 4, HSDPA DC 42.2 Mbps (Category 24), HSUPA 5.76 Mbps, EDGE Class 12, WiFi 802.11b/g/n (2.4GHz), Bluetooth Class 4.0 LE Version 4.1 LE+EDR, aGPS (assisted)/ AGPS (autonomous)/Galileo/GLONASS/sGPS (simultaneous)/ Standalone GPS

Form Factor: Touch Tablet
Size: 76.4 x 156.4 x 6.99 mm
Weight: 165g
Display: 5.5" AMOLED Full HD (1920x1080)
Battery: Embedded
Imagers: 16 MP with Rear Dual CCT LED Flash, Digital Zoom, Auto Focus, P2P 5 MP FF
Video: DivX; H.263; H.264; MPEG4; WMV v9
Audio: 3 Mics, FM Radio Receive, AAC, AAC+, AAC+ Enhanced, AMR NB, AMR WB, MIDI, MP3, RA v10, RA v9, WAV, WMA v10, WMA v9
Speaker: Mono front-facing
Messaging: Google Mail, MMS, SMS
Connectivity: 3.5mm, USB 2.0 HS and USB 3.0 SS, Over the Air Sync (OTA)
Sensors: Accelerometer, Proximity Sensor, Ambient Light Sensor, Gyroscope, Hall Effect Sensor, Fingerprint sensor, Magnetometer, Sensor Hub
Add'l Features: Embedded Battery, Sign of Charge LED
SIM Type: Nano SIM
SIM Config: Single and Dual SIM, (Dual SIM Dual Standby DSDS in select markets and channels)
Micro SD: 2TB Max
Browser: Chrome
Connectors: USB-C, 3.5mm

Integrated Circuits:

Baseband Proc: MSM8953
GPU: Adreno 506
Power Mgmt: PM8953/PMI8952
RF Chipset: WTR2965/WTR2965
Internal ROM: eMMC 16GB (NA and EMEA), 32GB (LATAM, APAC, China)
Internal RAM: 3GB DDR3
Bluetooth: WCN 3660B
WLAN (802.11): WCN 3660B
GPS: Qualcomm Gen 8C
NFC: NXP NQ210

1.5 Electrical Interface

1.5.1 Battery Connector

Pin #	Name	Description
1	CELL_MINUS_CONN	Battery Sense
2	BATT_RESET_CONN	Battery Reset
3	CELL_PLUS_CONN	Battery Sense
4	BATT_ID	Battery ID
G1, G2	BATT_PLUS_CONN	Battery Voltage
G3, G4	BATT_NEG_CONN	Battery Ground

1.5.2 USB-C Connector

Pin #	Name	Description
A4,A9,B4,B9	VBUS	USB Slave device power
A2	SS_TX1_DP	Super Speed differential pair 1, TX pos
A3	SS_TX1_DM	Super Speed differential pair 1, TX neg
B2	SS_TX2_DP	Super Speed differential pair 2, TX pos
B3	SS_TX2_DM	Super Speed differential pair 2, TX neg
B11	SS_RX1_DP	Super Speed differential pair 1, RX pos
B10	SS_RX1_DM	Super Speed differential pair 1, RX neg
A11	SS_RX2_DP	Super Speed differential pair 2, RX pos
A10	SS_RX2_DM	Super Speed differential pair 2, RX neg
A8,B8	SBU1,SBU2	Sideband Use
A7,B7	D-	USB-C 2.0 D- Data Line
A6,B6	D+	USB-C 2.0 D+ Data Line
A5,B5	CC1, CC2	Configuration Channels
A1,A12,B1,B12,G1,G2,G3,G4	Power Ground	GND

1.6 General Specifications and Test Conditions

1.6.1 Frequency of Operation

LTE Band 1 FDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

1920 to 1980 MHz
Frequency (MHz) = $1920 + 0.1(\text{EARFCN} - 18000)$
where: $18000 \leq \text{UARFCN} \leq 18599$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

2110 to 2170 MHz
Frequency (MHz) = $2110 + 0.1(\text{EARFCN} - 0)$
where: $0 \leq \text{UARFCN} \leq 599$

Uplink / Downlink center frequency separation = 190 MHz

LTE Band 2 FDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

1850 to 1910 MHz
Frequency (MHz) = $1850 + 0.1(\text{EARFCN} - 18600)$
where: $18600 \leq \text{UARFCN} \leq 19199$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

1930 to 1990 MHz
Frequency (MHz) = $1930 + 0.1(\text{EARFCN} - 600)$
where: $600 \leq \text{UARFCN} \leq 1199$

Uplink / Downlink center frequency separation = 80 MHz

LTE Band 3 FDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

1710 to 1785 MHz
Frequency (MHz) = $1710 + 0.1(\text{EARFCN} - 19200)$
where: $19200 \leq \text{UARFCN} \leq 19949$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

1805 to 1880 MHz
Frequency (MHz) = $1805 + 0.1(\text{EARFCN} - 1200)$
where: $1200 \leq \text{UARFCN} \leq 1949$

Uplink / Downlink center frequency separation = 95 MHz

LTE Band 4 FDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

1710 to 1755 MHz
Frequency (MHz) = $1710 + 0.1(\text{EARFCN} - 19950)$
where: $19950 \leq \text{UARFCN} \leq 20399$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

2110 to 2155 MHz
Frequency (MHz) = $2110 + 0.1(\text{EARFCN} - 1950)$
where: $1950 \leq \text{UARFCN} \leq 2399$

Uplink / Downlink center frequency separation = 400 MHz

LTE Band 5 FDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

824 to 849 MHz
Frequency (MHz) = $824 + 0.1(\text{EARFCN} - 20400)$
where: $20400 \leq \text{UARFCN} \leq 20649$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

869 to 894 MHz
Frequency (MHz) = $869 + 0.1(\text{EARFCN} - 2400)$
where: $2400 \leq \text{UARFCN} \leq 2649$

Uplink / Downlink center frequency separation = 45 MHz

LTE Band 7 FDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

2500 to 2570 MHz
Frequency (MHz) = $2500 + 0.1(\text{EARFCN} - 20750)$
where: $20750 \leq \text{UARFCN} \leq 21449$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

2620 to 2690 MHz
Frequency (MHz) = $2620 + 0.1(\text{EARFCN} - 2750)$
where: $2750 \leq \text{UARFCN} \leq 3449$

Uplink / Downlink center frequency separation = 120 MHz

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LTE Band 8 FDD (E-UTRA Operating Band)
Transmit / Reverse / Uplink

880 to 915 MHz

Frequency (MHz) = $880 + 0.1(\text{EARFCN} - 21450)$
where: $21450 \leq \text{UARFCN} \leq 21799$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

925 to 960 MHz

Frequency (MHz) = $925 + 0.1(\text{EARFCN} - 3450)$
where: $3450 \leq \text{UARFCN} \leq 3799$

Uplink / Downlink center frequency separation = 45MHz

LTE Band 9 FDD (E-UTRA Operating Band)
Transmit / Reverse / Uplink

1749.9 to 1784.9 MHz

Frequency (MHz) = $1749.9 + 0.1(\text{EARFCN} - 21800)$

where: $21800 \leq \text{UARFCN} \leq 22149$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

1844.9 to 1879.9 MHz

Frequency (MHz) = $1844.9 + 0.1(\text{EARFCN} - 3800)$

where: $3800 \leq \text{UARFCN} \leq 4149$

Uplink / Downlink center frequency separation = 95MHz

LTE Band 11 FDD (E-UTRA Operating Band)
Transmit / Reverse / Uplink

1427.9 to 1447.9 MHz

Frequency (MHz) = $1427.9 + 0.1(\text{EARFCN} - 22750)$

where: $22750 \leq \text{UARFCN} \leq 22949$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

1475.9 to 1495.9 MHz

Frequency (MHz) = $1475.9 + 0.1(\text{EARFCN} - 4750)$

where: $4750 \leq \text{UARFCN} \leq 4949$

Uplink / Downlink center frequency separation = 48MHz

LTE Band 12 FDD (E-UTRA Operating Band)
Transmit / Reverse / Uplink

699 to 716 MHz

Frequency (MHz) = $699 + 0.1(\text{EARFCN} - 23010)$
where: $23010 \leq \text{UARFCN} \leq 23179$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

729 to 746 MHz

Frequency (MHz) = $729 + 0.1(\text{EARFCN} - 5010)$
where: $5010 \leq \text{UARFCN} \leq 5179$

Uplink / Downlink center frequency separation = 30MHz

LTE Band 13 FDD (E-UTRA Operating Band)
Transmit / Reverse / Uplink

777 to 787 MHz

Frequency (MHz) = $777 + 0.1(\text{EARFCN} - 23180)$
where: $23180 \leq \text{UARFCN} \leq 23279$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

746 to 756 MHz

Frequency (MHz) = $746 + 0.1(\text{EARFCN} - 5180)$
where: $5180 \leq \text{UARFCN} \leq 5279$

Uplink / Downlink center frequency separation = -31 MHz

LTE Band 17 FDD (E-UTRA Operating Band)
Transmit / Reverse / Uplink

704 to 716 MHz

Frequency (MHz) = $704 + 0.1(\text{EARFCN} - 23730)$
where: $23730 \leq \text{UARFCN} \leq 23849$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

734 to 746 MHz

Frequency (MHz) = $734 + 0.1(\text{EARFCN} - 5730)$
where: $5730 \leq \text{UARFCN} \leq 5849$

Uplink / Downlink center frequency separation = 30 MHz



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LTE Band 19 FDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

830 to 845 MHz

Frequency (MHz) = $830 + 0.1(\text{EARFCN} - 24000)$
where: $24000 \leq \text{UARFCN} \leq 24149$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

875 to 890 MHz

Frequency (MHz) = $875 + 0.1(\text{EARFCN} - 6000)$
where: $6000 \leq \text{UARFCN} \leq 6149$

Uplink / Downlink center frequency separation = 45 MHz

LTE Band 20 FDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

832 to 862 MHz

Frequency (MHz) = $832 + 0.1(\text{EARFCN} - 24150)$
where: $24150 \leq \text{UARFCN} \leq 24449$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

791 to 821 MHz

Frequency (MHz) = $791 + 0.1(\text{EARFCN} - 6150)$
where: $6150 \leq \text{UARFCN} \leq 6449$

Uplink / Downlink center frequency separation = - 41 MHz

LTE Band 25 FDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

1850 to 1915 MHz

Frequency (MHz) = $1850 + 0.1(\text{EARFCN} - 26040)$
where: $26040 \leq \text{UARFCN} \leq 26689$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

1930 to 1995 MHz

Frequency (MHz) = $1930 + 0.1(\text{EARFCN} - 8040)$
where: $8040 \leq \text{UARFCN} \leq 8689$

Uplink / Downlink center frequency separation = 80 MHz

LTE Band 26 FDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

814 to 849 MHz

Frequency (MHz) = $814 + 0.1(\text{EARFCN} - 26690)$
where: $26690 \leq \text{UARFCN} \leq 27039$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

859 to 894 MHz

Frequency (MHz) = $859 + 0.1(\text{EARFCN} - 8690)$
where: $8690 \leq \text{UARFCN} \leq 9039$

Uplink / Downlink center frequency separation = 45 MHz

LTE Band 28 FDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

703 to 748 MHz

Frequency (MHz) = $703 + 0.1(\text{EARFCN} - 27210)$
where: $27210 \leq \text{UARFCN} \leq 27659$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Receive / Forward / Downlink

758 to 803 MHz

Frequency (MHz) = $758 + 0.1(\text{EARFCN} - 9210)$
where: $9210 \leq \text{UARFCN} \leq 9659$

Uplink / Downlink center frequency separation = 55 MHz

LTE Band 29 FDD (E-UTRA Operating Band)

Receive / Forward / Downlink

717 to 728 MHz

Frequency (MHz) = $717 + 0.1(\text{EARFCN} - 9660)$
where: $9660 \leq \text{UARFCN} \leq 9769$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 kHz

Uplink / Downlink center frequency separation = N/A



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LTE Band 34 TDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

2010 to 2025 MHz
Frequency (MHz) = $2010 + 0.1(\text{EARFCN} - 36200)$
where: $36200 \leq \text{UARFCN} \leq 36349$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 KHz

Receive / Forward / Downlink

2010 to 2025 MHz
Frequency (MHz) = $2010 + 0.1(\text{EARFCN} - 36200)$
where: $36200 \leq \text{UARFCN} \leq 36349$

Uplink / Downlink center frequency separation = 0 MHz

LTE Band 38 TDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

2570 to 2620 MHz
Frequency (MHz) = $2570 + 0.1(\text{EARFCN} - 37750)$
where: $37750 \leq \text{UARFCN} \leq 38249$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 KHz

Receive / Forward / Downlink

2570 to 2620 MHz
Frequency (MHz) = $2570 + 0.1(\text{EARFCN} - 37750)$
where: $37750 \leq \text{UARFCN} \leq 38249$

Uplink / Downlink center frequency separation = 0 MHz

LTE Band 39 TDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

1880 to 1920 MHz
Frequency (MHz) = $1880 + 0.1(\text{EARFCN} - 38250)$
where: $38250 \leq \text{UARFCN} \leq 38649$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 KHz

Receive / Forward / Downlink

1880 to 1920 MHz
Frequency (MHz) = $1880 + 0.1(\text{EARFCN} - 38250)$
where: $38250 \leq \text{UARFCN} \leq 38649$

Uplink / Downlink center frequency separation = 0 MHz

LTE Band 40 TDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

2300 to 2400 MHz
Frequency (MHz) = $2300 + 0.1(\text{EARFCN} - 38650)$
where: $38650 \leq \text{UARFCN} \leq 39649$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 KHz

Receive / Forward / Downlink

2300 to 2400 MHz
Frequency (MHz) = $2300 + 0.1(\text{EARFCN} - 38650)$
where: $38650 \leq \text{UARFCN} \leq 39649$

Uplink / Downlink center frequency separation = 0 MHz

LTE Band 41 TDD (E-UTRA Operating Band)

Transmit / Reverse / Uplink

2496 to 2690 MHz
Frequency (MHz) = $2496 + 0.1(\text{EARFCN} - 39650)$
where: $39650 \leq \text{UARFCN} \leq 41590$

Channel spacing = $(\text{BW}_{\text{Channel (1)}} + \text{BW}_{\text{Channel (2)}})/2$
Channel raster = 100 KHz

Receive / Forward / Downlink

2496 to 2690 MHz
Frequency (MHz) = $2496 + 0.1(\text{EARFCN} - 39650)$
where: $39650 \leq \text{UARFCN} \leq 41590$

Uplink / Downlink center frequency separation = 0 MHz

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1920 to 1980 MHz
Frequency (MHz) = $\text{UARFCN} \div 5$
where: $9612 \leq \text{UARFCN} \leq 9888$

Channel spacing = 5 MHz
Channel raster = 200 kHz

Receive / Forward / Downlink

2110 to 2170 MHz
Frequency (MHz) = $\text{UARFCN} \div 5$
where: $10562 \leq \text{UARFCN} \leq 10838$

Uplink / Downlink frequency separation = 190 MHz

WCDMA 1900 (Band 2)Transmit / Reverse / Uplink

1850 to 1910 MHz
Frequency (MHz) = $\text{UARFCN} \div 5$
where: $9262 \leq \text{UARFCN} \leq 9538$

Region 2 additional channels:

Frequency (MHz) = $\text{UARFCN} \div 5 + 1850.1$
Where: $\text{UARFCN}=12, 37, 62, 87, 112, 137, 162, 187, 212, 237, 262, 287$

Channel spacing = 5 MHz
Channel raster = 200 kHz

Receive / Forward / Downlink

1930 to 1990 MHz
Frequency (MHz) = $\text{UARFCN} \div 5$
where: $9662 \leq \text{UARFCN} \leq 9938$

Region 2 additional channels:

Frequency (MHz) = $\text{UARFCN} \div 5 + 1850.1$
Where: $\text{UARFCN}=412, 437, 462, 487, 512, 537, 562, 587, 612, 637, 662, 687$

Uplink / Downlink frequency separation = 80 MHz

WCDMA 1800 (Band 3)Transmit / Reverse / Uplink

1710 to 1785 MHz
Frequency (MHz) = $\text{UARFCN} \div 5$
where: $937 \leq \text{UARFCN} \leq 1288$

Channel spacing = 5 MHz
Channel raster = 200 kHz

Receive / Forward / Downlink

1805 to 1880 MHz
Frequency (MHz) = $\text{UARFCN} \div 5$
where: $1162 \leq \text{UARFCN} \leq 1513$

Uplink / Downlink frequency separation = 95 MHz

WCDMA 1700 (Band 4)Transmit / Reverse / Uplink

1710 to 1755 MHz
Frequency (MHz) = $\text{UARFCN} \div 5 + 1450$
where: $1312 \leq \text{UARFCN} \leq 1513$

Region 2 additional channels:

Frequency (MHz) = $\text{UARFCN} \div 5 + 1380.1$
Where: $\text{UARFCN}=1662, 1687, 1712, 1737, 1762, 1787, 1812, 1837, 1862$

Channel spacing = 5 MHz
Channel raster = 200 kHz

Receive / Forward / Downlink

2110 to 2155 MHz
Frequency (MHz) = $\text{UARFCN} \div 5 + 1805$
where: $1537 \leq \text{UARFCN} \leq 1738$

Region 2 additional channels:

Frequency (MHz) = $\text{UARFCN} \div 5 + 1735.1$
Where: $\text{UARFCN}=1887, 1912, 1937, 1962, 1987, 2012, 2037, 2062, 2087$

Uplink / Downlink frequency separation = 400 MHz

WCDMA 850 (Band 5 [N. America])Transmit / Reverse / Uplink

824 to 849 MHz
Frequency (MHz) = $\text{UARFCN} \div 5$
where: $4132 \leq \text{UARFCN} \leq 4233$

Region 5 additional channels:

Frequency (MHz) = $\text{UARFCN} \div 5 + 670.1$
Where: $\text{UARFCN}=782, 787, 807, 812, 837, 862$

Channel spacing = 5 MHz
Channel raster = 200 kHz

Receive / Forward / Downlink

869 to 894 MHz
Frequency (MHz) = $\text{UARFCN} \div 5$
where: $4357 \leq \text{UARFCN} \leq 4458$

Region 5 additional channels:

Frequency (MHz) = $\text{UARFCN} \div 5 + 670.1$
Where: $\text{UARFCN}=1007, 1012, 1032, 1037, 1062, 1087$

Uplink / Downlink frequency separation = 45 MHz



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WCDMA 850 (Band 6 [Japan])

Transmit / Reverse / Uplink

830 to 840 MHz

Frequency (MHz) = $\text{UARFCN} \div 5$
where: $4162 \leq \text{UARFCN} \leq 4188$

Region 6 additional channels:

Frequency (MHz) = $\text{UARFCN} \div 5 + 670.1$
Where: $\text{UARFCN} = 812, 837$

Channel spacing = 5 MHz
Channel raster = 200 kHz

Receive / Forward / Downlink

875 to 885 MHz

Frequency (MHz) = $\text{UARFCN} \div 5$
where: $4387 \leq \text{UARFCN} \leq 4413$

Region 6 additional channels:

Frequency (MHz) = $\text{UARFCN} \div 5 + 670.1$
Where: $\text{UARFCN} = 1037, 1062$

Uplink / Downlink frequency separation = 45 MHz

WCDMA 900 (Band 8 [Europe])

Transmit / Reverse / Uplink

880 to 915 MHz

Frequency (MHz) = $\text{UARFCN} \div 5 + 340$
Where: $2712 \leq \text{UARFCN} \leq 2863$

Channel spacing = 5 MHz
Channel raster = 200 kHz

Receive / Forward / Downlink

925 to 960 MHz

Frequency (MHz) = $\text{UARFCN} \div 5 + 340$
where: $2937 \leq \text{UARFCN} \leq 3088$

Uplink / Downlink frequency separation = 45 MHz

For UMTS/TD-LTE:

WCDMA 1500 (Band 11 [Japan])

Transmit / Reverse / Uplink

1427.9 to 1447.9 MHz

Frequency (MHz) = $\text{UARFCN} \div 5 + 733$
Where: $3487 \leq \text{UARFCN} \leq 3562$

Channel spacing = 5 MHz
Channel raster = 200 kHz

Receive / Forward / downlink

1475.9 to 1495.9 MHz

Frequency (MHz) = $\text{UARFCN} \div 5 + 736$
where: $3712 \leq \text{UARFCN} \leq 3787$

Uplink / Downlink frequency separation = 48 MHz

For all other products:

WCDMA 1500 (Band 11 [Japan])

Transmit / Reverse / Uplink

1427.9 to 1452.9 MHz

Frequency (MHz) = $\text{UARFCN} \div 5 + 733$
Where: $3487 \leq \text{UARFCN} \leq 3587$

Channel spacing = 5 MHz
Channel raster = 200 kHz

Receive / Forward / downlink

1475.9 to 1500.9 MHz

Frequency (MHz) = $\text{UARFCN} \div 5 + 736$
where: $3712 \leq \text{UARFCN} \leq 3812$

Uplink / Downlink frequency separation = 48 MHz

WCDMA 800 (Band 19 [Japan])

Transmit / Reverse / Uplink

1427.9 to 1452.9 MHz

Frequency (MHz) = $\text{UARFCN} \div 5 + 733$
Where: $3487 \leq \text{UARFCN} \leq 3587$

Channel spacing = 5 MHz
Channel raster = 200 kHz

Receive / Forward / downlink

1475.9 to 1500.9 MHz

Frequency (MHz) = $\text{UARFCN} \div 5 + 736$
where: $3712 \leq \text{UARFCN} \leq 3812$

Uplink / Downlink frequency separation = 48 MHz

Title: Factory Test Specification for LTE / 3GSM / CDMA / TD-SCDMA Handsets

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TD-SCDMA/TD-LTE Band 34 (A band)
Transmit / Reverse / Uplink

2010 to 2025 MHz

Frequency (MHz) = $\text{UARFCN} \div 5$

where: $10054 \leq \text{UARFCN} \leq 10121$

Channel spacing = 1.6MHz

Channel raster = 200 kHz

Receive / Forward / Downlink

2010 to 2025 MHz

Frequency (MHz) = $\text{UARFCN} \div 5$

where: $10054 \leq \text{UARFCN} \leq 10121$

Uplink / Downlink frequency separation = 0 MHz

TDS-CDMA Band 39 (F Band)
Transmit / Reverse / Uplink

1880 to 1920 MHz

Frequency (MHz) = $\text{UARFCN} \div 5$

where: $9404 \leq \text{UARFCN} \leq 9596$

Channel spacing = 1.6MHz

Channel raster = 200 kHz

Receive / Forward / Downlink

1880 to 1920 MHz

Frequency (MHz) = $\text{UARFCN} \div 5$

where: $9404 \leq \text{UARFCN} \leq 9596$

Uplink / Downlink frequency separation = 0 MHz

GSM 850
Transmit / Reverse / Uplink

824 to 849 MHz

Frequency (MHz) = $824.2 + 0.2 \times (n - 128)$

where: $128 \leq n \leq 251$

Channel spacing = 200 kHz

Channel raster = 200 kHz

Receive / Forward / Downlink

869 to 894 MHz

Frequency (MHz) = $869.2 + 0.2 \times (n - 128)$

where: $128 \leq n \leq 251$

Uplink / Downlink frequency separation = 45 MHz

E-GSM 900
Transmit / Reverse / Uplink

880 to 915 MHz

Frequency (MHz) = $890 + 0.2 \times n$

where: $0 \leq n \leq 124$

Frequency (MHz) = $890 + 0.2 \times (n - 1024)$

where: $975 \leq n \leq 1023$

Channel spacing = 200 kHz

Channel raster = 200 kHz

Receive / Forward / Downlink

925 to 960 MHz

Frequency (MHz) = $935 + 0.2 \times n$

where: $0 \leq n \leq 124$

Frequency (MHz) = $935 + 0.2 \times (n - 1024)$

where: $975 \leq n \leq 1023$

Uplink / Downlink frequency separation = 45 MHz

DCS 1800
Transmit / Reverse / Uplink

1710 to 1785 MHz

Frequency (MHz) = $1710.2 + 0.2 \times (n - 512)$

where: $512 \leq n \leq 885$

Channel spacing = 200 kHz

Channel raster = 200 kHz

Receive / Forward / Downlink

1805 to 1880 MHz

Frequency (MHz) = $1805.2 + 0.2 \times (n - 512)$

where: $512 \leq n \leq 885$

Uplink / Downlink frequency separation = 95 MHz

PCS 1900
Transmit / Reverse / Uplink

1850 to 1910 MHz

Frequency (MHz) = $1850.2 + 0.2 \times (n - 512)$

where: $512 \leq n \leq 810$

Channel spacing = 200 kHz

Channel raster = 200 kHz

Receive / Forward / Downlink

1930 to 1990 MHz

Frequency (MHz) = $1930.2 + 0.2 \times (n - 512)$

where: $512 \leq n \leq 810$

Uplink / Downlink frequency separation = 80 MHz



Material or Methods Specification

No. 12014512001

Title: **Factory Test Specification for LTE / 3GSM / CDMA / TD-SCDMA Handsets**

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CDMA BC0 800

Transmit / Reverse / Uplink

824.7 to 825 MHz

$f_{TX} = 0.03 * (\text{Channel} - 1023) + 825 \text{ (MHz)}$

where: $1013 \leq n \leq 1023$

825.03 to 848.31 MHz

$f_{TX} = 0.03 * \text{Channel} + 825 \text{ (MHz)}$

where: $1 \leq n \leq 777$

Channel spacing = 30 kHz

Receive / Forward / Downlink

869.7 to 870 MHz

$f_{RX} = 0.03 * (\text{Channel} - 1023) + 870 \text{ (MHz)}$

where: $1013 \leq n \leq 1023$

870.03 to 893.31 MHz

$f_{RX} = 0.03 * \text{Channel} + 870 \text{ (MHz)}$

where: $1 \leq n \leq 777$

Uplink / Downlink frequency separation = 45 MHz

CDMA BC1 1900

Transmit / Reverse / Uplink

1851.25 to 1908.75 MHz

$f_{TX} = 0.05 * \text{Channel} + 1850 \text{ (MHz)}$

where: $25 \leq n \leq 1175$

Channel spacing = 50 kHz

Receive / Forward / Downlink

1931.25 to 1988.75 MHz

$f_{RX} = 0.05 * \text{Channel} + 1930 \text{ (MHz)}$

where: $25 \leq n \leq 1175$

Uplink / Downlink frequency separation = 80 MHz

CDMA BC10 Secondary 800

Transmit / Reverse / Uplink

806 to 824 MHz

Frequency (MHz) = $0.025 * N + 806.000$

where: $0 \leq N \leq 719$

Channel spacing = 25 kHz

Transmit / Reverse / Uplink

896 to 901 MHz

Frequency (MHz) = $0.025 * (N - 720) + 896.000$

where: $720 \leq N \leq 919$

Channel spacing = 25 kHz

Receive / Forward / Downlink

851 to 869 MHz

Frequency (MHz) = $0.025 * N + 851.000$

where: $1 \leq N \leq 1299$

Uplink / Downlink frequency separation = 55 MHz

Receive / Forward / Downlink

935 to 940 MHz

Frequency (MHz) = $0.025 * (N - 720) + 935.000$

where: $720 \leq N \leq 919$

Uplink / Downlink frequency separation = 39 MHz

Bluetooth

Europe & USA

2400 to 2483.5 MHz

Frequency (MHz) = $2402 + n$

where: $0 \leq n \leq 78$

Spain

2445 to 2475 MHz

Frequency (MHz) = $2449 + n$

where: $0 \leq n \leq 22$

Channel spacing = 1MHz

Japan

2471 to 2497MHz

Frequency (MHz) = $2473 + n$

where: $0 \leq n \leq 22$

France

2446.5 to 2483.5 MHz

Frequency (MHz) = $2454 + n$

where: $0 \leq n \leq 22$

1.6.2 Low, Mid, and High Channels and Frequencies

LTE Band 1 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	5	18025	1922.5	25	2112.5
	10	18050	1925	50	2115
	15	18075	1927.5	75	2117.5
	20	18100	1930	100	2120
Mid Range	5/10/15/20	18300	1950	300	2140
High Range	5	18575	1977.5	575	2167.5
	10	18550	1975	550	2165
	15	18525	1972.5	525	2162.5
	20	18500	1970	500	2160

LTE Band 2 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	1.4	18607	1850.7	607	1930.7
	3	18615	1851.5	615	1931.5
	5	18625	1852.5	625	1932.5
	10	18650	1855	650	1935
	15	18675	1857.5	675	1937.5
	20	18700	1860	700	1940
Mid Range	1.4/3/5/10 15 /20	18900	1880	900	1960
High Range	1.4	19193	1909.3	1193	1989.3
	3	19185	1908.5	1185	1988.5
	5	19175	1907.5	1175	1987.5
	10	19150	1905	1150	1985
	15	19125	1902.5	1125	1982.5
	20	19100	1900	1100	1980

NOTE 1: Bandwidth for which a relaxation of the specified UE receiver sensitivity requirement (TS 36.101 [27] Clause 7.3) is allowed.

LTE Band 3 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	1.4	19207	1710.7	1207	1805.7
	3	19215	1711.5	1215	1806.5
	5	19225	1712.5	1225	1807.5
	10	19250	1715	1250	1810
	15	19275	1717.5	1275	1812.5
	20	19300	1720	1300	1815
Mid Range	1.4/3/5/10 15 /20	19575	1747.5	1575	1842.5
High Range	1.4	19943	1784.3	1943	1879.3
	3	19935	1783.5	1935	1878.5
	5	19925	1782.5	1925	1877.5
	10	19900	1780	1900	1875
	15	19875	1777.5	1875	1872.5
	20	19850	1775	1850	1870

LTE Band 4 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	1.4	19957	1710.7	1957	2110.7
	3	19965	1711.5	1965	2111.5
	5	19975	1712.5	1975	2112.5
	10	20000	1715	2000	2115
	15	20025	1717.5	2025	2117.5
	20	20050	1720	2050	2120
Mid Range	1.4/3/5/10/15/20	20175	1732.5	2175	2132.5
High Range	1.4	20393	1754.3	2393	2154.3
	3	20385	1753.5	2385	2153.5
	5	20375	1752.5	2375	2152.5
	10	20350	1750	2350	2150
	15	20325	1747.5	2325	2147.5
	20	20300	1745	2300	2145

LTE Band 5 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	1.4	20407	824.7	2407	869.7
	3	20415	825.5	2415	870.5
	5	20425	826.5	2425	871.5
	10	20450	829	2450	874
Mid Range	1.4/3/5/10	20525	836.5	2525	881.5
High Range	1.4	20643	848.3	2643	893.3
	3	20635	847.5	2635	892.5
	5	20625	846.5	2625	891.5
	10	20600	844	2600	889

LTE Band 7 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	5	20775	2502.5	2775	2622.5
	10	20800	2505	2800	2625
	15	20825	2507.5	2825	2627.5
	20	20850	2510	2850	2630
Mid Range	5/10/15/20	21100	2535	3100	2655
High Range	5	21425	2567.5	3425	2687.5
	10	21400	2565	3400	2685
	15	21375	2562.5	3375	2682.5
	20	21350	2560	3350	2680

LTE Band 8 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	1.4	21457	880.7	3457	925.7
	3	21465	881.5	3465	926.5
	5	21475	882.5	3475	927.5
	10	21500	885	3500	930
Mid Range	1.4/3/5 10	21625	897.5	3625	942.5
High Range	1.4	21793	914.3	3793	959.3
	3	21785	913.5	3785	958.5
	5	21775	912.5	3775	957.5
	10	21750	910	3750	955

LTE Band 9 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	5	21825	1752.4	3825	1847.4
	10	21850	1754.9	3850	1849.9
	15	21875	1757.4	3875	1852.4
	20	21900	1759.9	3900	1854.9
Mid Range	5/10 15 /20	21975	1767.4	3975	1862.4
High Range	5	22125	1782.4	4125	1877.4
	10	22100	1779.9	4100	1874.9
	15	22075	1777.4	4075	1872.4
	20	22050	1774.9	4050	1869.9

LTE Band 11 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	5	22775	1430.4	4775	1478.4
	10	22800	1432.9	4800	1480.9
Mid Range	5/10	22850	1437.9	4850	1485.9
High Range	5	22925	1445.4	4925	1493.4
	10	22900	1442.9	4900	1490.9

LTE Band 12 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	1.4	23017	699.7	5017	729.7
	3	23025	700.5	5025	730.5
	5	23035	701.5	5035	731.5
	10	23060	704	5060	734
Mid Range	1.4/3 5 /10	23095	707.5	5095	737.5
High Range	1.4	23173	715.3	5173	745.3
	3	23165	714.5	5165	744.5
	5	23155	713.5	5155	743.5
	10	23130	711	5130	741

LTE Band 13 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	5	23205	779.5	5205	748.5
	10	23230	782	5230	751
Mid Range	5 /10	23230	782	5230	751
High Range	5	23255	784.5	5255	753.5
	10	23230	782	5230	751

LTE Band 17 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	5	23755	706.5	5755	736.5
	10	23780	709	5780	739
Mid Range	5 /10	23790	710	5790	740
High Range	5	23825	713.5	5825	743.5
	10	23800	711	5800	741

LTE Band 19 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	5	24025	832.5	6025	877.5
	10	24050	835	6050	880
	15	24075	837.5	6075	882.5
Mid Range	5/10 /15	24075	837.5	6075	882.5
High range	5	24125	842.5	6125	887.5
	10	24100	840	6100	885
	15	24075	837.5	6075	882.5

LTE Band 20 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	5	24175	834.5	6175	793.5
	10	24200	837	6200	796
	15	24225	839.5	6225	798.5
	20	24250	842	6250	801
Mid Range	5/10/15/20	24300	847	6300	806
High range	5	24425	859.5	6425	818.5
	10	24400	857	6400	816
	15	24375	854.5	6375	813.5
	20	24350	852	6350	811

LTE Band 25 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	1.4	26047	1850.7	8047	1930.7
	3	26055	1851.5	8055	1931.5
	5	26065	1852.5	8065	1932.5
	10	26090	1855	8090	1935
	15	26115	1857.5	8115	1937.5
	20	26140	1860	8140	1940
Mid Range	1.4/3/5/10 15 /20	26365	1882.5	8365	1962.5
High Range	1.4	26683	1914.3	8683	1994.3
	3	26675	1913.5	8675	1993.5
	5	26665	1912.5	8665	1992.5
	10	26640	1910	8640	1990
	15	26615	1907.5	8615	1987.5
	20	26590	1905	8590	1985

LTE Band 26 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	1.4	26697	814.7	8697	859.7
	3	26705	815.5	8705	860.5
	5	26715	816.5	8715	861.5
	10	26750	820	8750	865
	15	26775	822.5	8775	867.5
Mid Range	1.4/3/5/10 15	26865	831.5	8865	876.5
High Range	1.4	27033	848.3	9033	893.3
	3	27025	847.5	9025	892.5
	5	27015	846.5	9015	891.5
	10	26990	844	8990	889
	15	26965	841.5	8965	886.5

LTE Band 28A FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	3	27225	704.5	9225	759.5
	5	27235	705.5	9235	760.5
	10	27260	708	9260	763
	15	27285	710.5	9285	765.5
	20	27310	713	9310	768
Mid Range	3/5/10/15/20	27360	718	9360	773
High Range	3	27495	731.5	9495	786.5
	5	27485	730.5	9485	785.5
	10	27460	728	9460	783
	15	27435	725.5	9435	780.5
	20	27410	723	9410	778

LTE Band 28B FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	5	27385	720.5	9385	775.5
	10	27410	723	9410	778
	15	27435	725.5	9435	780.5
	20	27460	728	9460	783
Mid Range	3/5/10/15/20	27510	733	9510	788
High Range	5	27635	745.5	9635	800.5
	10	27610	743	9610	798
	15	27585	740.5	9585	795.5
	20	27560	738	9560	793

LTE Band 29 FDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	3	NA	NA	9675	718.5
	5	NA	NA	9685	719.5
	10	NA	NA	9710	722
Mid Range	3 5 /10	NA	NA	9715	722.5
High Range	3	NA	NA	9755	726.5
	5	NA	NA	9745	725.5
	10	NA	NA	9720	723

NOTE 1: Restricted to E-UTRA operation when carrier aggregation is configured. The downlink operating band is paired with the uplink operating band (external) of the carrier aggregation configuration that is supporting the configured Pcell.

LTE Band 34 TDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	EARFCN	Frequency (UL and DL) [MHz]
Low Range	5	36225	2012.5
	10	36250	2015
	15	36275	2017.5
Mid Range	5/10/15	36275	2017.5
High Range	5	36325	2022.5
	10	36300	2020
	15	36275	2017.5

LTE Band 38 TDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	EARFCN	Frequency (UL and DL) [MHz]
Low Range	5	37775	2572.5
	10	37800	2575
	15	37825	2577.5
	20	37850	2580
Mid Range	5/10/15/20	38000	2595
High Range	5	38225	2617.5
	10	38200	2615
	15	38175	2612.5
	20	38150	2610

LTE Band 39 TDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	EARFCN	Frequency (UL and DL) [MHz]
Low Range	5	38275	1882.5
	10	38300	1885
	15	38325	1887.5
	20	38350	1890
Mid Range	5/10/15/20	38450	1900
High Range	5	38625	1917.5
	10	38600	1915
	15	38575	1912.5
	20	38550	1910

LTE Band 40 TDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	EARFCN	Frequency (UL and DL) [MHz]
Low Range	5	38675	2302.5
	10	38700	2305
	15	38725	2307.5
	20	38750	2310
Mid Range	5/10/15/20	39150	2350
High Range	5	39625	2397.5
	10	39600	2395
	15	39575	2392.5
	20	39550	2390



Material or Methods Specification

No. 12014512001

Title: **Factory Test Specification for LTE / 3GSM / CDMA / TD-SCDMA Handsets**

Revision Date: **22-June-2016**

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LTE Band 41 TDD (E-UTRA Operating Band)

Test Frequency ID	Bandwidth [MHz]	EARFCN	Frequency (UL and DL) [MHz]
Low Range	5	39675	2498.5
	10	39700	2501
	15	39725	2503.5
	20	39750	2506
Mid Range	5/10/15/20	40620	2593
High Range	5	41565	2687.5
	10	41540	2685
	15	41515	2682.5
	20	41490	2680

WCDMA 2100 (Band 1)

TX Band UARFCN		Frequency (MHz)
Low Channel	9612	1922.4
Mid Channel	9750	1950
High Channel	9888	1977.6
RX Band UARFCN		Frequency (MHz)
Low Channel	10562	2112.4
Mid Channel	10700	2140
High Channel	10838	2167.6

WCDMA 1900 (Band 2)

TX Band UARFCN		Frequency (MHz)
Low Channel	9262	1852.4
Mid Channel	9400	1880
High Channel	9538	1907.6
RX Band UARFCN		Frequency (MHz)
Low Channel	9662	1932.4
Mid Channel	9800	1960
High Channel	9938	1987.6

WCDMA 1800 (Band 3)

TX Band UARFCN		Frequency (MHz)
Low Channel	937	1712.4
Mid Channel	1112	1747.4
High Channel	1288	1782.6
RX Band UARFCN		Frequency (MHz)
Low Channel	1162	1807.4
Mid Channel	1337	1842.4
High Channel	1513	1877.6

WCDMA 1700 (Band 4)

TX Band UARFCN		Frequency (MHz)
Low Channel	1312	1712.4
Mid Channel	1413	1732.5
High Channel	1513	1752.6
RX Band UARFCN		Frequency (MHz)
Low Channel	1537	2112.4
Mid Channel	1638	2132.5
High Channel	1738	2152.6

WCDMA 850 (Band 5 [N. America])

<u>TX Band</u>	<u>UARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	4132	826.4
Mid Channel	4180	836
High Channel	4233	846.6
<u>RX Band</u>	<u>UARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	4357	871.4
Mid Channel	4405	881
High Channel	4458	891.6

WCDMA 850 (Band 6 [Japan])

<u>TX Band</u>	<u>UARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	4162	832.4
Mid Channel	4175	835
High Channel	4188	837.6
<u>RX Band</u>	<u>UARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	4387	877.4
Mid Channel	4400	880
High Channel	4413	882.6

WCDMA 900 (Band 8 [Europe])

<u>TX Band</u>	<u>UARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	2712	882.4
Mid Channel	2787	897.4
High Channel	2863	912.6
<u>RX Band</u>	<u>UARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	2937	927.4
Mid Channel	3012	942.4
High Channel	3088	957.6

For UMTS/TD-LTE:
WCDMA 1500 (Band 11 [Japan])

<u>TX Band</u>	<u>UARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	3487	1430.4
Mid Channel	3525	1438.0
High Channel	3562	1445.4
<u>RX Band</u>	<u>UARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	3712	1478.4
Mid Channel	3750	1486.0
High Channel	3787	1493.4

For all other products:
WCDMA 1500 (Band 11 [Japan])

<u>TX Band</u>	<u>UARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	3487	1430.4
Mid Channel	3537	1440.4
High Channel	3587	1450.4

<u>RX Band</u>	<u>UARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	3712	1478.4
Mid Channel	3762	1488.4
High Channel	3812	1498.4

WCDMA 800 (Band 19 [Japan])

<u>TX Band</u>	<u>UARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	3487	1430.4
Mid Channel	3537	1440.4
High Channel	3587	1450.4

<u>RX Band</u>	<u>UARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	3712	1478.4
Mid Channel	3762	1488.4
High Channel	3812	1498.4

GSM 850

<u>TX Band</u>	<u>ARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	128	824.2
Mid Channel	194	837.4
High Channel	251	848.8

<u>RX Band</u>	<u>ARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	128	869.2
Mid Channel	194	882.4
High Channel	251	893.8

E-GSM 900

<u>TX Band</u>	<u>ARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	975	880.2
Mid Channel	38	897.6
High Channel	124	914.8

<u>RX Band</u>	<u>ARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	975	925.2
Mid Channel	38	942.6
High Channel	124	959.8

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DCS 1800

<u>TX Band</u>	<u>ARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	512	1710.2
Mid Channel	699	1747.6
High Channel	885	1784.8

<u>RX Band</u>	<u>ARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	512	1805.2
Mid Channel	699	1842.6
High Channel	885	1879.8

PCS 1900

<u>TX Band</u>	<u>ARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	512	1850.2
Mid Channel	661	1880
High Channel	810	1909.8

<u>RX Band</u>	<u>ARFCN</u>	<u>Frequency (MHz)</u>
Low Channel	512	1930.2
Mid Channel	661	1960
High Channel	810	1989.8

CDMA 800 (BC0 - Cellular)
Ref Channel
306

Band Channels
1018, 46, 98, 150, 202, 254, 306, 358, 410, 462, 514, 566, 618, 670, 722, 774

CDMA 1900 (BC1 - PCS)
Ref Channel
563

Band Channels
25, 113, 188, 263, 338, 413, 488, 563, 638, 713, 788, 863, 938, 1013, 1088, 1175

CDMA Secondary 800 (BC10)
Ref Channel
476

Band Channels
476, 526, 576, 626, 680

1.6.3 Test Channels

Call test channels for GSM & EDGE call test are defined in bands. Any channel in the appropriate band is acceptable for call test. This is done to allow staggering of channels to minimize interference between stations, as well as local interference sources (local live networks).

These channels are selected based on BABT 340, Annex 7, Note 6.

1.6.3.1 GSM850 Channels to be tested:

Band	Channels	TX Frequency (MHz)	Rx Frequency (MHz)
Reference Channel	189	836.4 MHz	881.4 MHz
Band 1	128-139	824.2 – 826.4 MHz	869.2 – 871.4 MHz
Band 2	183-194	835.2 – 837.4 MHz	880.2 – 882.4 MHz
Band 3	240-251	846.6 – 848.8 MHz	891.6 – 893.8 MHz

1.6.3.2 GSM900 EGSM Channels to be tested:

Band	Channels	TX Frequency (MHz)	Rx Frequency (MHz)
Reference Channel	37	897.4	942.4
Band 1	975–986	880.2 – 882.4	925.2 – 927.4
Band 2	30–41	896.0 – 898.2	941.0 – 943.2
Band 3	113-124	912.6 – 914.8	957.6 – 959.8

1.6.3.3 GSM1800 Channels to be tested:

Band	Channels	TX Frequency (MHz)	Rx Frequency (MHz)
Reference Channel	700	1747.8	1842.8
Band 1	512-523	1710.2 – 1712.4	1805.2 – 1807.4
Band 2	693-704	1746.4 – 1748.6	1841.4 – 1843.6
Band 3	874-885	1782.6 – 1784.8	1877.6 – 1879.8

1.6.3.4 GSM1900 Channels to be tested:

Band	Channels	TX Frequency (MHz)	Rx Frequency (MHz)
Reference Channel	661	1880.0	1960.0
Band 1	512-523	1850.2 – 1852.4	1930.2 – 1932.4
Band 2	656-667	1879.0 – 1881.2	1959.0 – 1961.2
Band 3	799-810	1907.6 – 1909.8	1987.6 – 1989.8

NOTE: it is permissible to extend the frequency bands above to avoid high local interference levels.

1.6.3.5 BC0 Channels to be tested:

Band	Channels	TX Frequency (MHz)	Rx Frequency (MHz)
Low	1013 – 1023, 1 - 74	824.70 – 827.22	869.70 – 872.22
Mid	328 - 412	834.84 – 837.36	879.84 – 882.36
High	692 - 777	845.76 – 848.31	890.76 – 893.31

1.6.3.6 BC1 Channels to be tested:

Band	Channels	TX Frequency (MHz)	Rx Frequency (MHz)
Low	25 - 100	1851.25 – 1855.00	1931.25 – 1935.00
Mid	600 - 700	1880.00 – 1885.00	1960.00 – 1965.00
High	1100 - 1175	1905.00 – 1908.75	1985.00 – 1988.75

1.6.3.7 BC10 Channels to be tested:

Band	Channels	TX Frequency (MHz)	Rx Frequency (MHz)
Band Subclass 2 & 3	476 – 680	817.90 – 823.00	862.90 – 868.00

NOTE: it is permissible to extend the frequency bands above to avoid high local interference levels.

1.6.4 Test Cable Loss

The following table specifies typical cable loss factors for the RF test cable (**Motorola P/N 30-88588M01**).

Test Cable Loss within 3G Frequency Bands

Band	Frequency Range	Cable Loss, dB
EGSM Tx	880MHz – 915MHz	0.4
EGSM Rx	925MHz – 960MHz	0.4
DCS Tx	1710MHz – 1785MHz	0.6
DCS Rx	1805MHz – 1880MHz	0.7
PCS Tx	1850MHz – 1910MHz	0.7
PCS Rx	1930MHz – 1990MHz	0.6
WCDMA2100 Tx	1920MHz – 1980MHz	0.6
WCDMA2100 Rx	2110MHz – 2170MHz	0.8
WCDMA2100 VCO	2300MHz – 2360MHz	0.8
Bluetooth	2402MHz – 2483.5MHz	0.7
GPS	1575MHz	0.5

Table 1.6.4.1

The following table specifies typical cable loss factors for the RF test cables (**AMS P/N 5-00-F4-10000 or Murata P/N MXGS83RK3000**).

Test Cable Loss within 3G Frequency Bands

Band	Frequency Range	Cable Loss, dB
GSM Tx	824MHz – 849MHz	0.5
GSM Rx	869MHz – 894MHz	0.5
EGSM Tx	880MHz – 915MHz	0.5
EGSM Rx	925MHz – 960MHz	0.5
DCS Tx	1710MHz – 1785MHz	0.8
DCS Rx	1805MHz – 1880MHz	0.8
PCS Tx	1850MHz – 1910MHz	0.8
PCS Rx	1930MHz – 1990MHz	0.8
WCDMA2100 Tx	1920MHz – 1980MHz	0.8
WCDMA2100 Rx	2110MHz – 2170MHz	0.9

Table 1.6.4.2

1.6.5 Test Connector Loss

The following table specifies typical insertion loss factors for the SMA Murata Factory RF Probe MM206615. Please see appendix [12.17 Appendix Q: Murata Factory Antenna and RF Probe Insertion Loss and Isolation](#) for more detail.

Factory RF Probe MM206615 Insertion Loss			
Band	Uplink (UL) UE Transmit	Downlink (DL) UE Receive	Insertion Loss (dB)
GSM 850	824 to 849 MHz		-0.16
		869 to 894 MHz	-0.16
GSM 900	880 to 915 MHz		-0.16
		925 to 960 MHz	-0.17
GSM 1800	1710 to 1785 MHz		-0.25
		1805 to 1880 MHz	-0.26
GSM 1900	1850 to 1910 MHz		-0.26
		1930 to 1990 MHz	-0.27
WCDMA 800	880 to 950 MHz		-0.16
		925 to 960 MHz	-0.17
WCDMA 850	824 to 849 MHz		-0.16
		869 to 894 MHz	-0.16
WCDMA 900	880 to 915 MHz		-0.16
		925 to 960 MHz	-0.17
WCDMA 1800	1710 to 1785 MHz		-0.25
		1805 to 1880 MHz	-0.26
WCDMA 1900	1850 to 1910 MHz		-0.26
		1930 to 1990 MHz	-0.27
WCDMA 1700	1710 to 1755 MHz		-0.25
		2110 to 2155 MHz	-0.29
WCDMA 2100	1920 to 1980 MHz		-0.27
		2110 to 2170 MHz	-0.29
LTE 1	1920 to 1980 MHz		-0.27
		2110 to 2170 MHz	-0.29
LTE 2	1850 to 1910 MHz		-0.26
		1930 to 1990 MHz	-0.27
LTE 3	1710 to 1785 MHz		-0.25
		1805 to 1880 MHz	-0.26
LTE 4	1710 to 1755 MHz		-0.25
		2110 to 2155 MHz	-0.29
LTE 5	824 to 849 MHz		-0.16
		869 to 894 MHz	-0.16
LTE 7	2500 to 2570 MHz		-0.33
		2620 to 2690 MHz	-0.35
LTE 8	880 to 915 MHz		-0.16
		925 to 960 MHz	-0.17
LTE 9	1749.9 to 1784.9 MHz		-0.25
		1844.9 to 1879.9 MHz	-0.26
LTE 12	699 to 716 MHz		-0.14
		729 to 746 MHz	-0.15

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Factory RF Probe MM206615 Insertion Loss			
Band	Uplink (UL) UE Transmit	Downlink (DL) UE Receive	Insertion Loss (dB)
LTE 13	777 to 787 MHz		-0.15
		746 to 756 MHz	-0.15
LTE 17	704 to 716 MHz		-0.14
		734 to 746 MHz	-0.15
LTE 19	830 to 845 MHz		-0.16
		875 to 890 MHz	-0.16
LTE 20	832 to 862 MHz		-0.16
		791 to 821 MHz	-0.15
LTE 25	1850 to 1915 MHz		-0.26
		1930 to 1995 MHz	-0.27
LTE 26	814 to 849 MHz		-0.16
		859 to 894 MHz	-0.16
LTE 28	703 to 748 MHz		-0.14
		758 to 803 MHz	-0.15
LTE 29	--		--
		717 to 728 MHz	-0.14
LTE 30	2305 to 2315 MHz		-0.31
		2350 to 2360 MHz	-0.31
LTE 38	2570 to 2620 MHz		-0.34
		2570 to 2620 MHz	-0.34
LTE 39	1880 to 1920 MHz		-0.26
		1880 to 1920 MHz	-0.26
LTE 40	2300 to 2400 MHz		-0.31
		2300 to 2400 MHz	-0.31
LTE 41	2496 to 2690 MHz		-0.34
		2496 to 2690 MHz	-0.34
CDMA 800	824 to 849 MHz		-0.16
		869 to 894 MHz	-0.16
CDMA 1900	1850 to 1910 MHz		-0.26
		1930 to 1990 MHz	-0.27
CDMA AWS	1710 to 1755 MHz		-0.25
		2110 to 2155 MHz	-0.29
CDMA BC10	806 to 901 MHz		-0.16
		851 to 940 MHz	-0.16
WLAN b,g	2412 to 2472 MHz		-0.32
		2412 to 2472 MHz	-0.32
WLAN a	5180 to 5825 MHz		
		5180 to 5825 MHz	
Bluetooth	2400 to 2483 MHz		-0.32
		2400 to 2483 MHz	-0.32
TD-SCDMA 34	2010 to 2025 MHz		-0.28
		2010 to 2025 MHz	-0.28
TD-SCDMA 39	1880 to 1920 MHz		-0.26
		1880 to 1920 MHz	-0.26

Table 1.6.5

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1.6.6 Test Bench RF setup requirements

It has been found that on 3GSM products, a 3dB attenuator MUST be used and MUST be placed as close to the radio in the RF path as possible.

If this attenuator is missing it will cause reflected power to affect the HDET readings of the radio during phasing and cause higher power out than desired once the radio is put into a call.

This attenuator is very critical to the proper phasing of the radio and should not be overlooked.

1.6.7 Supply Voltage

Nominal Battery Voltage = 3.9V

Low Battery Voltage = 3.4V

High Battery Voltage = 4.1V

Absolute Min. Battery Voltage = 3.2V

Absolute Max. Battery Voltage = 4.3V

Nominal External Supply Voltage = 5.8V

Low External Supply Voltage = 4.6V

High External Supply Voltage = 5.9V

Absolute Min. External Voltage = 4.3V

Absolute Max. External Voltage = 6.2V

1.6.8 Power Supply Protection Requirements

To ensure a protected application of the VBAT voltage to the radio, the protection circuit as shown in Figure 1.6.8 shall be used. This circuit shall be used, at all test stations, analyze benches or any other location where power is applied to the battery terminals of the phone.

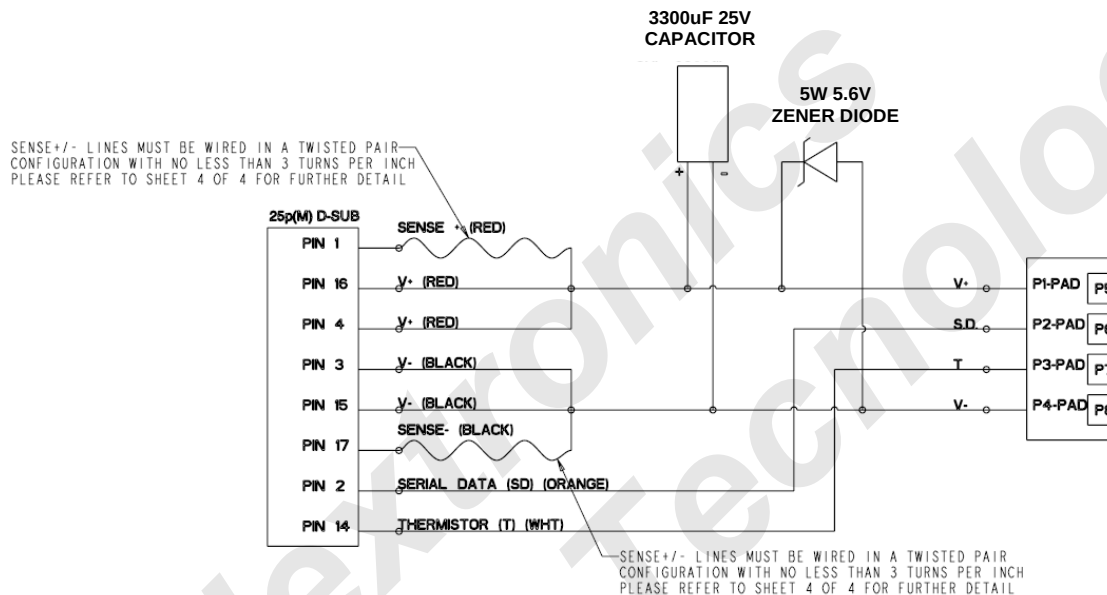


Figure 1.6.8
VBat Protection

The protection components shall consist of a 5W 5.6V zener diode (On-Semiconductor 1N5339B **ONLY**) in parallel with a 3300uF 25V Elect VR Radial capacitor (Nichicon P/N UVR1E332MHD **ONLY**).

1.6.9 Temperature Range

Nominal Temperature Operating Range:	-20°C to +50°C	(Motorola Internal Requirement – design guideline)
Extreme Temperature Test Range:	-30°C to +60°C	(FCC Requirement – for selected FCC tests only)
Extreme Temperature Operating Range:	-10°C to +55°C	(3GPP Requirement – for all factory thermal testing)

1.6.10 Common Test Command Data

1.6.10.1 FTM_SET_MODE

Band	Data
BC0	0x0500
BC1	0x0600
BC10	0x2100
BC14	0x1A00
BC15	0x2000

1.6.10.2 FTM BANDS – PHONE_MODE

Phone Mode	Dec	Hex
PHONE_MODE_CDMA_800	5	5
PHONE_MODE_CDMA_1900	6	6
PHONE_MODE_CDMA_1800	8	8
PHONE_MODE_J_CDMA	14	E
PHONE_MODE_CDMA_450	17	11
PHONE_MODE_CDMA_IMT	19	13
PHONE_MODE_CDMA_1900_EXT	26	1A
PHONE_MODE_CDMA_450_EXT	27	1B
PHONE_MODE_CDMA_800_SEC	33	21
PHONE_MODE_CDMA_BC15	32	20
PHONE_MODE_GSM_850	18	12
PHONE_MODE_GSM_900	10	A
PHONE_MODE_GSM_1800	11	B
PHONE_MODE_GSM_1900	12	C
PHONE_MODE_WCDMA_IMT	9	9
PHONE_MODE_WCDMA_1900A	15	F
PHONE_MODE_WCDMA_1900B	16	10
PHONE_MODE_WCDMA_800	22	16
PHONE_MODE_WCDMA_800A	23	17
PHONE_MODE_WCDMA_1800	25	19
PHONE_MODE_WCDMA_BC4	28	1C
PHONE_MODE_WCDMA_BC8	29	1D
PHONE_MODE_WCDMA_BC9	31	1F
PHONE_MODE_WCDMA_1500	40	28
PHONE_MODE_WCDMA_BC19	75	4B
PHONE_MODE_LTE_B1	34	22
PHONE_MODE_LTE_B2	43	2B
PHONE_MODE_LTE_B3	44	2C
PHONE_MODE_LTE_B4	42	2A
PHONE_MODE_LTE_B5	45	2D
PHONE_MODE_LTE_B6	46	2E
PHONE_MODE_LTE_B7	35	23
PHONE_MODE_LTE_B8	47	2F
PHONE_MODE_LTE_B9	48	30
PHONE_MODE_LTE_B10	49	31



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PHONE_MODE_LTE_B11	41	29
PHONE_MODE_LTE_B12	50	32
PHONE_MODE_LTE_B13	36	24
PHONE_MODE_LTE_B14	51	33
PHONE_MODE_LTE_B15	52	34
PHONE_MODE_LTE_B16	53	35
PHONE_MODE_LTE_B17	37	25
PHONE_MODE_LTE_B18	54	36
PHONE_MODE_LTE_B19	55	37
PHONE_MODE_LTE_B20	56	38
PHONE_MODE_LTE_B21	57	39
PHONE_MODE_LTE_B22	58	3A
PHONE_MODE_LTE_B23	59	3B
PHONE_MODE_LTE_B24	60	3C
PHONE_MODE_LTE_B25	62	3E
PHONE_MODE_LTE_B26	62	3E
PHONE_MODE_LTE_B27	63	3F
PHONE_MODE_LTE_B28	64	40
PHONE_MODE_LTE_B29	65	41
PHONE_MODE_LTE_B30	66	42
PHONE_MODE_LTE_B31	67	43
PHONE_MODE_LTE_B32	68	44
PHONE_MODE_LTE_B33	69	45
PHONE_MODE_LTE_B34	70	46
PHONE_MODE_LTE_B35	71	47
PHONE_MODE_LTE_B36	72	48
PHONE_MODE_LTE_B37	73	49
PHONE_MODE_LTE_B38	38	26
PHONE_MODE_LTE_B39	74	4A
PHONE_MODE_LTE_B40	39	27
PHONE_MODE_LTE_B41	76	4C
PHONE_MODE_LTE_B42	77	4D
PHONE_MODE_TDSCDMA_B34	90	5A
PHONE_MODE_TDSCDMA_B39	91	5B
PHONE_MODE_TDSCDMA_B40	92	5C

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1.7 Minimum Test System and UE Interface Requirements

The following requirements must be met for a test system to be considered compliant with this 12M document.

- 1.) A reference frequency standard (I.e., 10 MHz reference) must be connected to any test equipment that is used as an RF signal source or measurement device.
- 2.) All WCDMA uplink RF power measurements must be averaged over at least one timeslot with a measurement bandwidth of 5 MHz and RRC filter off.
- 3.) The UE supply voltage must remain stable under rapidly changing load conditions. Voltage droops can cause the UE to power-down. When the UE is connected to an External Supply at the accessory connector only (without a Battery Supply), and voltage droops are present in the test system, the External Supply Voltage may be increased up to but not exceeding the High External Supply Voltage level specified in section 1.6.7.
- 4.) When removing and re-applying power to the UE, set the power supply voltage to 0 Volts at the UE battery connector for at least 4 seconds prior to re-applying power. Removing and re-applying power to the UE for a shorter duration than this may cause the power-cut detection circuit to turn the UE on.
- 5.) All current measurements performed while the UUT is transmitting are to be made using Current Integration over 240 ms at the fastest sampling rate the power supply can sustain. If the power supply is not capable of Current Integration, then 20 samples over at least 2 seconds must be averaged

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2.0 Tables of Tests and Limits

Model Key

A = All Models
B = Vertex VZW
C = Vertex ROW
D = Vertex China

Sx = Sampling: See Section [12.20 Appendix T: Total Percent Sampler Details](#)

Sampling:

Sampling may be implemented for some models and tests to improve throughput and process flow. If sampling is allowed, the sampling rate will be specified in the "Models" column of the Limit table for the given model and test. Additional sampling requirements are specified in [12.20 Appendix T: Total Percent Sampler Details](#).

2.1 Common Test Limits

The test limits in this table apply to all measurements unless specified otherwise in the test procedure. These limits are the minimum test requirements specified by development engineering. The use of tighter limits is permissible as long as these minimum test requirements are met.

All tests with P/F units: 1 = Pass, other = Fail.*

***Note:** Some test systems may use different pass/fail identifiers; **Nextest** uses 0 as Pass and 1 as Fail.

Section	Test Name	Models		Parameter	Lower Limit	Upper Limit	Units	CPK
		GNPO	Lead Site					
3.1	In-Line Flash	A	A	In-line Flash	1	1	P/F	N
3.2	Turn On	A	A	Enumeration Status	1	1	P/F	N
3.3	On Current Drain	A	A	Suspend Current (w/o display) – board level	30	100	mA	Y
		A	A	Suspend Current (w/ display) - radio level	50	150	mA	Y
3.4	Turn Off	A	A	Turn Off	1	1	P/F	N
3.5	Off Current Drain	A	A	Off Current Drain	10	350	µA	Y
3.6	Test Mode	A	A	Suspend	1	1	P/F	N
3.7	Software Version	A	A	Version	1	1	P/F	N
3.8	HW and Revision Identifier	A	A	HW Device Product Name Vertex HW Device Product Name	addison	addison	ascii	N
		A	A	HW Rev Version Vertex P1A build (EVT1A) HW Rev Version	81A0	81A0	Hex	N
		A	A	P2A build (DVT1A) HW Rev Version	82A0	82A0	Hex	N
		A	A	P2A build (DVT1B) HW Rev Version	82B0	82B0	Hex	N
		A	A	P2C build (DVT1C) HW Rev Version	82C0	82C0	Hex	N
		A	A	P3 build (DVT2) HW Rev Version	8300	8300	Hex	N
		A	A	P4 build (DVT3) HW Rev Version	8400	8400	Hex	N
		A	A	P5 build (DVT3B) HW Rev Version	8500	8500	Hex	N
		A	A	P6 build (DVT3C) HW Rev Version	8600	8600	Hex	N
		A	A	P7 build (PVT) HW Rev Version	8700	8700	Hex	N
		A	A	HW Radio RF SKU ID Vertex HW Radio RF SKU ID UTAG(VZW)	VZW	VZW	Hex	N
		B	B					

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Section	Test Name	Models		Parameter	Lower Limit	Upper Limit	Units	CPK
		GNPO	Lead Site					
		A	A	REGION_1_MULTI_PIXEL_IGNORE	0	10	CNT	N
		A	A	REGION_1_SINGLE_PIXEL_IGNORE	0	500	CNT	N
		A	A	REGION_2_MULTI_PIXEL_IGNORE	0	100	CNT	N
		A	A	REGION_2_SINGLE_PIXEL_IGNORE	0	600	CNT	N
		A	--	Camera_Ext_Color_Ratio_Blue_Green_Max	0.7	1.2	%	N
		A	--	Camera_Ext_Color_Ratio_Blue_Green_Min	0.7	1.2	%	N
		A	--	Camera_Ext_Color_Ratio_Red_Green_Max	0.8	1.2	%	N
		A	--	Camera_Ext_Color_Ratio_Red_Green_Min	0.8	1.2	%	N
		A	A	CAMERA_EXT_CORNER_RELATIVE_UNIFORMITY	0	10	%	N
3.17	Internal Camera Blemish and Relative Uniformity Test	A	A	Internal Camera Blemish Image Analysis	0	0	P/F	N
		A	A	CAMERA_INT_DARK_REG_COL_RATIO	0.0	3.7	Ratio	N
		A	A	CAMERA_INT_DARK_REG_DETECT_CNT	0.0	0.0	CNT	N
		A	A	CAMERA_INT_DARK_REG_ROW_RATIO	0.0	3.0	Ratio	N
		A	--	Internal Camera Color Uniformity	0.0	10.0	Cab	N
		A	--	CAMERA_INT_MAX_COLOR_DIFFERENCE	0.0	30.0	Cab	N
		A	A	Internal Camera Horizontal Lines	0	0	CNT	N
		A	A	Int Camera Horizontal Line Blue Detect Lev	0	3.0	CNT	N
		A	A	Int Camera Horizontal Line Grn Detect Lev	0	3.0	CNT	N
		A	A	Int Camera Horizontal Line Red Detect Lev	0	3.0	CNT	N
		A	A	Internal Camera Image Height	1944	1944	CNT	N
		A	A	Internal Camera Image Width	2592	2592	CNT	N
		A	A	Internal Camera Vertical Lines	0	0	CNT	N
		A	A	Int Camera Vertical Line Blue Detect Lev	0	3.0	CNT	N
		A	A	Int Camera Vertical Line Green Detect Lev	0	3.0	CNT	N
		A	A	Int Camera Vertical Line Red Detect Lev	0	3.0	CNT	N
		A	A	Internal Relative Illumination	65	100	%	N
		A	A	Internal Number Large Blemishes Region 1	0	0	CNT	N
		A	A	Internal Number Large Blemishes Region 2	0	0	CNT	N
		A	A	Internal Number Small Blemishes Region 1	0	1	CNT	N
		A	A	Internal Number Small Blemishes Region 2	0	3	CNT	N
		A	A	CAMERA_INT_REGION_1_MULTI_PIXEL_IGNORE	0	50	CNT	N
		A	A	CAMERA_INT_REGION_2_MULTI_PIXEL_IGNORE	0	50	CNT	N
		A	A	CAMERA_INT_REGION_1_SINGLE_PIXEL_IGNORE	0	100	CNT	N
		A	A	CAMERA_INT_REGION_2_SINGLE_PIXEL_IGNORE	0	500	CNT	N
		A	A	Internal Average Image Luminance	96	160	CNT	N
		A	A	Internal Max Delta E	0	10	CNT	N
		A	A	Internal Relative Uniformity Delta Luminance	0	7.5	CNT	N
		A	--	Camera_Int_Color_Ratio_Blue_Green_Max	0.75	1.2	%	N
		A	--	Camera_Int_Color_Ratio_Blue_Green_Min	0.75	1.2	%	N
		A	--	Camera_Int_Color_Ratio_Red_Green_Max	0.75	1.5	%	N
		A	--	Camera_Int_Color_Ratio_Red_Green_Min	0.65	1.2	%	N
3.18	Camera APK - Read # of Cameras	A	A	ANDROID_READ_NUMBER_OF_CAMERA	2	2	CNTS	N
3.19	Camera Calibration Status Test	A	A	EXT_Camera Cal_Status	0	0	P/F	N
		A	A	INT_Camera Cal_Status	0	0	P/F	N
3.20	Camera LED Calibration Test - External	A	A	LED1_AvgLum	80	140	CNT	Y
		A	A	LED1_BG_RATIO	0.51	0.71	CNT	Y
		A	A	LED1_RG_RATIO	0.41	0.61	CNT	Y
		A	A	LED2_AvgLum	80	140	CNT	Y
		A	A	LED2_BG_RATIO	0.15	0.39	CNT	Y
		A	A	LED2_RG_RATIO	0.97	1.17	CNT	Y
		A	A	Both_LED_AvgLum	80	140	CNT	Y
		A	A	Both_LED_BG_RATIO	0.36	0.61	CNT	Y
		A	A	Both_LED_RG_RATIO	0.61	0.86	CNT	Y
3.21	Camera LED Calibration Test -Internal	A	A	FRONT_LED_AvgLum	80	140	CNT	Y
		A	A	FRONT_LED_BG_RATIO	0.48	0.73	CNT	Y
		A	A	FRONT_LED_RG_RATIO	0.48	0.68	CNT	Y
3.22	Camera Laser Focus Test	A	A	Stored_Focus_Offset	-100	100	Cnts	N
		A	A	Focus_Offset	-100	100	Cnts	N
		A	A	Focus_Offset_Delta	-30	30	Cnts	N
		A	A	Focus_Distance	2	42	mm	N
3.23	Camera VSYNC to Sensor Hub Line Test	A	A	CAMERA_VSYNC_LINE	1	1	P/F	N
3.25	Internal Battery Test P2K Method	A	A	Battery Thermistor Temp	360	460	tenths°C	Y
		A	A	Battery Voltage OCV	3727	4700	mV	N
		A	A	Battery ID Resistance	229541	256939	ohms	Y
	Internal Battery Test APK Method	A	A	Battery Present Check	1	1	P/F	N
		A	A	Battery Plugged State	1	1	P/F	N
		--	--	Battery Voltage Level	3727	--	mV	N



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		GNPO	Lead Site					
		--	--	Battery Charge Level	18	100	%	N
3.24	PA On No Flash	A	A	EXT_LUMINANCE_PA_ON_NOFLASH_GPIO_LOW	70	256	CNT	N
		A	A	EXT_LUMINANCE_PA_ON_NOFLASH_GPIO_HIGH	0	25	CNT	N
3.26	Battery/Charger/Thermistor Test	A	A	B+ Voltage	3400	3600	mV	Y
		A	A	VBAT Voltage	3200	3600	mV	Y
		A	A	Input Current	425	525	mA	Y
3.27	User Reset Test	A	A	BATT RESET Voltage HIGH	1	1	DEC	-
		A	A	BATT RESET Voltage LOW	0	0	DEC	-
		A	A	User Reset Before Assert	01	01	P/F	N
		A	A	User Reset After Assert	00	00	P/F	N
3.28	USB PMIC Line Check Test	A	A	No Charger Attached	0	0	bit	N
		A	A	Charger Attached	1	1	Bit	N
3.29	USB-C HS Reverse Plug In Test	A	A	USB-C HS Reverse Plug In	1	1	P/F	N
3.30	USB-C HS Power Cable/Accessory Test	A	A	USBC_CC_Line_Voltage	4.5	5.5	V	N
3.31	Bluetooth - P2K TCMD Method	A	A	Bluetooth CW Power	-5	5	dBm	N
		--	--	Bluetooth Connection	1	1	P/F	N
		--	--	Bluetooth TX Power	-10	10	dBm	Y
		--	--	Bluetooth BER	0	0.4	%	Y
3.32	Bluetooth RSSI Test	A	A	BT RSSI Power	-75	-65	dBm	N
		A	A	BT BER	0	0.1	%	N
3.33	WLAN Radiated Power	--	--	WLAN SCPC Mod Power (11b Chan A)	13	23	dBm	Y
		A	A	WLAN SCPC Mod Power (11b Chan B)	14	24	dBm	Y
		--	--	WLAN SCPC Mod Power (11b Chan C)	13	23	dBm	Y
		--	--	WLAN_2.4GHZ_RAD_CALC_TX_ONOFF_CURR_LCH01	180	360	mA	Y
		A	A	WLAN_2.4GHZ_RAD_CALC_TX_ONOFF_CURR_MCH06	180	380	mA	Y
		--	--	WLAN_2.4GHZ_RAD_CALC_TX_ONOFF_CURR_HCH13	220	420	mA	Y
		--	--	WLAN_2.4GHZ_RAD_CALC_MOD_DRAIN_LCH01	20	150	mA	Y
		A	A	WLAN_2.4GHZ_RAD_CALC_MOD_DRAIN_MCH06	35	55	mA	Y
		--	--	WLAN_2.4GHZ_RAD_CALC_MOD_DRAIN_HCH13	15	50	mA	Y
		A	A	WLAN 5.0 GHz TX Current Main Path (Chan E)	300	485	mA	Y
		A	A	WLAN 5.0 GHz Closed loop Mod Power Main Path (Chan E)	14	24	dBm	Y
		A	A	WLAN_MOD_DRAIN	45	60	mA	N
3.34	WLAN RSSI Test	A	--	WLAN_CH01_RSSI	-70	-60	P/F	N
		A	A	WLAN_CH06_RSSI	-70	-60	P/F	N
		A	--	WLAN_CH11_RSSI	-70	-60	P/F	N
3.35	GPS	A	A	GPS_Radiated_Carrier_To_Noise	50.5	60.5	dB/Hz	N
		A	A	GPS Radiated Freq Offset	94	106	kHz	N
3.36	NFC Antenna and SWP Line Self Test	A	A	NFC Antenna Self Test Status	00	00	Hex	N
		A	A	NFC Measured RF Power (radio level)	6,000,000	20,000,000	Cnts	N
		A	A	SWP Line Status - NFC SIM Inserted	00	00	Hex	N
3.37	NFC Tag Test	A	A	TAG_DETECTED	1	1	P/F	N
		A	A	PAYLOAD_DATA	1	1	P/F	N
3.38	Sensor Hub Test	A	A	Sensor Hub Firmware Test	1	1	P/F	N
3.39	Sensor Hub WAKE Lines	A	A	CCP_AP_WAKE	1	1	P/F	N
		A	A	AP_CCP_WAKE	1	1	P/F	N
		A	A	CDC_CCP_WAKE	1	1	P/F	N
		A	A	SENSORHUB_I2C	1	1	P/F	N
3.40	Accelerometer P2K TCMD Method	A	A	Accelerometer Self Test	1	1	P/F	N
		A	A	Accelerometer_Self_Cal_X_Offset	-51	51	Dec	N
		A	A	Accelerometer_Self_Cal_Y_Offset	-51	51	Dec	N
		A	A	Accelerometer_Self_Cal_Z_Offset	-51	51	Dec	N
		A	A	Accelerometer_Self_Cal_Storage_Writes	0	10	Dec	N
	Accelerometer	A	A	ACCELEROMETER_X	-1	1	M/S^2	N
	APK CommServer	--	--	MAX_ACCELEROMETER_X	1	99	M/S^2	N
	Method	--	--	MIN_ACCELEROMETER_X	-99	-1	M/S^2	N
		A	A	ACCELEROMETER_Y	-1	1	M/S^2	N
		--	--	MAX_ACCELEROMETER_Y	1	99	M/S^2	N
		--	--	MIN_ACCELEROMETER_Y	-99	-1	M/S^2	N
		A	A	ACCELEROMETER_Z (Display Face up)	8	12	M/S^2	N
		A	A	ACCELEROMETER_Z (Display Face Down)	-12	-8	M/S^2	N
		--	--	MAX_ACCELEROMETER_Z	1	99	M/S^2	N
		--	--	MIN_ACCELEROMETER_Z	-99	-1	M/S^2	N
		A	A	TOTAL_ACCELERATION	8	12	M/S^2	N
3.41	Gyroscope P2K TCMD Method	A	A	GYRO_SELF_TEST	1	1	P/F	Y
	Gyroscope	A	A	ANGULAR_SPEED_X	-0.4	0.4	rps	Y



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		GNPO	Lead Site					
	APK CommServer Method	A	A	ANGULAR_SPEED_Y	-0.4	0.4	rps	Y
		A	A	ANGULAR_SPEED_Z	-0.4	0.4	rps	Y
3.42	Magnetometer	A	A	Absolute Magnetic Field X Axis (APK)	-1000	1000	Dec.	N
	APK Method	A	A	Absolute Magnetic Field Y Axis (APK)	-1000	1000	Dec.	N
		A	A	Absolute Magnetic Field Z Axis (APK)	-1000	1000	Dec.	N
3.43	Proximity Sensor P2K TCMD Method (Board Level)	A	A	Proximity Sensor Board level (Covered)	600	2047	Dec.	N
3.43	Proximity Sensor P2K TCMD Method (Assembled Level)	A	A	Proximity Sensor Uncovered	0	600	Dec.	N
		--	--	Proximity Sensor Covered	1800	2047	Dec	N
	Proximity Sensor APK Method (Assembled Level)	--	--	Proximity Sensor Uncovered	100	100	Dec	N
		A	A	Proximity Sensor Covered	1	3	Dec	N
	Proximity Sensor APK Method (CQA Test)	--	--	CQA Sequential Test Result	1	1	P/F.	N
3.44	Light Sensors	A	A	Light Sensor Bright DAC	20	120	Dec	N
		A	A	Light Sensor Dark DAC	0	1	Dec	N
		A	A	Light Sensor Bright LUX	1600	4000	Dec	N
		--	--	Light Sensor Dark LUX	0	5	Dec	N
3.45	Light Sources	A	A	GP Single Color LED # (SOC/SOL)	1	1	P/F	N
		A	A	ROI_LED_SOC_LUM_AVG_DELTA_ROI_00	75	255	Lumens	Y
		A	A	ROI_LED_SOC_LUM_AVG_OFF_ROI_00	0	15	Lumens	Y
		A	A	ROI_LED_SOC_LUM_AVG_ON_ROI_00	90	255	Lumens	Y
		A	A	ROI_LED_SOC_BLUE_PERCENTAGE_ROI_00	23	43	%	Y
		A	A	ROI_LED_SOC_GREEN_PERCENTAGE_ROI_00	23	43	%	Y
		A	A	ROI_LED_SOC_RED_PERCENTAGE_ROI_00	23	43	%	Y
3.46	UTAG Write Front Housing Color Test	A	A	UTAG Write Front Housing Color	1	1	P/F	N
3.47	UTAG Read Front Housing Color Test	A	A	UTAG Read Front Housing Color	1	1	P/F	N
3.48	Fingerprint Sensor Test - P2K Method	A	A	FPS_SELFTEST	00	00	HEX	Y
		A	A	FPS_CHECKERBOARD_TEST	00	00	HEX	Y
	Fingerprint Sensor Test-APK Method	A	A	FPS_IMAGE_QUALITY_TEST	0	0	P/F	Y
3.49	Hall Effect Sensor Test	--	--	North Pole Detection	01	01	Hex	N
		A	A	South Pole Detection	02	02	Hex	N
		A	A	No Pole Detection	00	00	Hex	N
3.50	RF Connector Detection	A	A	LB_HB RF Connector Status - Connected	0	0	P/F	N
		A	A	LB_HB RF Connector Status - Disconnected	0	0	P/F	N
		A	A	ULB_MB RF Connector Status - Connected	0	0	P/F	N
		A	A	ULB_MB RF Connector Status - Disconnected	0	0	P/F	N
		A	A	DIV RF Connector Status - Connected	0	0	P/F	N
		A	A	DIV RF Connector Status - Disconnected	0	0	P/F	N
3.51	Thermistors and Temp Sensors	--	--	Battery Thermistor Temp (See sections 3.25 and 3.26)	--	--	tenths°C	N
		A	A	MSM_Thermistor_Temp	20	50	°C	N
		A	A	Charger_Thermistor_Temp	20	50	°C	N
		A	A	PA_THERM0	10	40	°C	N
3.52	Touchscreen	A	A	VERIFY_TOUCH_RESET_LINES	1	1	P/F	Y
		A	A	TOUCH_SYNAPTICS_FULL_RAW_CAPACITANCE	1	1	P/F	Y
		A	A	TOUCH_SYNAPTICS_HIGH_RESISTANCE	1	1	P/F	Y
		A	A	TOUCH_SYNAPTICS_TRX_GND_SHORTS	1	1	P/F	Y
		A	A	TOUCH_SYNAPTICS_TRX_TRX_SHORTS	1	1	P/F	Y
3.53	Manual Touchscreen APK Test	A	A	TOUCH_SCREEN_APK	1	1	P/F	N
3.54	Display Tests	A	A	White w/Black Border	1	1	P/F	N
		A	A	Black w/White Border	1	1	P/F	N
		A	--	Bright Color Bar Pattern	1	1	P/F	N
		A	A	DISPLAY_BW_BORDER_MISSING_PIXELS	0	0	CNT	N
		A	A	DISPLAY_BW_BORDER_STUCK_PIXELS	0	0	CNT	N
		A	A	DISPLAY_W_B_BORDER_LARGEST_STD_DEV	0	10	%	N
		A	A	DISPLAY_W_B_BORDER_LARGEST_LUM_DELTA	-30	30	CNTS	N
		A	A	DISPLAY_W_B_BORDER_RELATIVE_LUM	1	30	CNTS	N
		A	A	DISP_W_B_BORDER_RELAT_LUM_BLOCK_NUM	0	10000	CNTS	N
		A	A	W_B_BORDER_BLUE_DELTA_TO_AVG_KEY00	-15	15	%	N
		A	A	DISPLAY_WB_BORDER_MISSING_PIXELS	0	0	CNT	N
		A	A	DISPLAY_W_B_BORDER_LUM	100	200	CNTS	N
		A	A	DISPLAY_WB_BORDER_STUCK_PIXELS	0	0	CNT	N
		A	A	W_B_BORDER_GRN_DELTA_TO_AVERAGE_KEY00	-15	15	%	N
		A	A	W_B_BORDER_RED_DELTA_TO_AVERAGE_KEY00	-15	15	%	N
		A	A	DISPLAY_WB_BORDER_LINE_MISSING_PIXELS	0	0	CNT	N
		A	A	DISPLAY_WB_BORDER_LINE_STUCK_PIXELS	0	0	CNT	N
3.55	Display TE Line Test	A	A	TE_SIGNAL_DISPLAY_TO_PROCESSOR	1	1	P/F	Y



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3.56	Display Vendor ID Check Test	A	A	DISPLAY_VENDOR_ID (SDC)	010B	010B	HEX	Y
3.57	Audio Calibration	A	A	AUDIO_INTERNAL_CAL_TEMPERATURE	20	44	°C	Y
		A	A	AUDIO_INTERNAL_CAL_COUNT	16	16	DEC	Y
		A	A	AUDIO_INTERNAL_CAL_RAW_IMPEDANCE	270000	310000	DEC	Y
		A	A	AUDIO_INTERNAL_CAL_RAW_RES_FREQ	800	1600	DEC	Y
		A	A	AUDIO_INTERNAL_CAL_RAW_RES_DIFF	1300	3000	DEC	Y
3.58	Audio Interrupt Line Test	A	A	AOV Interrupt	00	00	HEX	Y
3.59	Alert Audio / Amplitude and Distortion Test	A	A	AUDIO_OUTPUT_ALERT_AMP_00140	60	85	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00160	60	89	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00180	62	92	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00200	65	95	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00224	68	98	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00250	70	101	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00280	73	103	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00315	75	105	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00355	78	106	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00400	80	108	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00450	81	110	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00500	81	111	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00560	82	112	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00630	83	112	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00710	84	113	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00800	85	114	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_00900	86	115		
		A	A	AUDIO_OUTPUT_ALERT_AMP_01000	87	115	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_01120	95	115	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_01250	88	116	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_01400	88	117	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_01600	89	118	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_01800	90	119	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_02000	90	120	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_02240	92	121	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_02500	90	120	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_02800	87	119	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_03150	85	115	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_03550	88	118	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_AMP_04000	88	118	dB SPL	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00140	0	40	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00160	0	35	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00180	0	25	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00200	0	20	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00224	0	15	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00250	0	12	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00280	0	10	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00315	0	8	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00355	0	6	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00400	0	5	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00450	0	4	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00500	0	5	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00560	0	5	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00630	0	3	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00710	0	2.75	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00800	0	2.5	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_00900	0	2	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_01000	0	2	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_01120	0	1.75	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_01250	0	1.5	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_01400	0	1.25	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_01600	0	1.25	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_01800	0	1	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_02000	0	1	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_02240	0	1	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_02500	0	1	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_02800	0	1	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_03150	0	1	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_03550	0	1	%	Y
		A	A	AUDIO_OUTPUT_ALERT_HOHD_04000	0	1	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00140	0	100	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00160	0	85	%	Y



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		GNPO	Lead Site					
		A	A	AUDIO_OUTPUT_ALERT_THD_00180	0	75	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00200	0	75	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00224	0	75	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00250	0	60	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00280	0	55	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00315	0	55	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00355	0	45	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00400	0	35	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00450	0	35	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00500	0	35	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00560	0	35	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00630	0	30	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00710	0	35	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00800	0	30	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_00900	0	13	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_01000	0	12	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_01120	0	11	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_01250	0	11	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_01400	0	10	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_01600	0	10	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_01800	0	15	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_02000	0	10	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_02240	0	9	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_02500	0	9	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_02800	0	8	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_03150	0	10	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_03550	0	6	%	Y
		A	A	AUDIO_OUTPUT_ALERT_THD_04000	0	5	%	Y
3.60	Microphone Test (Board Level)	A	A	AUD_SAMP_PRIMARY_MIC_BRD_LEVEL_AMPL	-84	0	dB	N
		A	A	AUD_SAMP_PRIMARY_MIC_BRD_LEVEL_FREQ	0	4000	Hz	N
		A	A	AUD_SAMP_SEC_MIC_BRD_LEVEL_AMPL	-84	0	dB	N
		A	A	AUD_SAMP_SEC_MIC_BRD_LEVEL_FREQ	0	4000	Hz	N
		A	A	AUD_SAMP_TERTIARY_MIC_BRD_LEVEL_AMPL	-84	0	dB	N
		A	A	AUD_SAMP_TERTIARY_MIC_BRD_LEVEL_FREQ	0	4000	Hz	N
	Microphone Test (APK method)	A	A	MIC_1_LEVEL_SPEC_00300	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_00350	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_00400	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_00450	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_00500	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_00550	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_00600	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_00650	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_00700	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_00750	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_00800	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_00850	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_00900	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_00950	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01000	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01050	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01100	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01150	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01200	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01250	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01300	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01350	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01400	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01450	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01500	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01550	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01600	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01650	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01700	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01750	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01800	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01850	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01900	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_01950	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02000	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02050	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02100	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02150	-38	-22	dB	N



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		GNPO	Lead Site					
		A	A	MIC_1_LEVEL_SPEC_02200	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02250	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02300	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02350	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02400	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02450	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02500	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02550	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02600	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02650	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02700	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02750	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02800	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02850	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02900	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_02950	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_03000	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_03050	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_03100	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_03150	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_03200	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_03250	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_03300	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_03350	-38	-22	dB	N
		A	A	MIC_1_LEVEL_SPEC_03400	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_00300	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_00350	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_00400	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_00450	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_00500	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_00550	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_00600	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_00650	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_00700	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_00750	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_00800	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_00850	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_00900	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_00950	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01000	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01050	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01100	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01150	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01200	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01250	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01300	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01350	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01400	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01450	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01500	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01550	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01600	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01650	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01700	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01750	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01800	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01850	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01900	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_01950	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02000	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02050	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02100	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02150	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02200	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02250	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02300	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02350	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02400	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02450	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02500	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02550	-38	-22	dB	N



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		GNPO	Lead Site					
		A	A	MIC_2_LEVEL_SPEC_02600	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02650	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02700	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02750	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02800	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02850	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02900	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_02950	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_03000	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_03050	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_03100	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_03150	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_03200	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_03250	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_03300	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_03350	-38	-22	dB	N
		A	A	MIC_2_LEVEL_SPEC_03400	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_00300	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_00350	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_00400	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_00450	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_00500	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_00550	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_00600	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_00650	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_00700	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_00750	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_00800	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_00850	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_00900	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_00950	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01000	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01050	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01100	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01150	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01200	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01250	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01300	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01350	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01400	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01450	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01500	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01550	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01600	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01650	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01700	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01750	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01800	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01850	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01900	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_01950	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02000	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02050	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02100	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02150	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02200	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02250	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02300	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02350	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02400	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02450	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02500	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02550	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02600	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02650	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02700	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02750	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02800	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02850	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02900	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_02950	-38	-22	dB	N



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		GNPO	Lead Site					
		A	A	MIC_3_LEVEL_SPEC_03000	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_03050	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_03100	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_03150	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_03200	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_03250	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_03300	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_03350	-38	-22	dB	N
		A	A	MIC_3_LEVEL_SPEC_03400	-38	-22	dB	N
3.61	Ultrasonic Test	A	A	AUDIO_OUTPUT_ULTRASONIC_AMP_01000	92.5	104.5	dB SPL	Y
3.62	Vibrator Test	A	A	Vibrator On Amplitude	-110	-65	dB	N
		A	A	Vibrator On Frequency	160	190	Hz	N
3.63	Keypad	A	A	Detect all key presses	1	1	P/F	N
3.64	Headset Interface	A	A	HEADSET_DETECT_INSERTION_STATUS	1	1	P/F	N
		A	A	HEADSET_DETECT_HAS_MIC_STATUS	1	1	P/F	N
		A	A	HEADSET_DETECT_PLUGGED_STATUS	1	1	P/F	N
		A	A	HEADSET_SENDEND_OVERALL_SPEC	1	1	P/F	N
		A	A	HEADSET_DETECT_REMOVAL_STATUS	1	1	P/F	N
	Headset Interface	A	A	SPKRRIGHT_1KHzONRIGHT_LEFTM	-30	-24	dB	N
	APK CommServer Method	A	A	SPKRLEFT_2KHzONLEFT_RIGHTM	-30	-24	dB	N
		A	A	SPKRLEFT_2KHzONRIGHT_LEFTM	-120	-70	dB	N
3.65	AMPS Connector Tests	A	A	APBA Test	0	0	P/F	N
		A	A	ANX78xx_INIT_Test	0	0	P/F	N
		A	A	CWIRE_MSM_GPIO44_HIGH	01	01	HEX	N
		A	A	CWIRE_MSM_GPIO44_LOW	00	00	HEX	N
		A	A	AMPS_AIN1_C1_VOLTAGE	1700000	1850000	uV	Y
		A	A	AMPS_AIN1_C1_GPIO_TEST	0	0	P/F	N
		A	A	AMPS_AIN1_C2_VOLTAGE	608000	640000	uV	Y
		A	A	AMPS_AIN1_C2_GPIO_TEST	0	0	P/F	N
		A	A	AMPS_AIN1_C3_VOLTAGE	182000	195000	uV	Y
		A	A	AMPS_AIN1_C3_GPIO_TEST	0	0	P/F	N
		A	A	AMPS_AIN1_C4_VOLTAGE	470000	495000	uV	Y
		A	A	AMPS_AIN1_C4_GPIO_TEST	0	0	P/F	N
		A	A	AMPS_AIN1_C5_VOLTAGE	1760000	1840000	uV	Y
		A	A	AMPS_AIN1_C5_GPIO_TEST	0	0	P/F	N
		A	A	AMPS_AIN1_C6_VOLTAGE	367000	387000	uV	Y
		A	A	AMPS_AIN1_C6_GPIO_TEST	0	0	P/F	N
		A	A	AMPS_AIN1_C7_VOLTAGE	96250	103500	uV	Y
		A	A	AMPS_AIN1_C7_GPIO_TEST	0	0	P/F	N
		A	A	AMPS_AIN1_C8_VOLTAGE	268000	290000	uV	Y
		A	A	AMPS_AIN1_C8_GPIO_TEST	0	0	P/F	N
		A	A	AMPS_AIN2_VOLTAGE	2700000	3400000	uV	Y
		A	A	AMPS_AIN3_VOLTAGE	3408000	3453000	uV	Y
		A	A	AMPS_I2S	0	0	P/F	N
		A	A	AMPS_SPI	0	0	P/F	N
3.66	EFS SYNC	A	A	File System Synchronization	0	0	P/F	N
3.68	Save to HOB	A	A	Save to HOB	0	0	P/F	N
3.69	Verify Save to HOB	A	A	Verify HOB	0	0	P/F	N
3.70	Phase File Storage	A	--	Phase File	1	1	P/F	N
3.71	NVM Setup	A	A	NVM Setup	1	1	P/F	N
3.72	NVM Default Programming	A	A	NVM Programming	1	1	P/F	N
3.73	Load RF NV	A	A	Load RF NV	1	1	P/F	N
3.73	Auto-cal Items	A	A	Autocal_RFNVM_MSM_SELF_CAL_I	0	0	P/F	N
		A	A	Autocal_RFNVM_RFDEVICE_HDET_AUTOCAL_I	0	0	P/F	N
		A	A	Autocal_NV_RC_PROCESS_ERRORS	0	0	P/F	N
3.74	XO Calibration	A	A	XO_CAL_INITIAL_COURSE_STATE_TEMP	0	45	°C	N
		A	A	XO Calibration	1	1	P/F	N



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2.2 WCDMA Test Limits

The complete set of limits that apply to this WCDMA test limits table can be found at the link below:

<https://drive.google.com/drive/#folders/0B7eiu1wwlowwV0JvZmwtQXlybzQ>

These limits are the minimum test requirements specified by development engineering. The use of tighter limits is permissible as long as these minimum test requirements are met.

Phasing targets are exactly between the upper and lower limits unless specified otherwise in the test procedure.

All tests with P/F units: 1 = Pass, other = Fail.*

***Note:** Some test systems may use different pass/fail identifiers; **Nextest** uses 0 as Pass and 1 as Fail.

Section	Test Name	Models		Parameter	Lower Limit	Upper Limit	Units	CPK
		GNPO	Lead Site					
4.1	WCDMA Internal DAC Calibration	A	A	NV_WCDMA_RC_PROC_ERRORS_OVERALL_STATUS	0	0	P/F	N
4.2	WCDMA RX, TX, HDET and DC Cal Phasing	A	A	WCDMA RX, TX, HDET and DC Cal Phasing	0	0	P/F	N
4.3	WCDMA Tes Mode Call Test	A	A	WCDMA Tes Mode Call Test	0	0	P/F	N



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2.3 GSM Test Limits

The complete set of limits that apply to this GSM test limits table can be found at the link below:

<https://drive.google.com/drive/#folders/0B7eiu1wwlowwV0JvZmwtQXlybzQ>

These limits are the minimum test requirements specified by development engineering. The use of tighter limits is permissible as long as these minimum test requirements are met.

Phasing targets are exactly between the upper and lower limits unless specified otherwise in the test procedure.

All tests with P/F units: 1 = Pass, other = Fail.*

***Note:** Some test systems may use different pass/fail identifiers; **Nextest** uses 0 as Pass and 1 as Fail.

Section	Test Name	Models		Parameter	Lower Limit	Upper Limit	Units	CPK
		GNPO	Lead Site					
5.1	GSM RSSI Chan Correction	A	A	GSM RSSI Chan Correction	0	0	P/F	N
5.2	GSM TX Phasing	A	A	GSM TX Phasing	0	0	P/F	N
5.3	GSM Non-Signaling Call	A	A	GSM Non-Signaling Call	0	0	P/F	N
5.4	8PSK Non-Signaling Data Connection Test	A	A	8PSK Non-Signaling Data Connection Test	0	0	P/F	N



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2.4 CDMA Test Limits

The complete set of limits that apply to this CDMA test limits table can be found at the link below:

<https://drive.google.com/drive/#folders/0B7eiu1wwlowwV0JvZmwtQXlybzQ>

These limits are the minimum test requirements specified by development engineering. The use of tighter limits is permissible as long as these minimum test requirements are met.

Phasing targets are exactly between the upper and lower limits unless specified otherwise in the test procedure.

All tests with P/F units: 1 = Pass, other = Fail.*

***Note:** Some test systems may use different pass/fail identifiers; **Nextest** uses 0 as Pass and 1 as Fail.

Section	Test Name	Models		Parameter	Lower Limit	Upper Limit	Units	CPK
		GNPO	Lead Site					
6.1	CDMA RX, TX and HDET Phasing	BD	BD	CDMA RX, TX and HDET Phasing	0	0	P/F	N
6.2	CDMA FTM Parametrics	BD	BD	CDMA FTM Parametrics	0	0	P/F	N



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2.5 LTE-FDD Test Limits

The complete set of limits that apply to this LTE test limits table can be found at the link below:

<https://drive.google.com/drive/#folders/0B7eiu1wwlowwV0JvZmwtQXlybzQ>

These limits are the minimum test requirements specified by development engineering. The use of tighter limits is permissible as long as these minimum test requirements are met.

Phasing targets are exactly between the upper and lower limits unless specified otherwise in the test procedure.

All tests with P/F units: 1 = Pass, other = Fail.*

***Note:** Some test systems may use different pass/fail identifiers; **Nextest** uses 0 as Pass and 1 as Fail.

Section	Test Name	Models		Parameter	Lower Limit	Upper Limit	Units	CPK
		GNPO	Lead Site					
7.1	LTE-FDD RX, TX and HDET Phasing	A	A	LTE-FDD RX, TX and HDET Phasing	0	0	P/F	N
7.2	LTE-FDD NS Parametric Tests	A	A	LTE-FDD Non-Signaling Parametric Tests	0	0	P/F	N

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2.6 LTE-TDD Test Limits

The complete set of limits that apply to this LTE test limits table can be found at the link below:

<https://drive.google.com/drive/#folders/0B7eiu1wvlowwV0JvZmwTQXlybzQ>

These limits are the minimum test requirements specified by development engineering. The use of tighter limits is permissible as long as these minimum test requirements are met.

Phasing targets are exactly between the upper and lower limits unless specified otherwise in the test procedure.

All tests with P/F units: 1 = Pass, other = Fail.*

***Note:** Some test systems may use different pass/fail identifiers; **Nextest** uses 0 as Pass and 1 as Fail.

Section	Test Name	Models		Parameter	Lower Limit	Upper Limit	Units	CPK
		GNPO	Lead Site					
8.1	LTE-TDD RX, TX and HDET Phasing	CD	CD	LTE-TDD RX, TX and HDET Phasing	0	0	P/F	N
8.2	LTE-TDD NS Parametric Tests	CD	CD	LTE-TDD Non-Signaling Parametric Tests	0	0	P/F	N

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2.7 TD-SCDMA Test Limits

The complete set of limits that apply to this TD-SCDMA test limits table can be found at the link below:
<https://drive.google.com/drive/#folders/0B7eiu1wvlowwV0JvZmwTQXlybzQ>

These limits are the minimum test requirements specified by development engineering. The use of tighter limits is permissible as long as these minimum test requirements are met.

Phasing targets are exactly between the upper and lower limits unless specified otherwise in the test procedure.

All tests with P/F units: 1 = Pass, other = Fail.*

*Note: Some test systems may use different pass/fail identifiers; Nextest uses 0 as Pass and 1 as Fail.

Section	Test Name	Models		Parameter	Lower Limit	Upper Limit	Units	CPK
		GNPO	Lead Site					
9.1	TD-SCDMA RX, TX and HDET Phasing	C	C	TD-CDMA RX, TX and HDET Phasing	0	0	P/F	N
9.2	TD-SCDMA Test Mode Call Test	C	C	TD-SCDMA Test Mode Call Test	0	0	P/F	N



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2.8 RF Antenna Test Limits

The complete set of limits that apply to this RF Antenna test limits table can be found at the link below:

<https://drive.google.com/drive/#folders/0B7eiu1wvlowwV0JvZmwtQXlybzQ>

These limits are the minimum test requirements specified by development engineering. The use of tighter limits is permissible as long as these minimum test requirements are met.

Phasing targets are exactly between the upper and lower limits unless specified otherwise in the test procedure.

All tests with P/F units: 1 = Pass, other = Fail.*

***Note:** Some test systems may use different pass/fail identifiers; **Nextest** uses 0 as Pass and 1 as Fail.

Section	Test Name	Models		Parameter	Lower Limit	Upper Limit	Units	CPK
		GNPO	Lead Site					
10.1	Main Antenna Test	A	A	RX Antenna Tests	0	0	P/F	N
		A	A	TX Antenna Tests	0	0	P/F	N
10.2	Diversity Antenna Test	A	A	RX Antenna Tests	0	0	P/F	N



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2.9 CFC Test Limits

The test limits in this table apply to all measurements unless specified otherwise in the test procedure.

These limits are the minimum test requirements specified by development engineering. The use of tighter limits is permissible as long as these minimum test requirements are met.

All tests with P/F units: 1 = Pass, other = Fail.*

***Note:** Some test systems may use different pass/fail identifiers; **Nextest** uses 0 as Pass and 1 as Fail.

Section	Test Name	Parameter	Lower Limit	Upper Limit	Units
11.1	Apk_USIM_Card_Detection	APK_USIM_Card_Detection	0	0	P/F
11.2	Apk_Verify_Talkback_Absence	GET_TALKBACK_STATUS	0	0	P/F
11.3	Aux_Engine_Ready_Ftm	AUX_ENGINE_READY_FTM	0	0	P/F
11.4	Battery_Sense_Voltage_Check	BATTERY_SENSE_VOLTAGE	3803	4390	mV
11.5	Battery_Thermistor_Temp_Check	ThermistorTempLevel	-10	55	°C
11.6	BP_Engine_Ready_FTM_VIA_ROUTE	AUX_ENGINE_READY_FTM_VIA_ROUTE	0	0	P/F
11.7	CHECK_FTI_NO_CHECKSUM_P2K	CHECK_FTI_NO_CHECKSUM_P2K	0	0	P/F
11.8	Clear_Nv_Ftm_Mode	CLEAR_NV_FTM_MODE	0	0	P/F
11.9	Fastboot_Fb_Mode_Clear	FASTBOOT_FB_MODE_CLEAR	0	0	P/F
11.10	Fastboot_Flash	FASTBOOT_FLASH	0	0	P/F
11.11	Fastboot_Oem_Prod_Fuse	FASTBOOT_OEM_PROD_FUSE	0	0	P/F
11.12	Fastboot_Query_Secure	FASTBOOT_QUERY_SECURE	0	0	P/F
11.13	Fastboot_Reboot_Bootloader	FASTBOOT_REBOOT_BOOTLOADER	0	0	P/F
11.14	Fastboot_Reboot	FASTBOOT_REBOOT	0	0	P/F
11.15	Fastboot_Set_Barcode_Utag	FASTBOOT_SET_BARCODE_UTAG	0	0	P/F
11.16	Fastboot_Unset_Bootmode	FASTBOOT_UNSET_BOOTNMODE	0	0	P/F
11.17	Fastboot_Verify_Unlock_Status	FASTBOOT_VERIFY_UNLOCK_STATUS	0	0	P/F
11.18	Full_Power_On_Check	FULL_POWER_ON_CHECK	0	0	P/F
11.19	IS_CDMA_MASTER_SUB_LOCKED	IS_CDMA_MASTER_SUB_LOCKED	0	0	P/F
11.20	Is_Subsidy_Lock_Required	IS_SUBSIDY_LOCK_REQUIRED	0	0	P/F
11.21	Is_Unit_Locked	IS_UNIT_LOCKED	0	0	P/F
11.22	Load_Wlan_Test_Firmware	LOAD_WLAN_TEST_FIRMWARE	0	0	P/F
11.23	MASTER_CLEAR	MASTER_CLEAR	0	0	P/F
11.24	Power_Down_Wlan_Radio	POWER_DOWN_WLAN_RADIO	0	0	P/F
11.25	POWER_OFF	POWER_OFF	0	0	P/F
11.26	PROGRAM_AKEY	PROGRAM_AKEY	0	0	P/F
11.27	Program_Cid_Data_Block	PROGRAM_CID_DATA_BLOCK	0	0	P/F
11.28	Program_CID_Via_Fastboot_IMEI / MEID	PROGRAM_CID_VIA_FASTBOOT_IMEI / MEID	0	0	P/F
11.29	Program_Imei_Datablock	PROGRAM_IMEI_DBS	0	0	P/F
11.30	PROGRAM_IMEI_FTM	PROGRAM_IMEI_FTM	0	0	P/F
11.31	Program_Master_Sdsy_Lck	PROGRAM_MASTER_SDSY_LCK	0	0	P/F
11.32	Program_Meid_Datablock	PROGRAM_MEID_DBS	0	0	P/F
11.33	PROGRAM_MEID_FTM	PROGRAM_MEID_FTM	0	0	P/F
11.34	PROGRAM_OBDM	PROGRAM_OBDM	0	0	P/F
11.35	Program_Subsidy_Lock	PROGRAM_SUBSIDY_LOCK	0	0	P/F
11.36	PROGRAM SUBSIDY NVM FILE FTM	PROGRAM SUBSIDY NVM FILE FTM	0	0	P/F
11.37	Program_Otk_Sdsy_Lck	PROGRAM_OTK_SDSY_LCK	0	0	P/F
11.38	PROGRAM_PKI_DATA	PROGRAM_PKI_DATA	0	0	P/F
11.39	Qcom_Efs_Sync_Clear_Ftm	QCOM_EFS_SYNC_CLEAR_FTM	0	0	P/F
11.40	READ_IMEI_FTM	READ_IMEI_FTM	0	0	P/F
11.41	READ_MEID_FTM	READ_MEID_FTM	0	0	P/F
11.42	READ_PROCESSOR_ID	READ_PROCESSOR_ID	0	0	P/F
11.43	READ_TRACK_ID	READ_TRACK_ID	0	0	P/F
11.44	RESTART	RESTART	0	0	P/F
11.45	SET_BP_RTC_FTM	SET_BP_RTC_FTM	0	0	P/F
11.46	SET_PHONE_SYSTEM_TIME	SET_PHONE_SYSTEM_TIME	0	0	P/F



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11.47	SUSPEND	SUSPEND	0	0	P/F
11.48	Switch_To_Fastboot_Flash_Mode	SWITCH_TO_FASTBOOT_FLASH_MODE	0	0	P/F
11.49	Unlock_Cid_Datablock_Imei	UNLOCK_CID_DATABLOCK_IMEI	0	0	P/F
11.50	Unlock_Cid_Datablock_Meid	UNLOCK_CID_DATABLOCK_MEID	0	0	P/F
11.51	UNLOCK_SUBSIDY_LOCK	PROGRAM_SUBSIDY_LOCK	0	0	P/F
11.52	Unlock_Cdma_Master_Lock	UNLOCK_CDMA_MASTER_LOCK	0	0	P/F
11.53	UTAG_CLEAR_DATECODE	UTAG_CLEAR_DATECODE	0	0	P/F
11.54	UTAG_CLEAR_ICCID	UTAG_CLEAR_ICCID	0	0	P/F
11.55	UTAG_CLEAR_IMEI	UTAG_CLEAR_IMEI	0	0	P/F
11.56	UTAG_CLEAR_MD5	UTAG_CLEAR_MD5	0	0	P/F
11.57	UTAG_CLEAR_MEID	UTAG_CLEAR_MEID	0	0	P/F
11.58	UTAG_CLEAR_SKU	UTAG_CLEAR_SKU	0	0	P/F
11.59	UTAG_CLEAR_TRACK_ID	UTAG_CLEAR_TRACK_ID	0	0	P/F
11.60	UTAG_VERIFY_BATTERY_ID	UTAG_VERIFY_BATTERY_ID	0	0	P/F
11.61	UTAG_VERIFY_BLUETOOTH	UTAG_VERIFY_BLUETOOTH	0	0	P/F
11.62	UTAG_VERIFY_BOOTMODE	UTAG_VERIFY_BOOTMODE	0	0	P/F
11.63	UTAG_VERIFY_DATECODE	UTAG_VERIFY_DATECODE	0	0	P/F
11.64	UTAG_VERIFY_ICCID	UTAG_VERIFY_ICCID	0	0	P/F
11.65	UTAG_VERIFY_IMEI	UTAG_VERIFY_IMEI	0	0	P/F
11.66	UTAG_VERIFY_MD5	UTAG_VERIFY_MD5	0	0	P/F
11.67	UTAG_VERIFY_MEID	UTAG_VERIFY_MEID	0	0	P/F
11.68	UTAG_VERIFY_MULTIPLE_WLAN	UTAG_VERIFY_MULTIPLE_WLAN	0	0	P/F
11.69	UTAG_VERIFY_MULTISIM	UTAG_VERIFY_MULTISIM	0	0	P/F
11.70	UTAG_VERIFY_SKU	UTAG_VERIFY_SKU	0	0	P/F
11.71	UTAG_VERIFY_TRACK_ID	UTAG_VERIFY_TRACK_ID	0	0	P/F
11.72	UTAG_WRITE_BATTERY_ID	UTAG_WRITE_BATTERY_ID	0	0	P/F
11.73	UTAG_WRITE_BLUETOOTH	UTAG_WRITE_BLUETOOTH	0	0	P/F
11.74	UTAG_WRITE_BOOTMODE_NORMAL	UTAG_WRITE_BOOTMODE_NORMAL	0	0	P/F
11.75	UTAG_WRITE_DATECODE	UTAG_WRITE_DATECODE	0	0	P/F
11.76	UTAG_WRITE_ICCID	UTAG_WRITE_ICCID	0	0	P/F
11.77	UTAG_WRITE_IMEI	UTAG_WRITE_IMEI	0	0	P/F
11.78	UTAG_WRITE_MD5	UTAG_WRITE_MD5	0	0	P/F
11.79	UTAG_WRITE_MEID	UTAG_WRITE_MEID	0	0	P/F
11.80	UTAG_WRITE_MULTIPLE_WLAN	UTAG_WRITE_MULTIPLE_WLAN	0	0	P/F
11.81	UTAG_WRITE_MULTISIM	UTAG_WRITE_MULTISIM	0	0	P/F
11.82	UTAG_WRITE_SKU	UTAG_WRITE_SKU	0	0	P/F
11.83	UTAG_WRITE_TRACK_ID	UTAG_WRITE_TRACK_ID	0	0	P/F
11.84	VALIDATE_HOB	VALIDATE_HOB	0	0	P/F
11.85	VERIFY_BATTERY_GAUGE	VERIFY_BATTERY_GAUGE	40	100	%
11.86	VERIFY_BP_RTC_FTM	VERIFY_BP_RTC_FTM	0	0	P/F
11.87	VERIFY_DHOB	VERIFY_DHOB	0	0	P/F
11.88	VERIFY_FSG_VERSION	VERIFY_FSG_VERSION	0	0	P/F
11.89	VERIFY_IMEI_FTM	VERIFY_IMEI	0	0	P/F
11.90	VERIFY_MBM_VERSION	VERIFY_MBM_VERSION	0	0	P/F
11.91	VERIFY_MEID_FTM	VERIFY_MEID	0	0	P/F
11.92	VERIFY_NV_FTM_MODE_CLEARED	VERIFY_NV_FTM_MODE_CLEARED	0	0	P/F
11.93	VERIFY_OBDM	VERIFY_OBDM	0	0	P/F
11.94	VERIFY_PESN	VERIFY_PESN	0	0	P/F
11.95	VERIFY_PHONE_CLEAN	VERIFY_PHONE_CLEAN	00	00	P/F
11.96	Verify_Phone_System_Time_Apk	VERIFY_PHONE_SYSTEM_TIME_APK	0	0	P/F
11.97	VERIFY_PKI_DATA	VERIFY_PKI_DATA	0	0	P/F
11.98	Verify_Ready_To_Ship_Efuse_Status	VERIFY_READY_TO_SHIP_EFUSE_STATUS	0	0	P/F
11.99	VERIFY_SBK	VERIFY_SBK	0	0	P/F
11.100	Verify_Setup_Wizard_Apk	VERIFY_SETUP_WIZARD_APK	1	1	P/F
11.101	Verify_Subsidy_Lock_Active	VERIFY_SUBSIDY_LOCK_ACTIVE	0	0	P/F
11.102	VERIFY_SW_VERSION	VERIFY_SW_VERSION	0	0	P/F
11.103	VERIFY_TRACK_ID	VERIFY_TRACK_ID	0	0	P/F
11.104	Verify_Values_Through_Apk	VERIFY_VALUES_THROUGH_APK	0	0	P/F
11.105	VERIFY_XCVR	VERIFY_XCVR	0	0	P/F

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3.0 Common Test Procedures

3.1 In-line Flash

Test Description

Program the proper software into the UE (first time). This procedure is for UEs with no AP or BP software loaded.

Test Dependencies:

Test Setup

Power Supply: 5.8V
Current Limit 1.5 Amps

The UE supply voltage may be supplied at the accessory connector only.

Note: Flashing will be done over the USB SS lines on the USB-C connector for phone models that support it. This serves two purposes:

- 1) Decrease flashing time of SW images
- 2) Verify the functionality of the USB Super Speed Lines on the USB-C connector

Both sets of Super Speed differential paired lines must be utilized during the In-Line flash procedure. This is achieved using "Cable Flip Orientation"

Note: The Cable Flip Orientation is the ability of the USB-C connector to function if plugged in inverted of its original plug in condition. This functionality is tested by doing an automated inversion of the connector during flashing of the SW images.

Test Procedure

SWITCH_TO_FASTBOOT_FLASH_MODE

Place radio in P2K command mode.

[3.6.5 P2K Command Mode](#)

Issue the FLASH command to switch the radio into fastboot flash mode.

<u>Command</u>	<u>Opcode/data</u>	
FLASH	0x000D 01	(Fastboot flash mode)

Wait for fastboot interface to enumerate

FASTBOOT_FLASH

The software image files are a package of many code files. Each of these code files have to be flashed sequentially using the fastboot flash command. Follow the instructions provided in fastboot xml file.

Execute "fastboot flash <code file>" to flash the particular code file.



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Once all the necessary files are flashed, check for the 'OKAY' response indicating success.
If 'FAIL' is received then fail the unit.

FASTBOOT_SET_BARCODE_UTAG

Execute "fastboot oem config barcode <unit's barcode>" to program barcode into UTAG memory partition.

Check for the 'OKAY' response.
If 'FAIL' is received then fail the unit.

FASTBOOT_LOG_PERSIST_MD5

Execute "fastboot oem md5 persist" to log MD5 (Message-Digest algorithm 5) checksums.

Check for the 'OKAY' response.
If 'FAIL' is received then fail the unit.

FASTBOOT_REBOOT

Send the following command to reboot the ME

Execute "fastboot reboot"

Check for the 'OKAY' response.
If 'FAIL' is received then fail the unit.

Wait for device to enumerate as blank network interface.

AUX_ENGINE_READY_FTM_VIA_ROUTE

Place radio in P2K command mode.

[3.6.5 P2K Command Mode](#)

Send the following test command to check if BP is ready for communication.

Command	Opcode/Data
TCMD_ROUTE	0x0FF0 044B 0B00 8041 8000 (Check if BP is ready)

Ensure the response is: 0x4B 0B00 8041 80FF A1E2 7E

PROGRAM_INITIAL_FTI

Use the STELEM command to program initial factory tracking info to the radio.

Command	Opcode/data
STELEM	0X002F 2726000100000080 + FTI Data (Write initial FTI)

Where:
FTI Data = 80 byte ASCII to Hex value of data that needs to be written.

VERIFY_TRACK_ID

Use the RDELEM command to read the unit Track ID.

Command	Opcode/data
RDELEM	0X0020 2726000100000080 (Read factory track ID)

Convert the data read from the radio from hex values to ASCII characters
Compare the number string to the expected number string of the Track ID.



SWITCH_TO_FASTBOOT_P2K

Issue the FLASH command to switch the radio into fastboot flash mode.

Command	Opcode/data
---------	-------------

FLASH	0x000D 01
-------	-----------

 (Fastboot flash mode)

Wait for fastboot interface to enumerate

Exit Test

3.2 Turn On

Test Description

Turn on the UE and verify that the turn-on initialization completes successfully.

Equipment:

Test controller (PC) with USB communications port.

Test Setup

Battery Power Supply Setup:

Voltage	See section 1.6.7 Supply Voltage
Current Limit	3 Amps
Output	Off

Accessory Power Supply Setup:

Voltage	See section 1.6.7 Supply Voltage
Current Limit	1.5 Amps
Output	Off

UE Setup:

Connect the Battery power supply output to the UE battery connector.
Connect the test system USB port to the UE USB interface.

Note: Unit must have the bootloader already configured for factory mode before Turn On Test.

Test Procedure

Monitor the UE supply current continuously throughout the following steps. Disable the power supply output if the UE supply current exceeds 1.0 Amp for longer than 0.5 seconds.

USB-C Connector - Board Level Method:

Apply Nominal Battery Voltage (See section 1.6.7) to Battery contacts.
Apply 5.6K resistor to Battery Thermistor line.
Connect USB HS D+ and USB HS D- to PC USB D+ and D- lines.
Apply Nominal Accessory Voltage (See section 1.6.7) to VBUS line via Charger Power Supply.
Monitor the USB enumeration status.
Verify that the UE enumerates and the Test Command Interface is present.
Measure the current on the Battery Power Supply.
Verify that the measured current is within the limits specified in section 2.1 Test Limits.

Use the PWR UP CHK command to verify if a pre-defined checkpoint in the device's power-up process has been reached.

Command	Opcode/data
---------	-------------

Command	Response
PWR UP CHK 0x009A 00	(Power Up Checkpoint Query)

Poll up to 120 seconds for the one byte test command response.

Verify that the one byte test command response is 0x01 (which means the checkpoint was reached).

Send the following test command to force phone current draw from the Charger Supply to the Battery Power Supply.



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Command	Opcode/data
BATT	0x0085 0601

Use the following command to disable the General Purpose Single Color LED #1 – WHITE SOC LED.

Command	Opcode/data
LEDS	0x003E 2000

 (Disable General Purpose Single Color LED #1 – WHITE SOC LED)

Radio is now turned on for factory use.

USB-C Connector - Assembled Level Method :

Connect USB HS D+ and USB-HS D- to PC USB D+ and D- lines

Apply Nominal Accessory Voltage (See section 1.6.7) to VBUS line via Charger Power Supply or PC

Monitor the USB enumeration status.

Verify that the UE enumerates and the Test Command Interface is present

Radio is now in factory mode

Use the PWR_UP_CHK command to verify if a pre-defined checkpoint in the device's power-up process has been reached.

Command	Opcode/data
PWR_UP_CHK	0x009A 00

 (Power Up Checkpoint Query)

Poll up to 120 seconds for the one byte test command response.

Verify that the one byte test command response is 0x01 (which means the checkpoint was reached).

Use the following command to disable the General Purpose Single Color LED #1 – WHITE SOC LED.

Command	Opcode/data
LEDS	0x003E 2000

 (Disable General Purpose Single Color LED #1 – WHITE SOC LED)

Exit Test

None



3.3 On Current Drain

Test Description

Measure the UE current drain while it is in a known operating state.

Equipment:
Power Supply

Test Dependencies:

[3.2 Turn On](#)

[3.6 Test Mode](#)

Test Setup

The UE must be in test mode as described in previous procedures. See test dependencies above.

Test Procedure

Measure the UE supply current.

Verify that the measured current is within the limits specified in section [2.1 Test Limits](#).

Exit Test



3.4 Turn Off

Test Description

Verify that the UE turns off completely.

Equipment:

Power Supply

Test Dependencies:

[3.2 Turn On](#)

1) Must be running off Battery Power (Switch to Battery)

Test Setup

Power Supply Setup:

Voltage	See section 1.6.7 Supply Voltage
Current Limit	2 Amps

UE Setup:

Connect the power supply output to the UE battery connector if testing at Board Test.

Test Procedure

Note the UE must be running off Battery Power Supply as indicated in test dependencies and is restated here for emphasis.

Send the following command to disable the RC (Resistor-Capacitor) oscillator calibration.

Command	Opcode/data
CTRL_RC_CAL	0x0010 0000

The following method should be used to turn the UE off.

PWR_OFF Command On/Off:

Use the PWR_OFF command to power down the UE.

Command	Opcode/data
PWR_OFF	0x001E (Power down)

Check for USB De-enumeration.

Disable VBUS +/- lines via Charger (Accessory) Power Supply.

Verification may be done by reading the UE current drain and checking it against the Off Current Drain.

Check against the TURN OFF limit in section [2.1 Test Limits](#).

Exit Test

Turn Off Charger (Accessory) Power Supply.



3.5 Off Current Drain

Test Description

Verify that the UE Off-Current Drain is within the specified tolerance.

Test Dependencies:

[3.4 Turn Off](#)

Equipment:

Power Supply

Test Setup

Power Supply Setup:

Voltage	See section 1.6.7 Supply Voltage
Current Limit	2 Amps
Output	ON

UE Setup:

The power supply for the off current measurements must be connected to the UE battery connector.

Test Procedure

If the UE is on, then turn the UE off as described in section [3.4 Turn Off](#).

Disconnect the Accessory power supply +/- lines powering VBUS from the radio VBUS +/- lines and apply a 100 ohm resistor from radio VBUS + to radio VBUS - within 2 seconds.

Monitor the UE supply current for up to 5 seconds until it drops to within the limit specified in section [2.1 Test Limits](#).
Record this value as **Off Current Drain**.

Exit Test



3.6 Test Mode

Test Description

For this platform, it is possible to switch between FTM and P2K suspend modes, however in order to do so, you must first put radio into FTM mode using section 3.6.1 then put radio into P2K suspend mode using sections 3.6.5 If the radio is put into P2K suspend mode first, it will not go into FTM mode.

Test Dependencies:

[3.2 Turn On](#)

Test Setup

Connect the test system USB port to the UE accessory connector.

Test Procedure

3.6.1 FTM Mode

Use the ChangeMode command to put the phone into FTM mode.

Command	Opcode/data	
CHANGEMODE	0x29 0300	(Enter FTM mode)

Once the radio has been changed to FTM mode, the radio must be put back into AMSS mode following the procedure in Section 3.6.2 to make a phone call. If FTM mode is left set, when you power cycle the radio, the radio will not be able to make a phone call.

3.6.2 DMSS/ONLINE Mode

Use the ChangeMode command with data 0x0400 to put the phone into DMSS/ONLINE mode.

Command	Opcode/data	
CHANGEMODE	0x29 0400	(Enter DMSS/ONLINE mode)

3.6.3 LOW POWER Mode

Use the ChangeMode command with data 0x0500 to put the phone into LOW POWER mode.

Command	Opcode/data	
CHANGEMODE	0x29 0500	(Enter LOW POWER mode)

3.6.4 Verify QCOM Mode

Send the following QCOM command to get the phone's QCOM mode status: 4B0F0000BB607E.

Read byte 8 of the returned data to get the QCOM mode.

0x01 = FTM Mode

0x04 = OFFLINE Mode

0x05 = ONLINE Mode

0x06 = LOW POWER Mode

Example:

If return data is 4B0F0000000000006, UE is in Low Power Mode

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3.6.5 P2K Command Mode

The P2K test commands are in the same format as P2K products, described in the Platform 2000 Test Commands document. Not all P2K test commands are available. See 1X P2K command document for details on which commands are supported.

Power up the phone. (Radio Defaults to 19.2 kbps baud rate)

Send raw data (ATMODE=8), or raw command data 0x41544D4F44453D380D, to the radio to put the radio in test command mode. (Note: if you are using the USB transport to send test commands you do not need to send the AT+MODE=8 command)

Now the radio should accept P2K test commands just like any other P2K unit. The radios baud rate remains at 19.2kbps baud rate.

3.6.6 P2K SUSPEND Mode

Follow the procedure in Section 3.6.5 to put the radio into P2K command mode.

Once in P2K command mode, use the SUSPEND command to place the UE into test mode.

Command	Opcode/data	
SUSPEND	0x0036 00	(Enter test mode)

Read the P2K05 test command response from the SUSPEND command.
Verify that the SUSPEND response data = 0x00.

Exit Test

3.7 Software Version

3.7.1 P2K TCMD Method

Test Description

The purpose of this test is to verify the BP bootloader version and AP software version (as applicable).

Test Dependencies:

[3.6 Test Mode](#)

Test Setup

N/A

Test Procedure

Use the VERSION command to read the software version from the UE.

Command	Opcode/data	
VERSION	0x0039 FFFF	(Read release label)

Command	Opcode/data	
VERSION	0x0039 1200	(Read BP Bootloader version)

Read the VERSION responses from the UE.

Verify that the reported release label is correct for the current model and build profile.

Exit Test

None

3.7.2 APK CommServer Method

Test Description

The purpose of this test is to verify the Software version(s) programmed in to the UE (as applicable).

Test Dependencies:

[3.2 Turn On](#)

Test Setup

UE must have the CommServer running and communication establish over BLAN Port 2631

Test Procedure

Use the VERSION Activity command to read the software version from the UE.

Command	Activity	
START	VERSION	(start the Version test activity)

Command	Activity	Data	Data	
TELL	VERSION	GET_VERSION	DISPLAY BOOTLOADER	(Read the Version info)

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Read the VERSION responses from the UE.

<u>Command</u>	<u>Activity</u>	
STOP	VERSION	(stop Version test activity)

Verify that the reported release label is correct for the current model and build profile.

Exit Test

None

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3.8 HW And Revision Identifier Test

Test Description

The purpose of this test is to verify the GPIO components of the HW Revision Identifier part. The sw kernel uses the GPIO information.

Test dependencies:

[3.2 Turn On](#)

[3.6.6 P2K SUSPEND Mode](#)

Test Setup

None

Test Procedure

Send the following test command to read the Device Product Name.

<u>Command</u>	<u>Opcode/data</u>	
VERSION	0x0039 1202	(Read Device Product Name)

Read byte 0 of the response to determine the byte length of the Hex Product Name string that follows after byte 0. Convert the Hex Product String to Ascii and record as **HW Device Product Name**.

Example of a possible HW Device Product Name response is the following:

Test Command Response Example: 0761646469736F6E

07 for 7 bytes to follow, and 61646469736F6E converts (from hex to ascii) to 'addison'.

Send the following test command to read the HW Revision Version.

<u>Command</u>	<u>Opcode/data</u>	
VERSION	0x0039 1203	(Read HW Revision Version)

Read byte 0 of the response to determine the byte length of the Hex HW Revision Version that follows after byte 0.

Record the Hex HW Revision Version as **HW Revision Version**.

Examples of some possible HW Revision Version responses are the following (where 8 represent the letter 'P'):

8000 (P000) for P0 build
8100 (P100) for P1 build
8101 (P101) for P1.1 build
81A0 (P1A0) for P1A build
81AD (P1AD) for P1AD build
8200 (P200) for P2 build

Send the following UTAG test command to read the HW Radio RF SKU ID_UTAG hex representation of ASCII string from UTAG memory partition.

<u>Command</u>	<u>Opcode/data</u>	
UTAG	0x0108 0000 1B	(Read HW Radio RF SKU ID UTAG)

Convert the data return from hex values to ASCII characters.

Where:

The response format is two bytes indicating the number of bytes in RF SKU ID UTAG followed by the actual HW Radio RF SKU.

Example:

Return data = 0003 524F57



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0x0003 - data length (number of bytes)
0x524F57 - ASCII "ROW"

Verify that **HW Device Product Name**, **HW Revision Version** and **HW Radio RF SKU ID UTAG** is within the limits specified in section [2.1 Test Limits](#).

Exit Test

None



3.9 Panic Test

Test Description

This test checks for panic codes stored in the phone. These codes could be present for several reasons: the UE may have panicked during an earlier test and is now being retested; the UE may have panicked during an earlier power down or power up; or a panic code could have been in the simulator used to generate the monster file (this applies to new software builds only).

Panics are usually caused by software bugs, but they may also be caused by process issues. For example, a SIM card that is not securely installed may come loose during testing, which can generate a SIM card contact panic.

Panic codes that appear very infrequently may not be worth investigating. The software team must be able to reproduce the panic and take a log to fix it. If there is no known way to reproduce the issue, and it does not occur often enough to be a significant impact on factory operation, then resources can probably be spent more productively elsewhere.

NOTE 1: THIS TEST IS TO BE DONE ONLY AT PROTO FACTORY TESTING. THIS TEST IS NOT INTENDED FOR HIGH VOLUME.

Test Dependencies:

[3.2 Turn On](#)

[3.6 Test Mode](#)

Radio Setup

N/A

Test Procedure

Retrieve Panic Data

Use the PANIC command to read the total number of panics and panic types of the UE.

Command	Opcode/data
PANIC	0x006E 0000

(Read the total number of panics and panic types)

Read the Panic data returned in the PANIC response. Verify that bytes 0-1 (counting from 0) of the data returned (PANIC_DT) are within specification (see section 2). Bytes 0-1 are the total number of panics found on the UUT. Any response for bytes 0-1 other than 0x0000 indicates that at least one Panic message exists **and the test fails**.

The panic response is understood as follows:

Bytes [0-1]: Total Panic Count

Bytes [2-3]: Panic Type 1

Bytes [4-5]: Panic Count 1 (1-n)

.

.

.

Bytes [2 bytes]: Panic Type x

Bytes [2 bytes]: Panic Count x (1-n)

(with there being a possibility of 4 Panic Types and the Panic Count being 0 based)

Panic Type	Amount of Panics Possible	Max bytes requested to retrieve data
0x0000 (BP Panic)	Multi	1024
0x0001 (AP Panic)	Multi	1024
0x0003 (Kernel Panic)	Multi	1024



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0x0004 (Java Panic)	Multi	1024
---------------------	-------	------

If bytes 0-1 are not within specifications (not 0x0000), use the PANIC command to log the panic data for all panics to a file for failure analysis **and fail the radio**.

NOTE: The number of times this command needs to be issued depends on how many and what type of panics were found using the previous command above.

Example 1: (2 Total Panics = All AP Panics)

Command	Opcode/data
---------	-------------

PANIC	0x006E 0002 0001 0000 0000 0000 0000 0400 (The data in bold represents AP Panic Count 0)
-------	---

PANIC	0x006E 0002 0001 0001 0000 0000 0000 0400 (The data in bold represents AP Panic Count 1)
-------	---

Example 2: (3 Total Panics = 1 BP Panic (Count 0) and 2 AP Panics (Count 0 and Count 1))

Command	Opcode/data
---------	-------------

PANIC	0x006E 0002 0000 0000 0000 0000 0000 01F4 (This command retrieves BP Panic Count 0)
-------	---

PANIC	0x006E 0002 0001 0000 0000 0000 0000 0400 (The data in bold represents AP Panic Count 0)
-------	---

PANIC	0x006E 0002 0001 0001 0000 0000 0000 0400 (The data in bold represents AP Panic Count 1)
-------	---

NOTE: The data requested in these examples for each panic is 1024 bytes. However, if data for a particular panic is greater than max bytes requested the PANIC command will need to be requested again and the offset updated. The following is an example for an AP panic:

Example 3: (1 Total Panic = 1 AP Panic (Count 0), Panic Data = 2157 bytes – the amount of bytes will be unknown but is used for example purposes)

Command	Opcode/data
---------	-------------

PANIC	0x006E 0002 0000 0000 0000 0000 0000 0400 (Retrieves 1024/2157 bytes)
-------	---

PANIC	0x006E 0002 0000 0000 0000 0400 0000 0400 (Retrieves 2048/2157 bytes)
-------	---

PANIC	0x006E 0002 0000 0000 0000 0800 0000 0400 (Retrieves 2157/2157 bytes)
-------	---

Erasing Panics

To erase all AP, BP, Kernel, and Java Panics send the following test command.

Command	Opcode/data
---------	-------------

PANIC	0x006E 0003 FFFF (Erase all AP, BP, Kernel, and Java Panics)
-------	--

Exit Test

The PANIC codes should be analyzed by the factory test engineers, PTEG, and the software development team. After a reasonable sample of radios has been produced, the logged panic codes should be investigated. Starting with the most frequent panic codes, the cause of each panic should be determined using the full panic data, analysis of the history of each failed radio, and the panic code itself.

After the cause of a panic is known, if the development team and/or PTEG determines the panic is not critical, that panic code may be added to an exception list. If a panic code is found that is on the exception list, the panic block should be erased and the UE is not a failure. Examples of panic codes that may be added to the exception list are panics determined to be caused by process issues and not radio faults; and panic codes caused by software issues where a fix is known but not integrated into the current manufacturing software version. *The exception list must be started over for each new product introduction.*

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After analyzing all panic codes seen, the development team, factory test engineers, and PTEG may jointly determine that the PANIC test can be moved to a prototype-only state. If this is done, the PANIC test should still be done when testing new software versions for acceptance.



3.10 USIM/SIM Card Interface

3.10.1 FTM TCMD Method

Test Description

This test verifies the card reader circuitry and also verifies the SIM Tray mechanical switch. It must be performed for each Smart Card interface (E-commerce products frequently have two).

Test Dependencies:

[3.6.1 FTM Mode](#)

Test Setup

Either a Test USIM card or a GSM Test SIM card may be used for this test.

Test Procedure

Note: The following steps may be performed at different test stations in any order, to facilitate factory process flow optimization.

Card Detection:

With the card/cards inserted, use the FTM_RAW_DATA commands to read the insertion state of the SIM/USIM card. Send the following FTM RAW DATA command to use UIMDIAG_MMGSDI_CLIENT_ID_AND EVT_REG_CMD to retrieve the Client ID

Command	Opcode/data
FTM_RAW_DATA	0x802100FA00000000 (Retrieve the client ID)

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 16,17,18,19 from the second response. Make this the 4 byte Client ID HIGH.

Send the following FTM RAW DATA command to use UIMDIAG_MMGSDI_GET_ALL_AVAILABLE_APPS_CMD for sim detect.

Command	Opcode/data
FTM_RAW_DATA	0x802148FA00000000 + Four byte Client ID HIGH (sim detect command)

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 12,13,14,15 from the second response and confirm it is 0x00000000. Then extract bytes 16,17,18,19 and confirm the 4byte value is greater than 0 which indicates a SIM card is present. Byte-swap the order of this 4byte value and call it NumApps. To confirm which card slot is detected the rest of the data is to be split into 200 byte chunks. The number of 200 byte chunks is equal NumApps. In each 200 byte chunk extract bytes 4, 5, 6, 7 and confirm this 4byte value is either 01000000 for slot 1 or 02000000 for slot 2. If the 4 bytes from each 200 byte chunk are all 01000000 then card is detected in slot 1. If the 4 bytes from each 200 byte chunk are all 02000000 then the card is detected in slot 2. If the 4 bytes from each 200 byte chunk are 01000000 and 02000000 then cards are detected in both slots 1 and 2.

Example of response when sim present in 1st slot :

[illegible]

Example of response when sims present in both 1st slot and second slot:

[illegible]

[illegible]

Send the following FTM RAW DATA command to use UIMDIAG_MMGSDI_CLIENT_ID_AND_EVT_DEREG_CMD to de-register the client.

Command	Opcode/data	
FTM_RAW_DATA	0x802101FA00000000 + Four byte Client ID HIGH	(de-register command)

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response to clear the buffer

With the card absent, use the FTM RAW DATA commands to read the insertion state of the SIM/USIM card.

Send the following FTM RAW DATA command to use UIMDIAG_MMGSDI_CLIENT_ID_AND EVT_REG_CMD to retrieve the Client ID

Command	Opcode/data
FTM RAW DATA	0x802100FA00000000 (Retrieve the client ID)

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 16,17,18,19 from the second response. Make this the 4 byte Client ID HIGH.

Send the following FTM RAW DATA command to use UIMDIAG_MMGSDI_GET_ALL_AVAILABLE_APPS_CMD for sim detect.

Command	Opcode/data
FTM RAW DATA	0x802148FA00000000 + Four byte Client ID HIGH (sim detect command)

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 12,13,14,15 from the second response and confirm it is 0x0B000000 indicating SIM Card not detected

Send the following FTM RAW DATA command to use UIMDIAG_MMGSDI_CLIENT_ID_AND_EVT_DEREG_CMD to de-register the client.

Command	Opcode/data
FTM RAW DATA	0x802101FA00000000 + Four byte Client ID HIGH (de-register command)

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response to clear the buffer

Verify P/F based on the limits specified in section 2.1 Test Limits.

Manual Method (Optional)

Insert a USIM card into the UUT and upon powering up, verify that the display is not showing 'Insert SIM' or 'Check SIM'.

Without a USIM card inserted into the UUT upon powering up, verify that the display is showing 'Insert SIM' or 'SIM Not Inserted' or 'Check SIM'.

Verify P/F based on the limits specified in section 2.1 Test Limits.



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Mechanical Switch Test

Insert a USIM card into the SIM Tray of the UUT before powering the UUT up.
Send the following test command to read the state of the USIM Mechanical Switch.

Command	Opcode/data
FTM_RAW_DATA	0x4B0B 0080 3347 0000 0000 (Read State of USIM Mechanical Switch)

Verify that byte 8 of the test command response is 0x00 and that **Mech Switch State – Tray Inserted** is within the limits specified in section [2.1 Test Limits](#).

Example Response: 0x4B0B 0080 3347 0000 **00**00 C03A 7E where byte 8 is the underlined and bolded byte.

Make sure there is not a SIM Tray card placed into the UUT (or it is in the ejected state) before powering the UUT up.
Send the following test command to read the state of the USIM Mechanical Switch GPIO.

Command	Opcode/data
FTM_RAW_DATA	0x4B0B 0080 3347 0000 0000 (Read State of USIM Mechanical Switch)

Verify that byte 8 of the test command response is 0x01 and that **Mech Switch State – Tray Removed** is within the limits specified in section [2.1 Test Limits](#).

Example Response: 0x4B0B 0080 3347 0000 **01**00 1823 7E where byte 8 is the underlined and bolded byte.

Exit Test

None

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3.10.2 APK CommServer Method

Test Description

This test verifies the card reader circuitry. It must be performed for each Smart Card interface (E-commerce products frequently have two).

Test Dependencies:

[3.2 Turn On](#) (Assembled Level)

[3.6.5 P2K Command Mode](#)

Test Setup

Either a Test USIM card or a GSM Test SIM card may be used for this test.

Send the following test command to set the preferred network mode to zero on the AP.

Command	Opcode/data	
PREF_MODE	0x0076 0000	(Set AP Side PREF_MODE to 0 (ie. 0x00))

Use the ChangeMode command with data 0x0400 to put the phone into DMSS/ONLINE mode.

Command	Opcode/data	
CHANGEMODE	0x29 0400	(Enter DMSS/ONLINE mode)

Execute EFS Sync

Refer to section [EFS_SYNC](#)

Note: AP_Pref Mode, DMSS/Online mode, and EFS Sync should be executed at the start of the test recipe on the current test station, no power cycles are required.

Test Procedure

Note: The following steps may be performed at different test stations in any order, to facilitate factory process flow optimization. If this is done, the SIM Card activity must be started and stopped at each station. If the activity is started and more SIM card tests are to be done then stopping is not necessary until complete, power off also stops activity.

Card Detection:

With the card absent, use the SIM_CARD command to read the insertion state of the SIM/USIM card.

Command	Activity	Data	
START	SIM_CARD		(Starts the SIM card Activity)
TELL	SIM_CARD	GET_SIM_CARD_STATUS	(Return Status of card interface)

Verify within the response SIM_STATE= ABSENT

With the card inserted, use the SIM_CARD command to verify connectivity of the card interface.

Command	Activity	Data	
START	SIM_CARD		(Starts the SIM card Activity)
TELL	SIM_CARD	GET_SIM_CARD_STATUS	(Return Status of card interface)

Verify within the response SIM_STATE= READY

Verify P/F based on the limits specified in section [2.1 Test Limits](#).

2nd SIM Slot for DUAL SIM Models ONLY

With the card absent from 2nd slot, use the SIM_CARD command to read the insertion state of the SIM/USIM card.

Command	Activity	Data	
START	SIM_CARD		(Starts the SIM card Activity)



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TELL SIM_CARD GET_SIM_CARD_STATUS (Return Status of card interface)
Verify within the response SIM_2_STATE= ABSENT

With the card inserted in the 2nd slot, use the SIM_CARD command to verify connectivity of the card interface.

Command	Activity	Data	
START	SIM_CARD		(Starts the SIM card Activity)
TELL	SIM_CARD	GET_SIM_CARD_STATUS	(Return Status of card interface)

Verify within the response SIM_2_STATE= READY

Verify P/F based on the limits specified in section [2.1 Test Limits](#)

Exit Test

Command	Activity	Data	
STOP	SIM_CARD		(Stops the SIM card Activity)



3.11 LPDDR3 Test

Test Description

The purpose of this test is to detect that the LPDDR3 memory part is placed and verify its vendor ID, and capacity.

Test Dependencies:

[3.2 Turn On](#)

[3.6 Test Mode](#) (SUSPEND)

Test Setup

N/A

Test Procedure

Note 1: This procedure may be performed at different test stations to facilitate factory test flow optimization.

Note 2: Limits for this test may differ per product. Please consult the limits table to see which limits apply to which products.

Use the TST_MSD command to read the LPDDR3 vendor ID, memory type, and capacity.

Command	Opcode/data
---------	-------------

TST_MSD	0x0C1B 0007 04
---------	----------------

 (Read LPDDR3 vendor ID, memory type, and capacity)

Convert the hexadecimal test command response to ASCII format.

Record and log the vendor ID only and not compare against any limits. We should NOT fail for vendor ID.

Recording or logging of memory type is **NOT** required.

However, make sure the memory capacity is correct per the limits table.

Example Test Command Response: Hynix:S8 SDRAM:1024MB

Where Hynix = Vendor ID, S8 SDRAM = memory type, and 1024MB = capacity.

Record the ASCII response for memory capacity as **LPDDR3 Memory Capacity**.

Verify that **LPDDR3 Memory Capacity** is within the limits specified in section [2.1 Test Limits](#).

Use the TST_MSD command to read LPDDR3 Revision ID1 Register.

Command	Opcode/data
---------	-------------

TST_MSD	0x0C1B 0007 02
---------	----------------

 (Read LPDDR3 Revision ID1 Register)

Convert the hexadecimal test command response to ASCII format.

Record and log the ASCII response as **LPDDR3 Revision ID1 Register**.

NOTE: This information is to be LOGGED only by request and not compared against any limits. This information has been proven to be beneficial to development engineers when looking at LPDDR3 part failures.

Use the TST_MSD command to read LPDDR3 Revision ID2 Register.

Command	Opcode/data
---------	-------------

TST_MSD	0x0C1B 0007 03
---------	----------------

 (Read LPDDR3 Revision ID2 Register)

Convert the hexadecimal test command response to ASCII format.

Record and log the ASCII response as **LPDDR3 Revision ID2 Register**.

NOTE: This information is to be LOGGED only by request and not compared against any limits. This information has been proven to be beneficial to development engineers when looking at LPDDR3 part failures.

Exit Test

None

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3.12 eMMC Test

Test Description

The purpose of this test is to verify connectivity of the eMMC.

Test Dependencies:

[3.2 Turn On](#)

[3.6 Test Mode \(Suspend\)](#)

Test Setup

N/A

Test Procedure

Note 1: This procedure may be performed at different test stations to facilitate factory test flow optimization.

Note 2: Limits for this test may differ per product. Please consult the limits table to see which limits apply to which products.

Use the TST_MSD command to detect eMMC.

<u>Command</u>	<u>Opcode/data</u>	
TST_MSD	0x0C1B 0003 0005	(detect eMMC)

Read the TST_MSD response.

Verify that the 1st byte of the response = 0x01 (for device connected). If it not true, then the unit fails.

Verify that the 2nd byte of the response = 0x00 (for MMC / eMMC device type). If it not true, then the unit fails.

Verify that the 3rd thru 6th bytes of the response = capacity in kbytes. Bytes 3-6 (Example: 0XXXXXXXX) should be converted to an integer and should be equal to the number of kilobytes of the eMMC. If this is not true, then the unit fails.

Use the TST_MSD command to read the eMMC Vendor ID.

<u>Command</u>	<u>Opcode/data</u>	
TST_MSD	0x0C1B 0008 00	(Read eMMC Vendor ID)

Record and log the 1st byte of the test command response (first byte being the MSB – most significant byte) as **eMMC Vendor ID**.

NOTE: Do not fail for eMMC Vendor ID. Please just log the data. The reason for this is that there could be multiple vendors used on that same product.

Exit Test

None

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3.13 External Memory Card Interface

3.13.1 P2K TCMD Method

Test Description

The purpose of this test is to verify connectivity of the memory card connector.

Test Dependencies:

[3.6 Test Mode](#)

Test Setup

Perform this test procedure on models that support the Tri-Flash interface.
Insert a Tri-Flash memory card into the UE.

Test Procedure

Note: This procedure may be performed at different test stations to facilitate factory test flow optimization.

Card Detect:

With card absent (*for external types only*):

Use the TST_MSD command to verify that there is no card detected.

Command	Opcode/Data
TST_MSD	0x0C1B 0003

(Detect card status)

Verify response = 0x00

With card inserted (*for external and internal types*):

Use the TST_MSD command to verify the card insertion detection.

Command	Opcode/Data
TST_MSD	0x0C1B 0003 0000

(Detect card status)

Verify that the first byte of the TST_MSD response = 0x01 (indicates that the device is connected.)

Parse the second byte to be the device type and the next four bytes to be the capacity of device in Kbytes. Please record and log the responses accordingly, for development reference.

Exit Test

Remove the Tri-Flash card from the UE.

**3.13.2 APK CommServer Method****Test Description**

The purpose of this test is to verify connectivity of the memory card connector.

Test Dependencies:

[3.2 Turn On](#) (Assembled Level)

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631

Perform this test procedure on models that support a removable Tri-Flash interface.

Test Procedure

Note: This procedure may be performed at different test stations to facilitate factory test flow optimization.

Removable Card Detect:

With card inserted:

Use the MMC APK command to verify the card insertion detection.

Command	Activity	Data	
START	MMC		(Start MMC card Activity)
TELL	MMC	GET_STORAGE_VOLUME_STATUS REMOVABLE	(Detect card status)

Verify within the response:

STORAGE_VOLUME_1_MOUNTED=mounted

STORAGE_VOLUME_1_SIZE=<Card memory size in bytes>

Unmount the removable card

Command	Activity	Data	
TELL	MMC	UNMOUNT_VOLUME REMOVABLE	(removes MMC Card from file system)
TELL	MMC	GET_STORAGE_VOLUME_STATUS REMOVABLE	(Detect card status)

Verify within the response:

STORAGE_VOLUME_1_MOUNTED=unmounted

STORAGE_VOLUME_1_SIZE=NOT_AVAILABLE

Remove MMC Card

Use the MMC APK command to verify that there is no card detected.

Command	Activity	Data	
START	MMC		(Start MMC card Activity)
TELL	MMC	GET_STORAGE_VOLUME_STATUS REMOVABLE	(Detect card status)

Verify within the response:

STORAGE_VOLUME_1_MOUNTED=removed

STORAGE_VOLUME_1_SIZE=NOT_AVAILABLE

Exit Test

Command	Activity	Data	
STOP	MMC		(Stop MMC Activity)



3.14 Camera Auto Focus Test - External

APK CommServer Method

Test Description

The purpose of this test is to verify the assembly and functionality of the external camera.

Test Dependencies:

The UE must be powered from the accessory connector.

The UE must be fully assembled.

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631

Setup for the Still Image Quality test:

Light box to provide controlled lighting for Reduced Focus Test Chart version 1060.22 at a simulated infinity distance in JOT.

Test Procedure

This procedure may be combined with other camera testing as needed to support factory flow optimization.

Note: It is advised that the phone must be in the focus test fixture prior to enabling the camera viewfinder.

Note: The commands for the "Camera Parameters" are case sensitive and should be entered as given in this document.

Camera Still Image Quality:

Position the camera with the appropriate camera lens facing directly towards the chart.

Use the CAMERA_FACTORY command to enable the external camera viewfinder.

If the CAMERA_FACTORY Activity has already been started, it does not need to be started again (skip the start command).

Start Activity

Command	Activity
START	CAMERA_FACTORY

Wait 1.5 seconds

Start External Viewfinder

Command	Activity	Data
TELL	CAMERA_FACTORY	OPEN_CAMERA CAMERA_FACING_BACK

Wait 0.5 second

Setup camera parameters to use during focus test with auto focus enabled

For external camera

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-enable-factorymode=1

Wait 0.15 second

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-manuallenspos=disable

Wait 0.15 second



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For 13MP Only

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_CAMERA_PARAMETERS picture-size=4160x3120 focus-mode=auto exposure-compensation=-1

For 16MP Only

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_CAMERA_PARAMETERS picture-size=4608x3456 focus-mode=auto exposure-compensation=-1

Wait 0.5 second

Capture Image

Command	Activity/Data	
TELL	CAMERA_FACTORY	TAKE_PICTURE

Read In Focus Score

Command	Activity/Data	
TELL	CAMERA_FACTORY	GET_FACTORY_CAMERA_PARAMETERS get-afscore

Read "Android Get In Focus Score" value from phone.

Verify that the "Android Get In Focus Score" result is within the limits specified in section [2.1 Test Limits](#).

Move Camera Lens To Macro

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_CAMERA_PARAMETERS focus-mode=fixed

Command	Activity/Data	
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-manuallenspos=enable,100

Delay for lens movement

Wait up to 1 second for the external camera.

Capture Image To Update Focus Score

Use the CAMERA FACTORY TAKE PICTURE command to force Auto Focus Update in Phone.

Command	Activity	
TELL	CAMERA_FACTORY	TAKE_PICTURE

Read Out of Focus Score

Command	Activity/data	
TELL	CAMERA_FACTORY	GET_FACTORY_CAMERA_PARAMETERS get-afscore

Read "Android Get Out of Focus Score" value from phone.

Verify that the "Android Get Out of Focus Score" result is within the limits specified in section [2.1 Test Limits](#).

Calculate the Focus Score Quotient from "Android Get In Focus Score" and "Android Get Out of Focus Score".
"Camera Focus Score Quotient" = "Android Get In Focus Score" / "Android Get Out of Focus Score"

Verify that the "Camera Focus Score Delta" result is within the limits specified in section [2.1 Test Limits](#).

Exit Test

Turn OFF Viewfinder

Command	Activity/Data
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TELL CAMERA_FACTORY RELEASE_CAMERA

Stop Activity

Command	Activity
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STOP	CAMERA_FACTORY
------	----------------

3.15 Internal Camera Focus Test

APK CommServer Method

Test Description

The purpose of this test is to verify the assembly and functionality of the internal camera focus.

Test Dependencies:

[3.2 Turn On](#) (Assembled Level)

The UE must be powered from the accessory connector.

The UE must be fully assembled.

Test Setup

UE must have the CommServer running and communication establish over BLAN Port 2631

Setup for the test using JOT Vision Camera bracket as target.

Test Procedure

Start Activity

Command	Activity
START	CAMERA_FACTORY

Wait 1.5 seconds

For Internal Viewfinder

Open Viewfinder

Command	Activity	Data
TELL	CAMERA_FACTORY	OPEN_CAMERA CAMERA_FACING_FRONT

Wait 0.5 seconds

Capture Image and Send to PC

Command	Activity	Data
TELL	CAMERA_FACTORY	TAKE_PICTURE SEND_FILE

Analyze Image

Capture the internal JOT box image with the JOT front blemish light on using the phone front camera. Analyze the area of the image that has the vision camera mounting bracket looking for high frequency changes in the area. Calculate the pseudo focus score. Record the result as CAMERA_INT_FOCUS_SCORE.

Verify that the results are within limits specified in section [2.1 Test Limits](#) .

Exit Test

Turn OFF Viewfinder

Command	Activity	Data
TELL	CAMERA_FACTORY	RELEASE_CAMERA

Stop Activity

Command	Activity
STOP	CAMERA_FACTORY

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3.16 External Camera Blemish-and Relative Uniformity Test

APK CommServer Method

Test Description

The purpose of this test is to verify the assembly and functionality of the external camera. Watermarks, spots or uneven luminance must not exceed the acceptable limits as defined in [2.1 Test Limits](#)

Test Dependencies:

[3.2 Turn On](#) (Assembled Level)

The UE must be powered from the accessory connector.

The UE must be fully assembled.

Test Setup

UE must have the CommServer running and communication establish over BLAN Port 2631

Setup for the Camera Blemish test with the diffused light source:

Use opal diffuser light test setup defined in 12.6 Appendix F: Camera Test Conditions. The camera will be looking at the diffused light in the fixture for test target.

For External Imager,

Blemish Analysis Inputs:

BLEMISH_SIZE_1_REG1_MAX_AREA = 15 pixels

BLEMISH_SIZE_1_REG2_MAX_AREA = 15 pixels

BLEMISH_THRESHOLD = 25 %

ENHANCED_RU_TEST = YES

NUMBER_OF_HORIZONTAL_ANALYSIS_REGIONS = 32

NUMBER_OF_VERTICAL_ANALYSIS_REGIONS = 24

LINE_DETECT_HIGH_PASS_FILTER = 16

PERFORM_LINE_DETECT_TEST = YES

DARK_REGION_HIGH_PASS_FILTER = 4

REGION_1_MAX_NOISE_SIZE = 5

REGION_1_PIXEL_NOISE_THRESHOLD = 100

REGION_2_MAX_NOISE_SIZE = 5

REGION_2_PIXEL_NOISE_THRESHOLD = 100

Test Procedure

Camera Blemish:

Position the camera with the appropriate camera lens facing directly towards diffused light source.

Note: The commands for the "Camera Parameters" are case sensitive and should be entered as given in this document.

Start Activity

Command	Activity
START	CAMERA_FACTORY

Wait 1.5 seconds

For External Viewfinder



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Open Viewfinder

Command	Activity	Data
TELL	CAMERA_FACTORY	OPEN_CAMERA CAMERA_FACING_BACK

Wait 0.5 second

Setup camera parameters to use during blemish test

FOR 13MP Only

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_CAMERA_PARAMETERS picture-size=4160x3120 exposure-compensation=-1

FOR 16MP Only

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_CAMERA_PARAMETERS picture-size=4608x3456 exposure-compensation=-1

Wait 0.15 second

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_CAMERA_PARAMETERS focus-mode=fixed
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-manuallenspos=enable,100

Wait 0.5 second

Capture Image and Send to PC

Command	Activity	Data
TELL	CAMERA_FACTORY	TAKE_PICTURE SEND_FILE

Analyze Image

Run Nextest Blemish Test Suite Software to analyze the captured JPEG file using image analysis settings specified in Appendix based on image resolution, diagonal field of view, and test chart being used. Verify that the results are within limits specified in section [2.1 Test Limits](#).

Exit Test

Turn OFF Viewfinder

Command	Activity	Data
TELL	CAMERA_FACTORY	RELEASE_CAMERA

Stop Activity

Command	Activity
STOP	CAMERA_FACTORY

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3.17 Internal Camera Blemish-and Relative Uniformity Test

APK CommServer Method

Test Description

The purpose of this test is to verify the assembly and functionality of the internal camera. Watermarks, spots or uneven luminance must not exceed the acceptable limits as defined in [2.1 Test Limits](#)

Test Dependencies:

[3.2 Turn On](#) (Assembled Level)

The UE must be powered from the accessory connector.

The UE must be fully assembled.

Test Setup

UE must have the CommServer running and communication establish over BLAN Port 2631

Setup for the Camera Blemish test with the diffused light source:

Use opal diffuser light test setup defined in [12.6 Appendix F: Camera Test Conditions](#). The camera will be looking at the diffused light in the fixture for test target.

For Internal Imager,

Blemish Analysis Inputs:

BLEMISH_SIZE_1_REG1_MAX_AREA = 15 pixels

BLEMISH_SIZE_1_REG2_MAX_AREA = 15 pixels

BLEMISH_THRESHOLD = 25 %

NUMBER_OF_HORIZONTAL_ANALYSIS_REGIONS = 32

NUMBER_OF_VERTICAL_ANALYSIS_REGIONS = 24

LINE_DETECT_HIGH_PASS_FILTER = 16

LINE_DETECT_TEST = YES

DARK_REGION_HIGH_PASS_FILTER = 4

REGION_1_MAX_NOISE_SIZE = 5

REGION_1_PIXEL_NOISE_THRESHOLD = 100

REGION_2_MAX_NOISE_SIZE = 5

REGION_2_PIXEL_NOISE_THRESHOLD = 100

Test Procedure

Camera Blemish:

Position the camera with the appropriate camera lens facing directly towards diffused light source.

Note: The commands for the "Camera Parameters" are case sensitive and should be entered as given in this document.

Start Activity

Command	Activity
START	CAMERA_FACTORY

Wait 1.5 seconds

For Internal Viewfinder

Open Viewfinder

Command	Activity	Data
TELL	CAMERA_FACTORY	OPEN_CAMERA CAMERA_FACING_FRONT

Wait 0.5 seconds

**Setup camera parameters to use during blemish test**

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_CAMERA_PARAMETERS picture-size=2592x1944 exposure-compensation=-2

Wait 0.5 second

Capture Image and Send to PC

Command	Activity	Data
TELL	CAMERA_FACTORY	TAKE_PICTURE SEND_FILE

Analyze Image

Run Nextest Blemish Test Suite Software to analyze the captured JPEG file using image analysis settings specified in Appendix based on image resolution, diagonal field of view, and test chart being used.

Verify that the results are within limits specified in section [2.1 Test Limits](#) .

Exit Test

Turn OFF Viewfinder

Command	Activity	Data
TELL	CAMERA_FACTORY	RELEASE_CAMERA

Stop Activity

Command	Activity
STOP	CAMERA_FACTORY

**3.18 Camera APK – Read # Of Cameras****Test Description**

The purpose of this test is to verify the assembly and functionality of the external camera.

Test Dependencies:

The UE must be powered from the accessory connector.

The UE must be fully assembled.

Test Setup

NA

Test Procedure

This procedure may be combined with other camera testing as needed to support factory flow optimization.

Note: It is advised that the phone must be in the focus test fixture prior to enabling the camera viewfinder.

Start Activity

Command	Activity
START	CAMERA_FACTORY

Read Number Of Available Cameras

Command	Activity	Data
TELL	CAMERA_FACTORY	GET_NUMBER_CAMERA

returned value would be NUMBER_OF_CAMERA=x

Verify that the result "x" is within the limits specified in section 2.1 Test Limits.

Stop Activity

Command	Activity
STOP	CAMERA_FACTORY

Exit Test



3.19 Camera Calibration Status Test

Test Description

The purpose of this test is to read out the data stored in the Camera Flash EEPROM to validate it was programmed.

Test Dependencies:

The UE must be powered from the accessory connector.

The UE must be fully assembled.

Test Setup

NA

Test Procedure

This procedure may be combined with other camera testing as needed to support factory flow optimization.

Start Activity

Command	Activity
START	CAMERA_FACTORY

Delay 1.5 sec

3.19.1 External Camera

Select Rear Facing Camera

Command	Activity	Data
TELL	CAMERA_FACTORY OPEN_CAMERA	CAMERA_FACING_BACK

Delay 0.5 sec

Read Internal Camera Calibration Status

Command	Activity	Data
TELL	CAMERA_FACTORY GET_FACTORY_CAMERA_PARAMETERS	get-calibrationstatus

Ensure that get-calibrationstatus=pass. Any other response is a failure.

Turn OFF Viewfinder

Command	Activity	Data
TELL	CAMERA_FACTORY RELEASE_CAMERA	

3.19.2 Internal Camera

Select Front Facing Camera

Command	Activity	Data
TELL	CAMERA_FACTORY OPEN_CAMERA	CAMERA_FACING_FRONT

Delay 0.5 sec

Read External Camera Calibration Status

Command	Activity	Data
TELL	CAMERA_FACTORY GET_FACTORY_CAMERA_PARAMETERS	get-calibrationstatus

Ensure that get-calibrationstatus=pass. Any other response is a failure.



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Turn OFF Viewfinder

Command	Activity	Data
TELL	CAMERA_FACTORY	RELEASE_CAMERA

Stop Activity

Command	Activity
STOP	CAMERA_FACTORY

Exit Test



3.20 Camera LED Calibration Text – External

Test Description

This test verifies the physical connection of the two Flash LEDs for the rear imager to the Flash Driver. Each LED will be calibrated separately in Strobe mode. Data will be stored to PDS.

Test Dependencies:

[3.2 Turn On](#)(Factory Cable Method – Assembled Level)

[3.6.5 P2K Command Mode](#)

NOTE: The UE must be fully assembled.

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631
This is accomplished by powering up the UUT with a factory cable.

Position the Unit Under Test (UUT) in a dark environment with the flash LED and camera exposed to a calibrated light pipe in which the strobe can illuminate it.

Test Procedure

Command	Activity
START	CAMERA_FACTORY

Wait 1.5 seconds

Command	Activity	Data
TELL	CAMERA_FACTORY	OPEN_CAMERA CAMERA_FACING_BACK

Wait 0.5 second

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_CAMERA_PARAMETERS focus-mode=infinity flash-mode=on

Set up LED 1 to test

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-strobe-currents=1000,0

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-enable-ledcal=1,wr:1,exp:0.00007,gain:1.0,exp_idx:92,lp-rg:1.0,lp-bg:1.0

Wait 100ms

Command	Activity	Data
TELL	CAMERA_FACTORY	TAKE_PICTURE IMAGE_ANALYSIS_OPTIONS=CalcImageLuminance

Verify that **LED1_avgLum** returned from the UE is within the limits specified under the limits specification section [2.1 Test Limits](#).

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-enable-ledcal=0

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Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-strobe-currents=0,1000

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-enable-ledcal=1,wr:1,exp:0.00009,gain:1,exp_idx:97,lp-rg:1.0,lp-bg:1.0

Wait 100ms

Command	Activity	Data
TELL	CAMERA_FACTORY	TAKE_PICTURE IMAGE_ANALYSIS_OPTIONS=CalcImageLuminance

Verify that **LED2_avgLum** returned from the UE is within the limits specified under the limits specification section [2.1 Test Limits](#).

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-enable-ledcal=0

Set up Both LED 1 and LED 2 to test

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-strobe-currents=750,750

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-enable-ledcal=1,wr:1,exp:0.000001,gain:1,exp_idx:75,lp-rg:1.0,lp-bg:1.0

Command	Activity	Data
TELL	CAMERA_FACTORY	TAKE_PICTURE IMAGE_ANALYSIS_OPTIONS=CalcImageLuminance

Verify that **Both_LED_avgLum** returned from the UE is within the limits specified under the limits specification section [2.1 Test Limits](#).

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-enable-ledcal=0

Command	Activity	Data
TELL	CAMERA_FACTORY	GET_FACTORY_CAMERA_PARAMETERS get-ledcal-pds

calvalid, sn,led1 1000, led2 0, rg_ratio, bg_ratio, led1 0, led2 1000, rg_ratio, bg_ratio, led1 750, led2 750, rg_ratio, and bg_ratio are returned by the UE. Record these as **LED1_RG_RATIO**, **LED1_BG_RATIO**, **LED2_RG_RATIO**, **LED2_BG_RATIO**, **BOTH_LED_RG_RATIO** and **BOTH_LED_BG_RATIO**.

Turn OFF Viewfinder

Command	Activity	Data
TELL	CAMERA_FACTORY	RELEASE_CAMERA

Stop Activity

Command	Activity
STOP	CAMERA_FACTORY

Exit Test



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3.21 Camera LED Calibration Test – Internal

Test Description

This test verifies the physical connection of the Flash LED for the front imager to the Flash Driver. LED will be calibrated in Strobe mode. Data will be stored to phone memory.

Test Dependencies:

[3.2 Turn On](#) (Factory Cable Method – Assembled Level)

[3.6.5 P2K Command Mode](#)

NOTE: The UE must be fully assembled.

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631
This is accomplished by powering up the UUT with a factory cable.

Position the Unit Under Test (UUT) in a dark environment with the flash LED and camera exposed to a calibrated light pipe in which the strobe can illuminate it.

Test Procedure

Command	Activity
START	CAMERA_FACTORY

Wait 1.5 seconds

Command	Activity	Data
TELL	CAMERA_FACTORY	OPEN_CAMERA CAMERA_FACING_FRONT

Wait 0.5 second

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_CAMERA_LEDCAL_VIEWFINDER_MODE OFF

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_CAMERA_PARAMETERS flash-mode=on

Set up LED to test

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-strobe-currents=75

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-enable-ledcal=1,wr:1,exp:0.0035,gain:1,exp_idx:125,lp-rg:1.0,lp-bg:1.0

Wait 0.1 second

Command	Activity	Data
TELL	CAMERA_FACTORY	TAKE_PICTURE IMAGE_ANALYSIS_OPTIONS=CalcImageLuminance

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Verify that **FRONT_LED_AvgLum** returned from the UE is within the limits specified under the limits specification section [2.1 Test Limits](#).

Command	Activity	Data
TELL	CAMERA_FACTORY	GET_FACTORY_CAMERA_PARAMETERS get-ledcal-pds

calvalid, sn, rg_ratio, and bg_ratio are returned by the UE. Record these as **FRONT_LED_RG_RATIO** and **FRONT_LED_BG_RATIO**.

Note: sn not required to be logged.

Verify that **FRONT_LED_RG_RATIO** and **FRONT_LED_BG_RATIO** returned from the UE are within the limits specified under the limits specification section [2.1 Test Limits](#).

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_CAMERA_LEDCAL_VIEWFINDER_MODE ON

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-enable-ledcal=0

Turn OFF Viewfinder

Command	Activity	Data
TELL	CAMERA_FACTORY	RELEASE_CAMERA

Stop Activity

Command	Activity
STOP	CAMERA_FACTORY

Exit Test



3.22 Camera Laser Focus Test

Test Description

The purpose of this test is to calibrate and store the laser focus offset and validate a distance read with the laser focus module.

Test Dependencies:

3.2 Turn On

Test Setup

Laser Target: 3M diffuser film on Acrylic Clear Plastic

Offset Calibration: Requires a target distance of 50mm

Distance Measurement: Requires a target distance less than 40mm.

Test Procedure

Offset Calibration:

Read the Device Stored Offset Calibration Value

Use the FOCUS_ASSIST command to read the Laser Focus Offset value in the device.

Command	Opcode/Data
FOCUS_ASSIST	0x0111 1103

(Read Stored Focus Offset Value)

The returned data will be 4 bytes in 2's complement format. Byte swap and convert the HEX value to decimal. This will be the **Stored_Focus_Offset**.

Example: Return data = 0x2B000000. The last byte is MSB = 0x0000002B. Converted to decimal = 43

Provide the Offset Calibration Target

Use the FOCUS_ASSIST command to calibrate the Laser Focus Offset

Command	Opcode/Data
FOCUS_ASSIST	0x0111 0101

(Calibrate and Store Focus Offset)

The returned data will be 4 bytes in 2's complement format. Byte swap and convert the HEX value to decimal. This will be the **Focus_Offset**.

Verify that the **Focus_Offset** result is within the limits specified in section 2.1 Test Limits

Example: Return data = 0x2B000000. The last byte is MSB = 0x0000002B. Converted to decimal = 43

Calculate the Delta between **Stored Focus Offset** and **Focus Offset** by subtracting **Focus Offset** from **Stored Focus Offset**. This will be the **Focus Offset Delta**.

Verify that the **Stored_Focus_Offset**, **Focus_Offset**, and **Focus_Offset_Delta** results are within the limits specified in section [2.1 Test Limits](#).

Read Distance:

Provide the Distance Measurement

Use the FOCUS_ASSIST command to read the distance measured by the laser focus module.

Command	Opcode/Data
FOCUS_ASSIST	0x0111 10

(Read Distance)



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The returned data will be 4 bytes containing a 32-bit unsigned integer. Convert the HEX value to decimal. The value is represented in mm. This will be the **Focus_Distance**.

Verify that the **Focus_Distance** result is within the limits specified in section [2.1 Test Limits](#).

Exit Test

None

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3.23 Camera VSYNC to Sensor Hub Line Test

APK CommServer Method

Test Description

The purpose of this test is to verify the CAM_R_VSYNC line to sensor hub.

Test Dependencies:

3.2 Turn On

The UE must be powered from the accessory connector.

The UE must be fully assembled.

The test is to be executed after all other camera test are completed.

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631

Test Procedure

Use the CAMERA_FACTORY command to enable the external camera viewfinder.

If the CAMERA_FACTORY Activity has already been started, it does not need to be started again (skip the start command).

Start Activity

Command	Activity
START	CAMERA_FACTORY

Wait 1.5 seconds

Start External Viewfinder

Command	Activity	Data
TELL	CAMERA_FACTORY	OPEN_CAMERA CAMERA_FACING_BACK

Wait 0.5 second

Send the following command to check the connectivity of the CAM_R_VSYNC Line.

Command	Opcode/data
SENSOR_HUB	0x009F 0403

Verify that the response is a success and there is no return data in the response (an empty string). If data is returned in the response then the data is a TCMD standard error message and the unit fails the test. Verify that it is within the limits specified in section [2.1 Test Limits](#).

Exit Test

Turn OFF Viewfinder

Command	Activity/Data
TELL	CAMERA_FACTORY RELEASE_CAMERA

Stop Activity

Command	Activity
STOP	CAMERA_FACTORY

**3.24 PA On No Flash Test****Test Description**

This test verifies the physical connection of the TX line (PA ON NOFLASH) routed into the LED Flash Driver. In terms of normal mode phone operation we are checking to make sure that when the user is in a call the flash driver is limited to only torch mode and prevents flash mode.

Test Dependencies:

[3.2 Turn On](#)(Factory Cable Method – Assembled Level)

[3.6.5 P2K Command Mode](#)

NOTE: The UE must be fully assembled.

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631
This is accomplished by powering up the UUT with a factory cable.

Position the Unit Under Test (UUT) in a dark environment with the flash LED and camera exposed to an open area in which the flash can illuminate it for the camera.

Test Procedure

Send the following commands.

Command	Activity
START	CAMERA_FACTORY

Wait 1.5 seconds

Command	Activity	Data
TELL	CAMERA_FACTORY	OPEN_CAMERA CAMERA_FACING_BACK

Wait 0.5 second

Send the following command to disable RF (set unit in airplane mode)

Command	Opcode/data
SWT_RF	0x006C 01

Send the following command to configure the PA ON NOFLASH GPIO to an output.

Command	Opcode/data
TST_CAMERA	0x0061 32FF 01 0200

Wait 0.5 second

Capture Image with Camera LED set to Strobe mode

Send the following command to enable the PA ON NOFLASH GPIO HIGH

Command	Opcode/data
TST_CAMERA	0x0061 32FF 01 0002

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-flash-led-reg=00,09,04

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Note: Sets flash inhibit function to be off, so neither torch or strobe will turn on if pin is high.

Command	Activity	Data
TELL	CAMERA_FACTORY	SET_CAMERA_PARAMETERS focus-mode=infinity flash-mode=on
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-strobe-currents=750,750
TELL	CAMERA_FACTORY	SET_FACTORY_CAMERA_PARAMETERS set-enable-ledcal=1,wr:0,exp:0.000001,gain:1,exp_idx:70,lp-rg:1.0,lp-bg:1.0

Wait 0.2 second

Command	Activity	Data
TELL	CAMERA_FACTORY	TAKE_PICTURE IMAGE_ANALYSIS_OPTIONS=CalcImageLuminance

Luminance results are returned by the UE within the "avgLum" response.

Verify that the luminance results returned from the UE are within the limits specified under "EXT_LUMINANCE_PA_ON_NOFLASH_GPIO_HIGH" of the limits specification section 2.1 Test Limits.

NOTE: The luminance when the GPIO is toggled high is LOWER since when the SKY81296 is engaged in a Flash event and the TX pin is pulled high.

Send the following command to enable the PA ON NOFLASH GPIO LOW

Command	Opcode/data
TST_CAMERA	0x0061 32FF 01 0000

Command	Activity	Data
TELL	CAMERA_FACTORY	TAKE_PICTURE IMAGE_ANALYSIS_OPTIONS=CalcImageLuminance

Luminance results are returned by the UE within the "avgLum" response.

Verify that the luminance results returned from the UE are within the limits specified under "EXT_LUMINANCE_PA_ON_NOFLASH_GPIO_LOW" of the limits specification section [2.1 Test Limits](#).

Turn OFF Viewfinder

Command	Activity	Data
TELL	CAMERA_FACTORY	RELEASE_CAMERA

Stop Activity

Command	Activity
STOP	CAMERA_FACTORY

Exit Test

Send the following command to enable RF.

Command	Opcode/data
SWT_RF	0x006C 00

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3.25 Internal Battery Test

3.25.1 P2K TCMD Method

Test Description

This test verifies internal battery connection, voltage level, thermistor, and batt_id at the assembled level.

Test Dependencies:

[3.2 Turn On](#)
[3.6 Test Mode](#)

NOTE: This test is an assembled level test.

Test Setup

Send the following test commands to select the power source to be the Accessory Power Supply.

<u>Command</u>	<u>Opcode/data</u>	
BATT*	0x0085 0600	(Select the power source to be the Accessory Power Supply)

**NOTE: This command closes the USBIN to USBOUT path. This ensures better accuracy when reading the battery voltage and thermistor temperature.*

Test Procedure

Battery Thermistor (Assembled Level):

Send the following test command to read the Thermistor value in tenths of degrees celsius.

<u>Command</u>	<u>Opcode/data</u>	
TEMP	0x008F 0002 0000	(Read the Battery Thermistor value) (Schematic Ref: BATT_THERM)

Convert the four byte response from **2's complement format** to decimal. The units in decimal are returned in tenths degrees celsius. Average 8 stable A/D readings and record it as **Battery Thermistor Temp**.

Verify that **Battery Thermistor Temp** is within the limits specified in section [2.1 Test Limits](#).

Battery Connection and Level (Assembled Level):

Send the following test command to read the VBAT voltage.

<u>Command</u>	<u>Opcode/data</u>	
BATT	0x0085 04 0008	(Read Gas Gauge – Battery Voltage OCV)

Convert the two byte hex response from **2's complement format** to decimal. The units in decimal are in millivolts. Record it as **Battery Voltage OCV**.

Verify that **Battery Voltage OCV** is within the limits specified in section [2.1 Test Limits](#).

Battery ID Resistance Test (Board or Assembled Level):

Use the BATT command to read the battery ID Resistance.

<u>Command</u>	<u>Opcode/data</u>	
BATT	0x0085 04 0007	(Read Gas Gauge Batt resistance id)



4 HEX bytes of data are returned in the BATT response.

Convert the 4 bytes to decimal.

Record the decimal number of the test command response as **Battery ID Resistance** and verify that it is within the limits specified in section 2.1 Test Limits.

Exit Test**3.25.2 APK CommServer Method**

This test verifies the Internal Battery connection at the assembled level.

Test Dependencies:

[3.2 Turn On](#) (Assembled Level)

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631

Test Procedure

Send the following command to run the BATTERY PRESENT CHECK test.

Command	Activity/data
START	BATTERYINFO
TELL	BATTERYINFO GET_BATTERY_PRESENT

The correct response returned back is TRUE for the internal battery being present.
Record the P/F result as **BATTERY PRESENT CHECK**

Send the following command to run the BATTERY PLUGGED STATE test.

Command	Activity/data
TELL	BATTERYINFO GET_PLUGGED_STATE

The correct response returned back is BATTERY_PLUGGED_USB.
Record the P/F result as **BATTERY PLUGGED STATE**.

Send the following command to run the GET BATTERY VOLTAGE test.

Command	Activity/data
TELL	BATTERYINFO GET_BATTERY_VOLTAGE

The response returned is battery voltage in Volts.
Record the result as **BATTERY VOLTAGE**.

Send the following command to run the GET BATTERY LEVEL test.

Command	Activity/data
TELL	BATTERYINFO GET_BATTERY_LEVEL

The response returned is battery charge percentage.

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Record the result as **BATTERY CHARGE LEVEL**.

Verify the responses for **BATTERY PRESENT CHECK**, **BATTERY PLUGGED STATE**, **BATTERY VOLTAGE** and **BATTERY CHARGE LEVEL** are within the limits specified in Section [2.1 Test Limits](#).

Exit Test

Send the following command to stop the Battery Information Activity.

Command	Activity/data
STOP	BATTERYINFO



3.26 Battery/Charger/Thermistor Test

Test Description

The purpose of this test is to verify connectivity and functionality of the battery, charger, and thermistor circuitry.

Test dependencies:

[3.2 Turn On](#) (EMU Style Connector Turn On)

[3.6 Test Mode](#)

Equipment:

2 power supplies with an accuracy of < 0.5% or calibrated to < 0.5%

Test Setup

Make sure that the Accessory Power Supply connected to the EMU connector is set to 5.0V (per the test dependency [3.2 Turn On](#)).

Send the following test commands to select the power source to be the Accessory Power Supply.

Command	Opcode/data
BATT*	0x0085 0600

(Select the power source to be the Accessory Power Supply)

*NOTE: This command closes the USBIN to USBOUT path.

Set the Battery Power Supply connected to the battery terminals to 3.4V.

Test Procedure

Battery/Thermistor/Charger Testing:

BATTERY

NOTE: This test is order dependant. This testing must be done in the order as stated below.

Send the following test commands to turn off the Battery FET.

Command	Opcode/data
BATT	0x0085 0701

(Turn off Battery FET)

Wait 500 ms for the Accessory Power Supply to settle.

Send the following test command to read the B+ voltage.

Command	Opcode/data
AD_CONV	0x0001 000A

(Read the B+ Voltage)

Convert the two byte hex response from **2's complement format** to decimal. The units in decimal are in millivolts. Acquire a stable reading and record it as **B+ Voltage**.

Send the following test command to read the VBAT voltage.

Command	Opcode/data
AD_CONV	0x0001 000C

(Read the VBAT Voltage)

Convert the two byte hex response from **2's complement format** to decimal. The units in decimal are in millivolts. Acquire a stable reading and record it as **VBAT Voltage**.

BATTERY THERMISTOR

This test is performed in the Internal Battery Test, Section **3.24.1**



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BATTERY CHARGER

Send the following test command to turn on the Battery FET.

Command	Opcode/data	
BATT	0x0085 0700	(Turn on Battery FET)

Send the following test command to set the USB input current limit to 500mA.

Command	Opcode/data	
BATT	0085 0801 0101 F4	(Set USB input current limit to 500mA)

Send the following test command to set the Battery input current limit to 1450mA.

Command	Opcode/data	
BATT	0085 0801 0005 AA	(Set Battery input current limit to 1450mA)

Measure the current sourcing from the Accessory power supply.
Acquire a stable current reading and record it as **Input Current**.

Verify that **B+ Voltage**, **VBAT Voltage**, and **Input Current** are within the limits specified in section [2.1 Test Limits](#).

Exit Test

Send the following test commands to turn off the Battery FET.

Command	Opcode/data	
BATT	0x0085 0701	(Turn off Battery FET)

Set the Battery Power Supply voltage to 3.9V.

Send the following test command to select the power source to be the Battery Power Supply.

Command	Opcode/data	
BATT*	0x0085 0601	(Select the power source to be the Battery Power Supply)

**NOTE: This command opens the USBIN to USBOUT path.*

Set the Battery Power Supply voltage to 3.9V.

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3.27 User Reset Test

Test Description

The purpose of this test is to verify connectivity and functionality of the user reset circuit.

Test Dependencies:

[3.2 Turn On](#) (EMU Style Connector Turn On – Board Level)
[3.6.6 P2K SUSPEND Mode](#)

Equipment:
N/A

NOTE: This test can be run in NORMAL mode (meaning UNSUSPENDED) or SUSPEND mode. For factory flow purposes at board level the phone is usually in the SUSPENDED state for most testing. For this reason, if the factory would like to run this test in SUSPENDED mode they should also run section 3.6 Test Mode (P2K SUSPEND MODE) after running section 3.2 Turn On.

Test Setup

Test Procedure

Connect the BATT_RESET test point to the Nextest system.

Measure the voltage state of the BATT_RESET test point and record it as **Batt Reset Voltage State HIGH**.

Send the following test command to read WARN line.

Command	Opcode/data
RDWR_IO	0x0C25 0000 7E

Record the one byte test command response as **User Reset Before Assert**.

Send the following test command for user reset software disable.

Command	Opcode/data
RDWR_IO	0x0C25 0100 7F01

Ground the power key for 7.33 seconds.

Note that grounding the power key for 6.6 to 7.33 seconds causes the WARN line to toggle state for approximately 2.5 seconds and Battery Reset to toggle for approximately 150 ms. Polling the WARN line and Battery Reset lines should begin no later than 6 seconds after the power key has been grounded.

Send the following test command to read WARN line while the line is in toggled state.

Command	Opcode/data
RDWR_IO	0x0C25 0000 7E

Record the one byte test command response as **User Reset After Assert**.

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Immediately measure the voltage state of the BATT_RESET test point until it drops to within the limit specified in section [2.1 Test Limits](#). Record it as **Batt Reset Voltage State LOW**.

Note: *This only tests the battery reset circuit only to the battery reset test point (TP_BATT_RESET). This does not test all the way through the battery connector (J510).*

Disconnect the power key from ground.

Send the following commands to reset the User Reset IC. (Must be toggled twice to reset)

Command	Opcode/data
RDWR_IO	0x0C25 0100 8201
Command	Opcode/data
RDWR_IO	0x0C25 0100 8200
Command	Opcode/data
RDWR_IO	0x0C25 0100 8201
Command	Opcode/data
RDWR_IO	0x0C25 0100 8200

Poll with the following test command for the WARN line to return to a high state after the line is toggled.

<u>Command</u>	<u>Opcode/data</u>
RDWR_IO	0x0C25 0000 7E

Note: Once the WARN line has switched back to a high state then Software User Reset can be enabled. This process should not take longer than 3 seconds.

Send the following test command to enable software for user reset.

<u>Command</u>	<u>Opcode/data</u>
RDWR_IO	0x0C25 0100 7F00

Verify that **Batt Reset Voltage State HIGH**, **Batt Reset Voltage State LOW**, **User Reset Before Assert**, and **User Reset After Assert** are within the limits specified in section [2.1 Test Limits](#).

Exit Test

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3.28 USB PMIC Line Check Test

Test Description

The purpose of this test is to verify connectivity and functionality of the USB lines at the PMIC.

Test dependencies:

[3.2 Turn On](#)

[3.6 Test Mode](#)

Test Setup

Must be able to control the USB port D+ and D- lines and be able to short them together.

Test Procedure

Send the following test command to disable the software factory kill.

Command	Opcode/data	
RDWR_IO	0x0C25 0100 8001	(DISABLE_SW_FACTORY_KILL)

Send the following test command to enable APSD.

Command	Opcode/data	
BATT	0x0085 1000	(ENABLE_APSD)

Delay until phone re-enumerates. Phone will briefly de-enumerate and then re-enumerate.

Send the following test command to read the status of the Charger circuit register DCP.

Command	Opcode/data
RDWR_REG	0x010F 0000 0002 1608

A 4 byte response will be returned. Record bit 6 of the fourth byte of the test command response as **No Charger Attached** and verify against the limits specified in section [2.1 Test Limits](#).

Send the following test command to run the DCP detection batch file.

Command	Opcode/data	
BATCH	0x0C2F 0000 0000 0001 + "DCPDetection.bat" + NULL	(in Unicode)

Ensure a successful start response (0x0000) for the BATCH command.

The following steps need to be executed with in 4 second of receiving the successful start response from the batch command.

Disable and Disconnect the Accessory Power Supply to the UE and apply a 10 ohm resistor from radio VBUS + to radio VBUS-

Disconnect the USB D+/D- lines from the hosting computer

Note: USB enumeration will go away

Short the D+/D- USB lines together on the UE side

Connect and enable Accessory Power Supply to the UE

Hold this state for the remainder of the time stated above plus 2 second, giving the batch file enough time to read the necessary register.

Disable and disconnect the accessory power supply to the UE and apply a 10 ohm resistor from radio VBUS + to radio VBUS -



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Un-short the D+/D- USB lines and connect them the host computer
Remove the 10 ohm resistor from radio VBUS + on radio VBUS
Connect and enable the Accessory Power Supply to the UE
Poll for enumeration

Send the following BATCH test command to run the DCP detection read file to get DCP register status.

Command	Opcode/data
BATCH	0x0C2F 0000 0000 0001 + "DCPDetectionReadFile.bat" + NULL (in Unicode)

Ensure a successful start response (0x0000) for the BATCH command. Also, ensure that 2 unsolicited responses are returned

The first unsolicited response will contain the DCP register status data.

Record the status of bit 6 of the fourth byte of the unsolicited response as **Charger Attached** and verify against the limits specified in section [2.1 Test Limits](#).

Ensure the second unsolicited response has a value of 0x8000 0000 which indicates that the BATCH command is complete and successful.

Note: If error detected, error information will be returned in the batch start return or unsolicited response.

Exit Test

Send the following test command to disable APSD.

Command	Opcode/data
BATT	0x0085 1001 (DISABLE_APSD)

Delay until phone re-enumerates. Phone will briefly de-enumerate and then re-enumerate.

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3.29 USB-C HS Reverse Plug In Test

Test Description

Verify that the UE enumerates when USB-C cable orientation is flipped.

Test Dependencies:

[3.2 Turn On](#)

Test Setup

UE Setup:

UE is connected to the test bench with USB-C and enumerated.

Test Procedure

Reverse the orientation of the USB-C connector either by manually inverting the cable connected to the UE or by switching the High Speed USB lines mechanically with the test bench fixture.

Verify the UE is enumerated. Fail the unit if it does not enumerate.

Check against the USB-C HS Reverse Plug In Test limit in section [2.1 Test Limits](#).

Exit Test

None

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3.30 USB-C Power Cable /Accessory Test

Test Description

The purpose of this test procedure is to verify the VCONN and power source connections to the USB-C connector. This will be done by routing VCONN voltage to the CC1 or CC2 line depending on connector insertion position. The voltage will be measured externally by the NexTest System.

Test Dependencies:

[3.2 Turn On](#)

Test Setup

N/A

Test Procedure

Send the following command to enable VCONN voltage on the CC line.

Command	Opcode/data	
HOST_MODE	0x0018 02	(Enable VCONN)

Read the VCONN voltage externally with the Nextest System and store the value as **USBC_CC_Line_Voltage**

Send the following command to disable VCONN voltage on the CC line.

Command	Opcode/data	
HOST_MODE	0x0018 03	(Disable VCONN)

Verify that the **USBC_CC_Line_Voltage** is within the limits specified in section [2.1 Test Limits](#).

Exit Test

None

3.31 Bluetooth

3.31.1 P2K TCMD Method

Test Description

The purpose of this test procedure is to verify that the Bluetooth module is connected and functioning properly.

Test Channels			Frequency	Site
	Decimal	Hex		
Channel 0	0	0	2402 MHz	All Sites
Channel 39	39	27	2442 MHz	GNPO Only
Channel 78	78	4E	2480 MHz	GNPO Only

3.31.1

Test Dependencies:

[3.6.6 P2K SUSPEND Mode](#)

Equipment:

Bluetooth Test Set or equivalent

Test Setup

Bluetooth test set setup:

Center frequency	Channel ± 5 MHz
Span	0 or lowest span setting for the particular test set in use
RBW	1MHz
Sweep	Single sweep

Connect the Bluetooth test set to an antenna fixture approx 1" away from the UE.

For the Bluetooth Connection Test, the Bluetooth RF path must be padded with enough attenuation such that the Bluetooth signal from the UE to the test set is 15 dB (± 5 dB) above the measured input sensitivity level of the test set.

Note: This path attenuation may not be suitable for Bluetooth Power measurements.

If the Bluetooth Connection, BER or Power tests are to be performed, the RF path must be free from interfering signals in the 2400 to 2483.5 MHz frequency range. WLAN channels 3 through 9 will cause interference in this frequency range. A minimum of 80 dB of attenuation on the interferer is required for these tests.

The following tests in the procedure must be executed in the order listed.

Test Procedure

Command	Opcode/data	
BT	0x0042 96	(Power off Bluetooth IC)
Command	Opcode/data	
BT	0x0042 94	(Start Bluetooth Test Firmware)

Bluetooth CW Power Out:



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Send the following command to set the BT chip to Continuous TX Mode.

Command	Opcode/data
BT	0x0042 12 + 1byte ChannelInHex (Continuous TX Mode at specified channel)

Measure the power at specified frequency and record this value as **BT CW Power**. Verify that **Bluetooth CW Power** is within the limits specified in section [2.1 Test Limits](#).

NOTE: If multiple channels are to be tested for Bluetooth CW Power Out, the entire sequence of test commands starting with "Power Off Bluetooth IC" must be performed for each channel tested.

Send the following command to disable the BT chip

Command	Opcode/data
BT	0x0042 96 (Power off Bluetooth IC)

BT DUT Mode Test (Optional for Factory):

Send the following command to enable the BT chip.

Command	Opcode/data
BT	0x0042 94 (Start Bluetooth Test Firmware)

NOTE: If this command has been sent previously and the BT chip has NOT been disabled, there is NO need to resend this command. If the phone has been power cycled then this command must be sent once again to enable the BT chip.

Send the following command to reset the BT module.

Command	Opcode/data
BT	0x0042 13 (reset BT module)

Send the following command to set DUT Mode.

Command	Opcode/data
BT	0x0042 14 (set DUT Mode)

NOTE: Allow up to 5 seconds to establish the connection and verify that a connection is established successfully.

At this point, we should be able to identify the Bluetooth MAC address via the Bluetooth test set once the connection is set up successfully. It should be a 6 byte address and it is NOT 0x000000000000 or 0xFFFFFFFFFFFF.

Measure TX Power while the device is connected. Record this value (**Bluetooth TX Power**)

Measure BER while the device is connected. Record this value (**Bluetooth RX BER**)

Send the following command to reset the BT module.

Command	Opcode/data
BT	0x0042 13 (reset BT module)

Send the following command to disable the BT chip

Command	Opcode/data
BT	0x0042 96 (Power off Bluetooth IC)

Exit Test

None



3.32 Bluetooth RSSI Test

Test Description

The purpose of this test procedure is to verify that the Bluetooth module is connected and functioning properly.

Test Channels			Frequency	Site
	Decimal	Hex		
Channel 0	0	0	2402 MHz	All Sites
Channel 39	39	27	2442 MHz	GNPO Only
Channel 78	78	4E	2480 MHz	GNPO Only

3.31.1

Test Dependencies:

[3.6.1 FTM Mode](#)[3.6.5 P2K Command Mode](#)

Equipment:

R&S CMW500 with ARB or Agilent EXT with ARB

Test Setup

This test uses both the FTM and Motorola test command interfaces.

Connect the test set to Bluetooth antenna port of the UE.

The following tests in the procedure must be executed in the order listed.

Note: description all other FTM commands used in this procedure can be found in Qualcomm HCI Vendor Specific Interface Control Document 80-VE132-7 Rev. F)

Test Procedure

Playback RF ARB (*BT_DH5_BD539CBD359C*) input signal at -70 dBm RF_TEST_LEVEL.

Send the following Motorola P2K test command to reset Bluetooth module.

Command	Opcode/data
---------	-------------

BT	0x0042 94	(Start Bluetooth Test Firmware)
----	-----------	---------------------------------

Bluetooth RX RSSI Test:

Send the following Motorola P2K test command to enable FTM daemon and FTM command functionality.

<u>Command</u>	<u>Opcode/data</u>	
FTM_PORT_ENABLE	0x0019 01	(Enable FTM daemon)

Retrieve ID Ranges, Get the Mask, and Set the Mask are used in order to enable DIAG logging (BT log code 0x1366)

Switch from Motorola test command interface to FTM test command interface.

Command	Opcode/data	
FTM TEST_LOGGING_REQUEST	0x7300 0000 0100 0000	(Retrieve ID Ranges)

Command	Opcode/data	
FTM_TEST_LOGGING_REQUEST	0x7300 0000 0400 0000 0100 0000	(Get Log Mask)

[illegible]

Command	Opcode/data
QC_FTM_BT_PROD_TEST_SUBCOMMAND_TEST_RX_BURST_CONFIG (Configure receiver)	0x4B 0B 01 04FC 03 07 000

Where,

- 0x04FC – Byte swapped op code (0xFC04)
- 0x03 – Parameter Total Length
- 0x07 – Command op code
- 0x0000 – Number of packets (range 0x0000 to 0xFFFF, 0x0000 = Rx burst continually on)

Command	Opcode/data
QC_FTM_BT_PROD_TEST_SUBCOMMAND_TEST_RX_BURST	0x4B0B 040000001900000001 04FC 15 06
28 28 28 28 04 0F 00 09 01 9C35BD9C35BD 00 5301 00	(Configure receiver)

Where,

- 0x04FC – Byte swapped op code (0xFC04)
- 0x15 – Parameter Total Length
- 0x06 – Command op code
- 0x28 – Five hopping channels used if limited hopping (All 5 channels are set to channel 40 (decimal) in this example)
- 0x04 – This value is ignored
- 0x0F – Packet type (0x0F = DH5 packets)
- 0x00 – Data whitening
- 0x09 – Transmit output power (0x00 to 0x09, 0x00 = lowest power, 0x09 = highest power for BTS4025, QSC/QTR, WCN2243 and WCN3660)
- 0x01 – Receiver Gain (0x01 – high gain)
- 0x9C35BD9C35BD – 6 bytes, byte swapped Bluetooth target device address (*BD539CBD359C*)

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0x00 – Hopping functionality (0 = device hops over all five hop channels, 1 = device starts at specified hop channel (0), then hops over the full range of channels)
0x5301 – Byte swapped payload length (0x0153 = 339 packets)
0x00 – Logical transport address

Send the following FTM command to retrieve RSSI values from the DUT. Please note that there will be a standard test command response followed by the un-solicited response. Please capture the unsolicited response and parse it as follows, beginning with 0xFFA3 and ending with RSSI value for channel 5.

Command	Opcode/data
QC FTM_BT_PROD_TEST_SUBCOMMAND_TEST_STATS	0x4B0B04000000005000000001 04FC 01 02 (retrieves accumulated test statistics on received packets and the number of errors)

Where,
0xFC04 – Event Code
0x01 – Length of the rest of the command
0x01 – Command opcode

Unsolicited response format

FF A3 04 5301	
03000000 03000000 00000000 00000000 00000000 00000000 00000000 B3FFFFFF	Channel 1 statistics
03000000 03000000 00000000 00000000 00000000 00000000 00000000 B3FFFFFF	Channel 2 statistics
03000000 03000000 00000000 00000000 00000000 00000000 00000000 B3FFFFFF	Channel 3 statistics
02000000 02000000 00000000 00000000 00000000 00000000 00000000 B3FFFFFF	Channel 4 statistics
02000000 02000000 00000000 00000000 00000000 00000000 00000000 B3FFFFFF	Channel 5 statistics

Where,
0xFF - Vendor-specific event
0xA3 - Length of remaining data
0x5301 - Byte swapped length of received packets (0x0153 = 339 packets)

Channels statistics:

- 1st 4 bytes - Number of packets received (no access code error)
- 2nd 4 bytes - Number of packet access code errors (not received)
- 3rd 4 bytes - Number of received packets with HEC errors
- 4th 4 bytes - Number of received packets with CRC errors
- 5th 4 bytes - Packet BIT error count
- 6th 4 bytes - Bit errors found in first half of payload
- 7th 4 bytes - Bit errors found in last half of payload
- 8th 4 bytes – Byte swapped average RSSI reading in 2's complement format (0xFFFFF3B3 represents -77 dBm)

Send the following command to disable log Bluetooth Log Mask (0x1366)

Command

Opcode/data

```
FTM_TEST_LOGGING_REQUEST 0x7300 0000 0300 0000 0100 0000 A408 0000 0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 4000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0090
35 7E                                     (Set Log Mask)
```

Parse RSSI power and BER at specified frequency and record this value as BT RSSI Power and BT BER.

Verify that Bluetooth RSSI Power and BT BER are within the limits specified in section [2.1 Test Limits](#).

NOTE: If multiple channels are to be tested for Bluetooth RSSI Power Out, the entire sequence of test commands starting with "Start Bluetooth Firmware" must be performed for each channel tested.

Exit Test

Switch from FTM test command interface to Motorola test command interface.

Send the following Motorola P2k test command to disable FTM daemon and FTM command functionality.

Command

Opcode/data

FTM PORT ENABLE

0x0019 02

(Disable FTM daemon)



3.33 WLAN Radiated Power

Test Description

The purpose of this test procedure is to verify that the WLAN module is connected and functioning properly.

WLAN Radiated Power							
802.11 Protocol	Band	Test Channels		Frequency	Duty Cycle	Data Rate	Site
b	2.4 GHz	Channel A	1	2412MHz	High	11 Mbps	GNPO Only
b	2.4GHz	Channel B	6	2437MHz	High	11 Mbps	All Sites
b	2.4 GHz	Channel C	13	2472MHz	High	11 Mbps	GNPO Only
a	5.0 GHz	Channel D	36	5180 MHz	High	6 Mbps	GNPO Only
a	5.0 GHz	Channel E	149	5745 MHz	High	6 Mbps	All Sites
a	5.0 GHz	Channel F	165	5825 MHz	High	6 Mbps	GNPO Only

Table 3.33

Test dependencies:

[3.6.1 FTM Mode](#)

[3.6.5 P2K Command Mode](#)

Equipment:

Power Meter or Spectrum Analyzer that supports up to 6GHz.

Test Setup

This test uses both the FTM and Motorola test command interfaces.

WLAN test set setup:

Channel: Channel A, D (B, C, D, E and F if needed depending on the site) (see Table 3.33)

Connect the Power Meter or Spectrum Analyzer to an antenna fixture approx 1" away from the UUT.

The WLAN RF path must not generate signals in the 2422 to 2453MHz frequency range (BT channels 20 through 51) if Bluetooth testing is taking place in the adjacent fixture as this range will cause interference. If you must use a channel in this range, the RF signal at the Bluetooth station must not be greater than -145 dB.

Test Procedure

The test should start in Motorola test command interface mode (P2K test command interface).

Send the following command to download the test firmware

Command Opcode/data

WLAN 0x0C1F 0001 0000 (Download Test Firmware)

Switch from Motorola test command interface to FTM test command interface.

2.4 GHz Max-index Power Out Testing (For 802.11b Test Channels)

Send the following command to start WLAN in FTM mode

Command Opcode/data

FTM_WLAN_GEN6_START 0x4B0B 1600 0000 0E00 0000 0200 0100

Wait 1 second for the module to enable.

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Measure the UUT supply current (**WLAN TX Off**).

Send the following command to set the RF channel (XX is channel number in Hex)

Command	Opcode/data
FTM_WLAN_GEN6_SET_CHANNEL	0x4B0B 1600 0000 1E00 0000 0200 0300 5030 1000 0000 0000 XX00 0000 0000 0000

Send the following command to set the internal 8960/36x0 configuration

Command	Opcode/data
FTM_WLAN_GEN6_ENABLE_CHAINS	0x4B0B 1600 0000 1A00 0000 0200 0300 4F30 0C00 0000 0000 0300 0000

Send the following command to enable SCPC mode.

Command	Opcode/data
FTM_WLAN_GEN6_OPEN_TPC_LOOP	0x4B0B 1600 0000 1A00 0000 0200 0300 C932 0C00 0000 0000 0100 0000

Send the following command to set the data rate, preamble, frame number, spacing and payload size (11Mbps, high duty cycle).

Command	Opcode/data
FTM_WLAN_GEN6_SET_TX_FRAME	0x4B0B 1600 0000 4200 0000 0200 0300 8130 3400 0000 0000 0000 0000 1400 0000 0300 0000 0200 0000 A00F A501 2222 5555 3333 2222 4444 3333 2222 AAAA 3333 0001 0400 0000

Send the following command to set TX Power (19dBm)

Command	Opcode/data
FTM_PTT_MSG_SET_TX_POWER	0x4B0B 1600 0000 1A00 0000 0200 0300 A530 0C00 0000 0000 6C07 0000

Send the following command to turn on TX

Command	Opcode/data
FTM_WLAN_GEN6_TX_PKT_START_STOP	0x4B0B 1600 0000 1A00 0000 0200 0300 8230 0C00 0000 0000 0100 0000

Measure the power at corresponding frequency from Table 3.33.

NOTE: On a spectrum analyzer you may have to use a peak or max hold type of measurement at the channel center frequency since the modulated signal may be still bursting.

Record this value as **WLAN 2.4GHZ Modulated Power**.

Verify that **WLAN 2.4GHZ Modulated Power** is within the limits specified in section [2.1 Test Limits](#).

Measure the UUT supply current (**WLAN TX On**).

Calculate the WLAN current drain:

WLAN TX Current = WLAN TX On - WLAN TX Off

Verify that the calculated **WLAN TX Current** is within the limit specified in Section [2.1 Test Limits](#).

Send the following command to turn off TX

Command	Opcode/data
FTM_WLAN_GEN6_TX_PKT_START_STOP	0x4B0B 1600 0000 1A00 0000 0200 0300 8230 0C00 0000 0000 0000 0000

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Repeat the above commands to perform the WLAN TX Power and Current on all test channels for the 2.4 GHz band in the Test Channel Table 3.33 (based on the site).

Send the following command to power down the WLAN module

Command	Opcode/data
FTM_WLAN_GEN6_STOP	0x4B0B 1600 0000 0E00 0000 0200 0200

Measure the UUT supply current (**WLAN Module Off**)

Calculate the WLAN Module Current drain:

WLAN Module Current drain = WLAN TX Off - WLAN Module Off

Verify that the calculated **WLAN Module Current drain** is within the limit specified in section [2.1 Test Limits](#).

*5.0 GHz Open-Loop Power Out testing

NOTE: The WLAN module needs to be turned on and FTM enabled before the following commands are executed.

Measure the UUT supply current (**WLAN TX Off**).

Send the following command to start WLAN in FTM mode

Command	Opcode/data
FTM_WLAN_GEN6_START	0x4B0B 1600 0000 0E00 0000 0200 0100

Send the following command to set the internal 8960Pro/WCN36X0 configuration

Command	Opcode/data
FTM_WLAN_GEN6_ENABLE_CHAINS	0x4B0B 1600 0000 1A00 0000 0200 0300 4F30 0C00 0000 0000 0300 0000

Send the following command to set the RF channel (XX is channel number in Hex)

Command	Opcode/data
FTM_WLAN_GEN6_SET_CHANNEL	0x4B0B 1600 0000 1E00 0000 0200 0300 5030 1000 0000 0000 XX00 0000 0000 0000

Send the following command to enable SCPC mode.

Command	Opcode/data
FTM_WLAN_GEN6_OPEN_TPC_LOOP	0x4B0B 1600 0000 1A00 0000 0200 0300 C932 0C00 0000 0000 0100 0000

Send the following command to set the PTT_MSG_SET_TX_POWER power target to **15dBm** for expected power for 5.0 GHz 6Mbps.

Command	Opcode/data
FTM_PTT_MSG_SET_TX_POWER	0x4B0B 1600 0000 1A00 0000 0200 0300 A530 0C00 0000 0000 DC05 0000

Send the following command to set TX FRAME for 6Mbps High duty cycle

Command	Opcode/data
FTM_WLAN_GEN6_SET_TX_FRAME	0x4B0B 1600 0000 4200 0000 0200 0300 8130 3400 0000 0000 0000 0000 1400 0000 0700 0000 0200 0000 A00F A501 2222 5555 3333 2222 4444 3333 2222 AAAA 3333 0001 0000 0000

Send the following command to turn on TX

Command	Opcode/data
FTM_WLAN_GEN6_TX_PKT_START_STOP	0x4B0B 1600 0000 1A00 0000 0200 0300 8230 0C00 0000 0000 0100 0000

Measure the power at corresponding frequency from Table 3.33.

NOTE: On a spectrum analyzer you may have to use a peak or max hold type of measurement at the channel center frequency since the modulated signal may be still bursting.



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Record this value as **WLAN 5GHZ Modulated Power**.

Verify that **WLAN 5GHZ Modulated Power** is within the limits specified in section [2.1 Test Limits](#)

Note: Wait for the FTM command to complete and the current to settle before taking a current measurement.

Measure the UUT supply current (**WLAN TX On**).

Calculate the WLAN current drain:

WLAN TX Current = WLAN TX On - WLAN TX Off

Verify that the calculated **WLAN TX Current** is within the limit specified in Section [2.1 Test Limits](#).

Send the following command to turn off TX

Command	Opcode/data
FTM_WLAN_GEN6_TX_PKT_START_STOP	0x4B0B 1600 0000 1A00 0000 0200 0300 8230 0C00 0000 0000 0000 0000

Repeat the above commands to perform the WLAN TX Power on all test channels for the 5.0 GHz band in the Test Channel Table 3.33. (based on the site).

Send the following command to power down the WLAN module

Command	Opcode/data
FTM_WLAN_GEN6_STOP	0x4B0B 1600 0000 0E00 0000 0200 0200

Exit Test



3.34 WLAN RSSI Test

Test Description

The purpose of this test procedure is to verify that the WLAN module is connected and functioning properly over the RX path.

WLAN RSSI TEST							
802.11 Protocol	Band	Test Channels		Frequency	Duty Cycle	Data Rate	Site
g	2.4 GHz	Channel	1	2412 MHz	High	54 Mbps	GNPO Only
g	2.4GHz	Channel	6	2437 MHz	High	54 Mbps	All Sites
g	2.4 GHz	Channel	11	2462 MHz	High	54 Mbps	GNPO Only
Table 3.34							

Test dependencies:

[3.6.1 FTM Mode](#)

[3.6.5 P2K Command Mode](#)

Equipment:

R&S CMW500 with ARB or Agilent EXT with ARB

Test Setup

This test uses both the FTM and Motorola test command interfaces.

Apply the WLAN RF input signal at a level that will apply -65 dBm at the channel(s) specified in Table 3.34 at the UUT antenna. The fixture antenna must be approx 1" away from the UUT.

Test Procedure

The test should start in Motorola test command interface mode (P2K test command interface).

Send the following command to download the test firmware

Command	Opcode/data
WLAN	0x0C1F 0001 0000 (Download Test Firmware)

Switch from Motorola test command interface to FTM test command interface.

Send the following command to start WLAN in FTM mode

Command	Opcode/data
FTM_WLAN_GEN6_START	0x4B0B 1600 0000 0E00 0000 0200 0100

Wait 1 second for the module to enable.

Send the following command to set the internal 8960/36x0 configuration

Command	Opcode/data
FTM_WLAN_GEN6_ENABLE_CHAINS	0x4B0B 1600 0000 1A00 0000 0200 0300 4F30 0C00 0000 0000 0200 0000



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Send the following command to set the RF channel (**XX** is channel number in Hex)

Command	Opcode/data
FTM_WLAN_GEN6_SET_CHANNEL	0x4B0B 1600 0000 1E00 0000 0200 0300 5030 1000 0000 0000 XX 00 0000 0000 0000

Send the following command to get the WLAN receiver RSSI value.

Command	Opcode/data
QLIB_FTM_WLAN_GEN6_GET_RX_RSSI_VAL	0x4B0B 1600 0000 1700 0000 0200 0300 D330 0C00 0900 0000 0000 00

Parse the RSSI response returned in 2's compliment. Take 3 receiver measurements and average them. Record this value as **WLAN_CHXX_RSSI**.

Ensure that the **WLAN_CHXX_RSSI** value meets the limits specified in Section [2.1 Test Limits](#) for the product being tested.

Repeat the above commands to perform the WLAN TX Power and Current on all test channels for the 2.4 GHz band in the Test Channel Table 3.34 (based on the site).

Send the following command to power down the WLAN module

Command	Opcode/data
FTM_WLAN_GEN6_STOP	0x4B0B 1600 0000 0E00 0000 0200 0200

Exit Test

3.35 GPS Test

3.35.1 GPS Test – Log Message Method

This test verifies the GPS module is connected and functioning properly for products. The phone must be in FTM mode to perform this test. There are no other prerequisites for this test.

Test Dependencies:

3.6.1 FTM Mode

Equipment:

RF generator that is able to produce a CW signal at 1575.52MHz with RF level accuracy better than +/- 1.2 dB and phase noise better than -121 dBc/Hz at 20 kHz offset from the carrier frequency

Test Setup

Parameter	All Products
RF_TEST_LEVEL (dBm)	-115
PATH_OFFSET (dBm)	0

Table 3.36.2

Test Procedure

Apply an unmodulated RF input signal at RF_TEST_LEVEL at 1575.52MHz to the input of the GPS receiver (Note: this may or may not be the same RF connector used for all other tests, depending on your product).

Note: Two different testing methods are listed below. Only one method is required for factory test. The limits are the same for either method used.

Existing Method

Send the following command to enable Stand Alone RF Verify Mode Switch

<u>Command</u>	<u>Opcode/data</u>	
FTM CGPS_SA_RF_VERIF_MODE_SWITCH	0x4B0D 6500 1401 00	(Enable SA RF Verify Mode Switch)

Retrieve ID Ranges, Get the Mask, and Set the Mask are used in order to enable DIAG logging (log code 0x1487)

Send the following command to Retrieve ID Ranges

<u>Command</u>	<u>Opcode/data</u>	
FTM TEST_LOGGING_REQUEST	0x7300 0000 0100 0000 0100 0000 0000 0000	(Retrieve ID Ranges)

Send the following command to Get Log Mask

<u>Command</u>	<u>Opcode/data</u>	
FTM_TEST_LOGGING_REQUEST	0x7300 0000 0400 0000 0100 0000 3606 0000	(Get Log Mask)

Send the following command to Set Log mask to enable log 'LOG_CGPS_WB_FFT_STATS_C' (0x1487)

<u>Command</u>	<u>Opcode/data</u>
----------------	--------------------

```
FTM_TEST_LOGGING_REQUEST      0x7300 0000 0300 0000 0100 0000 3606 0000 0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
8000 0000 0000 0000 0000 0000 0000                                     (Set Log Mask)
```

Send the following command to Start the Carrier/Noise and Freq Offset measurement

Command	Opcode/data
---------	-------------

FTM CGPS_START_IQ_TEST (Freq Offset measurement)	0x4B0D 6500 1632 0010 0000 0004 0100 0120 000A 0000 0000	(Start C/No and
---	--	-----------------

Poll for diagnostic log code 0x1487 (LOG_CGPS_WB_FFT_STATS_C)

Parse the diagnostic log response to obtain the Carrier/Noise and Freq Offset measurement.

Example response: 100015001500871429A443DB010000000012010000C3860100E3257E

Example response contains:

Header: 100015001500871429A443DB0100000000

4 bytes of Carrier to Noise (store byte swapped and in units of 0.1 dBHz): 12010000

4 bytes Frequency Offset (stored byte swapped and in units for Hz): C3860100

3 byte (2 byte checksum and 1 stop flag byte).

Carrier to Noise: 12010000 => 00000112 (byteswapped)

0x00000112 => 274 (hex to deci)

274 / 10 => 27.4 (divide by 10 as value stored in units of 0.1 dB/Hz)

27.4 dB/Hz

Freq Offset: C3860100 => 000186C3 (byteswapped)

0x000186C3 => 100035 (Hex to Deci)

100.035 kHz.

Send the following commands to Get the Mask and Set the Mask in order to turn off DIAG logging (log code 0x1487)

CommandOpcode/data

FTM TEST LOGGING REQUEST	0x7300 0000 0400 0000 0100 0000 3606 0000	(Get Log Mask)
--------------------------	---	----------------

Command Opcode/data

```
FTM_TEST_LOGGING_REQUEST 0x7300 0000 0300 0000 0100 0000 3606 0000 0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0080 0000 0000 0000 0000 0000 0000 0000
(Set Log Mask)
```

Send the following command to Disable Stand Alone RF Verify Mode Switch.

Command	Opcode/data
Send the following command to Disease Status Agents: All Verify media	

Command	Spsecedata	
FTM CGPS SA RF VERIF MODE SWITCH	0x4B0D 6500 1401 01	(Disable SA RF Verify Mode switch)

GEN8 Method - Optional

Send the following command to enable Stand Alone RF Verify Mode Switch

Command	Opcode/data
Send the following command to enable Stand Alone RT-Vonly mode	

Command	Speed/data	
FTM CGPS SA RF VERIF MODE SWITCH	0x4B0D 6500 1401 00	(Enable SA RF Verify Mode Switch)

Retrieve ID Ranges, Get the Mask, and Set the Mask are used in order to enable DIAG logging (log code 0x1487)

Send the following command to Retrieve ID Ranges

Command	Opcode/data	
FTM TEST_LOGGING_REQUEST	0x7300 0000 0100 0000 0100 0000 0000 0000	(Retrieve ID Ranges)

Send the following command to Get Log Mask

Command	Opcode/data	
FTM_TEST_LOGGING_REQUEST	0x7300 0000 0400 0000 0100 0000 3606 0000	(Get Log Mask)

Send the following command to Set Log mask to enable log 'LOG_CGPS_WB_FFT_STATS_C' (0x1487)

Command	Opcode/data
FTM_TEST_LOGGING_REQUEST	0x7300 0000 0300 0000 0100 0000 3606 0000 0000 0000 0000 0000 0000 0000 0000 0000
	0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
	0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
	0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
	8000 0000 0000 0000 0000 0000 0000
	(Set Log Mask)

Send the following command to Start the Carrier/Noise and Freq Offset measurement

Command	Opcode/data	
FTM CGPS_START_IQ_TEST Freq Offset measurement)	0x4B0D 6500 1632 0010 0000 0004 0100 0120 000A 0000 0000	(Start C/No and

Poll for diagnostic log code 0x1487 (LOG_CGPS_WB_FFT_STATS_C)

Parse the diagnostic log response to obtain the Carrier/Noise and Freq Offset measurement.

Example response: 100015001500871429A443DB010000000012010000C3860100E3257E

Example response contains:

Header: 100015001500871429A443DB0100000000

4 bytes of Carrier to Noise (store byte swapped and in units of 0.1 dBHz): 12010000

4 bytes Frequency Offset (stored byte swapped and in units for Hz): C3860100

3 byte (2 byte checksum and 1 stop flag byte).

Carrier to Noise: 12010000 => 00000112 (byteswapped)

0x00000112 => 274 (hex to dec)

274 / 10 => 27.4 (divide by 10 as value stored in units of 0.1 dB/Hz)

27.4 dB/Hz

Freq Offset: C3860100 => 000186C3 (byteswapped)

0x000186C3 => 100035 (hex to dec)

100.035 kHz.

Send the following commands to Get the Mask and Set the Mask in order to turn off DIAG logging (log code 0x1487)

Command	Opcode/data	
FTM TEST LOGGING REQUEST	0x7300 0000 0400 0000 0100 0000 3606 0000	(Get Log Mask)

Command	Opcode/data
FTM TEST_LOGGING_REQUEST	0x7300 0000 0300 0000 0100 0000 3606 0000 0000 0000 0000 0000 0000 0000 0000 0000
	0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
	0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
	0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
	0080 0000 0000 0000 0000 0000 0000
	(Set Log Mask)



Send the following command to Disable Stand Alone RF Verify Mode Switch.

<u>Command</u>	<u>Opcode/data</u>	
FTM CGPS_SA_RF_VERIF_MODE_SWITCH	0x4B0D 6500 1401 01	(Disable SA RF Verify Mode switch)

Ensure that the GPS RADIATED CARRIER TO NOISE and GPS RADIATED FREQUENCY OFFSET values meet the limits specified in Section 2.1 Test Limits for the product being tested.

Exit Test**3.35.2 GPS Test – Diag Command Method****Test Description**

This test verifies the GPS module is connected and functioning properly for products using the gpsOne Gen 8 solution. The phone must be in FTM mode to perform this test. There are no other prerequisites for this test.

Test Dependencies:

3.6.1 FTM Mode

Equipment:

RF generator that is able to produce a CW signal at 1575.52MHz with RF level accuracy better than +/- 1.2 dB and phase noise better than -121 dBc/Hz at 20 kHz offset from the carrier frequency.

Test Setup

Parameter	All Products
RF_TEST_LEVEL (dBm)	-115
PATH_OFFSET (dBm)	0

Table 3.36.2

Test Procedure

Apply an unmodulated RF input signal at RF_TEST_LEVEL at 1575.52MHz to the input of the GPS receiver (Note: this may or may not be the same RF connector used for all other tests, depending on your product).

Send the following command to enable Stand Alone RF Verify Mode Switch

<u>Command</u>	<u>Opcode/data</u>	
FTM CGPS_SA_RF_VERIF_MODE_SWITCH	0x4B0D 6500 1401 00	(Enable SA RF Verify Mode Switch)

Retrieve ID Ranges is used in order to enable log results.

Send the following command to Retrieve ID Ranges

<u>Command</u>	<u>Opcode/data</u>	
FTM TEST_LOGGING_REQUEST	0x7300 0000 0100 0000 0100 0000 0000 0000	(Retrieve ID Ranges)

Send the following command to initialize message counter of the results

<u>Command</u>	<u>Opcode/data</u>	
GNSS_SA_RF_GET_LATEST_WBIQ_FFT_RESULTS	0x4B0D 6500 1F	

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Message response: 4B0D65001F01wwxxxxxxxxxyyyyyyyzzzz7E

where ww is a message counter that is incremented every time new test results come in, xxxxxxxx is C/No, and yyyyyyyy is frequency offset, and zzzz is the CRC and 7E is the stop byte that marks the end of the message.

Parse for the message counter byte.

Send the following command to Start the Carrier/Noise and Freq Offset measurement

<u>Command</u>	<u>Opcode/data</u>	
FTM CGPS_START_IQ_TEST	0x4B0D 6500 1632 0010 0000 0004 0100 0120 000A 0000 0000	(Start C/No and Freq Offset measurement)

Delay 1 second

Send the following command to read the message counter, Carrier/Noise, and Freq Offset measurement

<u>Command</u>	<u>Opcode/data</u>
GNSS_SA_RF_GET_LATEST_WBIQ_FFT_RESULTS	0x4B0D 6500 1F

Message response: 4B0D65001F01wwxxxxxxxxxyyyyyyyzzzz7E

where ww is a message counter that is incremented every time new test results come in, xxxxxxxx is C/No, and yyyyyyyy is frequency offset, and zzzz is the CRC and 7E is the stop byte that marks the end of the message.

Verify that the message counter has incremented. If the message counter has not incremented then resend the command until it either increments or 5 seconds has elapsed. If the message counter fails to increment, fail the unit as the results are not valid.

Note: For multiple samples, start the next measurement with the FTM CGPS_START_IQ_TEST command again and continue through the test procedure. The message counter will increment for each valid measurement taken. The Carrier/Noise and Freq Offset measurement results should be collected and averaged.

Parse the diagnostic log response to obtain the Carrier/Noise and Freq Offset measurement.

Example response: 4B0D65001F0101410200005AA20100C2FC7E

Example response contains:

Header: 4B0D65001F01

1 byte for message counter: 0x01

4 bytes of Carrier to Noise (store byte swapped and in units of 0.1 dBHz): 0x41020000

4 bytes Frequency Offset (stored byte swapped and in units for Hz): 0x5AA20100

3 byte (2 byte checksum and 1 stop flag byte).

Carrier to Noise:

41020000 => 00000241 (byte swapped)

0x00000241 => 577 (hex to dec)

577 / 10 => 57.7 (divide by 10 as value stored in units of 0.1 dB/Hz)

57.7 dB/Hz

Freq Offset:

5AA20100 => 0001A25A (byte swapped)

0x0001A25A => 107098 (hex to dec)

107.098 kHz

Send the following command to Disable Stand Alone RF Verify Mode Switch.

<u>Command</u>	<u>Opcode/data</u>	
FTM CGPS_SA_RF_VERIF_MODE_SWITCH	0x4B0D 6500 1401 01	(Disable SA RF Verify Mode switch)

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Ensure that the GPS RADIATED CARRIER TO NOISE and GPS RADIATED FREQUENCY OFFSET values meet the limits specified in Section [2.1 Test Limits](#) for the product being tested.

Exit Test

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3.36 NFC Antenna and SWP Line Self Test

Test Description

The purpose of this test is to verify the overall connectivity and functionality of the NFC antenna matching components, the NFC antenna, and the SWP line connected to the NFC SIM.

Test dependencies:

[3.2 Turn On](#)

[3.6 Test Mode](#)

Note: When completing the NFC Antenna Self Test at radio level, make sure metal object are not near the phone, specifically NFC antenna. For instance a good position for a phone to be placed is face down so the antenna points up. This face down position assumes there is no metal in the fixture that is placed over the back of the phone.

Equipment:

NFC Supported SIM Card

Test Setup

Send the following test command to enable NFC test mode.

Command	Opcode/data	
NFC	0x0106 0002	(Enable NFC test mode)

Test Procedure

NOTE: The tests in this section can be run in any order. There is no order dependency.

3.36.1 NFC Antenna Self Test

Send the following test command to reset the NFC part.

Command	Opcode/data	
NFC*	0x0106 01	(Reset NFC part)

**NOTE: Every time antenna self test is run the NFC part must be reset again by the command above.*

Send the following test command to run the NFC antenna self test.

Command	Opcode/data	
NFC	0x0106 0200	(Run NFC antenna self test)

Record the 1st byte (Byte 0) of the test command response as **NFC Antenna Self Test Status** (where 0x00 = self test success and 0x01 = self test failed).

The 4th – 7th bytes (Byte 3 -6) is the data field for NFC Measured RF Power Reading. Byte swap the order of the value and convert from Hex to Decimal. Record as **NFC Measured RF Power**.

ExampleOnly:

75cd2d00 (4th – 7th bytes)

002DCD75 (Byte swap the order)

3001717 (Hex to Decimal conversion)

Verify that **NFC Antenna Self Test Status** and **NFC Measured RF Power** are within the limits specified in section [2.1 Test Limits](#).

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3.36.2 NFC Antenna SWP Line Test

Note: The NFC Antenna SWP Line Test is run on products that support it only. Please see table of limits for products that support it.

Insert an NFC supported SIM Card into the UE.

Send the following test command to reset the NFC part.

<u>Command</u>	<u>Opcode/data</u>	
NFC*	0x0106 01	(Reset NFC part)

**NOTE: Every time the SWP line test is run the NFC part must be reset again by the command above.*

Send the following test command to run the NFC SWP line test.

<u>Command</u>	<u>Opcode/data</u>	
NFC	0x0106 0201	(Run SWP line self test)

Record the 1st byte (Byte 0) of the test command response as **NFC SWP Line Status – NFC SIM Inserted** (where 0x00 = self test success and 0x01 = self test failed).

Verify that **NFC SWP Line Test Status – NFC SIM Inserted** is within the limits specified in section [2.1 Test Limits](#).

Exit Test

None

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3.37 NFC Tag Test

Test Description

The purpose of this test procedure is to verify that the physical hardware lines being used on the accelerometer are connected and functioning properly.

Test Dependencies:

3.2 Turn On (Factory Cable Method – Assembled Level)

*NOTE: The phone must be either manually unlocked and at home screen or the phone must have been powered up with the lock screen setting disabled. If the lock screen setting is enabled, the phone is locked and Android **will** disable NFC tag detection.*

Test Setup

Make sure upon powering up the phone that the phone is in the fixture with the NFC tag positioned 1.5 cm from the phone.

The UUT must have the CommServer running and communication established over BLAN Port 2631

This is accomplished by powering up the UE with a factory cable.

Connect to the CommServer.

Test Procedure

1) Send the following command to start the NFC activity.

Command	Activity/Data
START	NFC_TEST

2) Send the following command to get the state of the NFC adapter.

Command	Activity	Data
TELL	NFC_TEST	GET_NFC_STATE

The above command returns the state of the NFC adapter in the STATE key-value pair.

If the state of the NFC adapter is OFF proceed to step 5 otherwise proceed to step 3.

3) Turn off the NFC adapter by sending the following command:

Command	Activity	Data
TELL	NFC_TEST	DISABLE_NFC

4) Send the following command to get the state of the NFC adapter and wait until the adapter is OFF

Command	Activity	Data
TELL	NFC_TEST	GET_NFC_STATE

5) Send the following command to have Android route the discovery of the NFC tag to the NFC_TEST activity

Command	Activity	Data
TELL	NFC_TEST	ENABLE_FOREGROUND_DISPATCH

6) Send the following command to turn on the NFC adapter.



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Command	Activity	Data
TELL	NFC_TEST	ENABLE_NFC

7) Send the following command to get the state of the NFC adapter and wait until the adapter is ON.

Command	Activity	Data
TELL	NFC_TEST	GET_NFC_STATE

8) Send the following command to poll for the NFC tag detection.

Command	Activity	Data
TELL	NFC_TEST	READ_TAG

The above command returns the tag detection state in the **TAG_DETECTED** key-value pair.
Loop until **TAG_DETECTED** returns YES.

Verify that **TAG_DETECTED** is within the limits specified in section [2.1 Test Limits](#).

9) Send the following command to write data to the NFC tag:

Command	Activity	Data
TELL	NFC_TEST	WRITE_TAG PAYLOAD_DATA=<PHONE_BARCODE>

Where <PHONE_BARCODE> is the barcode of the UUT.

10) Send the following command to read back the newly written tag data:

Command	Activity	Data
TELL	NFC_TEST	READ_TAG

The above command returns the tag's payload data in the **PAYLOAD_DATA** key-value pair.
Verify that the **PAYLOAD_DATA** matches the barcode of the UUT.

Verify that **PAYLOAD_DATA** is within the limits specified in section [2.1 Test Limits](#).

11) Turn off the NFC adapter by sending the following command:

Command	Activity	Data
TELL	NFC_TEST	DISABLE_NFC

12) Send the following command to get the state of the NFC adapter and wait until the adapter is OFF.

Command	Activity	Data
TELL	NFC_TEST	GET_NFC_STATE

13) Send the following command to stop the NFC activity:

Command	Activity/Data
STOP	NFC_TEST

Exit Test

None

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3.38 Sensor Hub Test

Test Description

The purpose of this test is to verify communication with the Sensor Hub.

Test Dependencies:

[3.2 Turn On](#)

[3.6.6 P2K SUSPEND Mode](#)

Test Setup

None

Test Procedure

SENSOR HUB FIRMWARE VERSION TEST

Loop the following test command to read the Sensor Hub Firmware Version to ensure the firmware download is complete.

Command	Opcode/data
VERSION	0x0039 1005

(Read Sensor Hub Firmware Version)

Loop the command for up to 60 seconds until a successful response of 0x00 is returned for byte0 (confirming that the firmware download has completed).

Verify that **Sensor Hub Firmware Test** is within the limits specified in section [2.1 Test Limits](#).

Exit Test

None



3.39 Sensor Hub WAKE Lines Test

Test Description

The purpose of this test is to verify communication of the WAKE lines between the MSM and the Sensor Hub.

Test Dependencies:

[3.2 Turn On](#)

Test Setup

None

Test Procedure

Send the following command to check the connectivity of the CCP_AP_WAKE Line.

Command	Opcode/data
SENSOR_HUB	0x009F 0401

Verify that the response is a success and there is no return data in the response (an empty string). If data is returned in the response then the data is a TCMD standard error message and the unit fails the test. Verify that it is within the limits specified in section 2.1 Test Limits.

Send the following command to check the connectivity of the AP_CCP_WAKE Line.

Command	Opcode/data
SENSOR_HUB	0x009F 0402

Verify that the response is a success and there is no return data in the response (an empty string). If data is returned in the response then the data is a TCMD standard error message and the unit fails the test. Verify that it is within the limits specified in section 2.1 Test Limits.

Send the following command to check the connectivity of the CDC_CCP_WAKE Line.

Command	Opcode/data
AUDIO	0x0079 0005 00

Verify that one byte response is 0x00. If not, fail the unit. Record this response as **CDC_CCP_WAKE**. Verify that it is within the limits specified in section 2.1 Test Limits.

Send the following command to test Sensor Hub to Cirrus i2c line.

Command	Opcode/data
AUDIO	0x0079 0005 01

(Checks i2c line between Sensor Hub and Cirrus)

Verify that one byte response is 0x00. If not, fail the unit. Record this response as **SENSORHUB_I2C**. Verify that it is within the limits specified in section [2.1 Test Limits](#).

Exit Test

None



3.40 Accelerometer

3.40.1 P2K TCMD Method

Test Description

The purpose of this test procedure is to verify that the physical hardware lines being used on the accelerometer are connected and functioning properly.

Test Dependencies:

[3.2 Turn On](#)

[3.6 Test Mode](#)

Test Setup

None

Test Procedure

Send the following command to run the Accelerometer Self Test.

Command	Opcode/data
ACCEL	0x0E03 06

Verify that the response (ACCEL_SELF_TEST) is a success and there is no return data in the response (an empty string). If data is returned in the response then the data is a TCMD standard error message and the unit fails the test. Verify that it is within the limits specified in section [2.1 Test Limits](#).

Exit Test

3.40.1 P2K TCMD Method – Accelerometer Self Calibration

Test Description

The purpose of this test procedure is to calibrate the offset for each of the accelerometers three axis.

Test Dependencies:

[3.2 Turn On \(Assembled Level\)](#)

Test Setup

Unit must be tested assembled and on a level surface in the test station.

Test Procedure

Send the following command to run the Accelerometer Self Calibration Test.

Command	Opcode/data
ACCEL	0x0E03 0700

The Data returned will be:

Bytes[0-1]: X Offset

Bytes[2-3]: Y Offset

Bytes[4-5]: Z Offset

Bytes[6]: Number of previous writes to persistent storage

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Notes:

1. x, y, z values are integers in 2's complement format.
2. The x, y, z data is in 3.9 mg unit.

Convert each Offset to Decimal in 2's Complement. Record them as **Accelerometer_Self_Cal_X_Offset**, **Accelerometer_Self_Cal_Y_Offset**, and **Accelerometer_Self_Cal_Z_Offset**. Verify that the responses are within the limits specified in section 2.1 Test Limits. Read the number of previous writes to persistent storage. Record this value as **Accelerometer_Self_Cal_Storage_Writes**. Verify that the response is within the limits specified in section 2.1 Test Limits.

Note: If any OFFSET values are outside the limits or if Accelerometer_Self_Cal_Storage_Writes is currently reading 10 previous storage writes then Accelerometer Self Calibration Store command cannot be performed.

Send the following command to run the Accelerometer Self Calibration Store.

Command	Opcode/data
ACCEL	0x0E03 0702

Verify that the response (Accelerometer Self Calibration Store) is a success

Note: If Accelerometer Self Calibration Store is not performed then the Accelerometer Self Calibration Exit command listed below is required. The EXIT command is not required to be executed if the STORE command is performed.

Send the following command to run the Accelerometer Self Calibration Exit.

Command	Opcode/data
ACCEL	0x0E03 0703

Exit Test

3.40.2 APK CommServer Method

Test Description

The purpose of this test procedure is to verify that the physical hardware lines being used on the accelerometer are connected and functioning properly.

Test Dependencies:

[3.2 Turn On](#) (Assembled Level)

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631

This test must be done in a flat position.

Test Procedure

Send the following command to run the Accelerometer test.

Command	Activity/Data
START	SENSOR_ACCELEROMETER
TELL	SENSOR_ACCELEROMETER CLEAR_MAX_MIN_READINGS

Move the UUT, must be at least enough movement to produce a change in all axis's, then place it in to a still position. Note the position of UUT, Display face Up or Display face Down.

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Retrieve all the value measurements and the total acceleration value from the UUT

<u>Command</u>	<u>Activity/Data</u>
TELL	SENSOR_ACCELEROMETER GET_READING

Returns: ACCELEROMETER_X, MAX_ACCELEROMETER_X, MIN_ACCELEROMETER_X, ACCELEROMETER_Y, MAX_ACCELEROMETER_Y, MIN_ACCELEROMETER_Y, ACCELEROMETER_Z, MAX_ACCELEROMETER_Z, MIN_ACCELEROMETER_Z; the x,y,z values are in m/s² and TOTAL_ACCELERATION is composite value generated on the UUT

Verify that **ACCELEROMETER_X**, **ACCELEROMETER_Y** and **ACCELEROMETER_Z**, **TOTAL_ACCELERATION** are within the limits specified in section [2.1 Test Limits](#).

Exit Test

Send the following command to stop the Accelerometer activity.

<u>Command</u>	<u>Activity/Data</u>
STOP	SENSOR_ACCELEROMETER



3.41 GyroscopeTest

3.41.1 P2K TCMD Method

Test Description

The purpose of this test is to verify connectivity and functionality of the Gyroscope part.

Test dependencies:

[3.5 Test Mode](#)

Test Setup

None

Test Procedure

Send the following command to run the Gyroscope Self Test.

Command	Opcode/data
GYRO	0x0099 04

Verify that the response (GYRO_SELF_TEST) is a success and there is no return data in the response (an empty string). If data is returned in the response then the data is a TCMD standard error message and the unit fails the test. Verify that it is within the limits specified in section [2.1 Test Limits](#).

Exit Test

None

3.41.2 APK CommServer Method

Test Description

The purpose of this test is to verify connectivity and functionality of the Gyroscope part.

Test dependencies:

[3.2 Turn On \(Factory Cable Method – Assembled Level\)](#)

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631. This is accomplished by powering up the UE with a factory cable.

Test Procedure

Send the following command to run the Gyroscope test.

Command	Activity/Data
START	SENSOR_GYROSCOPE

Move the UE, must be at least enough movement to produce a change in all axis's.

Retrieve the Max value measurement from the UE while the UE is in a stationary position

Command	Activity/Data
TELL	SENSOR_GYROSCOPE GET_READING



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Returns: ANGULAR_SPEED_X, MAX_ANGULAR_SPEED_X, MIN_ANGULAR_SPEED_X, ANGULAR_SPEED_Y, MAX_ANGULAR_SPEED_Y, MIN_ANGULAR_SPEED_Y, ANGULAR_SPEED_Z, MAX_ANGULAR_SPEED_Z, MIN_ANGULAR_SPEED_Z; the x,y,z values are in radians per second

Verify that **ANGULAR_SPEED_X**, **ANGULAR_SPEED_Y**, **ANGULAR_SPEED_Z** are within the limits specified in section [2.1 Test Limits](#).

Exit Test

Send the following command to stop the Gyroscope activity.

Command	Activity/Data
STOP	SENSOR_GYROSCOPE



3.42 Magnetometer

APK CommServer Method

Test Description

This test verifies that all the lines used on the Magnetometer part are functional by reading out the absolute magnetic field data of the X, Y, and Z axes.

Test dependencies:

3.2 Turn On (Factory Cable Method – Assembled Level)

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631
This is accomplished by powering up the UE with a factory cable.

This test must be done in a flat position.

Test Procedure

Send the following command to run the Magnetometer test.

Command	Activity/data
START	SENSOR_MAGNETOMETER
TELL	SENSOR_MAGNETOMETER GET_READING

The response back is a measurement of the Magnetic fields, Magnetic_Field_X, Magnetic_Field_Y, Magnetic_Field_Z in micro-Tesla units

Record the values of **Absolute Magnetic Field X Axis** = Magnetic_Field_X, **Absolute Magnetic Field Y Axis** = Magnetic_Field_Y, **Absolute Magnetic Field Z Axis** = Magnetic_Field_Z

If All 3 axis report 0 then the unit fails.

Verify all 3 axis not zero and that the values for **Absolute Magnetic Field X Axis**, **Absolute Magnetic Field Y Axis**, and **Absolute Magnetic Field Z Axis** are within the limits specified in section [2.1 Test Limits](#).

Exit Test

Send the following command to stop the Magnetometer Activity.

Command	Activity/data
STOP	SENSOR_MAGNETOMETER



3.43 Proximity Sensor

3.43.1 Proximity Sensor Distance Test (P2K TCMD Method) Board Level

Test Description

The purpose of this Board Level test is to verify the connectivity of the Proximity sensor on the board.

Test Dependencies:

[3.6.5 P2K Command Mode](#)

Test Procedure

Note: The output of the query command indicates the digitized amount of light that is detected by the photo detector.

With the Prox Sensor covered, send the following PROXimity Sensor command to read the corresponding value.

Command	Opcode/data	
PROX	0x0084 00 01	(Set to Enable proximity sensor)
PROX	0x0084 01	(Query state of proximity sensor)

The first byte of the return response is the status byte indicating 0x00 for sensor disabled and 0x01 for sensor enabled. If a zero is being read, fail the unit. Average 5 stable readings. Verify that this value is within the limits specified in section [2.1 Test Limits](#).

To disable the Proximity Sensor send the following PROXimity Sensor command.

Command	Opcode/data	
PROX*	0x0084 00 00	(Set to disable proximity sensor)

Exit Test

3.43.2 Proximity Sensor Distance Test (P2K TCMD Method or APK Method)

Test Description

The purpose of this test is to verify that the Proximity sensor is functioning by ensuring proper operation when an object like a hand or face is in close proximity and not in close proximity.

Test Dependencies:

[3.2 Turn On \(Assembled Level\)](#)

Equipment:

3M Diffuser Film 3635-30

Test Setup

This test is performed on the radio level.

Test Procedure



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Note: The output of the query command indicates the digitized amount of light that is detected by the photo detector.

With the Proximity Sensor uncovered by the 3M Diffuser Film, send the following PROXimity Sensor command to read the corresponding value.

Command	Opcode/data	
PROX	0x0084 00 01	(Set to Enable proximity sensor)
PROX	0x0084 01	(Query state of proximity sensor)

The first byte of the return response is the status byte indicating 0x00 for sensor disabled and 0x01 for sensor enabled.

If a zero is being read, fail the unit. Average 5 stable readings. Verify that this value is within the limits specified in section [2.1 Test Limits](#).

With the Proximity Sensor covered by the 3M Diffuser Film, send the following PROXimity Sensor command to read the corresponding value.

Command	Opcode/data	
PROX	0x0084 01	(Query state of proximity sensor)

If a zero is being read, fail the unit. Average 5 stable readings. Verify that this value is within the limits specified in section [2.1 Test Limits](#).

To disable the Proximity Sensor send the following PROXimity Sensor command.

Command	Opcode/data	
PROX	0x0084 00 00	(Set to disable proximity sensor)

APK Mode

Send the following PROXimity Sensor commands to obtain a reading.

Command	Activity	Data	
START	SENSOR_PROXIMITY		(Start proximity sensor)
TELL	SENSOR_PROXIMITY	GET_READING	(Query state of proximity sensor)

The return response for "Sensor Proximity Get Reading" is PROXIMITY DISTANCE = 100 for sensor disabled (i.e. no object detected), 3 for sensor enabled (i.e. object detected) and 1 for sensor saturated (i.e. object detected so close readings saturated).

To disable the Proximity Sensor send the following PROXimity Sensor command.

Command	Activity	Data	
STOP	SENSOR_PROXIMITY		(Stop proximity sensor activity)

CQA Test Mode

Run the "CQA Test in Model Assembly Mode" and ensure that the CQA Sequential Test Result returns:

Proximity: PASS

Exit Test

N/A

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3.44 Light Sensor (ALS)

3.44.1 P2K TCMD Method

Test Description

The purpose of this test is to verify that the Light sensor is functioning by ensuring proper operation in low and high lighting conditions.

Test Dependencies:

[3.6.6 P2K SUSPEND Mode](#)

Equipment:

N/A

Test Setup

The ambient light surrounding the Light Sensor must be at least 5 lux or greater.

The test bench must also be capable of eliminating the ambient light either via automated, or manual methods to achieve at least 3 lux or less.

Test Procedure

Primary Light Sensor:

With the Light Sensor uncovered and exposed to the ambient light, send the following AD_CONV command to measure the corresponding DAC value.

<u>Command</u>	<u>Opcode/data</u>	
AD_CONV	0x0001 0030	(Read Light Sensor DAC)

Record this value as **LightSensorBrightDAC**.

Eliminate the ambient light that is exposed to the light sensor by manual or automated means.

Send the following AD_CONV command to measure the corresponding DAC value.

<u>Command</u>	<u>Opcode/data</u>	
AD_CONV	0x0001 0030	(Read Light Sensor DAC)

Record this value as **LightSensorDarkDAC**.

Verify that the calculated **LightSensorDarkDAC** and **LightSensorBrightDAC** are within the limit specified in section [2.1 Test Limits](#).

Exit Test

**3.44.1 APK CommServer Method****Test Description**

The purpose of this test is to verify that the Light sensor is functioning by ensuring proper operation in low and high lighting conditions.

Test Dependencies:

3.2 Turn On (Factory Cable Method – Assembled Level)

Equipment:

N/A

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631

The ambient light surrounding the Light Sensor must be at least 300 LUX or greater.

The test bench must also be capable of eliminating the ambient light either via automated, or manual methods to achieve at least 3 LUX or less.

Test Procedure

With the Light Sensor uncovered and exposed to the ambient light, send the following FRONTLIGHT command to measure the corresponding LUX value.

Command	Activity	Data
START	SENSOR_FRONTLIGHT	(Start Light sensor activity)
TELL	SENSOR_FRONTLIGHT GET_READING	(Read Light Sensor LUX)

Record this value as **LightSensorBrightLUX**.

Eliminate the ambient light that is exposed to the light sensor by manual or automated means.

Send the following FRONTLIGHT command to measure the corresponding LUX value.

Command	Opcode/data
TELL	SENSOR_FRONTLIGHT GET_READING (Read Light Sensor LUX)

Record this value as **LightSensorDarkLUX**.

Verify that the returned LUX values **LightSensorDarkLUX** and **LightSensorBrightLUX** are within the limit specified in section [2.1 Test Limits](#).

Exit Test

Stop FRONTLIGHT activity.

Command	Activity	Data
STOP	SENSOR_FRONTLIGHT	

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3.45 Light sources

3.45.1 Status LEDs

Test Description

This test must be performed on products with a Status LED

Test Dependencies:

[3.6.6 P2K SUSPEND Mode](#)

Test Setup

N/A

Test Procedure

General Purpose Single Color LEDs

NOTE: General purpose Single Color LED IDs are generic for all products and are controlled by the tcmd_leds.cfg file provided by the hardware teams.

General Purpose Single Color LED #1

General Purpose Single Color LED #1 – WHITE SOC LED Test

Use the following command to enable the General Purpose Single Color LED #1 – WHITE SOC LED.

Command Opcode/data

LEDS 0x003E 2001 (Enable General Purpose Single Color LED #1 – WHITE SOC LED)

Visually verify the General Purpose Single Color LED #1 – WHITE SOC LED.

Use the following command to disable the General Purpose Single Color LED #1 – WHITE SOC LED.

Command Opcode/data

LEDS 0x003E 2000 (Disable General Purpose Single Color LED #1 – WHITE SOC LED)

Exit Test



3.46 UTAG Write Front Housing Color Test

Test Description

This procedure writes the front housing color hex representation of ASCII string to UTAG memory partition.

Test Dependencies:

[3.6.5 P2K Command Mode](#)

Test Setup

N/A

Test Procedure

Issue the UTAG command to write the front housing color

Command	Opcode/Data
---------	-------------

UTAG	0x0108 01 0011 0005 + COLOR
------	------------------------------------

(Write front housing color)

Where:

0005 = Number of bytes to be written

COLOR = Alphanumeric color converted to hexadecimal

*Note: For COLOR, only **black** or **white** is used and text is not case sensitive.*

Example:

If **black** needs to be written, send 0x0108 01 0011 0005 626C61636B

Exit Test

None



3.47 UTAG Read Front Housing Color Test

Test Description

This procedure reads the front housing color hex representation of ASCII string from UTAG memory partition.

Test Dependencies:

[3.6.5 P2K Command Mode](#)

Test Setup

N/A

Test Procedure

Issue the UTAG command to read the front housing color

Command	Opcode/Data	
UTAG	0x0108 00 0011	(Read front housing color)

Convert the data return from hex values to ASCII characters.

Where:

The response format is two bytes indicating the number of bytes in the color string followed by the actual color value.

Example:

Return data = 0005 626C61636B

Front housing color = 626C61636B (hex values)

Front housing color = black (ASCII characters)

*Note: For front housing color, only **black** or **white** is used and text is not case sensitive.*

Exit Test

None

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3.48 Fingerprint Sensor Test

3.48.1 P2K TCMD Method

Test Description

The purpose of this test is to validate connections to the finger print sensor. This will be done through self test mode and a checkerboard test. Both tests are internal to the finger print sensor IC.

Test Dependencies:

3.2 Turn On

Test Setup

None

Test Procedure

Use the FPS command to execute the self test.

<u>Command</u>	<u>Opcode/data</u>	
FPS	0x008E 0000	(self test mode)

Read the FPS response.

Verify that the 1st byte of the response = 0x00. If not then the unit fails.

Use the FPS command to execute the checkerboard test.

<u>Command</u>	<u>Opcode/data</u>	
FPS	0x008E 0001	(checkerboard test)

Read the FPS response.

Verify that the 1st byte of the response = 0x00. If not then the unit fails.

Record and log any additional bytes for vendor error. This will be a variable hex response that converts to ASCII to read the error.

For example: for dead pixels error
016465616420706978656C73

01 – vendor error
dead pixels - 6465616420706978656C73

Record these HEX responses as **FPS_SELFTEST** and **FPS_CHECKERBOARD_TEST**. Verify the results are within the limits specified in section [2.1 Test Limits](#).

Exit Test

None

3.48.2 APK Method

Test Description

The purpose of this test is to validate finger print image detection for the sensor.



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Test Dependencies:

3.2 Turn On

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631

Test Procedure

Command	Activity	Data
---------	----------	------

START	Sensor_FingerPrintTest	(Start Fingerprint Test activity)
-------	------------------------	-----------------------------------

The following steps are order-dependent (the finger-present test must be executed last).

First, ensure the finger print sensor is not being touched (by a finger or otherwise.) Send the following command:

Command	Activity	Data
---------	----------	------

TELL	Sensor_FingerPrintTest	START_FPS_FINGERDETECTIONTEST
------	------------------------	-------------------------------

Note: The status response may take a few seconds to complete

Pass case is the following:

TEST_RESULT=FPC_FINGER_ABSENT

For any other response fail the unit.

Record this PASS/FAIL response as **FPS_IMAGE_QUALITY_TEST** and verify the response is within the limits specified in section [2.1 Test Limits](#)

Exit Test

Stop Fingerprint Sensor Test activity.

Command	Activity	Data
---------	----------	------

STOP	Sensor_FingerPrintTest	
------	------------------------	--



3.49 Hall Effect Test

P2K TCMD Method

Test Description

The purpose of this test is to verify that the physical hardware lines being used on the hall effect sensor are connected and functioning properly. Board level will test functionality of the Hall Effect sensor circuit with magnet present. Assembled level will test the functionality of the Hall Effect sensor circuit with magnet absent.

Test dependencies:

3.6 Test Mode

Equipment:

Disc magnet with strength of 170 Maxwells.

Test Setup

None

Test Procedure

Place magnet with the north pole facing the phone hall effect sensor from a distance of 5mm away.

Send the DOCK command to query the hall effect sensor.

Command	Opcode/data
DOCK	0x0C38 00

Record the response as **North Pole Detection**.

Place magnet with the south pole facing the phone hall effect sensor from a distance of 5mm away.

Send the DOCK command to query the hall effect sensor.

Command	Opcode/data
DOCK	0x0C38 00

Record the response as **South Pole Detection**.

Remove the magnet away from hall effect sensor completely.

Send the DOCK command to query the hall effect sensor.

Command	Opcode/data
DOCK	0x0C38 00

Record the response as **No Pole Detection**.

Verify that **North Pole Detection**, **South Pole Detection**, and **No Pole Detection** are within the limits specified in section [2.1 Tests Limits](#).

Exit Test

None



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3.50 RF Connector Detection

Test Description

The purpose of this test procedure is to verify that the RF Connector detection circuitry in the radio is connected and functioning properly. In order to avoid engaging and disengaging the RF connector at a single test position, the following method can be split between 2 test positions, one with the RF connector engaged and at a different test position where there is no RF connector present. The presence of the RF connector **MUST** be tested in both an engaged and disengaged state.

Test Dependencies:

[3.2 Turn On](#)
[3.6.1 FTM Mode](#)

Equipment:

The RF connector that attaches to the radio must be available to test for the connected condition. A test set is not needed for this test.

Test Setup

N/A

Test Procedure

NOTE: Please see limits table to determine which products support which RF connectors.

For LB_HB Band RF Connector:

With an RF connector connected to the LB_HB RF connector on the board send the following test command to configure the LB_HB RF connector GPIO.

<u>Command</u>	<u>Opcode/data</u>
FTM_GPIO_TEST	0x4B0B 0080 1280 0364 0001 0000 0000 0000 0000 (Configure LB_HB RF Connector GPIO)

Send the following test command to read the state of the LB_HB RF connector GPIO.

<u>Command</u>	<u>Opcode/data</u>
FTM_GPIO_TEST	0x4B0B 0080 1280 0064 0000 0000 0000 0000 0000 (Read LB_HB RF Connector GPIO)

Verify that byte 9 of the test command response is 0x01 and that **LB_HB RF Connector Status – Connected** is within the limits specified in section [2.1 Test Limits](#).

Example Response: 0x4B0B 0080 1280 0064 00**01** 0000 0000 0000 0000 where byte 9 is the underlined and bolded byte.

Disconnect the RF connector connected to the LB_HB RF connector on the board and send the following test command to read the state of the LB_HB RF connector GPIO.

<u>Command</u>	<u>Opcode/data</u>
FTM_GPIO_TEST	0x4B0B 0080 1280 0064 0000 0000 0000 0000 0000 (Read LB_HB RF Connector GPIO)

Verify that byte 9 of the test command response is 0x00 and that **LB_HB RF Connector Status – Disconnected** is within the limits specified in section [2.1 Test Limits](#).

Example Response: 0x4B0B 0080 1280 0064 00**00** 0000 0000 0000 0000 where byte 9 is the underlined and bolded byte.

For ULB_MB Band RF Connector:

With an RF connector connected to the ULB_MB RF connector on the board send the following test command to configure the ULB_MB RF connector GPIO.

<u>Command</u>	<u>Opcode/data</u>
FTM_GPIO_TEST	0x4B0B 0080 1280 0365 0001 0000 0000 0000 0000 (Configure ULB_MB RF Connector GPIO)



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Send the following test command to read the state of the ULB_MB RF connector GPIO.

Command	Opcode/data
FTM_GPIO_TEST	0x4B0B 0080 1280 0065 0000 0000 0000 0000 (Read ULB_MB RF Connector GPIO)

Verify that byte 9 of the test command response is 0x01 and that **ULB_MB RF Connector Status – Connected** is within the limits specified in section [2.1 Test Limits](#).

Example Response: 0x4B0B 0080 1280 0065 **0001** 0000 0000 0000 0000 where byte 9 is the underlined and bolded byte.

Disconnect the RF connector connected to the ULB_MB RF connector on the board and send the following test command to read the state of the ULB_MB RF connector GPIO.

Command	Opcode/data
FTM_GPIO_TEST	0x4B0B 0080 1280 0065 0000 0000 0000 0000 (Read ULB_MB RF Connector GPIO)

Verify that byte 9 of the test command response is 0x00 and that **ULB_MB RF Connector Status – Disconnected** is within the limits specified in section [2.1 Test Limits](#).

Example Response: 0x4B0B 0080 1280 0065 **0000** 0000 0000 0000 0000 where byte 9 is the underlined and bolded byte.

For DIV RF connector:

With an RF connector connected to the DIV RF connector on the board send the following test command to configure the DIV RF connector GPIO.

Command	Opcode/data
FTM_GPIO_TEST	0x4B0B 0080 1280 035F 0001 0000 0000 0000 (Configure DIV RF Connector GPIO)

Send the following test command to read the state of the DIV RF connector GPIO.

Command	Opcode/data
FTM_GPIO_TEST	0x4B0B 0080 1280 005F 0000 0000 0000 0000 (Read DIV RF Connector GPIO)

Verify that byte 9 of the test command response is 0x01 and that **DIV RF Connector Status – Connected** is within the limits specified in section [2.1 Test Limits](#).

Example Response: 0x4B0B 0080 1280 005F **0001** 0000 0000 0000 0000 where byte 9 is the underlined and bolded byte.

Disconnect the RF connector connected to the DIV RF connector on the board and send the following test command to read the state of the DIV RF connector GPIO.

Command	Opcode/data
FTM_GPIO_TEST	0x4B0B 0080 1280 005F 0000 0000 0000 0000 (Read DIV RF Connector GPIO)

Verify that byte 9 of the test command response is 0x00 and that **DIV RF Connector Status – Disconnected** is within the limits specified in section [2.1 Test Limits](#).

Example Response: 0x4B0B 0080 1280 005F **0000** 0000 0000 0000 0000 where byte 9 is the underlined and bolded byte.

Exit Test

None

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3.51 Thermistors and Temp Sensors Test

Test Description

The purpose of this test is to verify the functionality of the thermistor's and temperature sensor's circuitry.

Test Dependencies:

3.2 Turn On
3.6 Test Mode

Test Setup

None

Test Procedure

Battery Thermistor:

Send the following test command to read the Thermistor value in tenths of degrees Celsius.

Command Opcode/data

TEMP 0x008F 0002 0000 (Read the Battery Thermistor value) (Schematic Ref: BATT_THERM)

Note: Battery Thermistor testing is covered more in depth in sections 3.25 Internal Battery Test and 3.26 Battery/Charger/Thermistor Test.

Convert the four byte response from **2's complement format** to decimal. The units in decimal are returned in tenths degrees Celsius. Acquire a stable reading and record it as **Battery Thermistor Temp**.

MSM Thermistor:

Send the following test command to read the Thermistor value in degrees Celsius.

Command Opcode/data

TEMP 0x008F 0002 0003 (Read the MSM Thermistor value) (Schematic Ref: Analog Temp Sensor)

Convert the four byte response from **2's complement format** to decimal. The units in decimal are returned in degrees Celsius. Acquire a stable reading and record it as **MSM_Thermistor_Temp**.

Charger Thermistor:

Send the following test command to read the Thermistor value in degrees Celsius.

Command Opcode/data

TEMP 0x008F 0002 0001 (Read the Charger Thermistor value) (Schematic Ref: Analog Charger Temp Sensor)

Convert the four byte response from **2's complement format** to decimal. The units in decimal are returned in degrees Celsius. Acquire a stable reading and record it as **Charger Thermistor Temp**.

PA Thermistor 0:

Send the following test command to read the Thermistor value in degrees Celsius.

Command Opcode/data

TEMP 0x008F 0002 0004 (Read the PA Thermistor 0 value) (Schematic Ref: PA_THERM_0)

Convert the four byte response from **2's complement format** to decimal. The units in decimal are returned in degrees Celsius. Acquire a stable reading and record it as **PA_THERM0**.



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Verify that the results for **Battery Thermistor Temp**, **MSM_Thermistor_Temp**, **Charger Thermistor Temp**, and **PA_THERM0** are within the limits specified in section [2.1 Test Limits](#)

Exit Test

None

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3.52 Touchscreen Test

Test Description

The purpose of this test is to verify that the functionality of the Main Touch Panel and the physical connections to the MSM.

Test dependencies:

[3.2 Turn On](#)

Test Setup

Note: This test is to be done on assembled level.

Test Procedure

3.52.1 Reset Line Test

Send the following command to verify the touch IC reset line connection.

Command	Opcode/data	
TOUCH_SCREEN	0x0C19 0B02	(Verify Physical Connections Reset Line)

Verify that the response is a success and there is no return data in the response (an empty string). If data is returned in the response then the data is a TCMD standard error message and the unit fails the test. Verify **VERIFY_TOUCH_RESET_LINES** is within the limits specified in section [2.1 Test Limits](#).

3.52.2 Touch Report Type Tests

Use the following two TST_DISP commands to enable the display in order to keep the display and touch system powered.

Note if the display is already on from a previous test then the two commands can be skipped.

Use the TST_DISP command to select the main display:

Command	Opcode/Data	
TST_DISP	0x0037 0100	(Select main display)

Use the TST_DISP command to enable the main display.

Command	Opcode/Data	
TST_DISP	0x0037 0C	(Enable main display)

Wait 250ms for the display to fully turn on.

3.52.2.1 Full Raw Capacitance

Send the following command to read the Raw Capacitance of the touch panel.

Command	Opcode/data	
TOUCH_SCREEN	0x0C19 0A01 1400 0380	(Read Raw Capacitance)

Where 0x01 corresponds to the number of reports to collect,

0x14 corresponds to the Report Type,

0x00 indicates Test mode, i.e. Normal,

0x0380 represents the size to read in bytes (X lines * Y lines *2).

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Verify Full Raw Capacitance limits for each channel are within the limits specified in this link –
<https://drive.google.com/drive/#folders/0B7eiu1wwlowwbEQ1dktpWFR0TGc>

3.52.2.2 High Resistance

Send the following command to read the High Resistance of the touch panel

Command	Opcode/data	
TOUCH_SCREEN	0x0C19 0A01 0400 0006	(Read High Resistance)

Here is an example of the High Resistance data:

83004700DEFF (in Hex) where Byte 0 is the first byte from the left.

Byte 0 represents the LSB of the Max RX Offset, Byte 1 represents the MSB of the Max RX Offset, i.e. 0x0083 (131 fF)

Byte 2 represents the LSB of the Max TX Offset, Byte 3 represents the MSB of the Max TX Offset, i.e. 0x0047 (71 fF)

Byte 4 represents the LSB of the Min Pixel, Byte 5 represents the MSB of the Min Pixel, i.e. 0xFFDE (-34 fF)

Verify that the Raw Data limit for each channel is within the limits specified in this link –

<https://drive.google.com/drive/#folders/0B7eiu1wwlowwbEQ1dktpWFR0TGc>

3.52.2.3 TRx –GND Shorts

Send the following command to read the TRx-GND shorts of the touch panel

Command	Opcode/data	
TOUCH_SCREEN	0x0C19 0A01 1900 0008	(Read TRx-GND shorts)

*Note that the 8th byte of return data is not currently used in our products, thus can be ignored.

Here is an example of the TRx-GND data:

FFFF5B16BB063F00 (Hex) where Byte 0 is the first byte from the left.

Byte 0 TRx responses TRx_7:TRx_0
Byte 1 TRx responses TRx_15:TRx_8
Byte 2 TRx responses TRx_23:TRx_16
Byte 3 TRx responses TRx_31:TRx_24
Byte 4 TRx responses TRx_39:TRx_32
Byte 5 TRx responses TRx_47:TRx_40
Byte 6 TRx responses TRx_54:TRx_48
Byte 7 TRx responses TRx_64:TRx_56

Each of the 8 bytes contains 1 bit per TX: A value of '1' on all active transmitters means high resistance (no shorts). A value of '0' on any of the transmitters means there is low resistance (a short to GND exists).

Please refer to the link – <https://drive.google.com/drive/#folders/0B7eiu1wwlowwbEQ1dktpWFR0TGc>
for the active TRx channels that are used. Fail the unit if the result is otherwise.

3.52.2.4 TRx –TRx Shorts

Send the following command to read the TRx-TRx shorts of the touch panel



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<u>Command</u>	<u>Opcode/data</u>	
TOUCH_SCREEN*	0x0C19 0A01 1A00 0008	(Read TRx-TRx shorts)
*Note that the 8th byte of return data is not currently used in our products, thus can be ignored.		

Each of the 8 bytes contains 1 bit per TX: A value of '0' on all active transmitters means high resistance (no shorts). A value of '1' on any of the transmitters means there is low resistance (a short exists).

Please refer to the link – <https://drive.google.com/drive/#folders/0B7eiu1wvlwwbEQ1dktPWFR0TGc> for the active TRx channels that are used. Fail the unit if the result is otherwise.

Exit Test

[OPTIONAL] Send the following TST_DISP command to disable the main display if required for subsequent tests.

<u>Command</u>	<u>Opcode/Data</u>	
TST_DISP	0x0037 0B	(Disable main display)

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3.53 Manual Touchscreen APK Test

The purpose of this test is to verify that the functionality of the Touch screen.

Test dependencies:

[3.2 Turn On](#) (Assembled Level)

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631

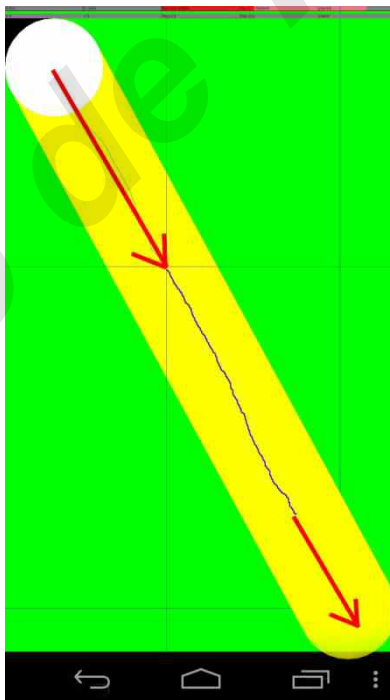
Test Procedure

Send the following command to start the Manual Touchscreen test:

Command	Activity	data
START	Touch_Config	
TELL	Touch_Config	SET_TOUCH_SETTINGS X_VERTICES=0,1000 Y_VERTICES=0,1000 TOUCH_WIDTH_DIVISOR=4

Without picking up your finger; touch the dot at the top left of the display and follow the path on the screen until completion. See figure 3.53 below for example.

Figure 3.53 Manual Touchscreen APK Test



(for reference only)



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If you complete this test, the display will go green and it will respond with an UNSOLICITED passed message, otherwise the user should press the down volume key to fail the test and will exit out.

Verify a passed case by sending the following command:

Command	Activity	Data
TELL	Touch_Config	GET_TOUCH_RESULT

The look for the following status response for pass and fail messages within the results returned.

Pass case is the following:

TOUCH_RESULT=PASS

Fail case is the following:

TOUCH_RESULT=FAILED

While test is being executed and not completed the status of the results response will be:

TOUCH_RESULTS=RUNNING

If operator terminates the test consider it a failure,

Send the following command to stop the Manual Touchscreen test:

Command	Activity
STOP	Touch_Config

Exit Test

None

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3.54 Display Tests

APK CommSever Method

Test Setup

Test dependencies:

3.2 Turn On (Assembled Level)

UUT must have the CommServer running and communication establish over BLAN Port 2631
 This is accomplished by powering up the UE with a factory cable

Note: UUT must have display on before further display tests are executed.

White with Black Border Pattern Test

Test Description

This test will find specific defects with the display such as missing rows, missing columns, dark sub-pixel defects, display uniformity and screen resolution defects.

Test Procedure

Note: All Display tests need an "All Pixels Off" reference point for visual verification test comparison.

Send the following command to simulate a PWR KEY press to blank out the display to visually acquire the "All Pixels Off" reference point.

<u>Command</u>	<u>Opcode/data</u>	
KEYS	0x0013 74	(KEYS command to press the PWR KEY)

Send the following command to force unit into CABC movie mode.

<u>Command</u>	<u>Opcode/data</u>	
TST_DISP	0x0037 1901 0155 0103	(Enable CABC Movie Mode)

Write the **White with Black Border Pattern** below (see figure Display Tests 3.54.1) on the main display using the **CommServer Command**.

<u>Command</u>	<u>Activity</u>
START	TESTPATTERN_WHITEWITHBLACKBORDER

Using the **White with Black Border Pattern** below as a reference:

Visually verify the display for any missing rows, columns, dark pixel, or wrong color defects.

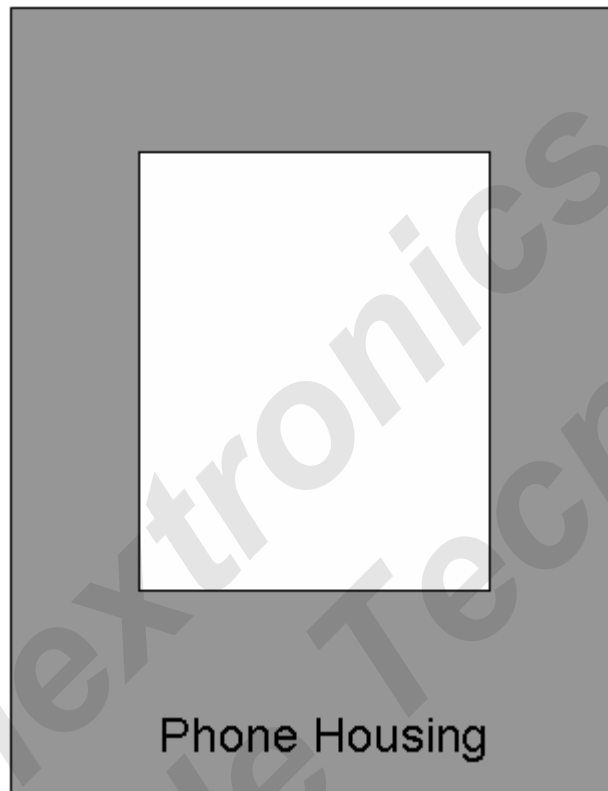
Verify that the black border is present on all four sides of image without any missing black pixels.

Fail units if any of these defects are visible on the display.

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Figure 3.54.1 White with Black Border Test Pattern (Main Display)



(for reference only)

Exit Test

Command	Activity
STOP	TESTPATTERN_WHITEWITHBLACKBORDER

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Black with White Border Pattern (with white borderline) Test

Test Description

This test will find specific defects with the display such as missing rows, missing columns, bright sub-pixel defects, and screen resolution defects.

Test Procedure

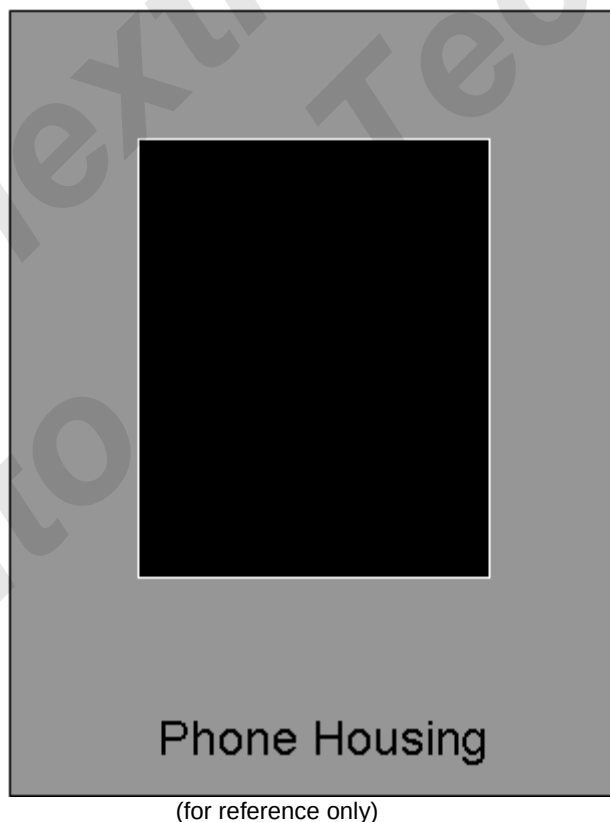
Write the **Black with White Border Pattern** below (see figure 3.54.2) on the main display using the **CommServer Command**.

Command	Activity
START	TESTPATTERN_BLACKWITHWHITEBORDER

Using the **Black with White Border Pattern** below as a reference:
 Visually verify the display for any missing rows, columns, bright pixel, or wrong color defects.
 Verify that the white border is present on all four sides of image without any missing white pixels.

Fail units if any of these defects are visible on the display.

Figure 3.54.2 Black with White Border Test Pattern (Main Display)



Exit Test

Command	Activity
STOP	TESTPATTERN_BLACKWITHWHITEBORDER

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Bright Color Bar Pattern Test

Test Description

This test will find specific defects with the display, such as irregular colors, faded displays, shorted columns.

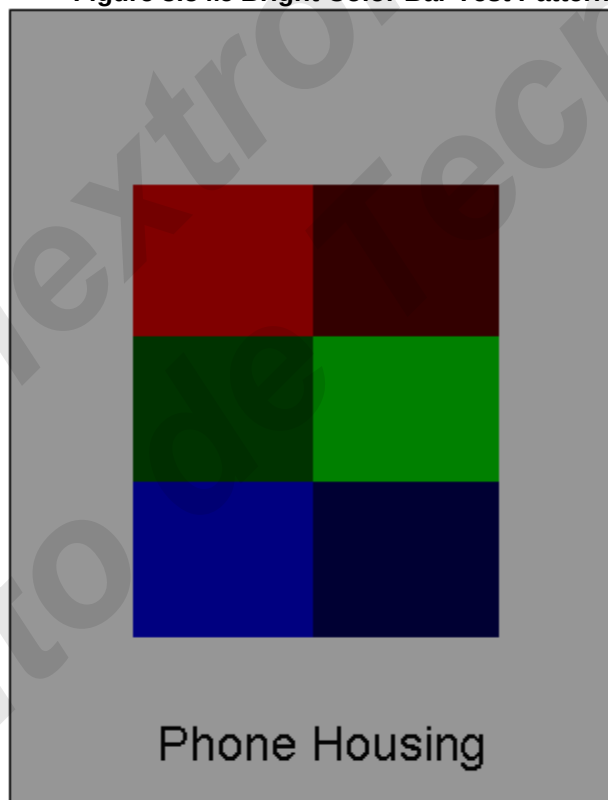
Test Procedure

Write the **Bright Color Bar Test Pattern** below (see figure 3.54.5) on the main display using the CommServer Command.

Command	Activity	
START	TESTPATTERN_COLOR_BRIGHT	(Displays bits 6,7 of R,G,B)

Visually verify the color appears as below.
Fail units if the color does not appear as below.

Figure 3.54.5 Bright Color Bar Test Pattern



(for reference only)

Exit Test

Command	Activity	
STOP	TESTPATTERN_COLOR_BRIGHT	

Command	Opcode/data	
TST_DISP	0x0037 1901 0155 0101	(Disable CABC Movie Mode to return to CABC UI Mode)

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3.55 Display TE Line Test

Test Description

The purpose of this test is to verify the Display TE functionality and circuitry. It verifies the display TE line connections between the display connector to the MSM.

Test dependencies:

[3.2 Turn On](#) (Factory Cable Turn On – Assembled Level)

[3.6.5 P2K Command Mode](#) (Non-Suspend Mode)

Test Setup

None

Test Procedure

This procedure may be combined with other camera testing as needed to support factory flow optimization.

Note: The display has to be active/on throughout this test hence the viewfinder.

Display Connector to MSM

Use the CAMERA_FACTORY command to enable the external camera viewfinder.

If the CAMERA_FACTORY Activity has already been started, it does not need to be started again (skip the start command).

Start Activity

Command	Activity
START	CAMERA_FACTORY

Start External Viewfinder

Command	Activity/Data
TELL	CAMERA_FACTORY OPEN_CAMERA CAMERA_FACING_BACK

Send the following command to verify TE line is working

Command	Opcode/data	
TST_DISP	0x00371602	(Verify Physical connection)

Make sure that the command executed without an error, if error fail Unit. This is a Pass or Fail test. Record a successful executed command as **TE_SIGNAL_DISPLAY_TO_PROCESSOR** and compare to limit in [2.1 Test Limits](#).

Exit Test

Turn OFF Viewfinder

Command	Activity/Data
TELL	CAMERA_FACTORY RELEASE_CAMERA

Stop Activity

Command	Activity
STOP	CAMERA_FACTORY

3.56 Display Vendor ID Check Test

Test Description

The purpose of this test is to ensure that the supplier ID panels are programmed correctly to avoid panel detection issues in the software.

Test dependencies:

[3.2 Turn On](#)

Test Setup

NOTE: This test requires the display panel to be connected to the unit.

Test Procedure

Use the TST_DISP command to select the main display:

Command	Opcode/Data	
TST_DISP	0x0037 0100	(Select main display)

Use the TST_DISP command to enable the main display.

Command	Opcode/Data	
TST_DISP	0x0037 0C	(Enable main display)

Wait 250ms for the display to fully turn on.

Send the following command to read the display vendor ID.

Command	Opcode/data
TST_DISP	0x0037 1900 01A1 02

Byte0 of the 3-byte long test command response string is 02 indicating two bytes of data back.

Record Byte1 and Byte2 as **DISPLAY_VENDOR_ID** and verify it is within the limits specified in 2.1 Test Limits.

Exit Test

[OPTIONAL] Send the following TST_DISP command to disable the main display if required for subsequent tests.

Command	Opcode/Data	
TST_DISP	0x0037 0B	(Disable main display)



3.57 Audio Calibration

Test Description

Audio Calibration must be done before any alert speaker tests. This is an assembled level procedure that requires the UE alert speaker presents at the time of audio calibration.

Test Dependencies:

[3.2 Turn On](#)

[3.6 Test Mode](#) (P2K Mode)

NOTE: This test must be run in P2K NORMAL mode (meaning **NOT** SUSPENDED).

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631

Test Procedure

Calibrate Audio:

Send the following APK CommServer commands to enable the Earpiece speaker audio path and play frequency tone list. If AUDIO_TONEGEN activity is already running, no need to send START AUDIO_TONEGEN command again.

Command	Activity	Data
START	AUDIO_TONEGEN	

Command	Activity	Data
TELL	AUDIO_TONEGEN	PLAY_MEDIA_FILE FILE_NAME audio_cal.wav

Send the following command to calibrate audio.

Command	Opcode/data	
AUDIO	0x0079 0006 04	(Calibrate Audio)

Convert byte 0 of the test command response from Hex to decimal and record as **AUDIO_INTERNAL_CAL_TEMPERATURE**.

Convert byte 1 of the test command response from Hex to decimal and record as **AUDIO_INTERNAL_CAL_COUNT**.

Convert bytes 2-5 of the test command response from Hex to decimal and record as **AUDIO_INTERNAL_CAL_RAW_IMPEDANCE**.

Convert bytes 6-9 of the test command response from Hex to decimal and record as **AUDIO_INTERNAL_CAL_RES_FREQ**.

Convert bytes 10-13 of the test command response from Hex to decimal and record as **AUDIO_INTERNAL_CAL_REF_DIFF**.

Command	Activity	Data
TELL	AUDIO_TONEGEN	STOP_MEDIA_FILE
STOP	AUDIO_TONEGEN	

Verify that **AUDIO_INTERNAL_CAL_TEMPERATURE**, **AUDIO_INTERNAL_CAL_COUNT**, **AUDIO_INTERNAL_CAL_RAW_IMPEDANCE**, **AUDIO_INTERNAL_CAL_RES_FREQ**, and **AUDIO_INTERNAL_CAL_REF_DIFF** are within the limits specified in section [2.1 Test Limits](#).

Exit Test

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3.58 Audio Interrupt Line Test

Test Description

This test verifies the connection for the DSP Interrupt line to AP for Always On Voice (AOV Line).

Test Dependencies:

[3.2 Turn On](#)

Test Setup

NA

Test Procedure

Send the following command to test the Always On Voice (AOV) Interrupt Line.

Command	Opcode/data	
AUDIO	0x0079 0005 02	(Checks AOV Interrupt Line)

Verify that one byte response is 0x00. If not, fail the unit. Record this response as **AOV_Interrupt**.

Ensure that the **AOV_Interrupt** is within the limit specified in section 2.1 Test Limits.

Exit Test

None

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3.59 Alert Audio / Amplitude and Distortion Test

3.59.1 APK CommServer Method

Test Description

Enable the UE test sound and measure the main transducer (speaker) output. Verify that the audio is within tolerance.

Test Dependencies:

[3.2 Turn On](#) (Factory Cable Method – Assembled Level)

[3.57 Audio Calibration](#)

Equipment:

- PC sound card w/ Analysis SW or equivalent Audio Analysis HW

- Measurement quality Mic calibrated in $\left(\frac{mV}{Pa} \right) = \text{MicCalFactor}$

Recommended Mic and preamp:

Mic-02 - <http://www.audiomatica.com/clieng.htm>

Pre-01 - <http://www.audiomatica.com/clieng.htm>

Alternative Mic:

SPL Meter with 3.5mm jack output

Test Setup

Place the UE into the product specific test fixture such that the UE display is facing up.

UUT must have the CommServer running and communication establish over BLAN Port 2631

This is accomplished by powering up the UE with a factory cable.

Test Procedure

Tone Generation Method:

Measure the Noise Level and ensure that it (AUDIO_OUTPUT_ALERT_NOISE_LIMIT) falls within the limit specified in section 2.1 Test Limits.

Send the following APK CommServer commands to enable the specified speaker audio path and play frequency tone list.

If AUDIO_TONEGEN activity is already running, no need to send START AUDIO_TONEGEN command again.

Command	Activity	Data
---------	----------	------

START	AUDIO_TONEGEN	
-------	---------------	--

TELL	AUDIO_TONEGEN	SET_AUDIO_TONE_SETTINGS TONE_LENGTH=2.56 SAMPLE_RATE=44100 NUMBER_OF_LOOPS=0 FREQ_LIST=4000,3550,3150,2800,2500,2240,2000,1800,1600,1400,1250,1120,1000,900,800,710,630,560,500,450,400,355,315,280,250,224,200,180,160,140 MULTI_FREQ_TYPE=STEPPED VOLUME=4300 (Set tone settings)
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TELL AUDIO_TONEGEN SET_AUDIO_PATH_SETTINGS
AUDIO_MODE=NORMAL SPEAKERPHONE=ON STREAM_TYPE=MUSIC
(Sets up the audio paths)

TELL AUDIO_TONEGEN SET_VOLUME_SETTINGS MUSIC_VOLUME=15 (Sets over all volume)

TELL AUDIO_TONEGEN TURN_ON_MONO_FREQ_LIST_TONE (Plays Tone List)

Wait 1 second

Command	Activity	data
TELL	AUDIO_TONEGEN	TURN_OFF_TONE (Stop Tone List)

Sample DUT audio using Nextest sound card and Clio mic through the audio fixture

Perform a FFT on the returned response and obtain the amplitude, THD and HOHD at the respective frequency indices.

Verify that the amplitude, THD and HOHD at the respective frequency indices are within the limits specified in section [2.1 Test Limits](#).

Exit Test

Command	Activity
STOP	AUDIO_TONEGEN (Stop activity)

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3.60 Microphone Test

3.60.1 P2K TCMD Method (Board Level)

Test Description

The purpose of this test is to verify that the microphones are functioning.

Test Dependencies:

[3.2 Turn On](#)

[3.6 Test Mode](#)

Test Setup

NOTE: Primary, Secondary and Tertiary Microphones are available on board level. This test is a go/no go test to check presence of microphones that are on board level

Test Procedure

Enable external stimulus.

Send the following commands to sample the audio at the microphones*:

Command	Opcode/data
AUDIO	0x0079 0004 0000 0001 01 (AP Sample at Primary Mic)
AUDIO	0x0079 0004 0300 0001 01 (AP Sample at Secondary Mic)
AUDIO	0x0079 0004 0800 0301 01 (AP Sample at Tertiary Mic)

**NOTE: The nomenclature for the microphones is different for every model. Check with respective DE Team*

Calculate the peak amplitude between the frequency-specified window. Record and verify the frequency and amplitude of each microphone tested are within limits specified in section [2.1 Test Limits](#).

Disable external stimulus.

Exit Test

3.60.2 Microphone Test APK CommServer Method (Assembled Level)

Test Description

The purpose of this test is to verify that the microphones are functioning properly and meets the product requirements.

Test Dependencies:

[3.2 Turn On](#) (Factory Cable Method – Assembled Level)

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631

This is accomplished by powering up the UE with a factory cable.

Test Procedure

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Microphone(Main Codec Path @Endo Level)

Generate a sine sweep with the following specifications:

Start Freq: 100Hz

Stop Freq: 3800Hz

Sweep Duration: 20ms (continuously looped)

Sweep Type: Linear

Send the following commands to sample the audio at the microphones:

Command	Activity/data	
START	AUDIO_SAMPLE	(Start AUDIO_SAMPLE activity)

To sample audio at Primary Mic

Command	Activity/data	
TELL	AUDIO_SAMPLE SET_SAMPLE_SETTINGS SAMPLE_RATE_HZ=8000 SAMPLE_LENGTH_MS=100 SAMPLE_DISCARD_TIME_MS=50 CAPTURE_TIMEOUT_MS=5000 INPUT_SOURCE= MIC INPUT_FORMAT=CHANNEL_IN_MONO MIC_SELECT=PRIMARY (Setup Primary Mic For Sampling)	
TELL	AUDIO_SAMPLE SAMPLE_AUDIO (Sample Primary Mic Audio)	

To sample audio at Secondary Mic:

Command	Activity/data	
TELL	AUDIO_SAMPLE SET_SAMPLE_SETTINGS SAMPLE_RATE_HZ=8000 SAMPLE_LENGTH_MS=100 SAMPLE_DISCARD_TIME_MS=50 CAPTURE_TIMEOUT_MS=5000 INPUT_SOURCE= MIC INPUT_FORMAT=CHANNEL_IN_MONO MIC_SELECT=SECONDARY (Setup Secondary Mic For Sampling)	
TELL	AUDIO_SAMPLE SAMPLE_AUDIO (Sample Secondary Mic Audio)	

To sample audio at Tertiary Mic:

Command	Activity/data	
TELL	AUDIO_SAMPLE SET_SAMPLE_SETTINGS SAMPLE_RATE_HZ=8000 SAMPLE_LENGTH_MS=100 SAMPLE_DISCARD_TIME_MS=50 CAPTURE_TIMEOUT_MS=5000 INPUT_SOURCE= MIC INPUT_FORMAT=CHANNEL_IN_MONO MIC_SELECT=TERTIARY (Setup Secondary Mic For Sampling)	
TELL	AUDIO_SAMPLE SAMPLE_AUDIO (Sample Tertiary Mic Audio)	

Send the following command to stop the audio sample activity.

Command	Activity/data	
STOP	AUDIO_SAMPLE	(Stop AUDIO_SAMPLE activity)

Take an FFT of the returned data for the microphone. Verify the Amplitude falls within the limits for the respective frequencies for uncovered readings specified in section [2.1 Test Limits](#).

Exit Test

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3.61 Ultrasonic Test

Test Description

Enable the UE test sound and measure the main transducer (speaker) output. Verify that the audio is within tolerance.

Test Dependencies:

3.2 Turn On (Factory Cable Method – Assembled Level)

3.6.6 P2K SUSPEND Mode

Equipment:

- PC sound card w/ Analysis SW or equivalent Audio Analysis HW

- Measurement quality Mic calibrated in $\left(\frac{mV}{Pa}\right) = \text{MicCalFactor}$

Recommended Mic and preamp:

Mic-02 - <http://www.audiomatica.com/clioeng.htm>

Pre-01 - <http://www.audiomatica.com/clioeng.htm>

Alternative Mic:

SPL Meter with 3.5mm jack output

Test Setup

Place the UE into the product specific test fixture such that the UE display is facing up.

Test Procedure

Send the following test commands to enable the ultrasonic path and enable the 1 KHz tone from Marley IC.

Command	Opcode/data
AUDIO	0x0079 0001 1000 0064 0101 007F FF (Enable 1KHz tone from Marley IC)

Wait 1 second

Sample DUT audio using Nextest sound card and Clio mic through the audio fixture

Perform a FFT on the returned response and obtain the amplitude at the respective frequency index.

Send the following commands to stop the tone.

Command	Opcode/data
AUDIO	0x0079 0000 (Stop Tone)

Verify that the amplitude at the respective frequency index is within the limits specified in section [2.1 Test Limits](#).

Exit Test

None

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3.62 Vibrator Test

3.62.1 P2K TCMD Method

Test Description

The purpose of this test is to verify that the all vibrators are physically connected electrically and functioning correctly. To test in an automated fashion, an external vibration sensor is used to measure the peak frequency and amplitude of the vibration.

Test Dependencies:

None

Equipment:

JOT Station External Vibration Sensor

Test Setup

None

Test Procedure

Use the VIBRATOR command to enable the vibrator.

Command	Opcode/data	
VIBRATOR	0x0096 0100	(Enable the vibrator)

Using and External Vibration Sensor, measure the peak frequency and amplitude of the vibration.

Ensure they are within the limits specified in section [2.1 Test Limits](#).

Use the VIBRATOR command to disable the vibrator.

Command	Opcode/data	
VIBRATOR	0x0096 0000	(Disable the vibrator)

Exit Test

None



3.63 Keypad

APK CommServer Method

Test Description

The purpose of this test is to verify that each key press is detected.

Test Dependencies:

[3.2 Turn On](#) (Assembled Level)

Test Setup

UE must have the CommServer running and communication establish over BLAN Port 2631

Test Procedure

Use the KEYTEST CommServer commands to read key press detection responses.

Command	Activity	Data
START	KEYTEST	(start up Keys test Activity)
TELL	KEYTEST	START_RECORD (capture key presses)

Press and release all desired keys as needed.

Enter the following commands to retrieve the results

Command	Activity	Data
TELL	KEYTEST	GET_KEYS_FOR_TEST (returns a list of keys pressed and released)

Verify that both a key press and release detection code is returned for each key. Returned data is in form of key code for each key detected.

Exit Test

Command	Activity	Data
STOP	KEYTEST	

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3.64 Headset Interface

Headset Interface (APK CommServer Method)

Test Description

The purpose of this test is to verify continuity on the headset connector jack.

Test Dependencies:

3.2 Turn On (Assembled Level)

Test Setup

UUT must have the CommServer running and communication establish over BLAN Port 2631
This is accomplished by powering up the UE with a factory cable

Test Procedure

Apply a 33 Ohm Load to the left and right pins of the headset connector.

Headset Detection

Send the following APK CommServer commands to read the headset detection status.

Command	Activity	Data
START	HEADSETINFO	
TELL	HEADSETINFO	GET_HEADSET_STATE

Poll with the same command to get the state change (Plugged). Once the headset detect test is done, stop the activity by issuing the following command:

Command	Activity	Data
STOP	HEADSETINFO	

Verify that **PLUG_STATE**, **HEADSET_HAS_MIC**, and **Headset Detection** are within the limits specified in section [2.1 Test Limits](#).

Headset Audio Loopback Test

1 KHz Tone on Right

Send the following APK CommServer commands to play a 1 KHz tone on the Right headset speaker channel

Command	Activity	Data
START	AUDIO_TONEGEN	
TELL	AUDIO_TONEGEN	SET_AUDIO_PATH_SETTINGS AUDIO_MODE=NORMAL SPEAKERPHONE=OFF STREAM_TYPE=MUSIC
TELL	AUDIO_TONEGEN	SET_VOLUME_SETTINGS MUSIC_VOLUME=15
TELL	AUDIO_TONEGEN	SET_AUDIO_TONE_SETTINGS NUMBER_OF_LOOPS=-1 TONE_LENGTH=1 LEFT_VOLUME=0 LEFT_START_FREQ=0 LEFT_STOP_FREQ=0 LEFT_LOG_SCALE=0 LEFT_MUTE=1 RIGHT_VOLUME=15000 RIGHT_START_FREQ=1000 RIGHT_STOP_FREQ=1000 RIGHT_LOG_SCALE=0 RIGHT_MUTE=0
TELL	AUDIO_TONEGEN	TURN_ON_STEREO_TONE

Headset Mic Via Audio Codec Path

Send the following APK CommServer commands to sample the routed audio at the headset mic input

Command	Activity	Data
START	AUDIO_SAMPLE	
TELL	AUDIO_SAMPLE	SET_SAMPLE_SETTINGS SAMPLE_RATE_HZ=8000 SAMPLE_LENGTH_MS=20 SAMPLE_DISCARD_TIME_MS=200 CAPTURE_TIMEOUT_MS=5000 INPUT_SOURCE=DEFAULT INPUT_FORMAT=CHANNEL_IN_MONO MIC_SELECT=headset
TELL	AUDIO_SAMPLE	SAMPLE_AUDIO

Take an FFT of the returned data. Verify that the largest frequency component of the returned signal is 1kHz. Verify that there are no other major frequency components present in the returned signal. Verify that **SPKRRIGHT_1KHzONRIGHT_LEFTM** is within the limits specified in section [2.1 Test Limits](#).

When all the above verification is done, send the following APK CommServer command to stop audio play media file

Command	Activity	Data
TELL	AUDIO_TONEGEN	TURN_OFF_TONE
STOP	AUDIO_SAMPLE	
STOP	AUDIO_TONEGEN	

2 KHz Tone on Left

Send the following APK CommServer commands to play a 2 KHz tone on the Left headset speaker channel

Command	Activity	Data
START	AUDIO_TONEGEN	
TELL	AUDIO_TONEGEN	SET_AUDIO_PATH_SETTINGS AUDIO_MODE=NORMAL SPEAKERPHONE=OFF STREAM_TYPE=MUSIC
TELL	AUDIO_TONEGEN	SET_VOLUME_SETTINGS MUSIC_VOLUME=15
TELL	AUDIO_TONEGEN	SET_AUDIO_TONE_SETTINGS NUMBER_OF_LOOPS=-1 TONE_LENGTH=1 LEFT_VOLUME=15000 LEFT_START_FREQ=2000 LEFT_STOP_FREQ=2000 LEFT_LOG_SCALE=0 LEFT_MUTE=0 RIGHT_VOLUME=0 RIGHT_START_FREQ=0 RIGHT_STOP_FREQ=0 RIGHT_LOG_SCALE=0 RIGHT_MUTE=1
TELL	AUDIO_TONEGEN	TURN_ON_STEREO_TONE

Headset Mic Via Audio Codec Path

Send the following APK CommServer commands to sample the routed audio at the headset mic input

Command	Activity	Data
START	AUDIO_SAMPLE	
TELL	AUDIO_SAMPLE	SET_SAMPLE_SETTINGS SAMPLE_RATE_HZ=8000 SAMPLE_LENGTH_MS=20 SAMPLE_DISCARD_TIME_MS=200 CAPTURE_TIMEOUT_MS=5000 INPUT_SOURCE=DEFAULT INPUT_FORMAT=CHANNEL_IN_MONO MIC_SELECT=headset

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TELL AUDIO_SAMPLE SAMPLE_AUDIO

Take an FFT of the returned data. Verify that the largest frequency component of the returned signal is 2kHz. Verify that there are no other major frequency components present in the returned signal. Verify that **SPKRLEFT_2KHzONLEFT_RIGHTM** is within the limits specified in section [2.1 Test Limits](#).

When all the above verification is done, send the following APK CommServer command to stop audio play media file

Command	Activity	Data
TELL	AUDIO_TONEGEN	TURN_OFF_TONE
STOP	AUDIO_SAMPLE	
STOP	AUDIO_TONEGEN	

2 KHz Tone on Right

Feed the audio output on the right speaker line to the audio input on the mic line.

Send the following APK CommServer commands to sample the routed audio at the headset mic input

Command	Activity	Data
START	AUDIO_TONEGEN	
TELL	AUDIO_TONEGEN	SET_AUDIO_PATH_SETTINGS AUDIO_MODE=NORMAL SPEAKERPHONE=OFF STREAM_TYPE=MUSIC
TELL	AUDIO_TONEGEN	SET_VOLUME_SETTINGS MUSIC_VOLUME=15
TELL	AUDIO_TONEGEN	SET_AUDIO_TONE_SETTINGS NUMBER_OF_LOOPS=-1 TONE_LENGTH=1 LEFT_VOLUME=0 LEFT_START_FREQ=0 LEFT_STOP_FREQ=0 LEFT_LOG_SCALE=0 LEFT_MUTE=1 RIGHT_VOLUME=15000 RIGHT_START_FREQ=2000 RIGHT_STOP_FREQ=2000 RIGHT_LOG_SCALE=0 RIGHT_MUTE=0
TELL	AUDIO_TONEGEN	TURN_ON_STEREO_TONE

Headset Mic Via Audio Codec Path

Send the following APK CommServer commands to sample the routed audio at the headset mic input

Command	Activity	Data
START	AUDIO_SAMPLE	
TELL	AUDIO_SAMPLE	SET_SAMPLE_SETTINGS SAMPLE_RATE_HZ=8000 SAMPLE_LENGTH_MS=20 SAMPLE_DISCARD_TIME_MS=200 CAPTURE_TIMEOUT_MS=5000 INPUT_SOURCE=DEFAULT INPUT_FORMAT=CHANNEL_IN_MONO MIC_SELECT=headset
TELL	AUDIO_SAMPLE	SAMPLE_AUDIO

Take an FFT of the returned data. Verify that the largest frequency component of the returned signal is 2kHz. Verify that **SPKRLEFT_2KHzONRIGHT_LEFTM** is within the limits specified in section [2.1 Test Limits](#).

When all the above verification is done, send the following APK CommServer command to stop audio play media file

Command	Activity	Data
TELL	AUDIO_TONEGEN	TURN_OFF_TONE
STOP	AUDIO_SAMPLE	
STOP	AUDIO_TONEGEN	

Headset Send/End Key

Note: This procedure tests Normally Open contact Headset key – Motorola headset type.



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Send the following APK CommServer commands to read the key press detection status

Command	Activity	Data
START	KEYTEST	
TELL	KEYTEST	START_RECORD

Short the mic line to ground by pressing and holding the Motorola send/end button on the headset. Then release the Motorola button

Command	Activity	Data
TELL	KEYTEST	GET_KEYS_FOR_TEST

Monitor the response while depressing and releasing of the press on the Send/End key. Verify the data returned for both the key press and release scenarios. For key press, it returns data as "KEYS = 4F00" and for key release it returns value as "KEYS=4F01 ". Verify that the **Headset Send/End Detect** is within the limits specified in section [2.1 Test Limits](#).

Then send the following APK CommServer commands to stop the key press detection test.

Command	Activity	Data
TELL	KEYTEST	STOP_RECORD
STOP	KEYTEST	

Headset Removal Detection

Remove the headset from the headset jack.

Send the following APK CommServer commands to read the headset detection indicator.

Command	Activity	Data
START	HEADSETINFO	
TELL	HEADSETINFO	GET_HEADSET_STATE

Ensure a successful response for the Unplugged status. Poll with the same command for a couple of times to make sure the state is stable.

Then stop the headset removal test by issuing the following APK CommServer command:

Command	Activity	Data
STOP	HEADSETINFO	

Check the data returned in CommServer to verify that the removal status is returned properly. Verify that **Headset Removal** is within the limits specified in section [2.1 Test Limits](#).

Exit Test

NOTE: After the headset is being removed, no audio path test command should be executed during any other subsequent tests until allowance is made for a proper headset detachment, i.e. debounce time.

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3.65 AMPS Connector Tests

Test Description

The purpose of this test is to verify connectivity and functionality of the AMPS connector lines.

Test dependencies:

3.2 Turn On

Test Setup

Must connect to the external AMPS test board on the Factory Test Bench.

Test Procedure

**APBA Testing (Board Level only)**

Send the following test command to enable APBA Factory Test Mode

Command	Opcode/data	
AMPS	0x0112 00	(Enable APBA Factory Test Mode)

Send the following test command to disable APBA Factory Test Mode

Command	Opcode/data	
AMPS	0x0112 10	(Disable APBA Factory Test Mode)

Verify there is not a test command failure. Record this as a PASS/FAIL response as **APBA_TEST** and verify it against the limits specified in section 2.1 Test Limits.

ANX78xx_INIT VERIFICATION Testing (Board Level only)

Send the following test command to read ANX I2C Probe Success at Power Up Init

Command	Opcode/data	
AMPS	0x0112 07	(ANX_INIT Verify)

Verify there is not a test command failure. Record this as a PASS/FAIL response as **ANX78xx_TEST** and verify it against the limits specified in section 2.1 Test Limits.

GPIO Testing

Note: GPIOs 0-3 are currently tested with Fast Functional testing as there is no current solution for automated testing.

Note: GPIOs 4-7 are covered with AINx ADC testing

CWIRE and ADC Testing

Send the following test command to run AMPS test:

Command	Opcode/data	
BATCH	0x0C2F 0000 0000 0001 + "AMPS_TEST.bat" + NULL	(in Unicode)

Ensure a successful response (0x0000) for the BATCH command and there will be no unsolicited responses.

Send the following test command to open /data/local/tmp/AMPS_TEST.bin

Command	Opcode/data	
FSAC	0x004a 0001 0000 0000 0000 2F64 6174 612F 6C6F 6361 6C2F 746D 702F 414D 5053 5F54 4553 542E 6269 6E00 00	

Send the following test command to read 48 bytes from /data/local/tmp/AMPS_TEST.bin

Command	Opcode/data	
FSAC	0x004a 0001 0001 0000 0030	

Send the following test command to close /data/local/tmp/AMPS_TEST.bin

Command	Opcode/data	
FSAC	0x004a 0001 0004	

Make sure 48 bytes are returned, otherwise fail the test.

Byte[0]: CWIRE MSM GPIO44 START LOW

Byte[1]: CWIRE MSM GPIO44 HIGH

Byte[2]: CWIRE MSM GPIO44 LOW

Byte[3-4]: AMPS_AIN1_C1_VOLTAGE

Byte[5]: MSM GPIO 60

Byte[6]: MSM GPIO 72

Byte[7]: MSM GPIO 59

Byte[8-9]: AMPS_AIN1_C2_VOLTAGE

Byte[10]: MSM GPIO 60
 Byte[11]: MSM GPIO 72
 Byte[12]: MSM GPIO 59
 Byte[13-14]: AMPS_AIN1_C3_VOLTAGE
 Byte[15]: MSM GPIO 60
 Byte[16]: MSM GPIO 72
 Byte[17]: MSM GPIO 59
 Byte[18-19]: AMPS_AIN1_C4_VOLTAGE
 Byte[20]: MSM GPIO 60
 Byte[21]: MSM GPIO 72
 Byte[22]: MSM GPIO 59
 Byte[23-24]: AMPS_AIN1_C5_VOLTAGE
 Byte[25]: MSM GPIO 60
 Byte[26]: MSM GPIO 72
 Byte[27]: MSM GPIO 59
 Byte[28-29]: AMPS_AIN1_C6_VOLTAGE
 Byte[30]: MSM GPIO 60
 Byte[31]: MSM GPIO 72
 Byte[32]: MSM GPIO 59
 Byte[33-34]: AMPS_AIN1_C7_VOLTAGE
 Byte[35]: MSM GPIO 60
 Byte[36]: MSM GPIO 72
 Byte[37]: MSM GPIO 59
 Byte[38-39]: AMPS_AIN1_C8_VOLTAGE
 Byte[40]: MSM GPIO 60
 Byte[41]: MSM GPIO 72
 Byte[42]: MSM GPIO 59
 Byte[43-44]: AMPS_AIN2_VOLTAGE
 Byte[45-46]: AMPS_AIN3_VOLTAGE
 Byte[47]: AMPS I2S

Data is explained as following:

CWIRE

A 1 byte response will be returned for each MSMGPIO_44 read. Record the hex response as **AMPS_MSM_GPIO44_HIGH (Byte[1])** and **AMPS_MSM_GPIO44_LOW (Byte[2])** and verify it against the limits specified in section 2.1 Test Limits.

AIN1

For AIN1 testing, the following configuration **Table 3.65** will be used.

	READ-GPIO			SET-GPIO			
Configuration	GPIO 60	GPIO 72	GPIO 59	GPIO 38	JOT_GPIO 5	JOT_GPIO 6	JOT_GPIO 7
C1	HIGH	HIGH	HIGH	HIGH	LOW	LOW	LOW
C2	LOW	HIGH	HIGH	HIGH	LOW	LOW	HIGH
C3	LOW	LOW	HIGH	HIGH	LOW	HIGH	LOW
C4	LOW	HIGH	HIGH	HIGH	HIGH	LOW	LOW
C5	HIGH	HIGH	HIGH	LOW	LOW	LOW	LOW
C6	LOW	HIGH	HIGH	LOW	LOW	LOW	HIGH



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C7	LOW	LOW	HIGH	LOW	LOW	HIGH	LOW
C8	LOW	HIGH	LOW	LOW	HIGH	LOW	LOW
Table 3.65							



Convert the two bytes(Byte[3-4]) from Hex to Decimal. Each unit (ADC step) = 125uV. Record this value as **AMPS_AIN1_C1_VOLTAGE**. Verify against the limits specified in section 2.1 Test Limits. Record the three GPIO (GPIO60 byte[5], GPIO72 byte[6], and GPIO59 byte[7]) reads as single Pass/Fail response, **AMPS_AIN1_C1_GPIO_TEST**. Each GPIO state should follow GPIO logic state listed in **Table 3.65** for **C1** Test configuration or the unit should be failed. Verify against the limits specified in section 2.1 Test Limits.

Convert the two bytes(Byte[8-9]) from Hex to Decimal. Each unit (ADC step) = 125uV. Record this value as **AMPS_AIN1_C2_VOLTAGE**. Verify against the limits specified in section 2.1 Test Limits. Record the three GPIO (GPIO60 byte[10], GPIO72 byte[11], and GPIO59 byte[12]) reads as single Pass/Fail response, **AMPS_AIN1_C2_GPIO_TEST**. Each GPIO state should follow GPIO logic state listed in **Table 3.65** for **C2** Test configuration or the unit should be failed. Verify against the limits specified in section 2.1 Test Limits.

Convert the two bytes(Byte[13-14]) from Hex to Decimal. Each unit (ADC step) = 125uV. Record this value as **AMPS_AIN1_C3_VOLTAGE**. Verify against the limits specified in section 2.1 Test Limits. Record the three GPIO (GPIO60 byte[15], GPIO72 byte[16], and GPIO59 byte[17]) reads as single Pass/Fail response, **AMPS_AIN1_C3_GPIO_TEST**. Each GPIO state should follow GPIO logic state listed in **Table 3.65** for **C3** Test configuration or the unit should be failed. Verify against the limits specified in section 2.1 Test Limits.

Convert the two bytes(Byte[18-19]) from Hex to Decimal. Each unit (ADC step) = 125uV. Record this value as **AMPS_AIN1_C4_VOLTAGE**. Verify against the limits specified in section 2.1 Test Limits. Record the three GPIO (GPIO60 byte[20], GPIO72 byte[21], and GPIO59 byte[22]) reads as single Pass/Fail response, **AMPS_AIN1_C4_GPIO_TEST**. Each GPIO state should follow GPIO logic state listed in **Table 3.65** for **C4** Test configuration or the unit should be failed. Verify against the limits specified in section 2.1 Test Limits.

MSM_Slice CC Control Test

Convert the two bytes(Byte[23-24]) from Hex to Decimal. Each unit (ADC step) = 125uV. Record this value as **AMPS_AIN1_C5_VOLTAGE**. Verify against the limits specified in section 2.1 Test Limits. Record the three GPIO (GPIO60 byte[25], GPIO72 byte[26], and GPIO59 byte[27]) reads as single Pass/Fail response, **AMPS_AIN1_C5_GPIO_TEST**. Each GPIO state should follow GPIO logic state listed in **Table 3.65** for **C5** Test configuration or the unit should be failed. Verify against the limits specified in section 2.1 Test Limits.

Convert the two bytes(Byte[28-29]) from Hex to Decimal. Each unit (ADC step) = 125uV. Record this value as **AMPS_AIN1_C6_VOLTAGE**. Verify against the limits specified in section 2.1 Test Limits. Record the three GPIO (GPIO60 byte[30], GPIO72 byte[31], and GPIO59 byte[32]) reads as single Pass/Fail response, **AMPS_AIN1_C6_GPIO_TEST**. Each GPIO state should follow GPIO logic state listed in **Table 3.65** for **C6** Test configuration or the unit should be failed. Verify against the limits specified in section 2.1 Test Limits.

Convert the two bytes(Byte[33-34]) from Hex to Decimal. Each unit (ADC step) = 125uV. Record this value as **AMPS_AIN1_C7_VOLTAGE**. Verify against the limits specified in section 2.1 Test Limits. Record the three GPIO (GPIO60 byte[35], GPIO72 byte[36], and GPIO59 byte[37]) reads as single Pass/Fail response, **AMPS_AIN1_C7_GPIO_TEST**. Each GPIO state should follow GPIO logic state listed in **Table 3.65** for **C7** Test configuration or the unit should be failed. Verify against the limits specified in section 2.1 Test Limits.

Convert the two bytes(Byte[38-39]) from Hex to Decimal. Each unit (ADC step) = 125uV. Record this value as **AMPS_AIN1_C8_VOLTAGE**. Verify against the limits specified in section 2.1 Test Limits. Record the three GPIO (GPIO60 byte[40], GPIO72 byte[41], and GPIO59 byte[42]) reads as single Pass/Fail response, **AMPS_AIN1_C8_GPIO_TEST**. Each GPIO state should follow GPIO logic state listed in **Table 3.65** for **C8** Test configuration or the unit should be failed. Verify against the limits specified in section 2.1 Test Limits.

AIN2

Convert the two bytes(Byte[43-44]) from Hex to Decimal. Each unit (ADC step) = 125uV. Record this value as **AMPS_AIN2_VOLTAGE**. Verify against the limits specified in section 2.1 Test Limits.

AIN3



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Convert the two bytes(Byte[45-46]) from Hex to Decimal. Each unit (ADC step) = 125uV. Record this value as **AMPS_AIN3_VOLTAGE**. Verify against the limits specified in section 2.1 Test Limits.

I2S Testing

A 1 byte(Byte[47]) response will be returned. Record the PASS/FAIL response as **AMPS_I2S** and verify it against the limits specified in section 2.1 Test Limits.

SPI Testing

Send the following test command to Verify SPI Loopback.

Command	Opcode/data
---------	-------------

AMPS	0x0112 04
------	-----------

 (Verify SPI Loopback)

A 1 byte response will be returned. Record the PASS/FAIL response as **AMPS_SPI** and verify it against the limits specified in section 2.1 Test Limits.

Exit Test

None

3.66 EFS_SYNC

Test Description

For this platform certain commands need to be followed up with synchronizing the file system to the backing store. The following procedure outlines this process.

Test Dependencies:

[3.2 Turn On](#)

[3.6.1 FTM Mode](#)

Test Setup

N/A

Test Procedure

More details on the specific commands referenced below can be gotten from Qualcomm document:

80-V1294-11_V_FS_Subsys_ICD.pdf

Issue the **EFS2_DIAG_SYNC_NO_WAIT** FTM command request message.

Command	OpCode/Data
EFS2_DIAG_SYNC_NO_WAIT	4B133000 xxxx 2F00
EFS2_DIAG_SYNC_NO_WAIT response	4B133000 xxxx yyyyyyyy zzzzzzzz – verify last 4 bytes are all zero (no error).

Ensure that the **SEQUENCE_NUMBER** (xxxx) field, the 5th byte of the input, of length 2 bytes, is always unique for each time the command is issued. Also ensure the response sequence number matches the request sequence number.

Query for the response and save the value of the **SYNC_TOKEN** (yyyyyyyy) field, the 7th byte of the response, of length 4 bytes, and save it as **TOKEN**.

Also ensure that the **ERRNO** (zzzzzzzz) field, the 11th byte of the response, of length 4 bytes, is 0, which implies a successful operation or 306 (ENOTHINGTOSYNC can be returned to specify that EFS is not dirty and there's nothing to synchronize).

Issue the **EFS2_DIAG_SYNC_GET_STATUS** FTM command request message.

Command	OpCode/Data
EFS2_DIAG_SYNC_GET_STATUS	4B133100 xxxx yyyyyyyy 2F00
EFS2_DIAG_SYNC_GET_STATUS response	4B133100 xxxx aa zzzzzzzz – verify SYNC_STATUS byte is 1 (sync complete) and last 4 bytes are all zero (no error).

Ensure that the **SEQUENCE_NUMBER** (xxxx) field, the 5th byte of the input, of length 2 bytes, is always unique for each time the command is issued. Also ensure the response sequence number matches the request sequence number.

Ensure that the **SYNC_TOKEN** (yyyyyyyy) field, the 7th byte of the input, of length 4 bytes, is populated with the value of **TOKEN** that was saved above.

Query for the response and check the value of the **SYNC_STATUS** field, the 7th byte of the response, of length 1 byte.

Continue to poll for up to 15 seconds, 250 ms between each poll, with the **EFS2_DIAG_SYNC_GET_STATUS** command as long as the **SYNC_STATUS** (aa) field is does not equal 1.

Also ensure that the **ERRNO** (zzzzzzzz) field, the 7th byte of the response, of length 4 bytes, is 0, which implies a successful operation or 306 (ENOTHINGTOSYNC can be returned to specify that EFS is not dirty and there's nothing to synchronize).

Exit Test

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3.67 Save to HOB

Test Description

This test will copy certain NV items into the HOB area for non-volatile backup. This data is then available during Flash upgrades or if the EFS is lost, so that certain key data can be restored. There are 2 pre-requisites for this test. 1) This test must be after all of the items in the HOB have been written to at least once. And 2) This test must be done BEFORE the Factory byte is set to OUT. The HOB test command will only work when the radio is set to IN factory mode.

Test Dependencies:

[3.2 Turn On](#)

[3.6 Test Mode](#)

All Items to HOB List must be initialized (written to at least once)

Unit must be "In Factory"

Radio Setup

N/A

Test Procedure

Send the following command to save the information to the HOB (Hand Over Block).

Command	Opcode	data
HOB	0x0D07	00

(To Save Data to HOB)

The return data from this command MUST be 0x00000000 or else the radio fails.

Exit Test

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3.68 Verify Save to HOB

Test Description

This test will verify the HOB. There are 2 pre-requisites for this test. 1) Save to HOB must have been run prior to this test. 2) The unit must be turned on in factory mode.

Test Dependencies:

[3.2 Turn On](#) (factory mode via EMU Style Connector method or Factory Cable method required)

[3.6 Test Mode](#)

[3.67 Save to HOB](#)

Radio Setup

N/A

Test Procedure

Send the following command to Verify HOB.

Command	Opcode	data	
HOB	0x0D07	04	(Verify HOB)

The return data from this command MUST be 0x0000 or else the radio fails.

Exit Test

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3.69 Phase File Storage

Test Description

Extract and save all phasing data in txt file in the NV format to a server on the network.

Test Dependencies:

[3.6 Test Mode](#)

All phasing procedures must be completed prior to execution

Test Setup

Power Supply:

Voltage	See section 1.6.7 Supply Voltage
Current Limit	1.5 Amps

The UE supply voltage may be supplied at either the battery contacts or the accessory connector.

Test Procedure

All phased NV Items and modified static NV items (from default file shall be saved in the proper format.

Example format of the product output file:

```
NV_ [label]
data
NV_ [Label]
data
```

Save the files to the root location: [\\il93brm06\Factory](#) in a sub-directory named by the first three characters of the unit ID barcode. If the directory does not exist, it must be created by the tool. Store files for multiple phasing stations in separate directories (ie GSM, UMTS, LTE). The phase data must be stored with the filename: "UNIT_ID.nvm".

These files can be restored using RadioComm FTM Common tab, "Write Cal File".

Exit Test

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3.70 NVM Setup

Test Description

NVM setup ensures the correct NVM information is stored in SEEM for successfully completing factory test. This includes factory tracking information and parameters that are different between models of the same product (such as a triband and quad-band model of the same product).

Any UE configuration that cannot be included in the manufacturing or customer flex is also performed. That may include settings for second-source displays or other components that cannot be automatically detected by software. The specific hardware installed is not readable after the unit is assembled so the correct information must be written in the factory. This should be a rare case and we will attempt to avoid it when we can, but sometimes it is needed.

NOTE: If any or all of the following SEEM elements are preset correctly in the manufacturing flex, they do not need to be re-written here.

Test Dependencies:

[3.2 Turn On](#)
[3.6 Test Mode](#)

Radio Setup

N/A

Test Procedure

Update the information in SEEM_FACTORY_TRACKING_INFO according to the requirements in factory tracking info document (see reference [\[4\]](#))

Command	Opcode	data
STELEM	0x002F	272600010000[Bytes][Factory Info] (Write factory tracking info)

Exit Test



3.71 Load NVM Defaults

Test Description

Program the UE SEEM tables with all necessary default values.

Test Dependencies:

[3.6 Test Mode](#)

Test Setup

N/A

Test Procedure

The required defaults for each model are included as part of the build request information. Updated default files may be found at the following web site: <https://drive.google.com/drive/folders/0B7sFdJVUO1x5VGUzQ3pWNEQxa3c> These files are for reference only.

Exit Test



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3.72 Load RF NV

Test Description

This operation is used to load RF NV from the NV to the RF driver immediately. This allows the test automation to skip the step of putting the mobile into Online mode only for the purpose of loading the RF NV properly.

This command is designed to be used in test stations where RF calibration is performed and nonsignaling (specification/performance) testing is performed immediately afterwards, without resetting the phone. If this step is not performed, then the RF NV is not loaded from the NV to the RF driver and the phone does not use the NV item changes that occurred since the last time the phone was powered up.

Test Dependencies:

[3.2 Turn On](#)

[3.6.1 FTM Mode](#)

Test Setup

N/A

Test Procedure

Send the following FTM command to load the phased NV data into RAM so it is applied during call testing:

Command	Data
FTM_LOAD_RF_NV	4B0B140083000000C00 (Load NV data to RAM)

Exit Test



3.73 Auto-cal Items

Test Description

The purpose of this test is to initialize the RFNV_MSM_SELF_CAL_I, RFNV_RFDEVICE_HDET_AUTOCAL_I and NV_RC_PROCESS_ERRORS items so that auto-calibration can occur correctly. This procedure needs to be run prior to CDMA/WCDMA TX calibration otherwise TX phasing will be inaccurate.

Test Dependencies:[3.6.1 FTM Mode](#)

Must be done prior to CDMA and WCDMA calibration

Equipment:

None

Test Setup

Place radio in FTM mode.

Test Procedure

For NV item: RFNV_MSM_SELF_CAL_I:

Use the following NV_WRITE command to initialize the NV item by setting it to 0x0000...

Command	Data
NV_WRITE	0x27C6590000000000000000 + fill with 0x00 until 128 bytes

For NV item: RFNV_RFDEVICE_HDET_AUTOCAL_I:

Use the following NV_WRITE command to initialize the NV item by setting it to 0x0000...

Command	Data
NV_WRITE	0x27D45C0000000000000000 + fill with 0x00 until 128 bytes

For NV item: NV_RC_PROCESS_ERRORS:

Use the following NV_WRITE command to initialize the NV item by setting it to 0x0000...

Command	Data
NV_WRITE	0x27CD1A0000000000000000 + fill with 0x00 until 128 bytes

EFS SYNC

Refer to section [3.66 EFS_SYNC](#)

Power cycle the radio.

Exit Test



3.74 XO Calibration

Coarse Calibration – WCDMA Method

Test Description

This calibration should be done in WCDMA. This procedure only needs to be once per phone, but care should be taken to apply the correct frequency offset as it is band dependent. The procedure should be performed prior to the TX being turned on in any technology.

Test Dependencies:

Coarse Cal section should be completed prior to the transmitter being turned on.

Equipment:

WCDMA Communications Test Set

Test Setup

N/A

Test Procedure

WCDMA Coarse Calibration

Note: The WCDMA XO procedure is done on WCDMA B5 DL Ch 4409(Ref. Freq 881800.0kHz).

Generate a CW waveform with a +200 kHz offset from the Ref. Frequency at an RF Power of -35 dBm.

Command	Data
FTM_SET_MODE	Mode (Select WCDMA band)

Where:

Band1	Mode = 0x0900, UL channel = (DL channel – 950)
Band2	Mode = 0x1000, UL channel = (DL channel – 400)
Band4	Mode = 0x1C00, UL channel = (DL channel – 225)
Band5/6	Mode = 0x1600, UL channel = (DL channel – 225)
Band8	Mode = 0x1D00, UL channel = (DL channel – 225)

Use the following FTM command to set secondary chain disable.

Command	Opcode/Data
FTM_SET_SECONDARY_CHAIN	4B0B1500790000007D5DC27E (Set Secondary Chain Disable)

Use the following FTM command to tune the UUT to the Reference Channel.

Command	Data
FTM_SET_CHAN	UL Ref Channel (Set UUT to UL channel)

Command	Data
FTM_WCDMA_SET_TX_OFF	none (Turn TX OFF)

Use the following FTM command to Set LNA Range to LNA_R2.

Command	Opcode/Data
FTM_SET_LNA_RANGE	4B0B01003A000200AB907E (Set LNA Range)



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Use the following command to perform a coarse cal:

Set the following parameters (example assumes calibration is done on WCDMA 850 band at channel 4184:

Command = CoarseCal
Version=3
Technology = WCDMA
Rx Freq (kHz) = 881800 (for WCDMA Band5 UL Ch 4184)
Freq Offset (Hz) = 200000
Temp Span (degree C) = 1024
Min FT Sample = 5
Max FT Sample = 100
Substage Time (ms) = 10
Timeout (ms) = 30000
UpdateNv=RecordToRamAndNv
TemperatureGradientTimeout (ms) = 10
CrystalPackage = TCXOMGR_XTAL_2016

For WCDMA Models that support WCDMA Band5 use the following FTM command to perform Coarse cal.

Command	Opcode/Data
---------	-------------

FTM_XO_CALIBRATION	4B 0B 1400 7002 0000 0C00 0100 03 01 88740D00 400D0300 00040000 0500 6400 0A00 3075 03 0000 0A00 01 6DC27E (perform Coarse cal for WCDMA Band5 UL Ch 4184)
--------------------	--

Ensure that the Coefficient State of the response is (11th Byte) = 01

Command	Data
FTM_WCDMA_SET_TX_OFF	none

(Turn TX OFF)

Exit Test

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4.0 WCDMA Test Procedures

The procedures in this section describe the methods to tune the WCDMA hardware control parameters and verify WCDMA functionality in the UE. The Reference phasing and the Band phasing must be done at the same time.

4.1 WCDMA Internal DAC Calibration

Test Description

This performs required internal device calibration for all devices on the current target 2 (e.g., VCO coarse tuning, baseband filter calibration, IQ resistor calibration, etc.). It performs the 3 calibrations, regardless of the state of the NV items. This is followed by a file system synchronization that latches all cached content into the underlying file system. ***This test is to be done before all regular WCDMA phasing tests.***

Test Dependencies:

[3.2 Turn On](#)

[3.6.1 FTM Mode](#)

[3.7.3 Auto-cal Items](#)

Test Setup

N/A

Test Procedure

Command	Opcode/Data	
FTM_DO_INTERNAL_DEVICE_CAL	0x4B0B1400450100000000	(Internal calibration)

Pass/Fail check the response packets 10th byte of length 2 bytes. 0x0000 is a pass and 0x0001 corresponds to a failure: 4B0B1400450100001000**0000**00000000 (check status byte shown in bold, 0-pass 1-fail)

For NV item: NV_RC_PROCESS_ERRORS:

Use the following NV_READ command to read the NV item

Command	Data
NV_READ	0x26CD1A000000000000000000 + fill with 0x00 until 128 bytes

Verify that at least the first byte of the return string is within the limits specified in section 2.1 Test Limits.

Ensure that first byte is NOT 0x00.

Exit Test

4.2 WCDMA RX, TX, HDET and DC Cal Phasing

Test Description

This APT characterization routine phases the receiver, transmitter and HDET (power detect circuit) by performing 3 sweeps. The sweeps consist of the following:

- Sweep 1 (Fixed Bias)
- Sweep 2 (APT Bias)
- Sweep 3 (RX, TX, HDET Freq and DC Cal Sweep)

A full description of the ESC V4 Sweeps can be found in the ESC V4 Software Design Guide; QC Document ID: 80-NG201-2.

Please note, all documents referred to in this section can be downloaded from <http://support.cdmatech.com> by searching for the document #.

Test Setup

For all products in this section phasing should be run at 3.9V.

Test Procedure

Pre-sweep

Use the following FTM command to select the WCDMA Mode for the desired band.

Command	Data
FTM_SET_MODE	Mode (Select WCDMA band)

Where:

WCDMA BAND: WCDMA 850/900/1500/1700/1900/2100

Use the following FTM command to disable the secondary path.

Command	Data
SET_SECONDARY_PATH	0x00 (Disable Secondary Path)

Use the following FTM command to tune the UE to the UL Reference Channel

Command	Data
FTM_SET_CHAN *	UL Ref Channel (Set UL channel)

Command	Data
FTM_SET_TX_ON	N/A (Set TX ON)

For each PA range, starting with the lowest and ending with the highest:

Use the FTM_SET_PA_RANGE command to set it to PA_Rx.

Read the following items out of the Default file:

MOTCONF_WCDMA_C0_(band)_TX_RISE_EXTENSION – list of 3 dB10 items
MOTCONF_WCDMA_C0_(band)_TX_FALL_EXTENSION – list of 3 dB10 items
MOTCONF_WCDMA_C0_(band)_TX_HIGH_RANGE_MAX_POWER – 1 dB10 item
MOTCONF_WCDMA_C0_(band)_TX_LOW_RANGE_MIN_POWER – 1 dB10 item
(band) variable above is one of the following: WCDMA 850/900/1500/1700/1900/2100

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Use these values to choose a start and stop TX_GAIN_INDEX such that the sweep will cover a range of powers which extents:

PA RANGE	POWER for Start Point	Power for Stop Point
Highest PA Range	HIGH_RANGE_MAX_POWER/10	High Range Fall Switchpoint - TX_FALL_EXTENSION[0]/10
Lowest PA Range	Low Range Rise Switchpoint + TX_RISE_EXTENSION[max range]/10	LOW_RANGE_MIN_POWER/10
All other PA Ranges	Range Rise Switchpoint + TX_RISE_EXTENSION[pa range index]/10	Range Fall Switchpoint - TX_FALL_EXTENSION[pa range index]/10

Table 4.2.1

Start at the lowest PA GAIN RANGE, find TX_GAIN_INDEX that gives you a power equal or below **"Power for Stop Point"** in table above for the associated PA Range. Call this "STOP_TX_GAIN_INDEX".

Start at the lowest PA GAIN RANGE, find TX_GAIN_INDEX that gives you a power equal or above **"Power for Start Point"** in table above for the associated PA Range. Call this "START_TX_GAIN_INDEX".

Note: Running average start/stop TX gain indexes can be used instead of static value to help optimize test time in production.

V4 Calibration

Sweep 1 (Fixed Bias)

This sweep will use a fixed PA bias (defined by the <Bias_List> parameter in the ESC_Param.xml file) to sweep from a starting RGI down to a stopping RGI measuring power at each step on reference channel. Verify that the start & stop RGIs will be sufficiently above/below the rise/fall switch points for each PA range before starting first sweep.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for sweep 1 calibration. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file for sweep type 0. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23

Arm the Test Set for initial TX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string of parameter list obtained from the ESC_Param.xml file.

Wait for test set to complete sweep operation.

Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured power values for each PA range from the test set.

There are 3 deliverables coming from this sweep:

- Rgi vs Power with fixed bias for TX_MULTI_LIN_V3 NV item
- Reference RGI vs Power levels for calculations on 2nd sweep
- HDET vs Power for high PA gain.

Save deliverables for processing at later points of the procedure.

Sweep 2 (APT Bias)

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This sweep will characterize the TX output power vs RGI using a variable Bias on each step calculated from the first sweep. This sweep is also done only on 1 reference channel. This sweep is identical to the first sweep except that the SMPS PA Bias for each step is calculated from the first sweep and HDET is not measured in this sweep.

The SMPS PA Bias is calculated by using the Power level for each RGI in the first sweep and looking that value up in the APT Characterization.xml file. This value is characterized by development to be a bias level that allows the PA to pass ACLR specs. Typically the maximum powers on a given band are close to max PA bias, and reduce as the rgi/power diminishes.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for sweep 2 calibration. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file for sweep type 1. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23. Note that the biases will come from the APT Characterization.xml file for this construction.

Arm the Test Set for second TX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string previously constructed for this sweep.

Wait for test set to complete sweep operation.

Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured power values from the test set.

There are 2 deliverables that come from this sweep:

- Rgi vs Power with APT Bias for TX_MULTI_LIN_V3 NV item
- RGI vs Power data used in sweep 3 for TX Linearizer

Verify that the maximum and minimum power measurements for each PA State exceed the respective switch points.

Save deliverables for processing at later points of the procedure.

Sweep 3 (RX, TX, HDET Freq & DC Cal Sweep)

This third APT (Average Power Tracking) sweep will take measurements across the band (typically 16 channels) in order to determine TX, HDET and RX versus frequency for the phone. After the 16 channels are measured, the sweep then performs 4 steps of DC RX Calibration. For each channel, there will be (5 + number of PA states) steps. Typically each step can do both RX and TX in parallel (unless the signal is TD (LTE-TD or TD-SCDMA) in which case the TX measurements are all done first and then the RX measurements are done, effectively requiring (5 + number of PA states + number of LNA states) steps. Table 4.2.2 below is a visual representation of a single channel in this sweep.

TX Measurement	RX Measurement
HDET (Target power + 2 RGI)	LNA 0 or VGA reading
HDET (Target power + 1 RGI)	LNA 1
HDET (@Target power)	LNA 2
HDET (Target power - 1 RGI)	LNA 3
HDET (Target power - 2 RGI)	LNA 4 (LTE only else do nothing)
TX Linearizer (typically at high PA state's fall SP power level)	LNA 5 (LTE only else do nothing)
TX Linearizer (typically at mid PA state's rise SP power level)	do nothing
TX Linearizer (typically at low PA state's rise SP power level)	do nothing

Table 4.2.2



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Secondary Chain calibration is done by default in parallel with primary chain. This requires the test bench to use a splitter across the primary/diversity paths. It is possible to disable the secondary chain calibration if diversity path cannot be connected in parallel. If secondary chain calibration is not done in parallel, it cannot be performed using a ESC V4 sweep.

DC Calibration will be done at the end of the frequency sweep.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for RX, TX, & HDET Freq Sweep. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23

Arm the Test Set for TX/RX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string previously constructed in this sweep.

Wait for test set to complete sweep operation.

Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured TX Freq Comp values from the test set.

There are 3 deliverables coming from this sweep:

- RX LNA (or DVGA) readings on 16 channels at each LNA state for NV calculation for both RX chains.
- HDET vs Freq readings for NV calculation
- TX Linearizer vs Freq (APT Mode) for NV calculation
- DVGA and 3 LNA readings for DC calibration

Process the retrieved data values gathered during the sweep. Verify that power measurements are close to expected from reference channel. Also check RX measurements for any outliers.

NV Calculation

For TX Linearizer Data

Store Rgi vs Power with fixed bias data from sweep 1, Store Rgi vs Power with APT bias data from sweep 2, and TX Linearizer vs Freq data from sweep 3 into the appropriate area of NV_<signal>_<band>_TX_MULTI_LIN_V3_DATA_I. Refer to QC Document ID: 80-N5420-170 section 4.3.2.1.

For HDET Data

Store HDET vs Level data from sweep 1 into the appropriate area of NV_<signal>_<band>_EXP_HDET_VS_AGC_V2. Store HDET vs Freq data from sweep 3 into the appropriate area of NV_<signal>_<band>_TX_LIMIT_VS_FREQ. Refer to QC Document ID: 80-N1622-62

The HDET Power targets phased in this procedure are listed in Table 4.2.3 below for reference

Parameter	All Bands
All Products	
HDET	
HDET_Trgt_15 (dBm)	30.8
HDET_Trgt_14 (dBm)	29.2
HDET_Trgt_13 (dBm)	27.6
HDET_Trgt_12 (dBm)	26

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Parameter	All Bands
HDET_Trgt_11 (dBm)	24.4
HDET_Trgt_10 (dBm)	22.8
HDET_Trgt_9 (dBm)	21.2
HDET_Trgt_8 (dBm)	19.6
HDET_Trgt_7 (dBm)	18
HDET_Trgt_6 (dBm)	16.4
HDET_Trgt_5 (dBm)	14.8
HDET_Trgt_4 (dBm)	13.2
HDET_Trgt_3 (dBm)	11.6
HDET_Trgt_2 (dBm)	10
HDET_Trgt_1 (dBm)	8.4
HDET_Trgt_0 (dBm)	6.8

Table 4.2.3

For RX Data

Store the RX Calibration data, RX DC Calibration data, and RX Channel list from sweep 3 into the "RFNV_WCDMA_<BAND>_RX_CAL_DATA_V2_I". Refer to document 80-NH377-170 H for structure and formatting.

Exit Test

4.3 WCDMA Test Mode Call Tests

Test Description

The purpose of this test is to verify that the UE is capable of establishing a WCDMA non-signaling connection and all parametrics are within tolerance. The WCDMA test mode call tests are not required if equivalent measurements are done in a normal WCDMA call test.

Test Dependencies:

All WCDMA phasing tests

Equipment:

WCDMA Communications Test Set

DL Test Channel

Band	Channel
W2100 (Band I)	10562, 10700, 10838
W1900 (Band II)	9671, 9800, 9938
W1700 (Band IV)	1638
W850/3800 (Bands V and VI)	4358, 4400, 4458
W900 (Band VIII)	3012
W1500 (Band XI) (for UMTS/TD-LTE)	3712, 3750, 3787
W1500 (Band XI) (for all other products)	3712, 3762, 3812

Table 4.3

Test Setup

Configure the test set for a 12.2kbps RMC non-signaling (test mode) call as specified in [3].

Call Type = Test Loop Mode 1

UARFCN = (see table 4.3).

Downlink RF Power Level for establishing connection = -65 dBm

Uplink Power Control = Alternating Pattern

Primary Scrambling Code = 0

DPCH Channel Code = 9

DPCH Secondary Scrambling Code = 0

Physical Channel Relative Power Levels (E_c/I_{or}):

CPICH = -3.3 dB

SCH = -5.3 dB

P-CCPCH = -5.3 dB

PICH = -8.3 dB

DPCH = -10.3 dB

OCNS = On

Test Procedure

4.3.1 WCDMA Non-signaling Connection

Use FTM CONFIGURE UL/DL TFCS command with data 000000000000020103 to set uplink TFCS mapping to 0,2,1,3.

Use FTM_CONFIGURE_UL/DL_TFCS command with data 000000000100020103 to set downlink TFCS mapping to 0,2,1,3.

Use FTM_LOG_ENABLE command with data 1400000006101400000001000000 to enable FTM log 0x1006 which is WCDMA non-signaling events.

Use **LOGGING CONFIGURATION** command with data

[illegible]



Use FTM_START_WCDMA_MODE_REQUEST command with data 00000000 to start WCDMA layer 1 tasks.

Poll for the FTM_START_WCDMA_MODE_CONFIRMATION event which is an unsolicited (asynchronous) response. Verify that this response is 10000F000F007C11<timestamp 8 bytes>061000. This event confirms that the phone has completed all internal tasks to go into WCDMA mode (i.e. start WCDMA layer 1 tasks).

Use the FTM_WCDMA_ACQUISITION_REQUEST with data 000000000001<channel 2 bytes>000000000000 to start a cell search acquisition procedure.

Poll for the FTM_WCDMA_ACQUIRE_CONFIRMATION event which is an unsolicited (asynchronous) response. Verify that this response is 1000140014007C11<timestamp 8 bytes>06100301<channel 2 bytes>0000. This event confirms that the phone has successfully acquired the WCDMA downlink pilot on the proper channel.

Use the FTM_WCDMA_RMC_RECONFIGURATION_REQUEST with data 00000000001C00000000000000000900080FC4FF0004000103<channel 2 bytes> to establish a RMC loopback non-signaling call.

Poll for the FTM_WCDMA_CPHY_SETUP_CONFIRMATION event which is an unsolicited (asynchronous) response. Verify that this response is 1000100010007C11<timestamp 8 bytes>06100401. This event confirms that the phone has processed the RMC reconfiguration request.

Poll for the FTM_WCDMA_PHYSICAL_CHANNEL_ESTABLISHED_CONFIRMATION event which is an unsolicited (asynchronous) response. Verify that this response is 1000100010007C11<timestamp 8 bytes>06100501. This event confirms that the phone has successfully established a WCDMA layer 1 physical channel connection.

Use LOGGING_CONFIGURATION command with data 000000000100000000000000 to turn off diagnostic log code 0x117C. Use FTM_LOG_DISABLE command with data 0C0000000610 to disable FTM log 0x1006.

Set the downlink RF power level to the following:

- Bands I, IV, and VI = -106.7 dBm (lor)
- Bands II and V = -104.7 dBm (lor)
- Band VIII = -103.7 dBm (lor)

Note: For models that support both Bands V and VI, the Band VI Power level applies only to the Band VI UARFCNs. Set the uplink power control to all up bits.

4.3.2 WCDMA TX Current

Measure the total current drain from the phone and verify the value is within the limits specified in section 2.2 [WCDMA Test Limits](#).

4.3.3 WCDMA TX Max Power

Measure the Tx power and verify the value is within the limits specified in section 2.2 [WCDMA Test Limits](#).

4.3.4 WCDMA TX Frequency Error

Measure the Tx frequency error and verify the value is within the limits specified in section 2.2 [WCDMA Test Limits](#).

4.3.5 WCDMA EVM

Measure the Tx EVM and verify the value is within the limits specified in section 2.2 [WCDMA Test Limits](#).

4.3.6 WCDMA ACLR

Measure the Tx adjacent channel leakage ratio (ACLR) and verify the value is within the limits specified in 2.2 [WCDMA Test Limits](#).

4.3.7 WCDMA 12.2K Single-ended BER

Use the FTM_SE_BER_CONFIGURE_REQUEST command with data 0000000001030014001E00 to properly describe the data for single-ended BER measurements.

Use the `FTM_SE_BER_START_LOG_REQUEST` command with data `0000000008` to enable the single-ended BER subsystem in the phone.

Use FTM_LOG_ENABLE command with data 1400000009101400000001000000 to enable FTM log 0x1009 which is SE BER generic log packets.

Use `LOGGING CONFIGURATION` command with data

[illegible]

Poll for the unsolicited SE BER log packet responses. Parse out the number of bits measured and number of bit errors. Keep a running total of the total bits measured and total bit errors. Calculate a running BER% which is the following equation:

$$\text{BER\%} = (\text{total bit errors} / \text{total bits measured}) \times 100\%$$

Quit measuring BER once a 95% confidence level is reached or the total measured bits is greater than or equal to 12200 bits. Verify the final BER% is within the limits specified in section 2.0 Tables of Tests and Limits.

Use LOGGING_CONFIGURATION command with data 000000000100000000000000 to turn off diagnostic log code 0x117C.

Use FTM_LOG_DISABLE command with data 0C0000000910 to disable FTM log 0x1009.

4.3.8 WCDMA Tx Open-loop Power and Rx RSSI

Set the downlink RF power level to -65dBm

Set the uplink power control to all up bits.

Set physical channel relative power levels (E_c/I_{or}) to the following:

CPICH = -3.9 dB

SCH = -8.3 dB

P-CCPCH = -8.3 dB

PICH = -8.3 dB

DPCH = -10.3 dB

OCNS = On

Downlink Test Power Levels (dBm)

Max Pwr	GNPO	Lead Site
11	-90	-90
1	-80	-80
-9	-70	-70
-19.1	-60	-60
-29.1	-50	-50
-39.1	-40	-40
-49.1	-30	-30

Table 4.3.8

Use FTM_LOG_ENABLE command with data 1400000006101400000001000000 to enable FTM log 0x1006 which is WCDMA non-signaling events.

Use `LOGGING CONFIGURATION` command with data

[illegible]

Use the FTM_WCDMA_RMC_RECONFIGURATION_REQUEST with data 0000000000<max pwr 2 bytes>0000000000000000900080FC4FF0004000103<channel 2 bytes> to set the next max power value listed in table 4.3.8 for the RMC loopback non-signaling connection.

Poll for the FTM_WCDMA_CPHY_SETUP_CONFIRMATION event which is an unsolicited (asynchronous) response. Verify that this response is 1000100010007C11<timestamp 8 bytes>06100401. This event confirms that the phone has processed the RMC reconfiguration request.

Poll for the FTM_WCDMA_PHYSICAL_CHANNEL_ESTABLISHED_CONFIRMATION event which is an unsolicited (asynchronous) response. Verify that this response is 1000100010007C11<timestamp 8 bytes>06100501. This event confirms that the phone has successfully established a WCDMA layer 1 physical channel connection.

Use LOGGING_CONFIGURATION command with data 000000000100000000000000 to turn off diagnostic log code 0x117C.
Use FTM_LOG_DISABLE command with data 0C0000000610 to disable FTM log 0x1006.

Use LOG_WCDMA_AGC_ENH command with data 14000000041014000000**1010**1400000001000000 to enable FTM log 0x1010 which is WCDMA AGC FTM log.

Use `LOGGING_CONFIGURATION` command with data

[illegible]

Set the downlink RF power level to the next downlink test power level in table 4.3.8.

Measure the Tx power and verify the value is within the limits specified in section 2.2 [WCDMA Test Limits](#).

Measure the Rx RSSI by polling for the unsolicited WCDMA AGC FTM log and parsing out the Rx AGC value. Verify the RSSI value is within limit specified in section 2.2 [WCDMA Test Limits](#).

Use LOGGING_CONFIGURATION command with data 000000000100000000000000 to turn off diagnostic log code 0x117C.
Use FTM_LOG_DISABLE command with data 0C0000000410 to disable FTM log 0x1004.

4.3.9 WCDMA Inner-loop Power

Set the downlink RF power level to -80dBm

Set physical channel relative power levels (E_c/I_{or}) to the following:

CPICH = -3,3 dB

SCH = -5.3 dB

P-CCPCH = -5.3 dB

PICH = -8.3 dB

DPCH = -10.3 dB

OCNS = On

Set the uplink power control to generate a test pattern E sequence according to [5] 3GPP TS 34.121.

Verify that all measured parameters are within the limits specified in section 2.0 Tables of Tests and Limits.



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Set the uplink power control to generate a test pattern F sequence according to [5] 3GPP TS 34.121.
Verify that all measured parameters are within the limits specified in section 2.0 Tables of Tests and Limits.

Exit Test

5.0 GSM Test Procedures

The procedures in this section describe the methods to tune the GSM hardware control parameters and verify GSM functionality in the UE.

5.1 GSM RSSI Channel Correction

Rx Gain Range Calibration measures and stores RSSI channel gain corrections into the UE for each band of operation to ensure that measurements and reporting of BCCH and TCH RF levels by the UT are accurate.

This routine will populate:

RFNV_GSM_Cx_y_RX_CAL_DATA

Where:

x = 0; 2

y = GSM850; GSM900; GSM1800; GSM1900

z = 850; 900; 1800; 1900

GSM RX Across Band Phasing Parameters		
Band	Model	Test Channels
GSM 850	A	128, 145, 163, 180, 198, 215, 233, 251
GSM 900	A	1, 31, 62, 92, 124, 975, 1000, 1023
DCS 1800	A	512, 537, 562, 587, 612, 637, 662, 687, 712, 737, 762, 787, 812, 837, 862, 885
PCS 1900	A	512, 532, 552, 572, 592, 612, 632, 652, 672, 692, 712, 732, 752, 772, 792, 810

Table 5.1.1

Read the MOTCONF_GSM_Cx_GSM_z_RX_CHANNELS item from the default file to retrieve the channel list.

Read the MOTCONF_GSM_Cx_GSM_z_PHASING_LEVELS item from the default file to retrieve the phasing levels.

Test Dependencies:

[3.2 Turn On](#)

[3.6.1 FTM Mode](#)

Equipment:

GSM Communications Test Set or equivalent

Test Setup

Test Set Setup:

BCH	See Table 5.1.1
Amplitude	-90 dBm
TCH	OFF

UE Setup:

Connect the UE RF Port to the test set output port.

Use the following FTM commands to setup the UE.

Command	Data	
FTM_GSM_SET_MODE	Mode	(Select the band mode)

Mode 0x0A00 = GSM 900
Mode 0x0B00 = GSM 1800
Mode 0x0C00 = GSM 1900
Mode 0x1200 = GSM 850

**Test Procedure**

Configure the test set to transmit a BCCH at -90 dBm on a channel in table 5.1.1 near the center of the band tested.

Tune the UE receiver to the first channel in table 5.1.1 for the band tested:

Command	Opcode/data
FTM_SET_CHAN	Channel (Select the channel)

Read averaged RSSI power measurements for LNA gain ranges 0 and 1:

Command	Opcode/data
FTM_RX_GAIN_RANGE_CAL	0x03030x (Read RSSI power)

Note: The last byte of the FTM_RX_GAIN_RANGE_CAL command toggles the receiver on and off. It should be set to 0x01 (TRUE) for the first channel measured, and 0x00 (FALSE) for all other channels.

R0_Ref_RSSI is the 12th through 15th bytes of the response, byte-swapped

R1_Ref_RSSI is the 16th through 19th bytes of the response, byte-swapped

Save the raw RSSI power measurements as **R0_Ref_RSSI** and **R1_Ref_RSSI**.

Configure the test set to transmit a BCCH at -50 dBm on the first channel in table 5.1.1 for the band tested.

Tune the UE receiver to the first channel in table 5.1.1 for the band tested:

Command	Opcode/data
FTM_SET_CHAN	Channel (Select the channel)

Read averaged RSSI power measurements for LNA gain ranges 2 & 3 :

Command	Opcode/data
FTM_RX_GAIN_RANGE_CAL	0x0C030x (Read RSSI power)

Note: The last byte of the FTM_RX_GAIN_RANGE_CAL command toggles the receiver on and off. It should be set to 0x01 (TRUE) for the last channel measured, and 0x00 (FALSE) for all other channels.

R2_Ref_RSSI is the 20th through 23rd bytes of the response, byte-swapped

R1_Ref_RSSI is the 24th through 27th bytes of the response, byte-swapped

Repeat measurements on each channel in table 5.1.1.

Save the raw RSSI power measurements as **R2_Ref_RSSI** and **R3_Ref_RSSI**.

Calculate RSSI correction factors for each channel calculated above, for all LNA gain ranges:

$$RxFreqComp_{Ri,CHj} = 16 * (10 * \log(Ri_Ref_RSSI_{CHj}) - BCCH_Level),$$

where **BCCH Level** is -90 dBm for R0-1 and -50 dBm for R2-3.

Store the calculated RSSI correction factors in the following NV items for the correct band:

RFNV_GSM_Cx_y_RX_CAL_DATA

rx_cal_chan_size

<datatype= uint 8><Size of = 1> number of channels used

rx_cal_chan_list

<datatype= int 16><Size of = 16> MOTCONF_GSM_Cx_GSM_z_RX_CHANNELS

rx_freq_comp_data

<datatype= int 16><Size of = 16> Measured data for Ina 1

rx_freq_comp_data

<datatype= int 16><Size of = 16> Measured data for Ina 2

rx_freq_comp_data

<datatype= int 16><Size of = 16> Measured data for Ina 3

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rx_freq_comp_data
<datatype= int 16><Size of = 16> Measured data for Ina 4

Exit Test

5.2 GSM TX Phasing

5.2.1 DA Calibration

Test Description

This procedure describes how to populate the following NV Items for each supported frequency band:

This routine will populate:

RFNV_GSM_Cx_y_TX_CAL_DATA

Where:

x = 0; 2

y = GSM850; GSM900; GSM1800; GSM1900

z = 850; 900; 1800; 1900

The purpose of GSM and low-power EDGE DA calibration is to construct information about RGI values vs. output TX power for all supported GSM/EDGE bands at each PA gain state and each calibration channel. DA calibration is done on three channels per band for all gain states. The PA Gain Ranges are listed by band in table 5.2.1.1.

PA Gain Range Table

Product	GSM850	GSM900	GSM1800	GSM1900
Lynx	0,1	0,1	0,2	0,2

Table 5.2.1.1

Read the MOTCONF_GSM_Cx_GSM_z_TX_CHANNELS item from the default file to retrieve the channel list

Read the MOTCONF_GSM_Cx_GSM_z_TX_SWEEP_RGI_STOP item from the default file to retrieve the STOP RGI

Read the MOTCONF_GSM_Cx_GSM_z_TX_SWEEP_RGI_DURATION item from the default file to retrieve the RGI duration

Read the MOTCONF_GSM_Cx_GSM_z_TX_SWEEP_MODULATION_ORDER item from the default file to determine the order of modulation transmit

Read the MOTCONF_GSM_Cx_GSM_z_TX_SWEEP_MEASUREMENT_LENGTH item from the default file to retrieve the segment measurement window (in uS)

Read the MOTCONF_GSM_Cx_GSM_z_TX_SWEEP_MEASUREMENT_OFFSET item from the default file to retrieve the offset from the rising edge to the first measurement window (in uS)

Test Dependencies:

[3.2 Turn On](#)

[3.74 XO Calibration](#)

Equipment:

GSM Communications Test Set

Test Procedure

Repeat the following procedures for each band:

Use the following FTM commands to setup the UE.

Command Opcode/data

FTM_GSM_SET_MODE Mode (Select the band mode)

Mode 0x0A00 = GSM 900

Mode 0x0B00 = GSM 1800

Mode 0x0C00 = GSM 1900

Mode 0x1200 = GSM 850

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Repeat the follow for each of the three calibration channel in each band:

<u>Command</u>	<u>Data</u>	
FTM_SET_CHANNEL	Channel	(Select the UT channel)

<u>Command</u>	<u>Data</u>
FTM_SET_TRANSMIT_BURST	00 0000 05 00000000 01 (iSlotNum = 0, iDataSource = 0, iTSCindex = 5, iNumBursts = 0, blsInfiniteDuration = 1)

<u>Command</u>	<u>Data</u>
FTM_SET_PDM	0000 0000 (iPDMtype = 0, iPDMvalue = 0)

<u>Command</u>	<u>Data</u>
FTM_SET_TX_OFF	

The channel to use for calibration should be read from the NV Table of the Radio. The channels are store in the following MOTOCONF item MOTCONF_GSM_Cx_GSM_z_TX_CHANNELS in a array low, mid, high channel.

Config the Testset to measure the sweep on the appropriate band and channel. The carrier frequency must be offset by + 67.7 kHz due to the CW waveform used during the sweep.

Send the following FTM command to trigger the measurement:

<u>Command</u>	<u>Data</u>
FTM_GSM_TX_SWEEP_CAL	4B0B11003601 + command length + 0000 + segment data (Trigger Tx power sweep)

Command length = hex[2 + (number of segments(129 for low bands, 193 for high bands) * 5)]

Segment Data is 5 bytes for each segment

Byte 1 & 2	= duration of segment (1)
Byte 3	= RGI Index setting
Byte 4	= Modulation type (1 – GSMK, 0 – 8PSK)
Byte 5	= PA Range setting

The first segment should be the trigger segment, which is a 1 symbol segment with RGI = MOTCONF_GSM_Cx_GSM_z_TX_SWEEP_RGI_STOP[0] and PA = 0.

The item MOTCONF_GSM_Cx_GSM_z_TX_SWEEP_RGI_STOP contains the highest RGI used by PA range, zero indicates the range in unused.

Next, step through the PA Ranges and RGIs, RGI 0 to the RGI STOP value, alternating the 2 modulation types at each RGI as specified by the MOTCONF_GSM_Cx_GSM_z_TX_SWEEP_MODULATION_ORDER item. The item MOTCONF_GSM_Cx_GSM_z_TX_SWEEP_RGI_DURATION contains the duration of segment by PA range, zero indicates the range in unused.

Read power amplitude measurements for each of the segments from the test set. Discard the power reading from the first segment (this segment is only used to trigger the measurement in the test set).

Sort the power measurements so that the powers are monotonic with each PA Range/modulation type. The powers should be converted to an integer by multiplying by 100 and rounding to the closest integer.

The QMSL provided function QMSL API QLIB_RFCAL_GSM_Tx_Cal_Results() can be used to process the data.

The RGI values are stored in RFNV_GSM_Cx_y_TX_CAL_DATA applies to both GMSK and 8PSK.

In the data type structure RFNV_DATA_RGI_LIST_TYPE

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Each RGI NV item consists of 32 elements per NV item that contains the RGI values used directly. The element values must be in the range of 0 to 31 (usually in ascending order) and no overlapping.

The corresponding TX measurement-related values are stored in RFNV_GSM_Cx_y_TX_CAL_DATA in the data type structure RFNV_DATA_PMEAS_LIST_TYPE.

5.2.2 Pre-distortion Calibration

Test Description

This procedure describes how to populate the following NV Items for each supported frequency band:

This routine will populate:

RFNV_GSM_Cx_y_TX_CAL_DATA

Where:

x = 0; 2

y = GSM850; GSM900; GSM1800; GSM1900

z = 850; 900; 1800; 1900

High-power EDGE operates at the nonlinear regions of the highest gain state of the PA. The required power and phase mapping tables are done in a fashion similar to polar calibration (AMAM/AMPM type setup).

Calibration sequence involves sending a known waveform (waveform where IQ samples are known), capturing the IQ measurements with the callbox and heavy data processing. The waveform generated is separated into two segments: frequency correction segment and pre-distortion calibration. The following sections provide details on waveform generation and NV calculations.

Read the MOTCONF_GSM_Cx_GSM_z_TX_CHANNELS item from the default file to retrieve the channel list

Pre-distortion Power Table

Product	GSM850	GSM900	GSM1800	GSM1900
All	27dbm	27dbm	26dbm	26dbm

Table 5.2.2.1

Test Dependencies:

[3.2 Turn On](#)

[5.2.1 DA Calibration](#)

Equipment:

GSM Communications Test Set or equivalent

Test Procedure

Repeat the following procedures for each band:

Prior to doing pre-distortion calibration, it is required to find the appropriate RGI values for each band based on the DA calibration measurement data.

The criteria for choosing the appropriate RGI is to choose the largest RGI value among the three calibration channels at the highest gain range (gain range 0) that will yield at least 27 dBm for G900/G850 or 26 dBm for G1800/G1900 the correct minimum level of pre-distortion output power per GSM band using 8PSK modulation. This is listed in table 5.2.2.1.

Store this RGI in rgi_for_pred for the given RFNV_GSM_Cx_y_TX_CAL_DATA.

Use the following FTM commands to setup the UE.

Command	Opcode/data	
FTM_GSM_SET_MODE	Mode	(Select the band mode)

Mode 0x0A00 = GSM 900

Mode 0x0B00 = GSM 1800

Mode 0x0C00 = GSM 1900

Mode 0x1200 = GSM 850

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Repeat the follow for each of the three calibration channel in each band:

<u>Command</u>	<u>Data</u>	
FTM_SET_CHANNEL	Channel	(Select the UT channel)

<u>Command</u>	<u>Data</u>
FTM_SET_TX_OFF	

<u>Command</u>	<u>Data</u>
FTM_SET_GSM_LINEAR_PA_RANGE	0000 0000 (iSlotNum = 0, iPaRange = 0)

The channel to use for calibration should be read from the NV Table of the Radio. The channels are store in the following MOTOCONF item MOTCONF_GSM_Cx_GSM_z_TX_CHANNELS in a array low, mid, high channel.

Config the Testset to measure the sweep on the appropriate band and channel. The carrier frequency must be offset by + 67.7 kHz due to the CW waveform used during the sweep.

Send the following FTM command to trigger the measurement:

<u>Command</u>	<u>Data</u>
FTM_TX_CFG2_AMAM_SWEEP_V2	4B0B1100350106000000 + seg_1_dur + seg_2_dur + RGI

SEG_1_DUR	Freq correction segment duration (1000)
SEG_2_DUR	Pre-distortion Cal segment duration (1000)
RGI	RGI value stored in rgi_for_pred from RFNV_GSM_Cx_y_TX_CAL_DATA

The command should respond with the following data:
4B0B1100350100000000 + MAX_VALUE (2 bytes) + STATUS (2 bytes)

STATUS should be 0x0000, if not the command has failed

MAX_VALUE is the maximum value of the PA ramp used during the sweep

If the command is successful, read power amplitude and phase measurements for the sweep. The power measurements should be in units of dBm, and phase measurements should be in degrees.

QMSL provides a function QLIB_RFCAL_GSM_Tx_Linear_Process_Swapped_PreDist_Cal_Results() which is used to process the data for the sweeps for the three channels of each GSM Band. The output of the function is the data to be stored in the associated NV Items for the band.

The procedures and formulas to manually calculate these NV Items are beyond the scope of this document, but can be found in detail in the Qualcomm document: 80-V9774-16_B_GSM_Linear_PA_Calib_Data_Process_NV_Gen.pdf. The function will return 3 AMAM arrays and 3 AMPM arrays, one set for each cal channel of the band, and a max power, which defines the dynamic range.

Each AMAM and AMPM arrays are 128 elements long.

Store the AMAM data in RFNV_GSM_Cx_y_TX_CAL_DATA in the data type structure RFNV_DATA_AMAM_LIST_TYPE data structure .

Store the AMPM data in RFNV_GSM_Cx_y_TX_CAL_DATA in the data type structure RFNV_DATA_AMPM_LIST_TYPE data structure

Store the Max Power in amam_max_dbm in RFNV_GSM_Cx_y_TX_CAL_DATA.

Also ensure that following parameters are set as specified for the respective FTM command arguments:



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Parameter Pre Dist:DC Transient duration - 60 (was hard coded in MDM procedure)
Parameter Pre Dist:Number of Transient Skip Symbols - 90 (was hard coded in MDM procedure)

Modem Constant Override: (Critical for proper phasing of the AMPM NV items for the 8x26 chipset)
4 items
Gain Unity = 64
Ramp Unity = 8192
Extension floor = 5200
Dac Scaling = 4

RFNV_GSM_Cx_y_TX_CAL_DATA

tx_cal_chan_size
<datatype= uint 8><Size of = 1> number of channels used, always 3
tx_cal_chan_list
<datatype= Uint 16><Size of = 3> MOTCONF_GSM_Cx_GSM_z_TX_CHANNELS
amam_max_dbm
<datatype= int 16><Size of = 1> Max Power
rgi_for_pred
<datatype= int 16><Size of = 1> Max Power
Variant_Marker
<datatype= uint 16><Size of = 1> version (only 0 is supported)
<datatype= uint 16><Size of = 1> num_of_element (count of RFNV_DATA_w_LIST_TYPE)
RFNV_DATA_w_LIST_TYPE structures as required in any order

RFNV_DATA_RGI_LIST_TYPE

<datatype= uint 8><Size of = 1> typeid=106
<datatype= uint 8><Size of = 1> mod
<datatype= uint 8><Size of = 1> pa_state
<datatype= uint 8><Size of = 1> valid_rgi_num (number of rgi_list indexes populated)
<datatype= uint 8><Size of = 32> rgi_list

RFNV_DATA_PMEAS_LIST_TYPE

<datatype= uint 8><Size of = 1> typeid=107
<datatype= uint 8><Size of = 1> mod
<datatype= uint 8><Size of = 1> channel_index (0 – low, 1 – mid. 2 – high)
<datatype= uint 8><Size of = 1> pa_state
<datatype= int 16><Size of = 32> pmeas_list

RFNV_DATA_AMAM_LIST_TYPE

<datatype= uint 8><Size of = 1> typeid=108
<datatype= uint 8><Size of = 1> amam_ampm_identifier (always 0)
<datatype= uint 8><Size of = 1> channel_index (0 – low, 1 – mid. 2 – high)
<datatype= int 16><Size of = 128> data_list

RFNV_DATA_AMPM_LIST_TYPE

<datatype= uint 8><Size of = 1> typeid=109
<datatype= uint 8><Size of = 1> amam_ampm_identifier (always 1)
<datatype= uint 8><Size of = 1> channel_index (0 – low, 1 – mid. 2 – high)
<datatype= int 16><Size of = 128> data_list

Exit Test



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5.3 GSM Non-Signaling Call Test

Test Description

This test measures the phone's ability to make a call and its performance in the call. This test must be performed on all prototype builds.

Test Dependencies:

[3.2 Turn On](#)

All GSM phasing tests

Equipment:

GSM Communications Test Set

Test Setup

Call Test Parameters		
	GSM850, GSM900	GSM1800, GSM1900
RF_Level_CAMP	-86 dBm (Production) -25 dBm (Prototype Builds)	
RF_level_CALL	-90 dBm	
RF_level_BER	-105 dBm	
MAX_LEV	5	0
MIN_LEV	19	15
Volts_Call	Nominal Voltage (See Section 1.6.7)	

5.3.1



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Channel and PCL Requirements for Each Test				
		Band 1	Band 2	Band 3
All Sites	Tx Power Timemask Peak, RMS Phase Error Frequency Error FBER RX_LEV	MAX_LEV	No Test	MIN_LEV
	Tx Current	MAX_LEV	No Test	No Test
	Modulation ORFS	MAX_LEV, MIN_LEV	No Test	MAX_LEV, MIN_LEV
	Switching ORFS	MAX_LEV, 7, 11, MIN_LEV	No Test	MAX_LEV, 7, 11, MIN_LEV
GNPO	Tx Power, Rx Level	GSM900: Channels 975, 997,1019,16,38,60,81,102,124 -- MAX LEV		
		GSM1800: Channels 512,566,620,663,699,760,828,885 -- MAX LEV		
		GSM1900: Channels 512,553,590,629,694,728,768,810 -- MAX LEV		
		GSM850: Channels 128,150,172,183,194,206,228,251 -- MAX LEV		
	Tx Power	Band 2: All valid power steps		
	Rx Level	Band 2: BCH Level = -30.-50.-70,-90 dBm		
	Tx Current	MAX_LEV	No Test	MAX_LEV
	BER, CRC, FBER	MAX_LEV	No Test	MIN_LEV, MAX_LEV
	Timing Error	Any (Perform Once Only)		

Table 5.3.2

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Equipment Setup:

Select non signaling mode for the serving cell/network setup.

Set up a BCH carrier in GSM900 **Band3** (see section 1.6) and enable a channel in GSM900 **Band1** on the BA allocation table (NOTE: These BCCH allocations may be swapped if necessary). Set BCH carrier to power level **RF_Level_CAMP**.

Set up a TCH carrier in GSM900 **Band1** at power level **RF_Level_CAMP**. (Setting up the TCH carrier at **RF_Level_BER** is also acceptable, at the factory's discretion.)

Set Test Set to appropriate **MS_TX_LEV**.

Ensure that the test set timing advance is set to 0T.

UE Setup:

Set the MS supply voltage at main battery terminals such that the voltage at the battery terminals is **Volts_Call** Volts at its lowest point during a TX burst at MAX Power.

Wait long enough for the capacitance in the battery terminals of the fixture to fully charge.

Apply 0 Volts over the charger terminal, or the charger terminal is floating.

Test Procedure

Associated with sending each of the setup commands to configure the UE for a non signaling connection there is a need to enable FTM logging to report acknowledgement and success of the sent commands. Details on the FTM logging can be reviewed in Qualcomm document **80-V9147-1**

Follow the sequence outlined below to accomplish this:

- a. *FTM log Enabling with command LOG_FTM2_LOG_GSM_BER*
- b. *Diagnostic log enabling the FTM log above, with log code 0x117C*
- c. *GSM SETUP command*
- d. *Scan event log expected for the respective GSM SETUP command event and parse to determine failure or success.*
- e. *Disable FTM log*
- f. *Disable Diagnostic logging with log code 0x117C*

Follow the sequence of GSM SETUP commands (in bold) below to configure the UE and get into a non signaling call.

1. Stop the GSM layer 1 services and return to the inactive state using the FTM command **STOP_GSM_MODE_REQ**
2. Initialize the GSM layer 1 subsystem and drivers using the FTM command **START_GSM_MODE_REQ**
3. Select the channel and band for the BCCH that is desired with FTM command **SELECT_SPECIFIC_BCCH_REQ**
4. Set the UE up for idle mode, where it is camped, using FTM **START_IDLE_MODE_REQ**
5. Transition from Idle mode to TCH mode on the specified timeslot and channel mode with FTM command **CHANNEL_ASSIGN_REQ**

**6. Configure loopback type during the connection using FTM command `CONFIG_LOOPBACK_TYPE_REQ`**

Refer to the Qualcomm document **80-V3951-1** for details on the specific parameters to populate for each field in the above commands.

Verify that the MS establishes a call.

Reduce BCH level of the Test Set to **RF_level_CALL**.

Reduce TCH level of the Test Set to **RF_level_BER**.

Ensure BCH RF output level of the Test Set has been at **RF_level_CALL** for at least 1 second.

Perform Tx, Rx, and ORFS tests below:

TX Tests

Measure burst average power out. Verify burst average power out is within limits (see section 2.0).

Verify TX signal falls within Time Mask template (see section 12.5 Appendix E: Time Mask Parameters).

Measure the worst case Frequency Error, Peak Phase Error, and RMS Phase Error over 20 bursts.

Convert frequency error to ppm:

$$ppm = \frac{Freq_Offset_{Hz}}{Freq_Channel},$$

where `Freq_Channel` is the measure channel frequency in MHz.

Verify that frequency error, peak phase error and RMS phase error are within limits (see section 2.0).

Set Test Set timing advance to 40T and measure timing error. Verify that timing error is within specification. Reset Test Set timing advance to 0T.

NOTE: The timing advance test only needs to be performed on one band. It can be performed on any band, at the factory's discretion. CMU-200 is not to be used for validating the Timing Advance Error. An HP8922M (or other test set that correlates to the HP8922) must be used. This test may be moved to audit at an off-line or FQA station.

Measure the B+ transmit current (with backlights off) and verify that the current drain is within specification (see section 2.0). (This test only needs to be done on bands tested at max power.)

RX Tests

Measure Class II BER and CRC ratio for 50k bits (prototype builds) or 10k bits (production builds). Verify that BER and CRC are within specifications (see section 2.0).

NOTE: FBER may be performed as a replacement for BER on production builds, as long as correlation has been established with the BER test on prototype builds. There may intermittent NTFs on some test sets if measuring with less than 30Kbits.

Measure `RX_LEV`. In a non signaling connection this cannot be queried from the test set as is usually the case. Instead, use the `DED_RPT` FTM event log report once in NST call to query for the `RX_LEV`. In this case this metric is strictly mobile measured. Verify that `RX_LEV` is within specification (see section 2.0).

OR,

Measure RSSI. Do this using the `RX_LEVEL_REQ` FTM log command. This is reported in dBm. Query for this parametric till 5 consecutive reads are within 1 dB of each other using a stability algorithm on the read values.

Choose either `RX_LEV` or RSSI as a receiver level measurement. For reference, RSSI is the faster method of validating the receiver.



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NOTE: Ensure at least 1.5 seconds, or 3 SACCH counts, have elapsed since the last TCH channel, timeslot, TX power or base station power change before measuring Rx level to ensure the Rx Level returned by the test set was sent by the UUT to the test set after the change occurred. This delay is required to allow time for the UT to report RxLev via standard GSM signalling protocols. There are cases where a Measurement report is not valid (NAN) after the 3rd SACCH count, query up to 3 additional reports until a valid report is retrieved, after the 3rd retry, the unit is a failure. If the RX Lev measurement is out-of-spec (per Section 2.0) on the 3rd SACCH count, remeasure and verify again on the 4th SACCH count. This '4th SACCH' test modification to the standard Motorola RX Lev test is due to a known QC hardware issue, and is done to maintain fastest possible test times. The 1.5 sec.delay and 3 (4) SACCH check is done to alleviate the chance of NTFs due to these known (ETSI qualified and QC HW) radio HW behaviors and apply to any change in the radio's TCH channel, timeslot, or base station power. Radios which report valid RX Lev values on the 4th SACCH instead of the 3rd SACCH may throw off the CPK for the RX Lev statistics. If this becomes an issue, the radio is to be validated on the 4th SACCH only (encompassing a test-time hit) and this document is to be updated as such.

In Band Output RF Spectrum Test

Note 1: Due to the inaccuracies (> 9 dB) of the HP8922G/H, this instrument may NOT be used for this test. Only CMD55, CMU200, HP8960, and FSE spectrum analyzer are authorized to perform this test.

Note 2: Refer to sections 12.4 Appendix D: GSM Output RF Spectrum Requirements for limits and result reporting requirements. The test setup and measurement requirements for this test are defined in detail in reference [\[2\] 3GPP TS 51.010](#).

Note 3: At least 0.5 seconds must have elapsed from the time the call is connected or handover is completed before performing ORFS testing. This delay is required for stable results.

Note 4: Since the fail code does not store the actual measured values then another data storage method should be used to assist in failure mode analysis (e.g. Factory Tracking Information Bytes – optional bytes).

Note 5: The test set must be configured to place the UT in loopback mode to loop back a pseudo-random data pattern (TCH LOOP B).

Insure that the Test Set timing advance is set to 0 (zero) T.

Spectrum Due to Modulation

Ensure that a minimum of 50 samples is achieved for each measurement point.

Trigger Output RF Spectrum Modulation measurement.

Verify measurement levels do not exceed test spec.

Note: Up to three (3) measurements in the range ≥ 600 KHz above and below the carrier may measure up to, but not exceeding, -39 dBm. Should this exception be exceeded then the PASS/FAIL flag is set to FAIL (1).

Spectrum Due to Switching Transients

Note: Ensure that a minimum of 10 samples is achieved for each measurement point.

Trigger Output RF Spectrum Switching measurement.

Verify measurement levels do not exceed test spec.

Should any measurements exceed the specification the PASS/FAIL flag is set to FAIL (1).

Note: No exceptions are allowed for Spectrum due to Switching Transients.

Calculate Fail Code

Map the frequency offset failures using the 4 byte measure code format shown on next page.

Bit	Freq Offset (kHz)	Pass/Fail	<u>Modulation / Transient Measurement</u>
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30 (msb)	- 1800	0/1	T
29	- 1200	0/1	T
28	- 600	0/1	T
27	- 400	0/1	T
26	+ 400	0/1	T
25	+ 600	0/1	T
24	+ 1200	0/1	T
23	+ 1800	0/1	T
22	PASS/FAIL Flag for Modulation		M
21	-1800	0/1	M
20	-1600	0/1	M
19	-1400	0/1	M
18	-1200	0/1	M
17	-1000	0/1	M
16	- 800	0/1	M
15	- 600	0/1	M
14	- 400	0/1	M
13	- 250	0/1	M
12	- 200	0/1	M
11	- 100	0/1	M
10	+ 100	0/1	M
9	+ 200	0/1	M
8	+ 250	0/1	M
7	+ 400	0/1	M
6	+ 600	0/1	M
5	+ 800	0/1	M
4	+ 1000	0/1	M
3	+ 1200	0/1	M
2	+ 1400	0/1	M
1	+ 1600	0/1	M
0 (lsb)	+1800	0/1	M



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Command the test set to change the PCL level of the UE and repeat Tx and Rx tests as called out in table 5.3.2

If necessary for reliable handovers, set BCH and TCH level of the Test Set to **RF_level_CALL** for the handover.

Perform intra-band handovers to all channels in the same band indicated in table 5.3.2. Verify that MS is at correct power level after the handover by querying the test set.

Perform Tx and Rx tests as described above, for all power steps as described in table 5.3.2.

Perform intra-band handover to test all bands indicated in table 5.3.2. Verify that MS is at correct power level after the handover by querying the test set.

Perform Tx and Rx tests as described above, for all power steps as described in table 5.3.2.

Exit Test

Disable the Test Set RF output.

5.4 8PSK Non-Signaling Data Connection Test

Test Description

The purpose of this test is to verify the ability of the UUT to establish and maintain an 8PSK connection, and to measure the performance of the UUT in the connection.

For prototype builds, the UE should be tested in the highest 8PSK class it supports. This will detect design issues, such as chip heat dissipation or timing issues, and it will help detect software bugs. For production builds, only the 8PSK class 4 test is required.

Test Dependencies:

[3.2 Turn On](#)

All GSM phasing tests

Equipment:

GSM Communications Test Set supporting:

- 8PSK non signaling connection
- Separate RF generators for BCH and TCH (all 8960, CMU with B95 option)

Call Test Parameters		
	GSM850, GSM900	GSM1800, GSM1900
RF_Level_CAMP	-86 dBm	
RF_level_CALL	-86 dBm	
RF_level_BER	-86 dBm	
MAX_LEV	8	2
MAX_LEV_MCS1	5	0
MIN_LEV	19	15
Volts_Call	Nominal Voltage (See Section 1.6.7)	

Table 5.4.1

Channel and PCL Requirements for Each Test				
		Band 1	Band 2	Band 3
Lead Site	Tx Power Timemask Peak EVM RMS EVM 95% EVM Frequency Error BER	MAX_LEV	No Test	MIN_LEV
	Tx Current	No Test	No Test	No Test
	Modulation ORFS	MAX_LEV	No Test	MAX_LEV
	Switching ORFS	MAX_LEV	No Test	MAX_LEV
GNPO	Tx Power Timemask Peak EVM RMS EVM 95% EVM Frequency Error BER	MAX_LEV MIN_LEV	No Test	MAX_LEV MIN_LEV
	Tx Power (MCS1) Timemask (MCS1) Peak, RMS Phase Error Frequency Error (MCS1)	MAX_LEV (Any channel)		
	Tx Current	MAX_LEV	No Test	MAX_LEV
	Modulation ORFS	MAX_LEV, 8 (1800/1900 only), 13 (850/900 only), MIN_LEV		
	Switching ORFS	MAX_LEV, 8 (1800/1900 only), 13 (850/900 only), MIN_LEV		

Table 5.4.2

Uplink and Downlink Configurations				
	8PSK Class 12	8PSK Class 10	8PSK Class 6	8PSK Class 4
Downlinks	1	3	2	1
Uplinks	4	2	2	1

Table 5.4.3

Test Setup

Equipment Setup:

BCH Table 5.4.2
 BCH RF Level **RF_Level_CAMP**
 TCH Table 5.4.2
 TCH RF Level **RF_Level_CALL**

(Start in GSM900 or GSM850 band)

Set Operating Mode to EGPRS non signaling

For Class 4 testing (High-volume production builds),

Set Connection Type to SRB (EGPRS Loopback sym. on CMU or SRB Loopback on 8960)

For non-class 4 testing (prototype builds),

Set Connection Type to ETSI Test Mode A.

Set the Modulation Coding Scheme to MCS9

Set the timeslot configurations according to Table 5.4.3

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Set Puncturing Scheme to PS1
Set PDU to 0

UE Setup:
Connect the UE RF port to the Test Set measurement port.
Set the power supply current limit to 2 Amps.
Apply 0 Volts over the charger terminal.

Associated with sending each of the setup commands to configure the UE for a non signaling connection there is a need to enable FTM logging to report acknowledgement and success of the sent commands. Details on the FTM logging can be reviewed in Qualcomm document 80-V9147-1

Follow the sequence outlined below to accomplish this:

- g. FTM log Enabling with command LOG_FTM2_LOG_GSM_BER
- h. Diagnostic log enabling the FTM log above, with log code 0x117C
- i. GSM SETUP command
- j. Scan event log expected for the respective GSM SETUP command event and parse to determine failure or success.
- k. Disable FTM log
- l. Disable Diagnostic logging with log code 0x117C

Follow the sequence of FTM commands (in bold) below to configure the UE and get into an 8PSK non signaling call.

- 7. Stop the GSM layer 1 services and return to the inactive state using the FTM command **STOP_GSM_MODE_REQ**
- 8. Initialize the GSM layer 1 subsystem and drivers using the FTM command **START_GSM_MODE_REQ**
- 9. Select the channel and band for the BCCH that is desired with FTM command **SELECT_SPECIFIC_BCCH_REQ**
- 10. Enter GPRS Idle mode by issuing **FTM_EGPRS_BER_START_GPRS_IDLE_MODE_REQ**
- 11. Configure the downlink TBF using **FTM_EGPRS_BER_CONFIGURE_DL_TBF**
- 12. Configure the uplink TBF using **FTM_EGPRS_BER_CONFIGURE_UL_TBF**
- 13. Initiate the TBFs and close the SRB loop with **FTM_EGPRS_BER_ESTABLISH_SRB_LOOPBACK**

Once the call is established, parametric data can be gathered on the channel of interest. To configure other channels when in the NS call, follow the steps below:

- 1. Configure the downlink TBF using **FTM_EGPRS_BER_CONFIGURE_DL_TBF**
- 2. Configure the uplink TBF using **FTM_EGPRS_BER_CONFIGURE_UL_TBF**
- 3. Initiate the TBFs and close the SRB loop with **FTM_EGPRS_BER_ESTABLISH_SRB_LOOPBACK**

To terminate the call once all required bands and channels have had the relevant parametrics measured on them:

- 1. Stop the GSM layer 1 services and return to the inactive state using the FTM command **STOP_GSM_MODE_REQ**

Refer to the Qualcomm document **80-V5529-1** for details on the specific parameters to populate for each field in the above commands.



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Test Procedure

Perform Tx, and ORFS tests in line with the table 5.4.2

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6.0 CDMA1X Test Procedures

The procedures in this section describe the methods to tune the CDMA hardware control parameters and verify CDMA functionality in the UE. The Reference phasing and the Band phasing must be done at the same time since they populate different data areas of the RFNV_CDMA_Cx_BCy_TX_MULTI_LIN_DATA NV item.

6.1 CDMA Rx, Tx and HDET phasing

Test Description

This APT characterization routine phases the receiver, transmitter and HDET (power detect circuit) by performing 3 sweeps. The sweeps consist of the following:

- Sweep 1 (Fixed Bias)
- Sweep 2 (APT Bias)
- Sweep 3 (RX, TX, & HDET Freq Sweep)

A full description of the ESC V4 Sweeps can be found in the ESC V4 Software Design Guide; QC Document ID: 80-NG201-2.

Please note, all documents referred to in this section can be downloaded from <http://support.cdmatech.com> by searching for the document #.

Test Setup

For all products in this section phasing should be run at 3.9V.

Test Procedure

Pre-sweep

Use the following FTM command to select the CDMA Mode for the desired band.

Command	Data
FTM_SET_MODE	Mode (Select CDMA band)

Where:

CDMA band is BC0 (CDMA800), BC1 (CDMA1900) or BC10

Use the following FTM command to disable the secondary path.

Command	Data
SET_SECONDARY_PATH	0x00 (Disable Secondary Path)
Or	
SET_SECONDARY_PATH	0x02 (Set for Simultaneous Voice Path)

Use the following FTM command to tune the UE to the UL Reference Channel

Command	Data
FTM_SET_CHAN *	UL Ref Channel (Set UL channel)

Command	Data
FTM_SET_TX_ON	N/A (Set TX ON)

For each PA range, starting with the lowest and ending with the highest:

Use the FTM_SET_PA_RANGE command to set it to PA_Rx.

Read the following items out of the Default file:

MOTCONF_CDMA_C(path)_(band)_TX_RISE_EXTENSION – list of 3 dB10 items

MOTCONF_CDMA_C(path)_(band)_TX_FALL_EXTENSION – list of 3 dB10 items

MOTCONF_CDMA_C(path)_(band)_TX_HIGH_RANGE_MAX_POWER – 1 dB10 item

MOTCONF_CDMA_C(path)_(band)_TX_LOW_RANGE_MIN_POWER – 1 dB10 item

(band) variable above is one of the following: BC0 (CDMA800) , BC1(CDMA1900), BC10 (CDMA 800 Extension)

(path) variable is either 0 (EVDO/Main path) or 1 (Simultaneous Voice path)

Use these values to choose a start and stop TX_GAIN_INDEX such that the sweep will cover a range of powers which extents:

PA RANGE	POWER for Start Point	Power for Stop Point
Highest PA Range	HIGH_RANGE_MAX_POWER/10	High Range Fall Switchpoint - TX_FALL_EXTENSION[0]/10
Lowest PA Range	Low Range Rise Switchpoint + TX_RISE_EXTENSION[max range]/10	LOW_RANGE_MIN_POWER/10
All other PA Ranges	Range Rise Switchpoint + TX_RISE_EXTENSION[pa range index]/10	Range Fall Switchpoint - TX_FALL_EXTENSION[pa range index]/10

Table 6.1.1

Start at the lowest PA GAIN RANGE, find TX_GAIN_INDEX that gives you a power equal or below **“Power for Stop Point”** in table above for the associated PA Range. Call this “STOP_TX_GAIN_INDEX”.

Start at the lowest PA GAIN RANGE, find TX_GAIN_INDEX that gives you a power equal or above **“Power for Start Point”** in table above for the associated PA Range. Call this “START_TX_GAIN_INDEX”.

Note: Running average start/stop TX gain indexes can be used instead of static value to help optimize test time in production.

V4 Calibration

Sweep 1 (Fixed Bias)

This sweep will use a fixed PA bias (defined by the <Bias_List> parameter in the ESC_Param.xml file) to sweep from a starting RGI down to a stopping RGI measuring power at each step on reference channel. Verify that the start & stop RGIs will be sufficiently above/below the rise/fall switch points for each PA range before starting first sweep.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for sweep 1 calibration. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file for sweep type 0. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23

Arm the Test Set for initial TX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string of parameter list obtained from the ESC_Param.xml file.

Wait for test set to complete sweep operation.

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Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured power values for each PA range from the test set.

There are 3 deliverables coming from this sweep:

- Rgi vs Power with fixed bias for TX_MULTI_LIN_V3 NV item
- Reference RGI vs Power levels for calculations on 2nd sweep
- HDET vs Power for high PA gain.

Save deliverables for processing at later points of the procedure.

Sweep 2 (APT Bias)

This sweep will characterize the TX output power vs RGI using a variable Bias on each step calculated from the first sweep. This sweep is also done only on 1 reference channel. This sweep is identical to the first sweep except that the SMPS PA Bias for each step is calculated from the first sweep and HDET is not measured in this sweep.

The SMPS PA Bias is calculated by using the Power level for each RGI in the first sweep and looking that value up in the APT Characterization.xml file. This value is characterized by development to be a bias level that allows the PA to pass ACLR specs. Typically the maximum powers on a given band are close to max PA bias, and reduce as the rgi/power diminishes.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for sweep 2 calibration. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file for sweep type 1. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23. Note that the biases will come from the APT Characterization.xml file for this construction.

Arm the Test Set for second TX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string previously constructed for this sweep.

Wait for test set to complete sweep operation.

Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured power values from the test set.

There are 2 deliverables that come from this sweep:

- Rgi vs Power with APT Bias for TX_MULTI_LIN_V3 NV item
- RGI vs Power data used in sweep 3 for TX Linearizer

Verify that the maximum and minimum power measurements for each PA State exceed the respective switch points.

Save deliverables for processing at later points of the procedure.

Sweep 3 (RX, TX, & HDET Freq Sweep)

This third APT (Average Power Tracking) sweep will take measurements across the band (typically 16 channels) in order to determine TX, HDET and RX versus frequency for the phone. For each channel, there will be (5 + number of PA states) steps. Typically each step can do both RX and TX in parallel (unless the signal is TD (LTE-TD or TD-SCDMA) in which case the TX measurements are all done first and then the RX measurements are done, effectively requiring (5 + number of PA states + number of LNA states) steps. Table 6.1.2 below is a visual representation of a single channel in this sweep.

TX Measurement	RX Measurement
HDET (Target power + 2 RGI)	LNA 0 or VGA reading
HDET (Target power + 1 RGI)	LNA 1
HDET (@Target power)	LNA 2
HDET (Target power - 1 RGI)	LNA 3
HDET (Target power - 2 RGI)	LNA 4 (LTE only else do nothing)
TX Linearizer (typically at high PA state's fall SP power level)	LNA 5 (LTE only else do nothing)
TX Linearizer (typically at mid PA state's rise SP power level)	do nothing
TX Linearizer (typically at low PA state's rise SP power level)	do nothing

Table 6.1.2

Secondary Chain calibration is done by default in parallel with primary chain. This requires the test bench to use a splitter across the primary/diversity paths. It is possible to disable the secondary chain calibration if diversity path cannot be connected in parallel. If secondary chain calibration is not done in parallel, it cannot be performed using a ESC V4 sweep.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for RX, TX, & HDET Freq Sweep. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23

Arm the Test Set for TX/RX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string previously constructed in this sweep.

Wait for test set to complete sweep operation.

Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured TX Freq Comp values from the test set.

There are 3 deliverables coming from this sweep:

- RX LNA (or DVGA) readings on 16 channels at each LNA state for NV calculation for both RX chains.
- HDET vs Freq readings for NV calculation
- TX Linearizer vs Freq (APT Mode) for NV calculation

Process the retrieved data values gathered during the sweep. Verify that power measurements are close to expected from reference channel. Also check RX measurements for any outliers.



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NV Calculation

For TX Linearizer Data

Store Rgi vs Power with fixed bias data from sweep 1, Store Rgi vs Power with APT bias data from sweep 2, and TX Linearizer vs Freq data from sweep 3 into the appropriate area of NV_<signal>_<band>_TX_MULTI_LIN_V3_DATA_I. Refer to QC Document ID: 80- N5420-170 section 4.3.2.1.

For HDET Data

Store HDET vs Level data from sweep 1 into the appropriate area of NV_<band>_EXP_HDET_VS_AGC_V2. Store HDET vs Freq data from sweep 3 into the appropriate area of NV_<band>_TX_LIM_VS_FREQ. Refer to QC Document ID: 80-N1622-62

The HDET Power targets phased in this procedure are listed in Table 6.1.3 below for reference

Parameter	BC0	BC1	BC10
All Products			
HDET			
HDET_Trgt_15 (dBm)	40.4	37.4	40.4
HDET_Trgt_14 (dBm)	38.8	35.8	38.8
HDET_Trgt_13 (dBm)	37.2	34.2	37.2
HDET_Trgt_12 (dBm)	35.6	32.6	35.6
HDET_Trgt_11 (dBm)	34	31	34
HDET_Trgt_10 (dBm)	32.4	29.4	32.4
HDET_Trgt_9 (dBm)	30.8	27.8	30.8
HDET_Trgt_8 (dBm)	29.2	26.2	29.2
HDET_Trgt_7 (dBm)	27.6	24.6	27.6
HDET_Trgt_6 (dBm)	26	23	26
HDET_Trgt_5 (dBm)	24.4	21.4	24.4
HDET_Trgt_4 (dBm)	22.8	19.8	22.8
HDET_Trgt_3 (dBm)	21.2	18.2	21.2
HDET_Trgt_2 (dBm)	19.6	16.6	19.6
HDET_Trgt_1 (dBm)	18	15	18
HDET_Trgt_0 (dBm)	16.4	13.4	16.4

Table 6.1.3

For RX Data

Store the RX Calibration data and RX Channel list from sweep 3 into the "RFNV_CDMA_<BAND>_RX_CAL_DATA_I". Refer to document 80-NH377-170 H for structure and formatting.

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6.2 CDMA1X FTM Parametric Tests

These tests are the same parametrics that are tested during call processing or in P2K Test Mode. These tests are meant to be faster and more reliable and replace call processing tests as well as not require a change to using P2K test commands at the end of testing. Call processing tests will be done on an audit basis when this test is performed in high volume testing. The radio must be completely phased and all NV items written to the phone prior to performing these tests. The radio must be in FTM mode to perform these tests. Section 6.2.1 has to be done once before any of the other sections can be performed.

6.2.1 FTM Parametric Setup

This test will put the radio into FTM Call Mode. This setup has to be done every time you need to change channel or band and must be after all phasing items are written to the phone. Any test in Section 6.2 can be performed provided they are conducted on the same band and channel with the exception of Open loop power which must be configured to ignore the reverse control bits.

Test Procedure

Setup the test set per the settings above in the desired band and on the desired testing channel.

For testing section 6.2.8 Radiated Diversity Receiver Test Only:

Use the FTM_SET_SECONDARY_CHAIN command with data 0x0100 to choose the secondary path.

Use the FTM_SET_SECONDARY_CHAIN_TEST_CALL_MODE command with data 0x01 to enable the call mode.

Use the FTM_ACQ_PILOT_FL command with data 0xAA000000BBBB00000000 where AA is either 0x00 for the CDMA band, 0x01 for the PCS band or 0x0F for AWS band and BBBB is the 2 byte, byte swapped value of the channel you are testing on.

Use the FTM_ACQ_SYNC_CHAN command to acquire the sync channel.

Use the FTM_ASSIGN_FFC command with data 0x020000000A000000000100000000000000 to assign the forward fundamental channel. Replace the 02 in this data string with the correct Radio Config value if using something other than RC2.

Use the FTM_ASSIGN_RFC command with data 0x02000000000000000001000000000000 to assign the reverse fundamental channel. Replace the 02 in this data string with the correct Radio Config value if using something other than RC2. If performing Open Loop Power, replace 01(respond to) with 00(ignore) reverse power control bits

Use the FTM_FC_LPBK command to enable loopback on the fundamental channel.

Use the FTM_DEASSIGN_ALL command to terminate the connection. This must be done prior to trying to reestablish a connection on another channel or band.

6.2.2 Open Loop Power Control in FTM Mode

Table 6.2.2 will specify which channels this procedure needs to be tested on as well as what RX levels. For tests where CHAN_X is called out, the table in this section will tell you what channel to be on to perform the test. For the CHANNEL tests, Table 6.2.2 will tell you which channels are required for each product.

Parameter	All Products	
All Products		
CHAN_X_800	350	X
CHAN_X_1900	650	--
CHAN_X_AWS	422	--
Table 6.2.2		

Test Procedure

Set test set basestation power to proper level for the test you are performing. (For tests that specify a RF level in the test name, use that RF level. For all other tests use, -73dBm for 800/BC10 and -76dBm for 1900).

Follow the procedure in Section 3.6 to put the radio into Phone Test Mode.

Use the FTM_SET_MODE command with data 0x0500 for BC0, 0x0600 for BC1, 0x1B00 for BC14 or 0x2000 for BC15.

Use the FTM_SET_CHAN command with the 2byte, byte swapped hex value of the channel to be tested.

Use the FTM_SET_TX_ON command to turn the carrier on.

Measure the CDMA Output Power in a 1.23MHz bandwidth.

The measured output power must be within the specifications listed in the appropriate table of Section [2.4 Test Limits](#) for the product being tested.

Use the FTM_SET_TX_OFF command to turn off the CDMA transmitter.

6.2.3 CDMA Maximum Current in FTM Mode

Test Procedure

While measuring CDMA Max Power in FTM Mode, measure the current the radio is drawing. Measured current must meet the specifications listed in Table 2.x for the product being tested.

6.2.4 Receiver Sensitivity (SER) and (SER AWGN) in FTM Mode

Parameter	BC0	BC1	BC10
13KDATA_TRAF_LEVEL	-15.6dB	-15.6dB	-15.6dB
Primary Path RX_LEVEL	-105dBm	-105dBm	-105dBm
Secondary Path RX_LEVEL	-104dBm	-104dBm	-104dBm
Thermal RX_LEVEL	-102dBm	-102dBm	-102dBm
RATE_SET (Radio Config)	1	1	1
BAUD_RATE	FULL	FULL	FULL
TRAFFIC_CHAN_WALSH	10	10	10
Max Frames	1000	1000	1000
RX_LEVEL_AWGN	-55dBm	-55dBm	-55dBm
AWGN_LEVEL	-54dBm	-54dBm	-54dBm
13KDATA_TRAF_LEVEL_AWGN	-15.6dB	-15.6dB	-15.6dB

Table 6.2.4

Test Procedure

Setup the test set per the settings above in the desired band and on the desired testing channel.

Use the FTM_ACQ_PILOT_FL command with data 0xAA000000BBBB00000000 where AA is either 00 for the CDMA band or 01 for the PCS band and BBBB is the 2 byte, byte swapped value of the channel you are testing on.

Use the FTM_ACQ_SYNC_CHAN command to acquire the sync channel.

Use the FTM_ASSIGN_FFC command with data 0x02000000A0000000001000000000000000 to assign the forward fundamental channel. Replace the 02 in this data string with the correct Radio Config value if using something other than RC2.

Use the FTM_ASSIGN_RFC command with data 0x02000000000000000000000000000000 to assign the reverse fundamental channel. Replace the 02 in this data string with the correct Radio Config value if using something other than RC2.

Use the FTM_FC_LPBK command to enable loopback on the fundamental channel.

Set the test set to 'ALLUP' control bits.
If testing AWGN, then turn on the AWGN level in the test set.

Enable logging of FER/SER messages.

Using mathematical calculations, ensure the confidence level is met when calculating the SER. Both Nextest and RadioComm have support for these complex calculations and those calculations will not be listed here. SER is currently the targeted value to meet confidence.

If the number of packets you have received reaches Max Frames specified in the table above and confidence is still not met, then simply fail the unit.

The Symbol Error Rate tests (SER) and (SER AWGN) must be within the specification listed in the appropriate table of Section [2.4 Test Limits](#) for the product being tested.

If so desired, at this time you can read the following items from the test set: Max Power, Freq Error, Time Error and Rho. If any of these items are read, they must be within the specification listed in the appropriate table of Section [2.4 Test Limits](#) for the product being tested.

Before moving on to the next test, perform the following items:
Disable logging of FER/SER messages.
Use the FTM_DASGN_ALL command.

6.2.5 CDMA Minimum Power in FTM Mode

Test Procedure

After measuring FER/SER from Section 6.2.4 perform the following:
Set the test set RF level to -24dBm.

Either set the test set to "ALLDOWN" control bits or use the FTM_WRITE_TX_GAIN_ACCUM command with data 0x7F to force the radio to use all down bits.

Measure the minimum power out of the radio.

Measured power must be within the specification listed in the appropriate table of Section [2.4 Test Limits](#) for the product being tested.

6.2.6 CDMA Maximum Power in FTM Mode

Test Procedure

Apply a CDMA,PCS or AWS RF input signal to the radio at -104dBm
Power up the phone.
Follow the procedure in Section 3.6 to put the radio into Phone Test Mode.
Use the FTM_SET_MODE command with data 0x0500 for BC0, 0x0600 for BC1, 0x1B00 for BC14 or 0x2000 for BC15.
Use the FTM_SET_CHAN command with the 2byte, byte swapped hex value of the channel to be tested.
Use the FTM_SET_TX_ON command to turn the carrier on.

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Measure the CDMA Output Power in a 1.23MHz bandwidth.

The measured output power must be within the specifications listed in the appropriate table of Section [2.4 Test Limits](#) for the product being tested.

Use the FTM_SET_TX_OFF command to turn off the CDMA transmitter.

6.2.7 Splatter at Maximum Output Power in FTM Mode

The phone must be phased before this test will function correctly. While measuring CDMA Max Power in FTM Mode, follow the procedure below to measure splatter.

Parameter	BC0	BC1	BC10
FREQUENCY_OFFSET_HIGH	+0.885MHz	+1.25MHz	+0.885MHz
FREQUENCY_OFFSET_LOW	-0.885MHz	-1.25MHz	-0.885MHz

Table 6.2.7

Test Procedure

Begin by measuring the total output power in a 1.23MHz Bandwidth.

The total output power is then REFERENCE_POWER.

Switch the power measurement instrument to measure in a 30kHz bandwidth.

Offset by FREQUENCY_OFFSET_HIGH and FREQUENCY_OFFSET_LOW and measure the power at each.

ADJ_CHANNEL_POWER is the higher of the two measured powers in dBm.

Calculate:

$SPLATTER = REFERENCE_POWER - ADJ_CHANNEL_POWER$

SPLATTER must be within the specification listed in the appropriate table of Section [2.4 Test Limits](#) for the product being tested.

6.2.8 Radiated Receiver Test
Test Procedure

The purpose of this test is to verify proper seating and electrical connection of the UE antenna. The test measures radiated power through an RF-coupled antenna fixture.

		Product Test Setup GNPO			
Test	Setup				
C800 Band RX Diversity	Channel	25	--	306	25,350,750
	RF level	-72	--	-72	-72
C1900 Band Rx Diversity	Channel	25	650	650	25,650,1150
	RF level	-72	-72	-72	-72

Table 6.2.8.1

		Product Test Setup Lead Site			
Test	Setup				
C800 Band RX Diversity	Channel	25	--	306	350
	RF level	-72	--	-72	-72
C1900 Band Rx Diversity	Channel	25	650	650	650
	RF level	-72	-72	-72	-72

Table 6.2.8.2

Parameter	All Products	
All Products		
CHAN_X_800	350	X
CHAN_X_1900	650	--
CHAN_X_AWS	422	--

Table 6.2.8.3

Test Procedure

All Products:

Setup:

Set the signal generator dBm output to RF level specified in tables 6.2.8.1 and 6.2.8.2

Use the following commands to set the RX reference channel (from tables 6.2.8.1 and 6.2.8.2) and read the RX AGC value.

Use the following command to set UE to the desired band (CDMA/PCS/AWS)

<u>Command</u>	<u>Opcode/data</u>
FTM_SET_MODE	0x4B0B 0200 0700 + Mode

Where:

BC0 Mode = 0x0500
 BC1 Mode = 0x0600
 BC14 Mode = 0x1B00
 BC15 Mode = 0x2000

For Primary Antenna:

<u>Command</u>	<u>Opcode/data</u>	
FTM_SET_CHAN	0x4B0B 0200 0800 + Channel	(Select CDMA/PCS/AWS channel)

For example where:
 Channel = 0x0002 (Channel 512 in hex, byte swiped)

<u>Command</u>	<u>Opcode/data</u>	
FTM_SET_LNA_RANGE	0x0100	(Set LNA Range to Range 1)
FTM_GET_RX_AGC		(Read the Rx AGC)

For Secondary Antenna:

<u>Command</u>	<u>Data</u>	
FTM_SET_SECONDARY_CHAIN	0x0100	(Enable Secondary Chain)
FTM_SET_CHAN	0x4B0B 0200 0800 + Channel	(Select CDMA/PCS/AWS channel)
FTM_SET_LNA_RANGE_SECPATH	0x0100	(Set LNA Range to Range 1)
FTM_GET_RX_AGC_SECPATH		(Read the Rx AGC)

Ensure that the RX_AGC is not outside the limits for RX_AGC specified in the appropriate section of Section [2.4 Test Limits](#) for the product being tested. If outside those limits, fail the test.

6.2.9 Conducted Spurious Emissions

While transmitting, the radio is required to limit its spurious emissions so as to avoid interference on neighboring frequencies. The following is a description of how the radios should be checked for assurance that it meets the conducted spurious emissions requirements. The phone must be phased before this test will function correctly. While measuring CDMA Max Power in Test Mode, follow the procedure below to measure spurs.

Parameter	BC0	BC1	BC10
RX_LEVEL	-120dBm	-120dBm	-120dBm
FREQ_OFFSET1	± .885MHz	± 1.25MHz	± .885MHz
FREQ_OFFSET2	± 1.98MHz	± 1.98MHz	± 1.98MHz
SPUR_LIMIT_FREQ_OFFSET1	-43dBc/30kHz	-43dBc/30kHz	-43dBc/30kHz
SPUR_LIMIT_FREQ_OFFSET2	-54dBc/30kHz	-50dBc/30kHz	-54dBc/30kHz

Table 6.2.9

Test Procedure

Adjust the RX level into the Radio to RX_LEVEL.

Measure the total output power of the CARRIER_FREQ in a 1.23MHz Bandwidth, REFERENCE_POWER.

Part A:

Sweep the frequency range between FREQ_OFFSET1 and FREQ_OFFSET2. Verify any spurs in this frequency range meet the limit SPUR_LIMIT_FREQ_OFFSET1: MEASURED_SPUR – REFERENCE_POWER must be less than SPUR_LIMIT_FREQ_OFFSET1.

Part B:

Sweep the frequency range past FREQ_OFFSET2 to ±12MHz. Verify any spurs in this frequency range meet the limit SPUR_LIMIT_FREQ_OFFSET2: MEASURED_SPUR – REFERENCE_POWER must be less than SPUR_LIMIT_FREQ_OFFSET2.

6.2.10 Channel Handoff

This procedure allows channel hard handoffs which in FTM mode. This eliminates the need to re-initiate the setup when changing bands, channel or reverse power control bits settings.

Test Procedure

Phone should be connected with the procedure in Section 6.2.1.

Change the channel on the Testset to a different band, channel and/or Reverse Control bits setting.

Use the FTM_HANDOFF command with data 0x00000000 + 4 bytes for the band class +2 bytes for the channel number + 1 byte for the control bits setting + 0x00.

Verify the phone remained in the connected.

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7.0 LTE-FDD Test Procedures

The procedures in this section describe the methods to tune the LTE hardware control parameters and verify LTE functionality in the UE.

7.1 LTE-FDD RX, TX and HDET Phasing

Test Description

This APT characterization routine phases the receiver, transmitter and HDET (power detect circuit) by performing 3 sweeps. The sweeps consist of the following:

- Sweep 1 (Fixed Bias)
- Sweep 2 (APT Bias)
- Sweep 3 (RX, TX, & HDET Freq Sweep)

A full description of the ESC V4 Sweeps can be found in the ESC V4 Software Design Guide; QC Document ID: 80-NG201-2.

Please note, all documents referred to in this section can be downloaded from <http://support.cdmatech.com> by searching for the document #.

Test Setup

For all products in this section phasing should be run at 3.9V.

Test Procedure

Pre-sweep

Use the following FTM command to select the LTE-FDD Mode for the desired band.

Command	Data
FTM_SET_MODE	Mode (Select LTE0-FDD band)

Use the following FTM command to disable the secondary path.

Command	Data
SET_SECONDARY_PATH	0x00 (Disable Secondary Path)

Use the following FTM command to tune the UE to the UL Reference Channel

Command	Data
FTM_SET_CHAN *	UL Ref Channel (Set UL channel)

Command	Data
FTM_SET_TX_ON	N/A (Set TX ON)

For each PA range, starting with the lowest and ending with the highest:

Use the FTM_SET_PA_RANGE command to set it to PA_Rx.

Read the following items out of the Default file:

MOTCONF_LTE_FDD_C0_B(band*)_TX_RISE_EXTENSION – list of 3 dB10 items

MOTCONF_LTE_FDD_C0_B(band*)_TX_FALL_EXTENSION – list of 3 dB10 items

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MOTCONF_LTE_FDD_C0_B(band*)_TX_HIGH_RANGE_MAX_POWER – 1 dB10 item

MOTCONF_LTE_FDD_C0_B(band*)_TX_LOW_RANGE_MIN_POWER – 1 dB10 item

* band variable above is one of the following: Band 01, 02, 03, 04, 05, 07, 08, 9, 11, 12, 13, 17, 19, 20, 25, 28

Use these values to choose a start and stop TX_GAIN_INDEX such that the sweep will cover a range of powers which extents:

PA RANGE	POWER for Start Point	Power for Stop Point
Highest PA Range	HIGH_RANGE_MAX_POWER/10	High Range Fall Switchpoint - TX_FALL_EXTENSION[0]/10
Lowest PA Range	Low Range Rise Switchpoint + TX_RISE_EXTENSION[max range]/10	LOW_RANGE_MIN_POWER/10
All other PA Ranges	Range Rise Switchpoint + TX_RISE_EXTENSION[pa range index]/10	Range Fall Switchpoint - TX_FALL_EXTENSION[pa range index]/10

Table 7.1.1

Start at the lowest PA GAIN RANGE, find TX_GAIN_INDEX that gives you a power equal or below **“Power for Stop Point”** in table above for the associated PA Range. Call this “STOP_TX_GAIN_INDEX”.

Start at the lowest PA GAIN RANGE, find TX_GAIN_INDEX that gives you a power equal or above **“Power for Start Point”** in table above for the associated PA Range. Call this “START_TX_GAIN_INDEX”.

Note: Running average start/stop TX gain indexes can be used instead of static value to help optimize test time in production.

V4 Calibration

Sweep 1 (Fixed Bias)

This sweep will use a fixed PA bias (defined by the <Bias_List> parameter in the ESC_Param.xml file) to sweep from a starting RGI down to a stopping RGI measuring power at each step on reference channel. Verify that the start & stop RGIs will be sufficiently above/below the rise/fall switch points for each PA range before starting first sweep.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for sweep 1 calibration. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file for sweep type 0. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23

Arm the Test Set for initial TX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string of parameter list obtained from the ESC_Param.xml file.

Wait for test set to complete sweep operation.

Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured power values for each PA range from the test set.

There are 3 deliverables coming from this sweep:

- Rgi vs Power with fixed bias for TX_MULTI_LIN_V3 NV item
- Reference RGI vs Power levels for calculations on 2nd sweep
- HDET vs Power for high PA gain.

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Save deliverables for processing at later points of the procedure.

Sweep 2 (APT Bias)

This sweep will characterize the TX output power vs RGI using a variable Bias on each step calculated from the first sweep. This sweep is also done only on 1 reference channel. This sweep is identical to the first sweep except that the SMPS PA Bias for each step is calculated from the first sweep and HDET is not measured in this sweep.

The SMPS PA Bias is calculated by using the Power level for each RGI in the first sweep and looking that value up in the APT Characterization.xml file. This value is characterized by development to be a bias level that allows the PA to pass ACLR specs. Typically the maximum powers on a given band are close to max PA bias, and reduce as the rgi/power diminishes.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for sweep 2 calibration. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file for sweep type 1. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23. Note that the biases will come from the APT Characterization.xml file for this construction.

Arm the Test Set for second TX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string previously constructed for this sweep.

Wait for test set to complete sweep operation.

Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured power values from the test set.

There are 2 deliverables that come from this sweep:

- Rgi vs Power with APT Bias for TX_MULTI_LIN_V3 NV item
- RGI vs Power data used in sweep 3 for TX Linearizer

Verify that the maximum and minimum power measurements for each PA State exceed the respective switch points.

Save deliverables for processing at later points of the procedure.

Sweep 3 (RX, TX, & HDET Freq Sweep)

This third APT (Average Power Tracking) sweep will take measurements across the band (typically 16 channels) in order to determine TX, HDET and RX versus frequency for the phone. For each channel, there will be (5 + number of PA states) steps. Typically each step can do both RX and TX in parallel (unless the signal is TD (LTE-TDD or TD-SCDMA) in which case the TX measurements are all done first and then the RX measurements are done, effectively requiring (5 + number of PA states + number of LNA states) steps. Table 7.1.2 below is a visual representation of a single channel in this sweep.

TX Measurement	RX Measurement
HDET (Target power + 2 RGI)	LNA 0 or VGA reading
HDET (Target power + 1 RGI)	LNA 1
HDET (@Target power)	LNA 2
HDET (Target power - 1 RGI)	LNA 3
HDET (Target power - 2 RGI)	LNA 4 (LTE only else do nothing)
TX Linearizer (typically at high PA state's fall SP power level)	LNA 5 (LTE only else do nothing)
TX Linearizer (typically at mid PA state's rise SP power level)	do nothing
TX Linearizer (typically at low PA state's rise SP power level)	do nothing
Table 7.1.2	

Secondary Chain calibration is done by default in parallel with primary chain. This requires the test bench to use a splitter across the primary/diversity paths. It is possible to disable the secondary chain calibration if diversity path cannot be connected in parallel. If secondary chain calibration is not done in parallel, it cannot be performed using a ESC V4 sweep.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for RX, TX, & HDET Freq Sweep. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23

Arm the Test Set for TX/RX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string previously constructed in this sweep.

Wait for test set to complete sweep operation.

Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured TX Freq Comp values from the test set.

There are 3 deliverables coming from this sweep:

- RX LNA (or DVGA) readings on 16 channels at each LNA state for NV calculation for both RX chains.
- HDET vs Freq readings for NV calculation
- TX Linearizer vs Freq (APT Mode) for NV calculation

Process the retrieved data values gathered during the sweep. Verify that power measurements are close to expected from reference channel. Also check RX measurements for any outliers.

NV Calculation

For TX Linearizer Data

Store Rgi vs Power with fixed bias data from sweep 1, Store Rgi vs Power with APT bias data from sweep 2, and TX Linearizer vs Freq data from sweep 3 into the appropriate area of NV_<signal>_<band>_TX_MULTI_LIN_V3_DATA_I. Refer to QC Document ID: 80-N5420-170 section 4.3.2.1.

For HDET Data

Store HDET vs Level data from sweep 1 into the appropriate area of NV_<signal>_<band>_EXP_HDET_VS_AGC_V2. Store HDET vs Freq data from sweep 3 into the appropriate area of NV_<signal>_<band>_TX_LIMIT_VS_FREQ. Refer to QC Document ID: 80-N1622-62

The HDET Power targets phased in this procedure are listed in Table 7.1.3 below for reference

Parameter	All Bands (non-LPM)	LPM
All Products		
HDET		
HDET_Trgt_15 (dBm)	30.8	20.9
HDET_Trgt_14 (dBm)	29.2	19.3
HDET_Trgt_13 (dBm)	27.6	17.7
HDET_Trgt_12 (dBm)	26	16.1
HDET_Trgt_11 (dBm)	24.4	14.5

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Parameter	All Bands (non-LPM)	LPM
HDET_Trgt_10 (dBm)	22.8	12.9
HDET_Trgt_9 (dBm)	21.2	11.3
HDET_Trgt_8 (dBm)	19.6	9.7
HDET_Trgt_7 (dBm)	18	8.1
HDET_Trgt_6 (dBm)	16.4	6.5
HDET_Trgt_5 (dBm)	14.8	4.9
HDET_Trgt_4 (dBm)	13.2	3.3
HDET_Trgt_3 (dBm)	11.6	1.7
HDET_Trgt_2 (dBm)	10	0.1
HDET_Trgt_1 (dBm)	8.4	-1.5
HDET_Trgt_0 (dBm)	6.8	-3.1

Table 7.1.3

For RX Data

Store the RX Calibration data and RX Channel list from sweep 3 into the "RFNV_LTE_<BAND>_RX_CAL_DATA_V2_I". Refer to document 80-NH377-170 H for structure and formatting.

Exit Test



7.2 LTE-FDD Non-signaling Parametric Tests

Test Description

The purpose of this test is to verify the LTE-FDD Tx waveform and spectrum quality and to verify LTE-FDD Rx quality.

Test Dependencies:

3.2 Turn On

7.1 LTE-FDD RX, TX and HDET Phasing

Equipment:

RF Signal generator with ARB playback capability.

RF Signal analyzer with LTE-FDD modulation analysis capability.

Test Setup

Signal Generator Setup:

Signal Type	LTE_FDD
Modulation	ARB
Band	See tables 7.2.1 - 7.2.6
Channel	See tables 7.2.1 - 7.2.6
Frequency	Test channel frequency (from tables 1.6.2)
DL Power Level	-65dBm
Baseband Mode	ARB
ARB Repetition	Continuous
Trigger	Autostart / Re-trigger

Signal Analyzer Setup:

Modulation	QPSK
Bandwidth	See tables 7.2.1 - 7.2.6
Channel	See tables 7.2.1 - 7.2.6
Frequency	Test channel frequency (from table 1.6.2)
Expected Pwr	24dBm

TX Test Channels for 10MHz BW products

Band	All Products	All Products
	GNPO (L / M / H)	Lead Site (L / M / H)
1	18050, 18300, 18550	18050, 18300, 18550
2	18650, 18900, 19150	18650, 18900, 19150
3	19250, 19575, 19900	19250, 19575, 19900
4	20000, 20175, 20350	20000, 20175, 20350
5	20450, 20525, 20600	20450, 20525, 20600
7	20800, 21100, 21400	20800, 21100, 21400
8	21500, 21625, 21750	21500, 21625, 21750
9	21850, 21975, 22100	21850, 21975, 22100
12	23050, 23090, 23130	23050, 23090, 23130
13	23230	23230
17	23780, 23790, 23800	23780, 23790, 23800
20	24200, 24300, 24400	24200, 24300, 24400
25	26090, 26365, 26640	26090, 26365, 26640
26	26750, 26865, 26990	26750, 26865, 26990
28A	27260, 27360, 27460	27260, 27360, 27460
28B	27410, 27510, 27610	27410, 27510, 27610
30	27710, 27710, 27710	27710, 27710, 27710

Table 7.2.1

TX Test Channels for 5MHz BW products

Band	All Products	All Products
	GNPO (L / M / H)	Lead Site (L / M / H)
1	18025, 18300, 18575	18025, 18300, 18575
2	18625, 18900, 19175	18625, 18900, 19175
3	19225, 19575, 19925	19225, 19575, 19925
4	19975, 20175, 20375	19975, 20175, 20375
5	20425, 20525, 20625	20425, 20525, 20625
7	20775, 21100, 21425	20775, 21100, 21425
8	21475, 21625, 21775	21475, 21625, 21775
9	21825, 21975, 22125	21825, 21975, 22125
12	23025, 23090, 23155	23025, 23090, 23155
13	---	---
17	23755, 23790, 23825	23755, 23790, 23825
20	24175, 24300, 24425	24175, 24300, 24425
25	26065, 26365, 26665	26065, 26365, 26665
26	26715, 26865, 27015	26715, 26865, 27015
28A	27235, 27360, 27485	27235, 27360, 27485
28B	27385, 27510, 27635	27385, 27510, 27635
30	27685, 27710, 27735	27685, 27710, 27735

Table 7.2.2



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	All Products	All Products
Band	GNPO (L / M / H)	Lead Site (L / M / H)
1	18075, 18300, 18525	18075, 18300, 18525
2	18675, 18900, 19125	18675, 18900, 19125
3	19275, 19575, 19875	19275, 19575, 19875
4	20025, 20175, 20325	20025, 20175, 20325
5	20475, 20525, 20575	20475, 20525, 20575
7	20825, 21100, 21375	20825, 21100, 21375
8	--	--
9	21875, 21975, 22075	21875, 21975, 22075
12	23075, 23090, 23105	23075, 23090, 23105
13	---	---
17	23805, 23790, 23775	23805, 23790, 23775
20	24225, 24300, 24375	24225, 24300, 24375
25	26115, 26365, 26615	26115, 26365, 26615
26	26765, 26865, 26965	26765, 26865, 26965
28A	27285, 27360, 27435	27285, 27360, 27435
28B	27435, 27510, 27585	27435, 27510, 27585
30	--	--

Table 7.2.3

RX Test Channels for 3MHz BW products

	All Products	All Products
Band	GNPO (L / M / H)	Lead Site (L / M / H)
29	9675, 9715, 9755	9675, 9715, 9755

Table 7.2.4

RX Test Channels for 5MHz BW products

	All Products	All Products
Band	GNPO (L / M / H)	Lead Site (L / M / H)
29	9685, 9715, 9745	9685, 9715, 9745

Table 7.2.5

RX Test Channels for 10MHz BW products

	All Products	All Products
Band	GNPO (L / M / H)	Lead Site (L / M / H)
29	9710, 9715, 9720	9710, 9715, 9720

Table 7.2.6

UE Setup:

Use the ChangeMode command to put the phone into FTM mode.

Command	Opcode/data
---------	-------------

CHANGEMODE	0x29 0300	(Enter FTM mode)
------------	-----------	------------------

Test Procedure

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7.2.1 LTE-FDD Non-signaling Connection

Setup RF signal generator to continuously execute ARB on the appropriate band and channel (refer to Test Setup section above). Use the following downlink vector ARB file: 'LTE_ProductionTest_1_v01.wv' for R&S CMW500 and 'LTE_downlink_BW10MHz_QC_BLER-SR15.36-RTS92-BFO0.wfm' for Keysight EXM and EXT (converted from virtual high level ARB file name 'LTE_downlink_QC_BLER_BW10MHz_50RB_ALLUP').

Set the downlink level to -65dBm.

Start LTE-FDD physical layer 1

Send the following command to start LTE physical layer 1 in non-signaling mode.

<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_NS_START	0x4B0B 1E00 F401 95F8 7E

Verify that the response status byte is 0 to indicate success of the command.

Acquiring LTE-FDD cell downlink

Send the following command to request LTE layer 1 to acquire the PSS, SSS, and MIB contained in the PBCH.

<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_NS_ACQ	0x4B0B 1E00 F501<band 1 byte><Rx channel 2 bytes><checksum 2 bytes>7E

Verify that the response status byte is 0 and response RF band and downlink channel match expected band and downlink channel to indicate success of the command.

Establishing LTE-FDD downlink/uplink data paths

Send the following command to establish uplink/downlink data paths.

<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_NS_START_DP	0x4B0B 1E00 F601 0000 0000 0E00 011E 0102 5252 7E

Verify that the response status byte is 0 to indicate success of the command.

Check LTE-FDD connection state

Send the following command to verify that the LTE connection has been established.

<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_IS_CONN	0x4B0B 1E00 F901 ED48 7E

Verify that the response status byte is 0 to indicate the phone is in a connection state.

Configure uplink waveform

Send the following command to configure the active resource blocks (Start RB Index and Number of RBs) and modulation type for the LTE uplink waveform.

<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_NS_CONFIG_UL_WAVEFORM	0x4B0B 1E00 FA01 0100 0032 00B2 4D7E

Verify that the response status byte is 0 to indicate success of the command.

Configure uplink power

Send the following command to override the open loop and closed loop power and explicitly configure the transmitter power in 1/10dBm units. Set the Tx level field in this command to 24dBm to drive the phone to this power level.

<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_NS_CONF_UL_POWER	0x4B0B 1E00 FB01 0000 <Tx level 2 bytes><checksum 2 bytes>7E

Example:

Send 0x4B0B1E00FB010000F00031777E to configure uplink power to 24dBm.

Verify that the response status byte is 0 to indicate success of the command.

7.2.2 LTE-FDD Tx Current

Measure the total current drain from the phone and verify the value is within the limits specified in section 2.5 [LTE-FDD Test Limits](#).

7.2.3 LTE-FDD TX Max Power

Measure the TX power and verify the value is within the limits specified in section 2.5 [LTE-FDD Test Limits](#).

Test condition	LTE BW	Modulation	UL RB number	UL RB offset	EARFCN
Max Power, MPR = 0dB	5 MHz	QPSK	1	0	low/mid
Max Power, MPR = 0dB	5 MHz	QPSK	1	24	high
Max Power, MPR = 0dB	10 MHz	QPSK	12	0	low/mid
Max Power, MPR = 0dB	10 MHz	QPSK	12	37	high
Max Power, MPR = 0dB	15 MHz	QPSK	1	0	low/mid
Max Power, MPR = 0dB	15 MHz	QPSK	1	74	high

7.2.4 LTE-FDD TX Frequency Error

Measure the TX frequency error and verify the value is within the limits specified in section 2.5 [LTE-FDD Test Limits](#).

Test condition	LTE BW	UL/DL Modulation	DL RB Number	UL RB Number	UL/DL RB offset	EARFCN
Max Power, MPR Active	10MHz	QPSK	50	50	0	mid

7.2.5 LTE-FDD TX EVM

Measure the Tx EVM and verify the value is within the limits specified in section 2.5 [LTE-FDD Test Limits](#).

Test condition	LTE BW	Modulation	UL RB number	UL RB offset	EARFCN	Site
Max Power, MPR Active	5 MHz	QPSK	25	0	mid	High volume
Max Power, MPR Active	5 MHz	16QAM	25	0	mid	High volume
Max Power, MPR = 0dB	5 MHz	QPSK	8	0	mid	Proto
Max Power, MPR Active	10 MHz	QPSK	50	0	mid	High volume
Max Power, MPR Active	10 MHz	16QAM	50	0	mid	High volume
Max Power, MPR = 0dB	10 MHz	QPSK	12	0	mid	Proto
Max Power, MPR Active	15 MHz	QPSK	75	0	mid	High volume
Max Power, MPR Active	15 MHz	16QAM	75	0	mid	High volume
Max Power, MPR = 0dB	15 MHz	QPSK	16	0	mid	Proto

7.2.6 LTE-FDD TX I/Q Origin Offset

Measure the Tx I/Q origin offset and verify the value is within the limits specified in section 2.5 [LTE-FDD Test Limits](#).

Test condition	LTE BW	Modulation	RB number	RB offset	EARFCN	Site
Max Power, MPR = 0dB	5 MHz	QPSK	8	0	mid	Proto
Max Power, MPR = 0dB	5 MHz	QPSK	8	16	mid	Proto
Max Power, MPR = 0dB	10 MHz	QPSK	12	0	mid	Proto
Max Power, MPR = 0dB	10 MHz	QPSK	12	37	mid	Proto



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Max Power, MPR = 0dB	15 MHz	QPSK	16	0	mid	Proto
Max Power, MPR = 0dB	15 MHz	QPSK	16	58	mid	Proto

7.2.7 LTE-FDD TX ACLR

Measure the TX adjacent channel leakage ratio (ACLR) and verify the value is within the limits specified in section 2.5 [LTE-FDD Test Limits](#).

Test condition	LTE BW	Modulation	RB number	RB offset	EARFCN
Max Power, MPR = 0dB	5 MHz	QPSK	8	0	low/mid/high
Max Power, MPR = 0dB	5 MHz	QPSK	8	16	low/mid/high
Max Power, MPR Active	5 MHz	QPSK	25	0	low/mid/high
Max Power, MPR = 0dB	10 MHz	QPSK	12	0	low/mid
Max Power, MPR = 0dB	10 MHz	QPSK	12	37	high
Max Power, MPR Active	10 MHz	QPSK	50	0	low/mid/high
Max Power, MPR = 0dB	15 MHz	QPSK	16	0	low/mid/high
Max Power, MPR = 0dB	15 MHz	QPSK	16	58	low/mid/high
Max Power, MPR Active	15 MHz	QPSK	75	0	low/mid/high

7.2.8 LTE-FDD RX BLER

Set the downlink level to -97dBm.

Use the following command to reset the BLER reporting in the phone.

Command	Opcode/data
FTM_LTE_NS_RESET_DL_BLER_REPORT	0x4B0B 1E00 FD01 8D2F 7E

Use the following command to read the BLER report from the phone.

Command	Opcode/data
FTM_LTE_NS_GET_DL_BLER_REPORT	0x4B0B 1E00 FE01 E505 7E

Parse out the response fields TOTAL_BLOCKS_RECEIVED and TOTAL_BLOCK_ERRORS. Add these values to cumulative total values as:

CUMULATIVE_TOTAL_BLOCKS_RECEIVED = CUMULATIVE_TOTAL_BLOCKS_RECEIVED + TOTAL_BLOCKS_RECEIVED

CUMULATIVE_TOTAL_BLOCK_ERRORS = CUMULATIVE_TOTAL_BLOCK_ERRORS + TOTAL_BLOCK_ERRORS.

Calculate cumulative BLER with the equation $BLER = (CUMULATIVE_TOTAL_BLOCK_ERRORS / CUMULATIVE_TOTAL_BLOCKS_RECEIVED) * 100\%$.

Keep reading BLER until 95% confidence level is met or have read 10000 blocks.

For LTE-FDD Band 29 Carrier Aggregation RX BLER Test

Enable DUT for carrier aggregation secondary component carrier on band BAND_29 and downlink channel 9715

Send the following FTM command to enable secondary cell.

Command	Opcode/data
FTM_LTE_NS_ENABLE_SCELL	0x4B0B 1E00 0702 1CF3 2500 0097 DD7E

Check LTE-FDD connection state

Send the following command to verify that the LTE connection has been established.

Command	Opcode/data
FTM_LTE_IS_CONN	0x4B0B 1E00 F901 ED48 7E

Verify that the response status byte is 0 to indicate the phone is in a connection state.

Set generator to carrier aggregation secondary component carrier.

Set the downlink level to -93dBm.

Configure uplink waveform

Send the following command to configure the active resource blocks (Start RB Index and Number of RBs) and modulation type for the LTE uplink waveform.

<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_NS_CONFIG_UL_WAVEFORM	0x4B0B 1E00 FA01 0100 0032 00B2 4D7E

Verify that the response status byte is 0 to indicate success of the command.

Configure uplink power

Send the following command to override the open loop and closed loop power and explicitly configure the transmitter power in 1/10dBm units. Set the *Tx level* field in this command to 24dBm to drive the phone to this power level.

<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_NS_CONF_UL_POWER	0x4B0B 1E00 FB01 0000 <Tx level 2 bytes><checksum 2 bytes>7E

Example:

Send 0x4B0B1E00FB010000F00031777E to configure uplink power to 24dBm.

Verify that the response status byte is 0 to indicate success of the command.

Use the following command to reset the BLER reporting in the phone.

<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_NS_RESET_DL_BLER_REPORT	0x4B0B 1E00 FD01 8D2F 7E

Send the following FTM command to read the BLER report from the phone.

<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_NS_GET_ALL_CARR_DL_BLER	0x4B0B 1E00 0902 7D5E 067E

Parse out the response fields for carrier #1: TOTAL_BLOCKS_RECEIVED_C1 and TOTAL_BLOCK_ERRORS_C1. Add these values to cumulative total values as:

CUMULATIVE_TOTAL_BLOCKS_RECEIVED = CUMULATIVE_TOTAL_BLOCKS_RECEIVED +
TOTAL_BLOCKS_RECEIVED

CUMULATIVE_TOTAL_BLOCK_ERRORS = CUMULATIVE_TOTAL_BLOCK_ERRORS + TOTAL_BLOCK_ERRORS.

Calculate cumulative BLER with the equation $BLER = (CUMULATIVE_TOTAL_BLOCK_ERRORS / CUMULATIVE_TOTAL_BLOCKS_RECEIVED) * 100\%$.

Keep reading BLER until 95% confidence level is met or have read 10000 blocks.

Send the following FTM command to disable secondary cell.

<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_NS_DISABLE_SCELL	0x4B0B 1E00 802A 61F7E

High Volume Test matrix

Band	Full CELL DL Power	LTE BW	Modulation (UL and DL)	RB DL number	RB DL offset	RB UL number	RB UL offset	EARFCN	UL NS	Site
1	-100	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
1	-97	10 MHz	QPSK	50	0	50	0	L/M/H	01	HV/Proto
1	-95.2	15 MHz	QPSK	75	0	75	0	L/M/H	01	HV/Proto
2	-98	5 MHz	QPSK	25	0	25	0	L/M/H	03	HV/Proto
2	-95	10 MHz	QPSK	50	0	50	0	L/M/H	03	HV/Proto
2	-93.2	15 MHz	QPSK	75	0	50	24	L/M/H	03	HV/Proto



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Band	Full CELL DL Power	LTE BW	Modulation (UL and DL)	RB DL number	RB DL offset	RB UL number	RB UL offset	EARFCN	UL NS	Site
3	-97	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
3	-94	10 MHz	QPSK	50	0	50	0	L/M/H	01	HV/Proto
3	-92.2	15 MHz	QPSK	75	0	50	24	L/M/H	01	HV/Proto
4	-100	5 MHz	QPSK	25	0	25	0	M	03	HV/Proto
4	-97	10 MHz	QPSK	50	0	50	0	M	03	HV/Proto
4	-95.2	15 MHz	QPSK	75	0	75	0	M	03	HV/Proto
5	-98	5 MHz	QPSK	25	0	50	0	L/M/H	01	HV/Proto
5	-95	10 MHz	QPSK	50	0	25	24	L/M/H	01	HV/Proto
7	-98	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
7	-95	10 MHz	QPSK	50	0	50	0	L/M/H	01	HV/Proto
7	-93.2	15 MHz	QPSK	75	0	75	0	L/M/H	01	HV/Proto
8	-97	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
8	-94	10 MHz	QPSK	50	0	25	0	L/M/H	01	HV/Proto
9	-99	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
9	-96	10 MHz	QPSK	50	0	25	0	L/M/H	01	HV/Proto
9	-94.2	15 MHz	QPSK	75	0	50	0	L/M/H	01	HV/Proto
12	-97	5 MHz	QPSK	25	0	20	4	L/M/H	06	HV/Proto
12	-94	10 MHz	QPSK	50	0	20	29	L/M/H	06	HV/Proto
13	-94	10 MHz	QPSK	50	0	20	0	M	06	HV/Proto
17	-97	5 MHz	QPSK	25	0	20	4	M	06	HV/Proto
17	-94	10 MHz	QPSK	50	0	20	29	M	06	HV/Proto
20	-97	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
20	-94	10 MHz	QPSK	50	0	20	0	L/M/H	01	HV/Proto
20	-91.2	15 MHz	QPSK	75	0	20	11	L/M/H	01	HV/Proto
25	-96.5	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
25	-93.5	10 MHz	QPSK	50	0	50	0	L/M/H	01	HV/Proto
25	-91.7	15 MHz	QPSK	75	0	50	24	L/M/H	01	HV/Proto
26	-97.5	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
26	-94.5	10 MHz	QPSK	50	0	25	0	L/M/H	01	HV/Proto
26	-92.7	15 MHz	QPSK	75	0	25	0	L/M/H	01	HV/Proto
28	-98.5	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
28	-95.5	10 MHz	QPSK	50	0	25	0	L/M/H	01	HV/Proto
28	-93.7	15 MHz	QPSK	75	0	25	0	L/M/H	01	HV/Proto
29	-97	5 MHz	QPSK	25	0	N/A	N/A	L/M/H	N/A	HV/Proto
29	-94	10 MHz	QPSK	50	0	N/A	N/A	L/M/H	N/A	HV/Proto
30	-99	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
30	-96	10 MHz	QPSK	50	0	25	0	L/M/H	01	HV/Proto

Note: The values in the table above have been derived from (3GPP TS 36.521-1 V12.5.0 (2015-03) Table 7.3.3-1: Reference sensitivity QPSK PREFSENS).

Band 13 Verizon Supplementary Test matrix (UL RB = 6)

Test condition	LTE BW	Modulation	RB DL number	RB DL offset	RB UL number	RB UL offset	EARFCN	Site
RSSI = -94 dBm	10 MHz	QPSK	6	0	6	0	mid	Proto
RSSI = -94 dBm	10 MHz	QPSK	6	22	6	0	mid	Proto
RSSI = -94 dBm	10 MHz	QPSK	6	44	6	0	mid	Proto

Verify the cumulative BLER value is within the limits specified in section 2.5 [LTE-FDD Test Limits](#).

7.2.9 LTE-FDD Rx RSSI

Signal Generator Setup:

Modulation	QPSK
Bandwidth	10 MHz
Frequency	Mid channel (from table 1.6.2)
Level RMS	See table below
RB Offset	0
#RBs	50

Downlink Test Power Levels

GNPO	Lead Site
-70	-70
-57	-57
-50	-50
-38	-38
-28	-28

Table 7.2.9

To measure RSSI, use the following command to read the downlink level reported by the phone.

Command	Opcode/data
FTM_LTE_NS_GET_DL_LEVEL	0x4B0B 1E00 FC01 0066 F57E

Get the downlink level from the command response field RX_LEVEL which is a signed 2 byte field in units of 1/10dBm. Multiply the value by 10 to get units of dBm.

Measure the RSSI for each downlink level defined in Table 7.2.9. Verify the values are within the limits specified in section 2.5 [LTE-FDD Test Limits](#).

7.2.10 LTE-FDD Tx Channel Power

Signal Analyzer Setup:

Modulation	QPSK
Bandwidth	10 MHz
Frequency	Test channel frequency (from table 1.6.2)
Level RMS	See table below
RB Offset	0
#RBs	12

Uplink Test Power Levels (dBm)

GNPO	Lead Site
-20	-20
5	5
20.5	20.5

Table 7.2.10

Configure uplink power

To force the phone to each uplink power level, Send the following command to override the open loop and closed loop power and explicitly configure the transmitter power in 1/10dBm units. Set the *Tx level* field in this command to each value in table 7.2.10 to drive the phone to each power level.

Command	Opcode/data
FTM_LTE_NS_CONF_UL_POWER	0x4B0B 1E00 FB01 0000 <Tx level 2 bytes><checksum 2 bytes>7E

Example:

Send 0x4B0B1E00FB010001CD00372B7E to configure uplink power to 20.5dBm.

Verify that the response status byte is 0 to indicate success of the command.

Measure the Tx channel power for each uplink level defined in Table 7.2.10. Verify the values are within the limits specified in section 2.5 [LTE-FDD Test Limits](#).

7.2.11 LTE-FDD Rx RSSI Across Band

Test condition	LTE BW	Modulation	RB number	RB offset	EARFCN
RSSI = -50dBm	10 MHz	QPSK	50	0	Low/High

To measure RSSI, use the following command to read the downlink level reported by the phone.

Command	Opcode/data
FTM_LTE_NS_GET_DL_LEVEL	0x4B0B 1E00 FC01 0066 F57E

Get the downlink level from the command response field RX_LEVEL which is a signed 2 byte field in units of 1/10dBm. Multiply the value by 10 to get units of dBm.

Measure the Rx receiver signal strength indicator (RSSI) for each supported band and channels defined in tables 7.2.1 - 7.2.6. Measure the RSSI for each downlink level defined in Table 7.2.9. Verify the values are within the limits specified in section 2.5 [LTE-FDD Test Limits](#).

7.2.12 LTE-FDD Tx Delta Current Drain

Measure UE supply current (TX ON) and calculate difference between TX ON current and TX OFF current (section 3.3) and verify the value is within the limits specified in section 2.5 [LTE-FDD Test Limits](#).

Test condition	LTE BW	Modulation	DL RB number	DL RB offset	UL RB number	UL RB offset	EARFCN
Max Power, MPR = 0dB	10 MHz	QPSK	50	0	12	0	Mid

Exit Test

Use the following command to stop LTE physical layer 1 to end the LTE non-signaling connection.

Command	Opcode/data
FTM_LTE_NS_STOP	0x4B0B 1E00 F801 3551 7E

Verify that the response status byte is 0 to indicate success of the command.

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8.0 LTE-TDD Test Procedures

The procedures in this section describe the methods to tune the LTE hardware control parameters and verify LTE-TDD functionality in the UE.

8.1 LTE-TDD RX, TX and HDET Phasing

Test Description

This APT characterization routine phases the receiver, transmitter and HDET (power detect circuit) by performing 3 sweeps. The sweeps consist of the following:

- Sweep 1 (Fixed Bias)
- Sweep 2 (APT Bias)
- Sweep 3 (RX, TX, & HDET Freq Sweep)

A full description of the ESC V4 Sweeps can be found in the ESC V4 Software Design Guide; QC Document ID: 80-NG201-2.

Please note, all documents referred to in this section can be downloaded from <http://support.cdmatech.com> by searching for the document #.

Test Setup

For all products in this section phasing should be run at 3.9V.

Test Procedure

Pre-sweep

Use the following FTM command to select the LTE-TDD Mode for the desired band.

Command	Data
FTM_SET_MODE	Mode (Select LTE-TDD band)

Use the following FTM command to disable the secondary path.

Command	Data
SET_SECONDARY_PATH	0x00 (Disable Secondary Path)

Use the following FTM command to tune the UE to the UL Reference Channel

Command	Data
FTM_SET_CHAN *	UL Ref Channel (Set UL channel)

Command	Data
FTM_SET_TX_ON	N/A (Set TX ON)

For each PA range, starting with the lowest and ending with the highest:

Use the FTM_SET_PA_RANGE command to set it to PA_Rx.

Read the following items out of the Default file:

MOTCONF_LTE_TDD_C0_B(band*)_TX_RISE_EXTENSION – list of 3 dB10 items
MOTCONF_LTE_TDD_C0_B(band*)_TX_FALL_EXTENSION – list of 3 dB10 items
MOTCONF_LTE_TDD_C0_B(band*)_TX_HIGH_RANGE_MAX_POWER – 1 dB10 item
MOTCONF_LTE_TDD_C0_B(band*)_TX_LOW_RANGE_MIN_POWER – 1 dB10 item
* band variable above is one of the following: Band 34, 38, 39, 40, 41:

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Use these values to choose a start and stop TX_GAIN_INDEX such that the sweep will cover a range of powers which extents:

PA RANGE	POWER for Start Point	Power for Stop Point
Highest PA Range	HIGH_RANGE_MAX_POWER/10	High Range Fall Switchpoint - TX_FALL_EXTENSION[0]/10
Lowest PA Range	Low Range Rise Switchpoint + TX_RISE_EXTENSION[max range]/10	LOW_RANGE_MIN_POWER/10
All other PA Ranges	Range Rise Switchpoint + TX_RISE_EXTENSION[pa range index]/10	Range Fall Switchpoint - TX_FALL_EXTENSION[pa range index]/10

Table 8.1.1

Start at the lowest PA GAIN RANGE, find TX_GAIN_INDEX that gives you a power equal or below **“Power for Stop Point”** in table above for the associated PA Range. Call this “STOP_TX_GAIN_INDEX”.

Start at the lowest PA GAIN RANGE, find TX_GAIN_INDEX that gives you a power equal or above **“Power for Start Point”** in table above for the associated PA Range. Call this “START_TX_GAIN_INDEX”.

Note: Running average start/stop TX gain indexes can be used instead of static value to help optimize test time in production.

V4 Calibration

Sweep 1 (Fixed Bias)

This sweep will use a fixed PA bias (defined by the <Bias_List> parameter in the ESC_Param.xml file) to sweep from a starting RGI down to a stopping RGI measuring power at each step on reference channel. Verify that the start & stop RGIs will be sufficiently above/below the rise/fall switch points for each PA range before starting first sweep.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for sweep 1 calibration. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file for sweep type 0. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23

Arm the Test Set for initial TX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string of parameter list obtained from the ESC_Param.xml file.

Wait for test set to complete sweep operation.

Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured power values for each PA range from the test set.

There are 3 deliverables coming from this sweep:

- Rgi vs Power with fixed bias for TX_MULTI_LIN_V3 NV item
- Reference RGI vs Power levels for calculations on 2nd sweep
- HDET vs Power for high PA gain.

Save deliverables for processing at later points of the procedure.

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Sweep 2 (APT Bias)

This sweep will characterize the TX output power vs RGI using a variable Bias on each step calculated from the first sweep. This sweep is also done only on 1 reference channel. This sweep is identical to the first sweep except that the SMPS PA Bias for each step is calculated from the first sweep and HDET is not measured in this sweep.

The SMPS PA Bias is calculated by using the Power level for each RGI in the first sweep and looking that value up in the APT Characterization.xml file. This value is characterized by development to be a bias level that allows the PA to pass ACLR specs. Typically the maximum powers on a given band are close to max PA bias, and reduce as the rgi/power diminishes.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for sweep 2 calibration. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file for sweep type 1. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23. Note that the biases will come from the APT Characterization.xml file for this construction.

Arm the Test Set for second TX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string previously constructed for this sweep.

Wait for test set to complete sweep operation.

Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured power values from the test set.

There are 2 deliverables that come from this sweep:

- Rgi vs Power with APT Bias for TX_MULTI_LIN_V3 NV item
- RGI vs Power data used in sweep 3 for TX Linearizer

Verify that the maximum and minimum power measurements for each PA State exceed the respective switch points.

Save deliverables for processing at later points of the procedure.

Sweep 3 (RX, TX, & HDET Freq Sweep)

This third APT (Average Power Tracking) sweep will take measurements across the band (typically 16 channels) in order to determine TX, HDET and RX versus frequency for the phone. For each channel, there will be (5 + number of PA states + number of LNA states) steps. Typically each step can do both RX and TX in parallel (unless the signal is TD (LTE-TDD or TD-SCDMA) in which case the TX measurements are all done first and then the RX measurements are done, effectively requiring (5 + number of PA states + number of LNA states) steps. Table 8.1.2 below is a visual representation of a single channel in this sweep.

Channel Sweep Steps	
TX Measurement	HDET (Target power + 2 RGI)
	HDET (Target power + 1 RGI)
	HDET (@Target power)
	HDET (Target power - 1 RGI)
	HDET (Target power - 2 RGI)
	TX Linearizer (typically at high PA state's fall SP power level)
	TX Linearizer (typically at mid PA state's rise SP power level)
	TX Linearizer (typically at low PA state's rise SP power level)
RX Measurement	LNA 0 or VGA reading
	LNA 1
	LNA 2
	LNA 3
	LNA 4 (LTE only else do nothing)
	LNA 5 (LTE only else do nothing)

Table 8.1.2

Secondary Chain calibration is done by default in parallel with primary chain. This requires the test bench to use a splitter across the primary/diversity paths. It is possible to disable the secondary chain calibration if diversity path cannot be connected in parallel. If secondary chain calibration is not done in parallel, it cannot be performed using a ESC V4 sweep.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for RX, TX, & HDET Freq Sweep. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23

Arm the Test Set for TX/RX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string previously constructed in this sweep.

Wait for test set to complete sweep operation.

Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured TX Freq Comp values from the test set.

There are 3 deliverables coming from this sweep:

- RX LNA (or DVGA) readings on 16 channels at each LNA state for NV calculation for both RX chains.
- HDET vs Freq readings for NV calculation
- TX Linearizer vs Freq (APT Mode) for NV calculation

Process the retrieved data values gathered during the sweep. Verify that power measurements are close to expected from reference channel. Also check RX measurements for any outliers.

NV Calculation

For TX Linearizer Data

Store Rgi vs Power with fixed bias data from sweep 1, Store Rgi vs Power with APT bias data from sweep 2, and TX Linearizer vs Freq data from sweep 3 into the appropriate area of NV_<signal>_<band>_TX_MULTI_LIN_V3_DATA_I. Refer to QC Document ID: 80- N5420-170 section 4.3.2.1.

For HDET Data

Store HDET vs Level data from sweep 1 into the appropriate area of NV_<signal>_<band>_EXP_HDET_VS_AGC_V2. Store HDET vs Freq data from sweep 3 into the appropriate area of NV_<signal>_<band>_TX_LIMIT_VS_FREQ. Refer to QC Document ID: 80-N1622-62

The HDET Power targets phased in this procedure are listed in Table 8.1.3 below for reference

Parameter	All Bands (non-LPM)	LPM
All Products		
HDET		
HDET_Trgt_15 (dBm)	30.8	20.9
HDET_Trgt_14 (dBm)	29.2	19.3
HDET_Trgt_13 (dBm)	27.6	17.7
HDET_Trgt_12 (dBm)	26	16.1
HDET_Trgt_11 (dBm)	24.4	14.5
HDET_Trgt_10 (dBm)	22.8	12.9
HDET_Trgt_9 (dBm)	21.2	11.3
HDET_Trgt_8 (dBm)	19.6	9.7
HDET_Trgt_7 (dBm)	18	8.1
HDET_Trgt_6 (dBm)	16.4	6.5
HDET_Trgt_5 (dBm)	14.8	4.9
HDET_Trgt_4 (dBm)	13.2	3.3
HDET_Trgt_3 (dBm)	11.6	1.7
HDET_Trgt_2 (dBm)	10	0.1
HDET_Trgt_1 (dBm)	8.4	-1.5
HDET_Trgt_0 (dBm)	6.8	-3.1

Table 8.1.3

For RX Data

Store the RX Calibration data and RX Channel list from sweep 3 into the "RFNV_LTE_<BAND>_RX_CAL_DATA_V2_I". Refer to document 80-NH377-170 H for structure and formatting.

Exit Test



8.2 LTE-TDD Non-signaling Parametric Tests

Test Description

The purpose of this test is to verify the LTE-TDD Tx waveform and spectrum quality and to verify LTE-TDD Rx quality.

Test Dependencies:

3.2 Turn On

8.1 LTE-TDD RX, TX and HDET Phasing

Equipment:

RF Signal generator with ARB playback capability.

RF Signal analyzer with LTE-TDD modulation analysis capability.

TX Test Channels for 10MHz BW products

Band	All Products	All Products
	GNPO (L / M / H)	Lead Site (L / M / H)
34	36250, 36275, 36300	36250, 36275, 36300
38	37800, 38000, 38200	37800, 38000, 38200
39	38300, 38450, 38600	38300, 38450, 38600
40	38700, 39150, 39600	38700, 39150, 39600
41(Full/US)	39700, 40620, 41540	39700, 40620, 41540
41 (aXGP)	40190, 40690, 41090	40190, 40690, 41090
41 (PRC)	40290, 40740, 41190	40290, 40740, 41190

Table 8.2.1

Test Setup

Signal Generator Setup:

Signal Type	LTE_TDD
Modulation	ARB
Band	See table 8.2.1
Channel	See table 8.2.1
Frequency	Test channel frequency (from table 1.6.2)
DL Power Level	-65dBm
Baseband Mode	ARB
ARB Repetition	Continuous
Trigger	Autostart / Re-trigger

Signal Analyzer Setup:

Modulation	QPSK
Bandwidth	See table 8.2.1
Channel	See table 8.2.1
Frequency	Test channel frequency (from table 1.6.2)
Expected Pwr	24dBm

UE Setup:

Use the ChangeMode command to put the phone into FTM mode.

Command	Opcode/data
---------	-------------

CHANGEMODE	0x29 0300	(Enter FTM mode)
------------	-----------	------------------

TX Test Channels for 5MHz BW products

Band	All Products	All Products
	GNPO (L / M / H)	Lead Site (L / M / H)
34	36225, 36275, 36325	36225, 36275, 36325
38	37775, 38000, 38225	37775, 38000, 38225
39	38275, 38450, 38625	38275, 38450, 38625
40	38675, 39150, 39625	38675, 39150, 39625
41(Full/US)	39675, 40620, 41565	39675, 40620, 41565
41(aXGP)	40165, 40690, 41115	40165, 40690, 41115
41(PRC)	40265, 40740, 41215	40265, 40740, 41215

Table 8.2.2

TX Test Channels for 15MHz BW products

Band	All Products	All Products
	GNPO (L / M / H)	Lead Site (L / M / H)
34	36275, 36275, 36275	36275, 36275, 36275
38	37825, 38000, 38175	37825, 38000, 38175
39	38325, 38450, 38575	38325, 38450, 38575
40	38725, 39150, 39575	38725, 39150, 39575
41(Full/US)	39725, 40620, 41515	39725, 40620, 41515
41(aXGP)	40215, 40690, 41065	40215, 40690, 41065
41(PRC)	40315, 40740, 41165	40315, 40740, 41165

Table 8.2.3

**Test Procedure****8.2.1 LTE-TDD Non-signaling Connection**

Setup RF signal generator to continuously execute ARB on the appropriate band and channel (refer to Test Setup section above). Use the following downlink vector ARB file: 'LTE_ProductionTest_TDD_1_OCNG_CRNT1100.wv' for R&S CMW500 and 'LTETDD_downlink_BW10MHz_50RB_ALLUP_QC.wfm' for Keysight EXM and EXT (converted from virtual high level ARB file name 'LTETDD_downlink_BW10MHz_50RB_ALLUP_QC').

Set the downlink level to -65dBm.

Start LTE-TDD physical layer 1

Send the following command to start LTE physical layer 1 in non-signaling mode.

Command	Opcode/data
FTM_LTE_NS_START	0x4B0B 1E00 F401 95F8 7E

Verify that the response status byte is 0 to indicate success of the command.

Acquiring LTE-TDD cell downlink

Send the following command to request LTE layer 1 to acquire the PSS, SSS, and MIB contained in the PBCH.

Command	Opcode/data
FTM_LTE_NS_ACQ	0x4B0B 1E00 F501<band 1 byte><Rx channel 2 bytes><checksum 2 bytes>7E

Verify that the response status byte is 0 and response RF band and downlink channel match expected band and downlink channel to indicate success of the command.

Establishing LTE-TDD downlink/uplink data paths

Send the following command to establish uplink/downlink data paths.

Command	Opcode/data
FTM_LTE_NS_START_DP	0x4B0B 1E00 F601 0000 0000 6400 011E 0102 EDFC 7E

Verify that the response status byte is 0 to indicate success of the command.

Check LTE-TDD connection state

Send the following command to verify that the LTE connection has been established.

Command	Opcode/data
FTM_LTE_IS_CONN	0x4B0B 1E00 F901 ED48 7E

Verify that the response status byte is 0 to indicate the phone is in a connection state.

Configure uplink waveform

Send the following command to configure the active resource blocks (Start RB Index and Number of RBs) and modulation type for the LTE uplink waveform.

Command	Opcode/data
FTM_LTE_NS_CONFIG_UL_WAVEFORM	0x4B0B 1E00 FA01 0100 0032 00B2 4D7E

Verify that the response status byte is 0 to indicate success of the command.

Configure uplink power

Send the following command to override the open loop and closed loop power and explicitly configure the transmitter power in 1/10dBm units. Set the Tx level field in this command to 24dBm to drive the phone to this power level.

Command	Opcode/data
FTM_LTE_NS_CONF_UL_POWER	0x4B0B 1E00 FB01 0000 <Tx level 2 bytes><checksum 2 bytes>7E

Example:

Send 0x4B0B1E00FB010000F00031777E to configure uplink power to 24dBm.

Verify that the response status byte is 0 to indicate success of the command.

8.2.2 LTE-TDD Tx Current

Measure the total current drain from the phone **integrated over 1 LTE-TDD uplink sub-frame (1msec) only**. This requires a pulse current measurement which triggers on the rise of the uplink pulse so that the current measurement only captures when the uplink is on. verify the value is within the limits specified in section 2.6 [LTE-TDD Test Limits](#).

8.2.3 LTE-TDD TX Max Power

Measure the TX power and verify the value is within the limits specified in section 2.6 [LTE-TDD Test Limits](#).

Test condition	LTE BW	Modulation	UL RB number	UL RB offset	EARFCN
Max Power, MPR = 0dB	5 MHz	QPSK	1	0	low/mid
Max Power, MPR = 0dB	5 MHz	QPSK	1	24	high
Max Power, MPR = 0dB	10 MHz	QPSK	12	0	low/mid
Max Power, MPR = 0dB	10 MHz	QPSK	12	37	high
Max Power, MPR = 0dB	15 MHz	QPSK	1	0	low/mid
Max Power, MPR = 0dB	15 MHz	QPSK	1	74	high

8.2.4 LTE-TDD TX Frequency Error

Measure the TX frequency error and verify the value is within the limits specified in section 2.6 [LTE-TDD Test Limits](#).

Test condition	LTE BW	UL/DL Modulation	DL RB Number	UL RB Number	UL/DL RB offset	EARFCN
Max Power, MPR Active	10MHz	QPSK	50	50	0	mid

8.2.5 LTE-TDD TX EVM

Measure the Tx EVM and verify the value is within the limits specified in section 2.6 [LTE-TDD Test Limits](#).

Test condition	LTE BW	Modulation	UL RB number	UL RB offset	EARFCN	Site
Max Power, MPR Active	5 MHz	QPSK	25	0	mid	High volume
Max Power, MPR Active	5 MHz	16QAM	25	0	mid	High volume
Max Power, MPR = 0dB	5 MHz	QPSK	8	0	mid	Proto
Max Power, MPR Active	10 MHz	QPSK	50	0	mid	High volume
Max Power, MPR Active	10 MHz	16QAM	50	0	mid	High volume
Max Power, MPR = 0dB	10 MHz	QPSK	12	0	mid	Proto
Max Power, MPR Active	15 MHz	QPSK	75	0	mid	High volume
Max Power, MPR Active	15 MHz	16QAM	75	0	mid	High volume
Max Power, MPR = 0dB	15 MHz	QPSK	16	0	mid	Proto

8.2.6 LTE-TDD TX I/Q Origin Offset

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Measure the Tx I/Q origin offset and verify the value is within the limits specified in section 2.6 [LTE-TDD Test Limits](#).

Test condition	LTE BW	Modulation	RB number	RB offset	EARFCN	Site
Max Power, MPR = 0dB	5 MHz	QPSK	8	0	mid	Proto
Max Power, MPR = 0dB	5 MHz	QPSK	8	16	mid	Proto
Max Power, MPR = 0dB	10 MHz	QPSK	12	0	mid	Proto
Max Power, MPR = 0dB	10 MHz	QPSK	12	37	mid	Proto
Max Power, MPR = 0dB	15 MHz	QPSK	16	0	mid	Proto
Max Power, MPR = 0dB	15 MHz	QPSK	16	58	mid	Proto

8.2.7 LTE-TDD TX ACLR

Measure the TX adjacent channel leakage ratio (ACLR) and verify the value is within the limits specified in section 2.6 [LTE-TDD Test Limits](#).

Test condition	LTE BW	Modulation	RB number	RB offset	EARFCN
Max Power, MPR = 0dB	5 MHz	QPSK	8	0	low/mid/high
Max Power, MPR = 0dB	5 MHz	QPSK	8	16	low/mid/high
Max Power, MPR Active	5 MHz	QPSK	25	0	low/mid/high
Max Power, MPR = 0dB	10 MHz	QPSK	12	0	low/mid
Max Power, MPR = 0dB	10 MHz	QPSK	12	37	high
Max Power, MPR Active	10 MHz	QPSK	50	0	low/mid/high
Max Power, MPR = 0dB	15 MHz	QPSK	16	0	low/mid/high
Max Power, MPR = 0dB	15 MHz	QPSK	16	58	low/mid/high
Max Power, MPR Active	15 MHz	QPSK	75	0	low/mid/high

8.2.8 LTE-TDD RX BLER

Set the downlink level to -97dBm.

Use the following command to reset the BLER reporting in the phone.

<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_NS_RESET_DL_BLER_REPORT	0x4B0B 1E00 FD01 8D2F 7E

Use the following command to read the BLER report from the phone.

<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_NS_GET_DL_BLER_REPORT	0x4B0B 1E00 FE01 E505 7E

Parse out the response fields TOTAL_BLOCKS_RECEIVED and TOTAL_BLOCK_ERRORS. Add these values to cumulative total values as:

CUMULATIVE_TOTAL_BLOCKS_RECEIVED = CUMULATIVE_TOTAL_BLOCKS_RECEIVED +
TOTAL_BLOCKS_RECEIVED

CUMULATIVE_TOTAL_BLOCK_ERRORS = CUMULATIVE_TOTAL_BLOCK_ERRORS + TOTAL_BLOCK_ERRORS.

Calculate cumulative BLER with the equation $BLER = (CUMULATIVE_TOTAL_BLOCK_ERRORS / CUMULATIVE_TOTAL_BLOCKS_RECEIVED) * 100\%$.

Keep reading BLER until 95% confidence level is met or have read 10000 blocks.

High Volume Test matrix

Band	Full CELL DL Power	LTE BW	Modulation (UL and DL)	RB DL number	RB DL offset	RB UL number	RB UL offset	EARFCN	UL NS	Site
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Band	Full CELL DL Power	LTE BW	Modulation (UL and DL)	RB DL number	RB DL offset	RB UL number	RB UL offset	EARFCN	UL NS	Site
34	-100	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
34	-97	10 MHz	QPSK	50	0	50	0	L/M/H	01	HV/Proto
34	-95.2	15 MHz	QPSK	75	0	75	0	L/M/H	01	HV/Proto
38	-100	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
38	-97	10 MHz	QPSK	50	0	50	0	L/M/H	01	HV/Proto
38	-95.2	15 MHz	QPSK	75	0	75	0	L/M/H	01	HV/Proto
39	-100	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
39	-97	10 MHz	QPSK	50	0	50	0	L/M/H	01	HV/Proto
39	-95.2	15 MHz	QPSK	75	0	75	0	L/M/H	01	HV/Proto
40	-100	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
40	-97	10 MHz	QPSK	50	0	50	0	L/M/H	01	HV/Proto
40	-95.2	15 MHz	QPSK	75	0	75	0	L/M/H	01	HV/Proto
41	-98	5 MHz	QPSK	25	0	25	0	L/M/H	01	HV/Proto
41	-95	10 MHz	QPSK	50	0	50	0	L/M/H	01	HV/Proto
41	-93.2	15 MHz	QPSK	75	0	75	0	L/M/H	01	HV/Proto

Note: The values in the table above have been derived from (3GPP TS 36.521-1 V10.1.0 (2012-03) Table 7.3.3-1: Reference sensitivity QPSK PREFSENS).

Band 13 Verizon Supplementary Test matrix (UL RB = 6)

Test condition	LTE BW	Modulation	RB DL number	RB DL offset	RB UL number	RB UL offset	EARFCN	Site
RSSI = -94 dBm	10 MHz	QPSK	6	0	6	0	mid	Proto
RSSI = -94 dBm	10 MHz	QPSK	6	22	6	0	mid	Proto
RSSI = -94 dBm	10 MHz	QPSK	6	44	6	0	mid	Proto

Verify the cumulative BLER value is within the limits specified in section 2.6 [LTE-TDD Test Limits](#).

8.2.9 LTE-TDD Rx RSSI
Signal Generator Setup:

Modulation	QPSK
Bandwidth	10 MHz
Frequency	Mid channel (from table 1.6.2)
Level RMS	See table below
RB Offset	0
#RBs	50

Downlink Test Power Levels

GNPO	Lead Site
-70	-70
-57	-57
-50	-50
-38	-38
-28	-28

Table 8.2.9

To measure RSSI, use the following command to read the downlink level reported by the phone.

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<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_NS_GET_DL_LEVEL	0x4B0B 1E00 FC01 0066 F57E

Get the downlink level from the command response field RX_LEVEL which is a signed 2 byte field in units of 1/10dBm. Multiply the value by 10 to get units of dBm.

Measure the RSSI for each downlink level defined in Table 8.2.9. Verify the values are within the limits specified in section 2.6 [LTE-TDD Test Limits](#).

8.2.10 LTE-TDD Tx Channel Power

Signal Analyzer Setup:

Modulation	QPSK
Bandwidth	10 MHz
Frequency	Test channel frequency (from table 1.6.2)
Level RMS	See table below
RB Offset	0
#RBs	12

Uplink Test Power Levels (dBm)

GNPO	Lead Site
-20	-20
5	5
20.5	20.5

Table 8.2.10

Configure uplink power

To force the phone to each uplink power level, Send the following command to override the open loop and closed loop power and explicitly configure the transmitter power in 1/10dBm units. Set the *Tx level* field in this command to each value in table 8.2.10 to drive the phone to each power level.

<u>Command</u>	<u>Opcode/data</u>
FTM_LTE_NS_CONF_UL_POWER	0x4B0B 1E00 FB01 0000 <Tx level 2 bytes><checksum 2 bytes>7E

Example:

Send 0x4B0B1E00FB010001CD00372B7E to configure uplink power to 20.5dBm.

Verify that the response status byte is 0 to indicate success of the command.

Measure the Tx channel power for each uplink level defined in Table 8.2.10. Verify the values are within the limits specified in section 2.6 [LTE-TDD Test Limits](#).

8.2.11 LTE-TDD Rx RSSI Across Band

Test condition	LTE BW	Modulation	RB number	RB offset	EARFCN
RSSI = -50dBm	10 MHz	QPSK	50	0	Low/High

To measure RSSI, use the following command to read the downlink level reported by the phone.

<u>Command</u>	<u>Opcode/data</u>
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FTM_LTE_NS_GET_DL_LEVEL

0x4B0B 1E00 FC01 0066 F57E

Get the downlink level from the command response field RX_LEVEL which is a signed 2 byte field in units of 1/10dBm. Multiply the value by 10 to get units of dBm.

Measure the Rx receiver signal strength indicator (RSSI) for each supported band and channels defined in tables 7.2.1 - 7.2.6. Measure the RSSI for each downlink level defined in Table 8.2.9. Verify the values are within the limits specified in section 2.6 [LTE-TDD Test Limits](#).

8.2.12 LTE-TDD Tx Delta Current Drain

Measure UE supply current (TX ON) and calculate difference between TX ON current and TX OFF current (section 3.3) and verify the value is within the limits specified in section 2.6 [LTE-TDD Test Limits](#).

Test condition	LTE BW	Modulation	DL RB number	DL RB offset	UL RB number	UL RB offset	EARFCN
Max Power, MPR = 0dB	10 MHz	QPSK	50	0	12	0	Mid

Exit Test

Use the following command to stop LTE physical layer 1 to end the LTE non-signaling connection.

Command	Opcode/data
FTM_LTE_NS_STOP	0x4B0B 1E00 F801 3551 7E

Verify that the response status byte is 0 to indicate success of the command.

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9.0 TD-SCDMA Test Procedures

The procedures in this section describe the methods to tune the TD-SCDMA hardware control parameters and verify TD-SCDMA functionality in the UE.

9.1 TD-SCDMA RX, TX and HDET Phasing

Test Description

This APT characterization routine phases the receiver, transmitter and HDET (power detect circuit) by performing 3 sweeps. The sweeps consist of the following:

- Sweep 1 (Fixed Bias)
- Sweep 2 (APT Bias)
- Sweep 3 (RX, TX, & HDET Freq Sweep)

A full description of the ESC V4 Sweeps can be found in the ESC V4 Software Design Guide; QC Document ID: 80-NG201-2.

Please note, all documents referred to in this section can be downloaded from <http://support.cdmatech.com> by searching for the document #.

Test Setup

For all products in this section phasing should be run at 3.9V.

Test Procedure

Use the following FTM command to select the WCDMA Mode for the desired band.

Command	Data
FTM_SET_MODE	Mode (Select TD-SCDMA band)

Where:

TD-SCDMA BAND: 34, 39

Use the following FTM command to disable the secondary path.

Command	Data
SET_SECONDARY_PATH	0x00 (Disable Secondary Path)

Use the following FTM command to tune the UE to the UL Reference Channel

Command	Data
FTM_SET_CHAN *	UL Ref Channel (Set UL channel)

Command	Data
FTM_SET_TX_ON	N/A (Set TX ON)

For each PA range, starting with the lowest and ending with the highest:

Use the FTM_SET_PA_RANGE command to set it to PA_Rx.

Read the following items out of the Default file:

MOTCONF_TD_SCDMA_C0_(band)_TX_RISE_EXTENSION – list of 3 dB10 items
MOTCONF_TD_SCDMA_C0_(band)_TX_FALL_EXTENSION – list of 3 dB10 items
MOTCONF_TD_SCDMA_C0_(band)_TX_HIGH_RANGE_MAX_POWER – 1 dB10 item
MOTCONF_TD_SCDMA_C0_(band)_TX_LOW_RANGE_MIN_POWER – 1 dB10 item

*(band) variable above is one of the following: Band 34, 39:

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Use these values to choose a start and stop TX_GAIN_INDEX such that the sweep will cover a range of powers which extents:

PA RANGE	POWER for Start Point	Power for Stop Point
Highest PA Range	HIGH_RANGE_MAX_POWER/10	High Range Fall Switchpoint - TX_FALL_EXTENSION[0]/10
Lowest PA Range	Low Range Rise Switchpoint + TX_RISE_EXTENSION[max range]/10	LOW_RANGE_MIN_POWER/10
All other PA Ranges	Range Rise Switchpoint + TX_RISE_EXTENSION[pa range index]/10	Range Fall Switchpoint - TX_FALL_EXTENSION[pa range index]/10

Table 9.1.1

Start at the lowest PA GAIN RANGE, find TX_GAIN_INDEX that gives you a power equal or below "Power for Stop Point" in table above for the associated PA Range. Call this "STOP_TX_GAIN_INDEX".

Start at the lowest PA GAIN RANGE, find TX_GAIN_INDEX that gives you a power equal or above "Power for Start Point" in table above for the associated PA Range. Call this "START_TX_GAIN_INDEX".

Note: Running average start/stop TX gain indexes can be used instead of static value to help optimize test time in production.

V4 Calibration

Sweep 1 (Fixed Bias)

This sweep will use a fixed PA bias (defined by the <Bias_List> parameter in the ESC_Param.xml file) to sweep from a starting RGI down to a stopping RGI measuring power at each step on reference channel. Verify that the start & stop RGIs will be sufficiently above/below the rise/fall switch points for each PA range before starting first sweep.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for sweep 1 calibration. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file for sweep type 0. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23

Arm the Test Set for initial TX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string of parameter list obtained from the ESC_Param.xml file.

Wait for test set to complete sweep operation.

Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured power values for each PA range from the test set.

There are 3 deliverables coming from this sweep:

- Rgi vs Power with fixed bias for TX_MULTI_LIN_V3 NV item
- Reference RGI vs Power levels for calculations on 2nd sweep
- HDET vs Power for high PA gain.

Save deliverables for processing at later points of the procedure.

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Sweep 2 (APT Bias)

This sweep will characterize the TX output power vs RGI using a variable Bias on each step calculated from the first sweep. This sweep is also done only on 1 reference channel. This sweep is identical to the first sweep except that the SMPS PA Bias for each step is calculated from the first sweep and HDET is not measured in this sweep.

The SMPS PA Bias is calculated by using the Power level for each RGI in the first sweep and looking that value up in the APT Characterization.xml file. This value is characterized by development to be a bias level that allows the PA to pass ACLR specs. Typically the maximum powers on a given band are close to max PA bias, and reduce as the rgi/power diminishes.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for sweep 2 calibration. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file for sweep type 1. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23. Note that the biases will come from the APT Characterization.xml file for this construction.

Arm the Test Set for second TX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string previously constructed for this sweep.

Wait for test set to complete sweep operation.

Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured power values from the test set.

There are 2 deliverables that come from this sweep:

- Rgi vs Power with APT Bias for TX_MULTI_LIN_V3 NV item
- RGI vs Power data used in sweep 3 for TX Linearizer

Verify that the maximum and minimum power measurements for each PA State exceed the respective switch points.

Save deliverables for processing at later points of the procedure.

Sweep 3 (RX, TX, & HDET Freq Sweep)

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This third APT (Average Power Tracking) sweep will take measurements across the band (typically 16 channels) in order to determine TX, HDET and RX versus frequency for the phone. For each channel, there will be (5 + number of PA states) steps. Typically each step can do both RX and TX in parallel (unless the signal is TD (LTE-TDD or TD-SCDMA) in which case the TX measurements are all done first and then the RX measurements are done, effectively requiring (5 + number of PA states + number of LNA states) steps. Table 9.1.2 below is a visual representation of a single channel in this sweep.

Channel Sweep Steps	
TX Measurement	HDET (Target power + 2 RGI)
	HDET (Target power + 1 RGI)
	HDET (@Target power)
	HDET (Target power - 1 RGI)
	HDET (Target power - 2 RGI)
	TX Linearizer (typically at high PA state's fall SP power level)
	TX Linearizer (typically at mid PA state's rise SP power level)
	TX Linearizer (typically at low PA state's rise SP power level)
RX Measurement	LNA 0 or VGA reading
	LNA 1
	LNA 2
	LNA 3
	LNA 4 (LTE only else do nothing)
	LNA 5 (LTE only else do nothing)

Table 9.1.2

Secondary Chain calibration is done by default in parallel with primary chain. This requires the test bench to use a splitter across the primary/diversity paths. It is possible to disable the secondary chain calibration, but it is not possible to do secondary chain calibration independently.

Read the ESC_Param.xml file to obtain all of the variables (i.e. tokens, settings) needed for RX, TX, & HDET Freq Sweep. Refer to QC Document ID: 80-NG201-2 sections 4.2.1 – 4.2.7 for ESC V4 Cal input files parameters description.

Construct the FTM_CFG_AND_EXEC_CAL_SWEEP test command using the parameter list obtained from the ESC_Param.xml file. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.23

Arm the Test Set for TX/RX sweep

Send the FTM_CFG_AND_EXEC_CAL_SWEEP test command with data string previously constructed in this sweep.

Wait for test set to complete sweep operation.

Send the FTM_GET_CAL_SWEEP_RESULTS command to provide the calibration sweep results. Refer to QC Document ID: 80-VA888-1 rev. T section 2.1.24.

Also retrieve the full set of measured TX Freq Comp values from the test set.

There are 3 deliverables coming from this sweep:

- RX LNA (or DVGA) readings on 16 channels at each LNA state for NV calculation for both RX chains.
- HDET vs Freq readings for NV calculation
- TX Linearizer vs Freq (APT Mode) for NV calculation

Process the retrieved data values gathered during the sweep. Verify that power measurements are close to expected from reference channel. Also check RX measurements for any outliers.

NV Calculation

For TX Linearizer Data

Store Rgi vs Power with fixed bias data from sweep 1, Store Rgi vs Power with APT bias data from sweep 2, and TX Linearizer vs Freq data from sweep 3 into the appropriate area of NV_<signal>_<band>_TX_MULTI_LIN_V3_DATA_I. Refer to QC Document ID: 80- N5420-170 section 4.3.2.1.

For HDET Data

Store HDET vs Level data from sweep 1 into the appropriate area of NV_<signal>_<band>_EXP_HDET_VS_AGC_V2. Store HDET vs Freq data from sweep 3 into the appropriate area of NV_<signal>_<band>_TX_LIMIT_VS_FREQ. Refer to QC Document ID: 80-N1622-62

The HDET Power targets phased in this procedure are listed in Table 9.1.3 below for reference

Parameter	All Bands
All Products	
HDET	
HDET_Trgt_15 (dBm)	30.8
HDET_Trgt_14 (dBm)	29.2
HDET_Trgt_13 (dBm)	27.6
HDET_Trgt_12 (dBm)	26
HDET_Trgt_11 (dBm)	24.4
HDET_Trgt_10 (dBm)	22.8
HDET_Trgt_9 (dBm)	21.2
HDET_Trgt_8 (dBm)	19.6
HDET_Trgt_7 (dBm)	18
HDET_Trgt_6 (dBm)	16.4
HDET_Trgt_5 (dBm)	14.8
HDET_Trgt_4 (dBm)	13.2
HDET_Trgt_3 (dBm)	11.6
HDET_Trgt_2 (dBm)	10
HDET_Trgt_1 (dBm)	8.4
HDET_Trgt_0 (dBm)	6.8

Table 9.1.3

For RX Data

Store the RX Calibration data and RX Channel list from sweep 3 into the "RFNV_TD_SCDMA_<BAND>_RX_CAL_DATA_V2_I". Refer to document 80-NH377-170 H for structure and formatting.

Exit Test

9.2 TD-SCDMA Test Mode Call Test

Test Description

The purpose of this test is to verify that the UE is capable of establishing a TD-SCDMA non-signaling connection and all parametrics are within tolerance. The TD-SCDMA test mode call tests are not required if equivalent measurements are done in a normal TD_SCDMA call test.

Test Dependencies:

9.1 TD-SCDMA RX, TX and HDET Phasing

Equipment:

TD_SCDMA Communications Test Set

TD-SCDMA Test Channels and Frequencies

Band	Channel	Channel	Frequency
34	Low	10054	2010.8 MHz
	Mid	10088	2017.6 MHz
	High	10121	2024.2 MHz
39	Low	9404	1880.8 MHz
	Mid	9500	1900 MHz
	High	9596	1919.2 MHz

Table 9.2

Test Setup

Signal Type = TD_SCDMA

Modulation = ARB

Waveform = TD-SCDMA downlink 12.2kbps RMC non-signaling (test mode) call with power control pattern all up bits

Band = (see table 9.2)

Channel = (see table 9.2)

Frequency = (see table 9.2)

Downlink RF Power Level = -65 dBm

First Active Uplink Timeslot = TS2

Duplex Switch Point Timeslot = TS2

Expected Power = 24 dBm

Measurement Coordination = Synchronous

Test Procedure

9.2.1 TD-SCDMA Non-signaling Connection

Setup test set for connection (refer to Test Setup section above)

Start TD-SCDMA physical layer 1

Send the following FTM command to start TD-SCDMA physical layer 1 tasks:

<u>Command</u>	<u>Data</u>
FTM_TDSCDMA_START_MODE_REQ	0x4B0B2C0000000000000000

Send the following FTM command and poll for the mode confirmation event which is an unsolicited response:

<u>Command</u>	<u>Data</u>
FTM_TDSCDMA_BER_START_TDSCDMA_MODE_CNF	0x4B0B2C0008000000000000088130000

Verify that the response is 0x4B0B2C00021008000000000005B127E. This event confirms that the phone has completed all internal tasks to go into TD_SCDMA mode (i.e. start TD-SCDMA layer 1 tasks).

Acquire TD-SCDMA cell downlink

Send the following FTM command to start a cell search acquisition procedure.

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Verify that the response is 0x4B0B2C0002100800000000005B127E. This event confirms that the phone has successfully established a TD-SCDMA physical channel connection.

This completes the hard handoff to required TD_SCDMA_BAND and downlink channel.

Setup test set for measurement (refer to Test Setup section above)

Configure uplink power to uplink power control state 'Off' and level '24'...

Send the following FTM command to configure uplink to power level 24 and power control state to off:

Command	Data
FTM_TDSCDMA_BER_SET_UL_POWER	0x4B0B2C000900000000000000F000

Verify that the response matches the expected string. This completes the configuration of uplink power.

9.2.3 TD-SCDMA 12.2K Single-ended BER

Test Setup:

Signal Type = TD_SCDMA

Modulation = ARB

Waveform = TD-SCDMA downlink 12.2kbps RMC non-signaling (test mode) call with power control pattern all up bits

Band = (see table 9.2)

Channel = (see table 9.2)

Frequency = (see table 9.2)

Downlink RF Power Level = -108 dBm

First Active Uplink Timeslot = TS2

Duplex Switch Point Timeslot = TS2

Expected Power = 24 dBm

Measurement Coordination = Synchronous

Send the following FTM command to configure and start single-ended BER reporting.

Command	Data
FTM_SE_BER_CONFIGURE_REQUEST	0x4B0B1B00000000000000001030014001E00

Send the following FTM command to clear single-ended BER sync status:

Command	Data
FTM_RAW_DATA	0x4B0B1B00050000000000

Measure BER

Send the following FTM command to get single-ended BER sync status:

Command	Data
FTM_RAW_DATA	0x4B0B1B00060000000000

Continue to send command until a 95% confidence level is reached or the total measured bits is greater than or equal to 12200 bits.

Parse out the number of bits measured and number of bit errors. Keep a running total of the total bits measured and total bit errors. Calculate a running BER% which is the following equation: BER% = (total bit errors / total bits measured) x 100%

Quit measuring BER once a 95% confidence level is reached or the total measured bits is greater than or equal to 10000 bits. Verify the final BER% is within the limits specified in section 2.7 TD-SCDMA Test Limits

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9.2.4 TD-SCDMA TX Current

Test Setup:

Signal Type = TD_SCDMA

Modulation = ARB

Waveform = TD-SCDMA downlink 12.2kbps RMC non-signaling (test mode) call with power control pattern all up bits

Band = (see table 9.2)

Channel = (see table 9.2)

Frequency = (see table 9.2)

Downlink RF Power Level = -108 dBm

First Active Uplink Timeslot = TS2

Duplex Switch Point Timeslot = TS2

Expected Power = 24 dBm

Measurement Coordination = Synchronous

Measure the total current drain from the phone and verify the value is within the limits specified in section 2.7 TD-SCDMA Test Limits.

9.2.5 TD-SCDMA TX Max Power

Test Setup:

Signal Type = TD_SCDMA

Modulation = ARB

Waveform = TD-SCDMA downlink 12.2kbps RMC non-signaling (test mode) call with power control pattern all up bits

Band = (see table 9.2)

Channel = (see table 9.2)

Frequency = (see table 9.2)

Downlink RF Power Level = -108 dBm

First Active Uplink Timeslot = TS2

Duplex Switch Point Timeslot = TS2

Expected Power = 24 dBm

Measurement Coordination = Synchronous

Measure the Tx power and verify the value is within the limits specified in section 2.7 TD-SCDMA Test Limits.

9.2.6 TD-SDMA TX Frequency Error

Test Setup:

Signal Type = TD_SCDMA

Modulation = ARB

Waveform = TD-SCDMA downlink 12.2kbps RMC non-signaling (test mode) call with power control pattern all up bits

Band = (see table 9.2)

Channel = (see table 9.2)

Frequency = (see table 9.2)

Downlink RF Power Level = -108 dBm

First Active Uplink Timeslot = TS2

Duplex Switch Point Timeslot = TS2

Expected Power = 24 dBm

Measurement Coordination = Synchronous

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Measure the Tx frequency error and verify the value is within the limits specified in section 2.7 TD-SCDMA Test Limits

9.2.7 TD-SCDMA EVM

Test Setup:

Signal Type = TD_SCDMA

Modulation = ARB

Waveform = TD-SCDMA downlink 12.2kbps RMC non-signaling (test mode) call with power control pattern all up bits

Band = (see table 9.2)

Channel = (see table 9.2)

Frequency = (see table 9.2)

Downlink RF Power Level = -108 dBm

First Active Uplink Timeslot = TS2

Duplex Switch Point Timeslot = TS2

Expected Power = 24 dBm

Measurement Coordination = Synchronous

Measure the Tx EVM and verify the value is within the limits specified in section 2.7 TD-SCDMA Test Limits

9.2.8 TD-SCDMA Power vs Time

Test Setup:

Signal Type = TD_SCDMA

Modulation = ARB

Waveform = TD-SCDMA downlink 12.2kbps RMC non-signaling (test mode) call with power control pattern all up bits

Band = (see table 9.2)

Channel = Mid Channel (see table 9.2)

Frequency = Mid Channel (see table 9.2)

Downlink RF Power Level = -108 dBm

First Active Uplink Timeslot = TS2

Duplex Switch Point Timeslot = TS2

Expected Power = 24 dBm

Measurement Coordination = Synchronous

Measure the Power vs Time and verify the value is within the limits specified in section 2.7 TD-SCDMA Test Limits

9.2.9 TD-SCDMA ACLR

Test Setup:

Signal Type = TD_SCDMA

Modulation = ARB

Waveform = TD-SCDMA downlink 12.2kbps RMC non-signaling (test mode) call with power control pattern all up bits

Band = (see table 9.2)

Channel = (see table 9.2)

Frequency = (see table 9.2)

Downlink RF Power Level = -108 dBm

First Active Uplink Timeslot = TS2

Duplex Switch Point Timeslot = TS2

Expected Power = 24 dBm

Measurement Coordination = Synchronous

Measure the Tx adjacent channel leakage ratio (ACLR) and verify the value is within the limits specified in section 2.7 TD-SCDMA Test Limits

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9.2.10 TD-SCDMA Closed Loop Power

Test Setup:

Signal Type = TD_SCDMA

Modulation = ARB

Waveform = TD-SCDMA downlink 12.2kbps RMC non-signaling (test mode) call with power control pattern segments B and C

Band = (see table 9.2)

Channel = Mid Channel (see table 9.2)

Frequency = (see table 9.2)

Downlink RF Power Level = -80 dBm

First Active Uplink Timeslot = TS2

Duplex Switch Point Timeslot = TS2

Expected Power = 24 dBm

Measurement Coordination = Synchronous

Perform hard handoff to required TD-SCDMA band and downlink mid channel (see section 9.2.2).

Configure uplink power to uplink power control state 'On' and level '24'...

Send the following FTM command to configure uplink to power level 24 and power control state to on:

Command	Data
FTM_TDSCDMA_BER_SET_UL_POWER	0x4B0B2C00090000000000000001F000

Verify that this response matches the expected string. This completes the configuration of uplink power.

Measure the TD-SCDMA Closed Loop Power for segment B

Start power = 23 dBm

Stop power = -50 dBm

Measure the power in each active time slot for power control pattern segment B and verify the relative powers are within spec.

Measure the TD-SCDMA Closed Loop Power for segment C

Start power = -50 dBm

Stop power = 23 dBm

Measure the power in each active time slot for power control pattern segment C and verify the relative powers are within spec.

9.2.11 TD-SCDMA TX Open-loop Power and RX RSSI

Test Setup:

Signal Type = TD_SCDMA

Modulation = ARB

Waveform = TD-SCDMA downlink 12.2kbps RMC non-signaling (test mode) call with power control pattern all up bits

Band = (see table 9.2)

Channel = Mid Channel (see table 9.2)

Frequency = (see table 9.2)

Downlink RF Power Level = Refer to table 9.2.11

First Active Uplink Timeslot = TS2

Duplex Switch Point Timeslot = TS2

Expected Power = 24 dBm

Measurement Coordination = Synchronous

Open Loop Tx Pwr	Downlink RF Power Levels (dBm)	
	GNPO	Lead Site
20	-90	-90
10	-80	-80
0	-70	-70
-10	-60	-60
-20	-50	-50
-30	-40	-40
Table 9.2.11		

Perform hard handoff to required TD-SCDMA band and downlink channel if required (see section 9.2.2).

Configure uplink power to uplink power control state 'Off' and next level (refer to table Table 9.2.11)

Send the following FTM command to configure uplink to power level in table 9.2.11 and power control state to off:

Example of configuring uplink to power level 20:

Send the following FTM command to configure uplink to power level 20.

<u>Command</u>	<u>Data</u>
FTM_TDSCDMA_BER_SET_UL_POWER	0x4B0B2C000900000000000000C800

Verify that this response matches the expected string. This completes the configuration of uplink power.

Measure RSSI

Send the following FTM command to measure RX RSSI:

<u>Command</u>	<u>Data</u>
FTM_TDSCDMA_BER_RX_LEVEL_REQ	0x4B0B2C000A00000000000000

Verify the RSSI value is within limits specified in section 2.7 TD-SCDMA Test Limits

Measure the TD-SCDMA Open Loop Power

Verify the measured values are within the limits specified in section 2.7 TD-SCDMA Test Limits

Repeat for each level as indicated in the "Downlink RF Power Levels" table 9.2.11.

Exit Test



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10.0 RF Antenna Tests

10.1 Main Antenna Tests

10.1.1 RX Antenna Test

Test Description

The purpose of this test is to verify proper seating and electrical connection of the external antenna to the C0 internal paths and by extension the robustness of the internal path routing as well. The test measures radiated power through an RF-coupled antenna fixture.

Test Dependencies:

[3.2 Turn On](#)

[3.6.1 FTM Mode](#)

Equipment:

LTE / WCDMA signal Generator or CW Signal Generator

RX/TX Test Channels, Paths, and Modes

Model	Mode	Band	Path (determines Mode ID)	Mode ID (dec)	DL Test Channel Low	DL Test Channel Mid	DL Test Channel High	UL Test Channel Low	UL Test Channel Mid	UL Test Channel High
B,C,D	WCDMA	BC01	C0	1	--	--	10838	--	--	--
B,C	WCDMA	BC05	C0	1	--	--	--	4133	--	--
B,C,D	WCDMA	BC08	C0	1	--	--	3088	--	--	--
B,C,D	LTE	BC03	C0	29	--	--	--	19250	--	--
B,C	LTE	BC07	C0	29	--	--	--	20800	--	--
B,C,D	LTE	BC07	C0	29	--	--	3400	--	--	--
C	LTE	BC12	C0	29	--	--	--	23060	--	--
B	LTE	BC13	C0	29	5184	--	--	--	--	23255
D	LTE	BC26	C0	29	--	--	9039	26740	--	--
C	LTE	BC28B	C0	29	--	--	9610	--	--	--

Table 10.1.1

Test Setup



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Signal Generator Setup:

Frequency If using a CW signal generator add, a 0.5 MHz Offset to the Test Channel specified in table 10.1.1

Amplitude -60 dBm

Modulation CW or WCDMA / LTE (Mode being tested. Refer to table 10.1.1)

Connect the test equipment to an antenna coupler. Place the coupled antenna parallel to the UE antenna.

NOTE: The test bench should be calibrated for power for each channel (for RX, use Channel + "frequency offset").

UE Setup:

Verify that the UE RF Ports are disconnected from RF cables as this is a radiated test and not a conducted test.

Test Procedure

Note: RX/TX tests are required to be done before any CDMA TX testing.

RX Power Method:

Set the signal generator output to RF specified in the Test Setup.

Use the following commands to set the RX reference channel (from table 10.1.1) and read the RX AGC value.

The Mode ID in the QC test command header is different for the paths, all other command data is identical.

**NOTE: Be certain that the correct mode id is used inside the QC test command header for all commands in the procedure. See table 10.1.1.*

Use the following commands to setup the UE.

Command	Data
FTM_SET_MODE*	0x4B 0B '1D00' 0700 2A00 (Select the phone mode)

Where: 1D00 is the 2-byte byte-swapped **Mode ID** and 2A00 is the 2-byte byte-swapped **Mode** selected.

Where **Mode** =

WCDMA

0x0900 for W2100(Band I)
0x0F00 for W1900(Band II)
0x1900 for W1800(Band III)
0x1C00 for W1700(Band IV)
0x1D00 for W900(Band VIII)
0x1600 for W850(Band V)

LTE

0x2200 for Band 1
0x2B00 for Band 2
0x2C00 for Band 3
0x2A00 for Band 4
0x2D00 for Band 5
0x2300 for Band 7
0x2F00 for Band 8
0x3200 for Band 12
0x2400 for Band 13
0x2500 for Band 17
0x3800 for Band 20
0x3D00 for Band 25
0x3E00 for Band 26
0x4000 for Band 28

CDMA

0x0500 for BC0 (CDMA800)
0x0600 for BC1 (CDMA1900)

Command	Data
FTM_SET_SECONDARY_CHAIN	0x4B 0B23007900 + CHAIN (Set Secondary Chain for the path being tested)

Where CHAIN =
0x0000 for C0



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0x0100 for C1

For CMDA:

Command	Data
FTM_SET_CHAN*	0x4B 0B 00000800 + CHAN (Set Channel from table 10.1.1)

Where CHAN is 2-byte byte-swapped value.

For WCDMA:

Command	Data
FTM_SET_CHAN*	0x4B 0B 01000800 + CHAN (Set Channel from table 10.1)

Where CHAN is 2-byte byte-swapped value.

For LTE:

Command	Data
FTM_LTE_SET_RX_BANDWIDTH	0x4B 0B 1D00910103 (Set RX Bandwidth to 10 MHz)
FTM_LTE_SET_TX_BANDWIDTH	0x4B 0B 1D00900103 (Set TX Bandwidth to 10 MHz)
FTM_LTE_SET_CHAN	0x4B 0B 1D000800 + CHAN (Set Channel from table 10.1)

Where CHAN is 2-byte byte-swapped value.

(Continue)

Command	Data
SET_LNA_STATE_DBM	0x4B 0B 1D00B900A8FD

Where:

Input power = -60dbm

Multiplied by 10 = -600

Convert to hex = 0xFDA8 (2's complement)

Byte swap = 0xA8FD

Set the test set to -60dbm (Test Setup)

Loop the following command until a stable reading is obtained. Store the stable reading as dBm.

Command	Data
GET_RX_LEVEL_DBM	0x4B0B 1D00B30000000000

Parse out the RSSI data from the response:

0x4B0B1D00B3009BFD9BFD

Where command response is 0x9BFD:

Byte swap = 0xFD9B

Convert to dec = -613 (2's complement)

Divide by 10 = -61.3dbm

Verify that the **QC_MODE_band_path_RX_RAD_RSSI_chan** Power Reading Value is within the limits specified in section [2.7 Test Limits](#).

Exit Test

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10.1.2 Tx Antenna Tests

Test Description

The purpose of this test is to verify connectivity and TX performance of the CDMA, WCDMA and LTE Antennas.

Test Dependencies:

[3.2 Turn On](#)

[3.6.1 FTM Mode](#)

Equipment:

RF Analyzer (Power Meter or equivalent)

Test Setup

RF Analyzer Setup:

Frequency: Set the analyzer to the Test Channel specified in table 10.1.1

Connect the test equipment to an antenna coupler. Place the coupled antenna parallel to the UE antenna.

NOTE: The test bench should be calibrated for power for each channel (for RX, use Channel + "frequency offset").

UE Setup:

Verify that the UE RF Ports are disconnected from rf cables as this is a radiated test and not a conducted test

Test Procedure

Note: Rx/Tx tests are required to be done before any CDMA Tx testing.

TX Power Method:

Set the RF analyzer to the channel required in table 10.1.1

The Mode ID in the QC test command header is different for the paths, all other command data is identical.

**NOTE: Be certain that the correct mode id is used inside the QC test command header for all commands in the procedure. See table 10.1.1.*

Use the following commands to setup the UE.

Command	Data	
FTM_SET_MODE*	0x4B0B '1D00' 07002A00	(Select the phone mode)

Where: 1D00 is the 2-byte byte-swapped **Mode ID** and 2A00 is the 2-byte byte-swapped **Mode** selected.

Where **Mode** =

WCDMA

LTE

CDMA



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0x0900 for W2100(Band I)	0x2200 for Band 1	0x0500 for BC0 (CDMA800)
0x0F00 for W1900(Band II)	0x2B00 for Band 2	0x0600 for BC1 (CDMA1900)
0x1900 for W1800(Band III)	0x2C00 for Band 3	
0x1C00 for W1700(Band IV)	0x2A00 for Band 4	
0x1D00 for W900(Band VIII)	0x2D00 for Band 5	
0x1600 for W850(Band V)	0x2300 for Band 7	
	0x2F00 for Band 8	
	0x3200 for Band 12	
	0x2400 for Band 13	
	0x2500 for Band 17	
	0x3800 for Band 20	
	0x3D00 for Band 25	
	0x3E00 for Band 26	
	0x4000 for Band 28	

For WCDMA:

Command	Data
FTM_SET_SECONDARY_CHAIN	0x4B 0B15007900 + CHAIN (Set Secondary Chain for the path being tested)
Where CHAIN =	
	0x0000 for C0
	0x0100 for C1

Command	Data
FTM_SET_CHAN*	0x4B 0B01000800 + CHAN (Set Channel from table 10.1.1)

Where CHAN is 2-byte byte-swapped value.

For LTE:

Command	Data
FTM_SET_SECONDARY_CHAIN	0x4B 0B23007900 + CHAIN (Set Secondary Chain for the path being tested)
Where CHAIN =	
	0x0000 for C0
	0x0100 for C1

Command	Opcode/Data
FTM_LTE_SET_RX_BANDWIDTH	0x4B 0B1D00910103 (Set RX Bandwidth to 10 MHz)
FTM_LTE_SET_TX_BANDWIDTH	0x4B 0B1D00900103 (Set TX Bandwidth to 10 MHz)
FTM_LTE_SET_CHAN	0x4B 0B1D000800 + CHAN (Set Channel from table 10.1)

Where CHAN is 2-byte byte-swapped value.

(Continue)

Command	Data
FTM_SET_TX_ON	0x4B0B '1D00' 0200 (Start Tx)

Where 1D00 is the 2-byte byte-swapped **Mode ID**.



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For LTE only:

<u>Command</u>	<u>Data</u>	
FTM_LTE_CONFIGURE_TX_WAVEFORM	0x4B0B '1D00' 9301 + Waveform	(Set waveform)

Where **Waveform** =

0x010C0013 (for Waveform LTE_PUSCH, numRBsPUSCH=12, numRBsPUCCH=0, PUSCHStartPBIndex =19)

<u>Command</u>	<u>Data</u>
LTE_SET_NET_SIG_VALUE	0x4B0B '1D00' B10001

(Continue)

<u>Command</u>	<u>Data</u>	
SET_TX_POWER_DBM	0x4B0B '1D00' B000 '01' + Power Level times ten	(Set TX Power)

Where:

0x01 = Enable TX Power

Verify that the **QC_MODE_band_path_TX_RAD_POW_chan** Power Reading Value is within the limits specified in section [2.7 Test Limits](#).

<u>Command</u>	<u>Data</u>
SET_TX_POWER_DBM	0x4B0B '1D00' B000 '00' 0000

Where:

0x00 = Disable TX Power

<u>Command</u>	<u>Data</u>	
FTM_SET_TX_OFF	0x4B0B '1D00' 0300	(Stop Tx)

After all channels required to be tested for a given band are cycled through, repeat for the next band.

Exit Test

10.2 Diversity Antenna Test

10.2.1 Rx Antenna Test

Test Description

The purpose of this test is to verify proper seating and electrical connection of the external antenna to the C1 internal paths and by extension the robustness of the internal path routing as well. The test measures radiated power through an RF-coupled antenna fixture.

Test Dependencies:

[3.2 Turn On](#)

[3.6.1 FTM Mode](#)

Equipment:

LTE / WCDMA signal Generator or CW Signal Generator

RX Test Channels, Paths, and Modes

Model	Mode	Band	Path (determines Mode ID)	Mode ID (dec)	DL Test Channel Low	DL Test Channel Mid	DL Test Channel High	UL Test Channel Low	UL Test Channel Mid	UL Test Channel High
B,C,D	WCDMA	BC01	C1	21	--	--	10838	--	--	--
B,C	WCDMA	BC05	C1	21	--	--	4458	--	--	--
D	WCDMA	BC08	C1	21	--	--	3088	--	--	--
C,D	LTE	BC03	C1	35	1250	--	--	--	--	--
B,C	LTE	BC07	C1	35	2800	--	--	--	--	--
B,C	LTE	BC07	C1	35	--	--	3400	--	--	--
B	LTE	BC13	C1	35	5184	--	--	--	--	--
D	LTE	BC26	C1	35	--	--	9039	--	--	--

Table 10.2.1

Test Setup

Signal Generator Setup:

Frequency If using a CW signal generator add, a 0.5 MHz Offset to the Test Channel specified in table 10.2.1
 Amplitude -60 dBm
 Modulation CW or WCDMA / LTE (Mode being tested. Refer to table 10.2.1)

Connect the test equipment to an antenna coupler. Place the coupled antenna parallel to the UE antenna.

NOTE: The test bench should be calibrated for power for each channel (for RX, use Channel + "frequency offset").

UE Setup:

Verify that the UE RF Ports are disconnected from RF cables as this is a radiated test and not a conducted test.

Test Procedure



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Note: RX/TX tests are required to be done before any CDMA TX testing.

RX Power Method:

Set the signal generator output to RF specified in the Test Setup.

Use the following commands to set the RX reference channel (from table 10.2.1) and read the RX AGC value.

The Mode ID in the QC test command header is different for the paths, all other command data is identical.

**NOTE: Be certain that the correct mode id is used inside the QC test command header for all commands in the procedure. See table 10.2.1.*

Use the following commands to setup the UE.

Command	Data
FTM_SET_MODE	0x4B0B '1D00' 0700 2A00 (Select the phone mode)

Where: 1D00 is the 2-byte byte-swapped **Mode ID** and 2A00 is the 2-byte byte-swapped **Mode** selected.

Where **Mode** =

WCMDA

0x0900 for W2100(Band I)
0x0F00 for W1900(Band II)
0x1900 for W1800(Band III)
0x1C00 for W1700(Band IV)
0x1D00 for W900(Band VIII)
0x1600 for W850(Band V)

LTE

0x2200 for Band 1
0x2B00 for Band 2
0x2C00 for Band 3
0x2A00 for Band 4
0x2D00 for Band 5
0x2300 for Band 7
0x2F00 for Band 8
0x3200 for Band 12
0x2400 for Band 13
0x2500 for Band 17
0x3800 for Band 20
0x3D00 for Band 25
0x3E00 for Band 26
0x4000 for Band 28

CDMA

0x0500 for BC0 (CDMA800)
0x0600 for BC1 (CDMA1900)

Command	Data
FTM_SET_SECONDARY_CHAIN	0x4B 0B23007900 + CHAIN (Set Secondary Chain for the path being tested)

Where CHAIN =

0x0000 for C0
0x0100 for C1

For CMDA:

Command	Data
FTM_SET_CHAN*	0x4B 0B 00000800 + CHAN (Set Channel from table 10.2.1)

Where CHAN is 2-byte byte-swapped value.

For WCMDA:

Command	Data
FTM_SET_CHAN*	0x4B 0B 01000800 + CHAN (Set Channel from table 10.2.1)

Where CHAN is 2-byte byte-swapped value.

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Command	Data	
FTM_LTE_SET_RX_BANDWIDTH	0x4B 0B 1D00910103	(Set RX Bandwidth to 10 MHz)
FTM_LTE_SET_TX_BANDWIDTH	0x4B 0B 1D00900103	(Set TX Bandwidth to 10 MHz)
FTM_LTE_SET_CHAN	0x4B 0B 1D000800 + CHAN	(Set Channel from table 10.2.1)

Where CHAN is 2-byte byte-swapped value.

(Continue)

Command	Data
SET_LNA_STATE_DBM	0x4B 0B 1D00B900 A8FD

Where:

Input power = -60dbm

Multiplied by 10 = -600

Convert to hex = 0xFDA8 (2's complement)

Byte swap = 0xA8FD

Set the test set to -60dbm (Test Setup)

Loop the following command until a stable reading is obtained. Store the stable reading as dBm.

Command	Data
GET_RX_LEVEL_DBM	0x4B0B 1D00B30000000000

Parse out the RSSI data from the response:

0x4B0B1D00B3009BFD**9BFD**

Where command response is 0x9BFD:

Byte swap = 0xFD9B

Convert to dec = -613 (2's complement)

Divide by 10 = -61.3dbm

Verify that the **QC_MODE_band_path_RX_RAD_RSSI_chan** Power Reading Value is within the limits specified in section [2.7 Test Limits](#).**Exit Test**

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11.0 CFC Test Procedures

Note: The methods listed below do NOT dictate process flow or specific recipes. CFC Test Group is solely responsible for gathering Carrier requirement from TAMs for the purpose of putting recipes together. The Global Test CFC Portal maintained by CFC specifies which methods to perform and in what order: <http://mqs.mot.com:8000/nextest2010cfcportal/default.aspx>. The tests below include all the methods available and are listed in alphabetical order for reference purposes only. Change requests must be submitted against the 12M via JIRA IKTM when additions / deletions or changes to the current test methods are required.

11.1 APK_USIM_Card_Detection

This procedure determines if a USIM card is present in the radio.

Test Dependencies:

Radio must be turned on in factory mode

Test Procedures:

With the card inserted prior to power-up, use the SIM_CARD command to verify connectivity of the card interface.

Refer to section [3.10 USIM/SIM Card Interface](#) sub section [3.10.2 APK CommServer Method, Card, Card Detection](#).

Verify within the response SIM_STATE= READY

Verify within the response SIM_OPERATOR_NAME= OPERATOR_NAME

Where:

OPERATOR_NAME = **Verizon Wireless, Verizon, Vodafone NL** (for Verizon units)
U.S. Cellular (for USC units)
Chameleon (for Sprint units)
AT&T (for ATT units)
 Etc.....

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11.2 APK_Verify_Talkback_Absence

This test verifies if talkback process is running on the phone.

Test Dependencies:

Radio must be turned on in factory mode

Setup:

UUT must have the CommServer running and communication establish over BLAN Port 2631.

Test Procedures:

Send the following command to run the Setup Wizard Status test.

Command	Activity/data
START	ServiceInfo
TELL	ServiceInfo GET_RUNNING_SERVICE_INFO
TELL	ServiceInfo GET_RUNNING_APP_INFO

Record the returning strings and parse for "talkback" string. Absence of the string signifies passing DUT. If string is detected in the response then fail the test.

Send the following command to stop the ServiceInfo Activity.

Command	Activity/data
STOP	ServiceInfo

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11.3 AUX_Engine_Ready_FTM

This procedure will check if BP is ready for communication.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Test Procedures:

Send the following test command to check if BP is ready.

Command	Opcode/Data
FTM_RAW_DATA	0x4B 0B00 8041 8000 (Check if BP is ready)

Ensure the response is: 0x4B 0B00 8041 80FF A1E2 7E

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11.4 Battery_Sense_Voltage_Check

This procedure verifies the battery charge value in actual units of millivolts.

Test Dependencies:

- Radio must be in P2K Mode. (Sec 3.6.5)
- Radio must be in Suspend mode. (Sec 3.6.6)

Test Procedures:

Send the following test command to acquire a stable A/D reading of the BATT+ line.

Command	Opcode/data
AD_CONV	0x0001 000C

Convert the two byte hex response to decimal. The units in decimal are in millivolts. Average 8 stable A/D readings and record it as the battery charge voltage value (refer to example).

Ensure battery voltage level is between 40% and 100% as specified in section [2.1 Test Limits](#).

Example 1:

Below is an example of the response data of a sample reading after sending the **AD_CONV** command

Response data in hex: 82370001000600000000000020F06

The last two bytes here are 0F06, whose decimal value will be 3846 which is the battery charge voltage value in millivolts.

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11.5 Battery_Thermistor_Temp_Check

This procedure verifies the battery thermistor(temperature).

Test Dependencies:

- Radio must be in P2K Mode. (Sec 3.6.5)
- Radio must be in Suspend mode. (Sec 3.6.6)
- Radio must have factory software in it as CFC SW does not contain batch files.

Test Procedures:

Send the following test command to acquire a stable A/D reading of the Thermistor battery line.
Average 8 stable A/D readings.

Command	Opcode/data	
TEMP	0x008F 0002 0000	(Read the Battery Thermistor value)

Convert the four byte response from **2's complement format** to decimal. The units in decimal are returned in tenths of degrees Celsius.

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11.6 BP_Engine_Ready_FTM_Via_Route

This procedure will check if BP is ready for communication.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Send the following test command to check if BP is ready.

Command	Opcode/Data
TCMD_ROUTE	0x0FF0 044B 0B00 8041 8000 (Check if BP is ready)

Ensure the response is: 0x4B 0B00 8041 80FF

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11.7 Check_FTI_No_Checksum_P2K

This procedure is used to ensure the radio has passed all factory test stations. The factory information element area used to store factory test results contains 10 bytes (bytes 91 – 100). The first nibble of each byte indicates the results of the last time the radio was tested at the station; the second nibble indicates the number of times the radio has been tested at that station. The station each byte represents varies based on product and factory.

Test Dependencies:

- Radio must be in P2K Mode. (Sec 3.6.5)
- Radio must be in Suspend mode. (Sec 3.6.6)

Test Procedures:

Issue RDELEM (0X020) command with data 0x2726000100000080 to read the 10 bytes of data which contain the factory test information. Compare each byte to see if the data is 0x8X or 0xFF, where X is do not care. Any other value indicates an invalid test station result.

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11.8 Clear_NV_FTM_Mode

This procedure writes 0x00 to this NV Item for software to boot up in AMSS mode (normal phone operation).

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Test Procedures:

Use the following NV_WRITE command to clear NV_FTM_MODE.

Command	Opcode/Data	
NV_WRITE	0x27 C501 + fill with 0x00 until 128 bytes	(Clear NV_FTM_MODE)

Ensure a successful response

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11.9 Fastboot_FB_Mode_Clear

This procedure will clear the forced fastboot mode and allow phone to reboot normally.

Test Dependencies:

Radio must be in Fastboot flash mode.(Sec 0)

Test Procedures:

Issue the "fastboot oem fb_mode_clear" command in fastboot flash mode to exit the fastboot loop.

Check for the 'OKAY' response.

If 'FAILED' is received then fail the unit.

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11.10 Fastboot_Flash

This procedure is for flashing phone software using the fastboot flashing method.

Test Dependencies:

Radio must be in Fastboot flash mode (Sec 0)

Test Procedures:

The software image files are a package of many code files. Each of these code files have to be flashed sequentially using the fastboot flash command, as follows:

Execute "fastboot flash <code file>" to flash the particular code file.

Following are some examples of fastboot flash commands for flashing AP files:

fastboot flash logo logo.bin

fastboot flash boot boot.img

fastboot flash system system.img

Once all the necessary files are flashed, check for a "success" response.

In most cases the ME will need to be rebooted.

Send the following command to reboot the ME

Execute "fastboot reboot"

There is no response back other than the unit rebooting.

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11.11 Fastboot_OEM_Prod_Fuse

This procedure will program the AP efuses necessary to move the phone into production state. Since the phone is resigning the BCT and the bootloader, it can take up to 65 seconds for this procedure to complete. Please note that this procedure blows fuses in the radio software. This procedure is irreversible.

Test Dependencies:

Radio must have AP signed software.

Test Procedures:

Execute the fastboot command "oem" with option "prod_fuse" to program AP/BP efuses into production state.

Check for the 'OKAY' response.

If 'FAILED' is received then fail the unit.

Execute the fastboot command "fastboot reboot" to reboot the radio and complete the programming of AP/ BP efuses into production state.

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11.12 Fastboot_Query_Secure

This procedure will determine if the AP is secure, fused.

Test Dependencies:

Radio must be in Fastboot flash mode.(Sec 0)

Test Procedures:

Issue the FastBoot getvar command with data "secure"

Fastboot getvar secure

Data returned:

secure: no

secure: yes

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11.13 Fastboot_Reboot_Bootloader

This procedure will reboot device into bootloader mode.

Test Dependencies:

Radio must be in Fastboot flash mode.(Sec 0)

Test Procedures:

Issue the "fastboot reboot-bootloader" command in fastboot flash mode using the Android fastboot tool.

Check for the 'OKAY' response.

If 'FAILED' is received then fail the unit.

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11.14 Fastboot_Reboot

This procedure will reboot device into normal mode.

Test Dependencies:

Radio must be in Fastboot flash mode.(Sec 0)

Test Procedures:

Issue the "fastboot reboot" command in fastboot flash mode using the Android fastboot tool.

Check for the 'OKAY' response.

If 'FAILED' is received then fail the unit.

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11.15 Fastboot_Set_Barcode_UTAG

This procedure programs the barcode (track ID) into UTAG memory partition.

Note: This procedure can only be done on non-secured units (NS Flash and SE Flash software allowed).

Test Dependencies:

Radio must be in Fastboot flash mode.(Sec 0)

Test Procedures:

Execute "fastboot oem config barcode <unit's barcode>" to program barcode (track ID) into UTAG memory partition.

Check for the 'OKAY' response.

If 'FAILED' is received then fail the unit.

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11.16 Fastboot_Unset_Bootmode

This procedure sets radio's boot mode to out of factory status.

Test Dependencies:

Radio must be in fastboot flash mode(Sec 11.48)

Test Procedures:

Execute "fastboot oem config unset bootmode" to set radio's bootmode to out of factory status.

Check for the 'OKAY' response.

If 'FAILED' is received then fail the unit.

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11.17 Fastboot_Verify_Unlock_Status

This procedure checks the unlock status of the bootloader.

Test Dependencies:

Radio must be in Fastboot flash mode.(Sec 0)

Test Procedures:

Execute "fastboot getvar unlocked" to verify if bootloader is unlocked.

Data returned:

unlocked: yes

unlocked: no

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11.18 Full_Power_On_Check

This procedure verifies the turn-on (power up) initialization completes successfully

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Use the PWR_UP_CHK command to verify if a pre-defined checkpoint in the device's power-up process has been reached.

<u>Command</u>	<u>Opcode/data</u>	
PWR_UP_CHK	0x009A 00	(Power Up Checkpoint Query)

Poll up for at least 2 minutes for the one byte test command response.

Verify that the one byte test command response is 0x01 (which means the checkpoint was reached).

Wait for the UE to fully power up.

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11.19 Is_CDMA_Master_Sub_Locked

This procedure will determine the state of the phone locks. For CDMA, the possible locks are the Service Password, Master Subsidy and One Time Keypad Lock. The One Time Keypad Lock will not be checked, since its locking mechanism is controlled by the Master Subsidy Lock. For GSM, the possible locks are the Service Password, NWSCP and SSCP. The SSCP will not be checked, since its locking mechanism is controlled by the NWSCP.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Use the SBSDY_LCK command to check the status of the Master Subsidy lock.

Command	Opcode/data
SBSDY_LCK	0x0035 00

The data byte return will indicate the state of the Master Subsidy lock.
(00 – Unlocked, 01 – Locked)

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11.20 Is_Subsidy_Lock_Required

This procedure will determine if the Subsidy lock must be programmed.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Test Procedures:

Issue the NV_READ (0X26) command with data 0x7220.

Note: For NV_Read's it is necessary to append 0x00 to the data in order to send 128 bytes of data.

The first byte of the returned data will indicate if the Subsidy lock must be programmed.

0x00 – No Programming Required

0x01 – Programming Required – Follow the steps below.

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11.21 Is_Unit_Locked

This procedure will check if unit is subsidy locked.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Test Procedures:

Send the following FTM RAW DATA command to register a client ID.

<u>Command</u>	<u>Opcode/Data</u>	
FTM_RAW_DATA	0x8021 00FA 00000000	(Register a client ID)

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 16,17,18,19 from the second response. Make this the 4 byte **Client ID HIGH**.

Send the following FTM RAW DATA Command to use to open a session.

<u>Command</u>	<u>Opcode/Data</u>
FTM_RAW_DATA	0x8021 02FA00000000 + Four byte Client ID HIGH + 00000000 + 12bytes of 0x00

Example of sent data: 0x802102FA 00000000 **748A3629** 00000000 00000000 00000000 00000000

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 20,21,22,23 from the second response. Make this the 4 byte **Session ID LOW**.

Send the following FTM RAW DATA Command to check if subsidy lock is active.

<u>Command</u>	<u>Opcode/Data</u>
FTM_RAW_DATA	0x8021 17FA 00000000 + Four byte Session ID LOW + 00000000

Example of sent data: 0x8021 17FA00000000 **993C1C16** 00000000

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Check byte 16 from the second response. This is the radio's Network Indicator Status.

Where NW_IND_STATUS:

0x00 = inactive

0x01 = active

If Network Indicator Status is active, unit is subsidy locked.

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11.22 Load_WLAN_Test_Firmware

This procedure downloads the test firmware.

Test dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Send the following commands to download test firmware.

Command	Opcode/data	
WLAN	0x0C1F 0001 0000	(Download Test Firmware)

Ensure a successful response

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11.23 Master_Clear

The INVM of the radio must be cleared before leaving the CFC in order to ensure the proper default values are loaded into various phone locations.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the INVM (0x0012) command with data 0x01.

When the test command returns a response, then the command is completed. Check that the response of the test command is a success.

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11.24 Power_Down_WLAN_Radio

This procedure powers down the WLAN module.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Send the following command to power down the WLAN module.

Command Opcode/data

WLAN 0x0C1F 000B (Power Down WLAN Module)

Ensure a successful response

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11.25 Power_Off

This procedure will power down the radio.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Use the PWR_OFF command to power down the UE.

Command Opcode/data

PWR_OFF 0x001E (Power down)

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11.26 Program_AKEY_FTM

This procedure programs the AKEY into NAM 1 and NAM 2. If dual AKEY programming is required, two AKEYS must be specified, since the AKEY in NAM 1 and NAM 2 are not necessarily identical.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Phone must have valid MEID or ESN already programmed. (This is currently done in the factory and is not defined in this document)

Radio must be unlocked for Master Subsidy Lock. (Sec 11.52)

Data:

A-KEY1 ddddddddddddddddddddddd (26-digit decimal number)

A-KEY2 ddddddddddddddddddddddd (26-digit decimal number)

Test Procedures:

Use the NVWRITE command to program the A-KEY into the phone.

Command	Opcode/data
NVWRITE	0x27 190000 + <A-KEY>

Where <A-KEY> is a 26 digit decimal number which is converted to a hexadecimal and stored in the phone. It is stored as an 8 byte number.

A-KEY Data Example:

Below is an example of 26 digit decimal number which is converted to a 16 digit hexadecimal and used as A-KEY.

Decimal value: 15808127962130412968220777

Hexadecimal: DB61C0A062B3F5A8

This hexadecimal number is the A-KEY that needs to be used with the data when issuing the NVWRITE command.

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11.27 Program_CID_Data_Block_IMEI / MEID

This method will program the various Data Blocks into the radio.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Data:

DB_TYPE - Refer to the Test Command Document to get valid Data Block Types.

Test Procedures:

Issue the DATABLK_SIGN (0x0C31) command with data 0x00 + DB_TYPE to request Data Block Data.

Verify the first byte of the returned data is 0x00, and then use the remaining bytes of returned data to obtain DB_DATA.

Issue DATABLK_SIGN (0x0C31) command with data 0x01 + DB_TYPE + DB_DATA to store Data Block data.

Verify that the returned data is 0x00.

Example:

Below is an illustration of programming a CID (Customer Identifier) data block using IMEI/MEID.

Issue the DATABLK_SIGN (0x0C31) command with data 0x00 + 0x00F0 to request Data Block Data.

Command	Opcode/Action
DATABLK_SIGN	0x0C31 00 00F0

Where 0x00 represents the "request" data block provision type.

0x00F0 represents "CID" data block type

Verify that the first byte of the returned data is 0x00, and then use the remaining bytes of returned data (DB_DATA) to program the CID data block.

Note: IMEI/MEID and ULMA must be updated in the datablock before getting it signed. For IMEI, the rule is imei[0]A and remaining byte swapped with luhn digit being 0. For MEID, make sure it is byte swapped. For ULMA, if there is no ULMA (bluetooth address) then use all 0s as the value.

Issue DATABLK_SIGN (0x0C31) command with data 0x01 (store) + 0x00F0 (DB_TYPE) + IMEI + DB_DATA to store the CID data block using IMEI. Verify that the returned data is 0x00.

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11.28 Program_CID_Via_Fastboot_IMEI / MEID

This method will program the CID Data Blocks into the radio using fastboot when the device cannot boot into Android to program and run Test Commands.

Test Dependencies:

Radio must be in Fastboot flash mode (Sec 0)

Data:

IMEI/MEID (Primary Serial Number) is required for the PKI Server to process the information coming from NextTest.

Test Procedures:

Issue the "fastboot oem cid_prov_req" command in fastboot flash mode using the Android fastboot tool

Verify that the information returned is OKAY as opposed to FAILED.

NextTest will take the information posted here and have the information signed by the PKI Server. After which it will flash a special CID partition with the appropriate data.

Example:

Below is an illustration of programming a CID (Customer Identifier) data block using IMEI/MEID. This example is based on MSM8960PRO.

NextTest will issue the "fastboot oem cid_prov_req" command

The message from the handset will be parsed and updated with appropriate fields and sent to the PKI server to be signed. NextTest will require the IMEI/MEID in advance before programming.

Note: IMEI/MEID and ULMA must be updated in the datablock before getting it signed. For IMEI, the rule is imei[0]A and remaining byte swapped with luhn digit being 0. For MEID, make sure it is byte swapped. For ULMA, if there is no ULMA (bluetooth address) then use all 0s as the value.

When the signed information comes back from the PKI Server, this information will be flashed into a special "fake" CID partition using the "fastboot flash" command that gets parsed and validated before the final CID information is stored in a persistent flash area.

11.29 Program_IMEI_Datablock

This method will program the IMEI Data Block into the radio.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Radio must not have IMEI programmed

Data:

DB_TYPE - Refer to the Test Command Document to get valid Data Block Types.

The ENCODED_IMEI data bytes are determined as follows:

Byte1: 0x08 → Upper nibble = '0', Lower nibble = '8'
 Byte2: 0xA → Upper nibble = TAC 1, Lower nibble = 'A'
 Byte3: 0xX → Upper nibble = TAC 3, Lower nibble = TAC 2
 Byte4: 0xX → Upper nibble = TAC 5, Lower nibble = TAC 4
 Byte5: 0xX → Upper nibble = FAC 1, Lower nibble = TAC 6
 Byte6: 0xX → Upper nibble = Serial 1, Lower nibble = FAC 2
 Byte7: 0xX → Upper nibble = Serial 3, Lower nibble = Serial 2
 Byte8: 0xX → Upper nibble = Serial 5, Lower nibble = Serial 4
 Byte9: 0x0X → Upper nibble = '0' (LUHN Digit), Lower nibble = Serial 6

Definitions:

FAC – Factory Code

LUHN – Check digit for the IMEI

TAC – Type Approval Code

Test Procedures:

Issue the DATABLK_SIGN (0x0C31) command with data 0x00 + 0x000F to request Data Block Data.

Command	Opcode/Action
DATABLK_SIGN	0x0C31 00 000F

Where 0x00 represents the "request" data block provision type.
 0x000F represents "IMEI" data block type

Verify that the first byte of the returned data is 0x00, and then use the remaining bytes of returned data (DB_DATA) to program the IMEI data block.

Note: IMEI/MEID and ULMA must be updated in the datablock before getting it signed. For IMEI, the rule is imei[0]A and remaining byte swapped with luhn digit being 0. For MEID, make sure it is byte swapped. For ULMA (bluetooth address), the value shall be set to all 0s.

Issue DATABLK_SIGN (0x0C31) command with data 0x01 + 0x000F + DB_DATA to store the IMEI data block.

Command	Opcode/Action
DATABLK_SIGN	0x0C31 01 000F + DB_DATA

Where 0x01 represents the "store" data block provision type.
 0x000F represents "IMEI" data block type

Verify that the returned data is 0x00.

11.30 Program_IMEI_FTM

This method describes how to program the IMEI into a phone.

Note: This command can only be sent once. The IMEI CANNOT be reprogrammed.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Data:

The IMEI data bytes are determined as follows:

Byte1: 0x08 → Upper nibble = '0', Lower nibble = '8'
 Byte2: Upper nibble = TAC 1, Lower nibble = 'A'
 Byte3: Upper nibble = TAC 3, Lower nibble = TAC 2
 Byte4: Upper nibble = TAC 5, Lower nibble = TAC 4
 Byte5: Upper nibble = FAC 1, Lower nibble = TAC 6
 Byte6: Upper nibble = Serial 1, Lower nibble = FAC 2
 Byte7: Upper nibble = Serial 3, Lower nibble = Serial 2
 Byte8: Upper nibble = Serial 5, Lower nibble = Serial 4
 Byte9: Upper nibble = '0' (LUHN Digit), Lower nibble = Serial 6

Definitions:

FAC – Factory Code

LUHN – Check digit for the IMEI

TAC – Type Approval Code

Test Procedures:

To determine if the IMEI has been programmed, send the following FTM command with data 0x0226.

Command	Opcode/data
NVRead	0x26 2602 + 128 bytes of 0x00

The response will be of the form 0x262602 + 128 bytes of data + 2 bytes of status

If the 2 bytes of status are 0x0500, then the IMEI has not yet been programmed. If the 2 bytes of status are 0x0000, then IMEI programming is not possible.

Issue the NVWrite command with data 0x0226 + 9 bytes of IMEI data to store the IMEI. Pad the data with 0x00 until data length equals 128 bytes

Command	Opcode/data
NVWrite	0x27 2602 + 9 bytes of IMEI data + 0x00 [Padded data = 128bytes]

The 15 decimal digits IMEI must be converted to GSM BCD format to be programmed correctly. For example, if the IMEI value is 354981050003535, then the IMEI to use with the NVWRITE command is 0x083A45890105005303.

Example:

If IMEI value 354981050003535 needs to be written, send the following NVWRITE command with data.

[illegible]



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11.31 Program_Master_Sdsy_Lck

This procedure programs the Master Subsidy Lock. This lock protects the NAM elements.

Test Dependencies:

- Radio must be in FTM Mode. (Sec 3.6.1)
- Radio must be unlocked for Master Subsidy Lock. (Sec 11.52)

Data:

Master Subsidy Lock Passcode (MSL) dddddd (6-digit decimal number)

Test Procedures:

Use the NVWRITE command to program the MSL passcode.

Command	Opcode/data
NVWRITE	0x27 5500 + <MSL>

Where MSL is the hex converted value of the lock passcode.

For example, if the lock code value is 141346, then its corresponding hex value, 313431333436 will be the MSL value to be used in the command.

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11.32 Program_MEID_Datablock

This method will program the MEID Data Block and pseudo-ESN into the radio. The pESN gets automatically programmed once the MEID is programmed.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Data:

DB_TYPE - Refer to the Test Command Document to get valid Data Block Types.

The MEID is a 7 byte (14 digit) hexadecimal number.

The pESN is a 4 byte (8 digit) hexadecimal number.

Test Procedures:

Issue the DATABLK_SIGN (0x0C31) command with data 0x00 + 0x000F to request Data Block Data.

Command	Opcode/Action
DATABLK_SIGN	0x0C31 00 000F

Where 0x00 represents the "request" data block provision type.

0x000F represents "MEID" data block type

Verify that the first byte of the returned data is 0x00, and then use the remaining bytes of returned data (DB_DATA) to program the MEID data block.

Note: IMEI/MEID and ULMA must be updated in the datablock before getting it signed. For IMEI, the rule is imei[0]A and remaining byte swapped with luhn digit being 0. For MEID, make sure it is byte swapped. For ULMA (bluetooth address), the value shall be set to all 0s.

Issue DATABLK_SIGN (0x0C31) command with data 0x01 + 0x000F + DB_DATA to store the MEID data block.

Command	Opcode/Action
DATABLK_SIGN	0x0C31 01 000F + DB_DATA

Where 0x01 represents the "store" data block provision type.

0x000F represents "MEID" data block type

Verify that the returned data is 0x00.

After MEID has been programmed, power cycle the radio to trigger phone software to automatically calculate and program pESN into the radio.

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11.35 Program_Subsidy_Lock

This procedure describes the programming of the NWSCP Lock.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

IS_SUBSIDY_LOCK_REQUIRED (Sec 3.25)

Data:

NWSCP = dddddd (8 digit decimal number)

KEY_DATA = NWSCP (Encoded KEY_DATA is the 8 byte ASCII to Hex value of the NWSCP Number)

Example:

Unlock code: 12345678

ASCII to HEX: 12345678 => 3132333435363738

[8Bytes unlock code] is 3132333435363738

Note: non-production (test, development, and prototype) units should always be programmed with a default password of "12345678"

Test Procedures:

Send the following FTM RAW DATA command to register a client ID.

Command	Opcode/Data	
FTM_RAW_DATA	0x8021 00FA 00000000	(Register a client ID)

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 16,17,18,19 from the second response. Make this the 4 byte **Client ID HIGH**.

Send the following FTM RAW DATA Command to use to open a session.

Command	Opcode/Data	
FTM_RAW_DATA	0x8021 02FA00000000 + Four byte Client ID HIGH + 00000000 + 12bytes of 0x00	

Example of sent data: 0x802102FA 00000000 **748A3629** 00000000 00000000 00000000 00000000

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 20,21,22,23 from the second response. Make this the 4 byte **Session ID LOW**.

Send the following FTM RAW DATA Command to check if subsidy lock is active.

Command	Opcode/Data	
FTM_RAW_DATA	0x8021 17FA 00000000 + Four byte Session ID LOW + 00000000	

Example of sent data: 0x8021 17FA00000000 **993C1C16** 00000000

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Check byte 16 from the second response. This is the radio's Network Indicator Status.

Where NW_IND_STATUS:

0x00 = inactive

0x01 = active

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If Network Indicator Status is active, fail the radio.

Send the following FTM RAW DATA Command to set up the carrier codes (personalization data).

Command Opcode/Data

FTM_RAW_DATA 0x8021 18FA00000000 + Four byte **Session ID LOW** + four byte **Perso Feature**

Where PERSO_FEATURE:

0x00000000 = network feature

0x02000000 = service provider feature

Example of sent data: 0x8021 18FA 00000000 **993C1C16 00000000** (using network feature)

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 20,21,22,23 from the second response. Make this the 4 byte **Session ID LOW**.

Send the following FTM RAW DATA Command to query the pass code.

Command Opcode/Data

FTM_RAW_DATA 0x8021 21FA00000000 + Four byte **Session ID LOW** + Four byte **Perso feature** + four byte **Key type**

Where KEY_TYPE:

0x00000000 = info for all keys

0x01000000 = info for unblock keys

Example of sent data: 0x8021 21FA 00000000 **993C1C16 00000000 00000000**

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Extract byte 16 and bytes 60, 61, 62, 63, 64, 65, 66, 67 from the second response.

Where:

Byte 16 = PERSO_FEATURE

Bytes 60, 61, 62, 63, 64, 65, 66, 67 = KEY_DATA (Unlock code in ASCII string form)

Send the following FTM RAW DATA Command to activate subsidy lock.

Command Opcode/Data

FTM_RAW_DATA 0x8021 15FA 00000000 + Four byte **Session ID LOW** + Four byte **Perso feature** + four byte **Key length** + n bytes **Key data**

Where:

KEY_LENGTH = Key length string

KEY_DATA = Unlock code data (typically 8 bytes)

Example of sent data: 0x8021 15FA 00000000 **993C1C16 00000000 08000000 3132333435363738**

Read the first FTM command response. Check byte 16 where 0x00 = NWSCP enabled

If NWSCP is enabled, poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 16, 20, and 24 from the second response.

Where:



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Byte 16 = PERSO_ACTION
0x00 = Activates personalization
0x01 = Deactivates personalization

Byte 20 = PERSO_FEATURE
0x00 = Network feature
0x02 = Service provider feature

Byte 24 = NUM_RETRIES (Number of unlock attempts allowed)

Send the following FTM RAW DATA Command to lock down subsidy lock pass code requests.

Command Opcode/Data

FTM_RAW_DATA 0x8021 27FA 00000000 + Four byte **Session ID LOW** + Four byte **Action Locked**

Where:

ACTION_LOCKED = 0x00000000

Example of sent data: 0x8021 27FA 00000000 **993C1C16** 00000000

Read the first FTM command response. Check that bytes 4, 5, 6, 7 are 0x00 indicating success.

Poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 16, 17, 18, 19 from the second response.

Where:

Bytes 16, 17, 18, 19 = 0x00000000 (pass code requests locked down)

Send the following FTM RAW DATA Command to query the pass code and verify that the subsidy lock pass code request has been lock down.

Command Opcode/Data

FTM_RAW_DATA 0x8021 21FA00000000 + Four byte **Session ID LOW** + Four byte **Perso feature** + four byte **Key type**

Where KEY_TYPE:

0x00000000 = info for all keys

0x01000000 = info for unblock keys

Example of sent data: 0x8021 21FA 00000000 **993C1C16** 00000000 00000000

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 12, 13, 14, 15 from the second response. Verify that byte 12 is 0x04 indicating that MMGSDI access has been denied.

11.36 Program_Subsidy_NVM_File_FTM

This procedure programs the SLCF (Subsidy Lock Config File) into NV prior to locking the device.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Test Procedures:

Use the NVWRITE command to program the SLCF into the phone.

Command	Opcode/data
NVWRITE	0x27 + <SLCF>

Where SLCF is the Subsidy Lock Config File file extracted from the fastboot zip package.

Example:

To write a typical SLCF default file into NV, send the following NVWRITE commands with data.

Command Opcode/data

[illegible]Command Opcode/data

```
NVWRITE 0x27      7220 00
```

Command Opcode/data

[illegible]

Command Opcode/data

```
NVWRITE 0x27 7420 00
```

Command Opcode/data

[illegible]

Command Opcode/data

```
NVWRITE 0x27    9120 00
```

Command Opcode/data

[illegible]

Command Opcode/data

```
NVWRITE 0x27      CF20 00
```

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11.37 Program_OTK_SDSY_LCK

This procedure programs the One Time Subsidy Lock. This lock protects the NAM elements.

Test Dependencies:

- Radio must be in FTM Mode. (Sec 3.6.1)
- Radio must be unlocked for Master Subsidy Lock. (Sec 11.52)

Data:

One Time Keypad Subsidy Lock Passcode (OTKSL) dddddd (6-digit decimal number)

Test Procedures:

Use the NVWRITE command to program the OTKSL passcodes.

Command	Opcode/data
NVWRITE	0x27 DB00 + <OTKSL>

Where OTKSL is the hex converted value of the lock passcode.

For example, if the lock code value is 167163, then its corresponding hex value, 313637313633 will be the OTKSL value to be used in the command.



11.38 Program_PKI_Data

This method will program the PKI (Private Key Information) Data into the radio.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Radio must already be programmed with IMEI.

Data:

SEQ_NUMBER	SEQ_DATA
0x01	First 3998 bytes of PKI_DATA
0x02	Next 3998 bytes of PKI_DATA
...	...
0x0X	Remaining bytes of PKI_DATA

Test Procedures:

Issue the PKI_DATA_PROVISION (0x06A) command with data 0x05 and PKI Keys, as follows:

Command	Opcode/Action	
PKI_DATA_PROVISION	0x006A 05 01 XXXX	(PKI type data request)

Where 05 is the request for PKI data type;

01 represents normal operation &

XXXX represents the PKI Keys (Array of 2 byte PKI types for which to retrieve the data request).

Verify the first byte of the returned data is 0x00, and then use the other bytes of returned data to obtain PKI_DATA.
Divide the PKI_DATA obtained from the server into 3998 byte portions.

Issue PKI_DATA_PROVISION (0x06A) command with data 0x01 + SEQ_NUMBER + SEQ_DATA for each portion of the PKI_DATA.

Verify that the returned data for each portion of PKI_DATA, except the last, is SEQ_NUMBER + 0x0B.

Verify that the returned data for the last portion of PKI_DATA is SEQ_NUMBER + 0x00.

NOTE: If the length of the PKI_DATA is less than 3998 bytes, then the whole data can be sent in one shot in a single sequence.

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11.39 Qcom_EFS_SYNC_Clear_FTM

For this platform certain commands need to be followed up with synchronizing the file system to the backing store. The following procedure outlines this process.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Test Procedures:

More details on the specific commands referenced below can be gotten from Qualcomm document:

80-V1294-11_V_FS_Subsys_ICD.pdf

Issue the **EFS2_DIAG_SYNC_NO_WAIT** FTM command request message.

Command	OpCode/Data
EFS2_DIAG_SYNC_NO_WAIT	4B133000 xxxx 2F00
EFS2_DIAG_SYNC_NO_WAIT response	4B133000 xxxx yyyyyyyy zzzzzzzz – verify last 4 bytes are all zero (no error).

Ensure that the *SEQUENCE_NUMBER* (xxxx) field, the 5th byte of the input, of length 2 bytes, is always unique for each time the command is issued. Also ensure the response sequence number matches the request sequence number.

Query for the response and save the value of the *SYNC_TOKEN* (yyyyyyy) field, the 7th byte of the response, of length 4 bytes, and save it as **TOKEN**.

Also ensure that the *ERRNO* (zzzzzzzz) field, the 11th byte of the response, of length 4 bytes, is 0, which implies a successful operation.

Issue the **EFS2_DIAG_SYNC_GET_STATUS** FTM command request message.

Command	OpCode/Data
EFS2_DIAG_SYNC_GET_STATUS	4B133100 xxxx yyyyyyyy 2F00
EFS2_DIAG_SYNC_GET_STATUS response	4B133100 xxxx aa zzzzzzzz – verify SYNC_STATUS byte is 1 (sync complete) and last 4 bytes are all zero (no error).

Ensure that the *SEQUENCE_NUMBER* (xxxx) field, the 5th byte of the input, of length 2 bytes, is always unique for each time the command is issued. Also ensure the response sequence number matches the request sequence number.

Ensure that the *SYNC_TOKEN* (yyyyyyy) field, the 7th byte of the input, of length 4 bytes, is populated with the value of **TOKEN** that was saved above.

Query for the response and check the value of the *SYNC_STATUS* field, the 7th byte of the response, of length 1 byte.

Continue to poll for up to 15 seconds, 250 ms between each poll, with the **EFS2_DIAG_SYNC_GET_STATUS** command as long as the *SYNC_STATUS* (aa) field is does not equal 1.

Also ensure that the *ERRNO* (zzzzzzzz) field, the 7th byte of the response, of length 4 bytes, is 0, which implies a successful operation.

11.40 Read_IMEI_FTM

This procedure reads the IMEI from the radio and compares it to the expected value.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Data:

The IMEI data bytes are determined as follows:

Byte1: 0x08 → Upper nibble = '0', Lower nibble = '8'
 Byte2: Upper nibble = TAC 1, Lower nibble = 'A'
 Byte3: Upper nibble = TAC 3, Lower nibble = TAC 2
 Byte4: Upper nibble = TAC 5, Lower nibble = TAC 4
 Byte5: Upper nibble = FAC 1, Lower nibble = TAC 6
 Byte6: Upper nibble = Serial 1, Lower nibble = FAC 2
 Byte7: Upper nibble = Serial 3, Lower nibble = Serial 2
 Byte8: Upper nibble = Serial 5, Lower nibble = Serial 4
 Byte9: Upper nibble = '0' (LUHN Digit), Lower nibble = Serial 6

Definitions:

FAC – Factory Code

LUHN – Check digit for the IMEI

TAC – Type Approval Code

Test Procedures:

Issue the NV Read (0X26) command with data 0x2602 + 128 bytes of 0x00.

[illegible]

The response will be of the form 0x262602 + 128 bytes of data + 2 bytes of status.

If the 2 bytes of status are 0x0500,

The IMEI has not yet been programmed and the data is not valid.

If the 2 bytes of status are 0x0000,

The IMEI is the first 9 bytes of the data.

Byte swap the returned data to get the correct IMEI.

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11.42 Read_Processor_ID

This procedure reads the processor ID from the radio.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Use the RD_ID command to read the processor ID information.

Command	Opcode/data	
RD_ID	0x0025 0004	(Read processor ID)

Compare the returned value with expected value.

Example:

Returned value = 1094A3D5030C00010000000000000000

Where: Byte [0] = how many bytes returned (data length)
Bytes [1-n] = returned data in hex of the specified type

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11.43 Read_Track_ID

This procedure reads the factory tracking number stored in the radio.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Test Procedures:

Use the RDELEM command to read the unit Track ID.

Command	Opcode/data
RDELEM	0X0020 2726000100000080

Convert the data read from the radio from hex values to ASCII characters

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11.44 Restart

This procedure performs a soft reset of the radio software.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the RESTART (0x022) command.

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11.45 Set_BP_RTC_FTM

This procedure is to program the radio's system time.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Test Procedures:

Send the following FTM RAW DATA command to program the radio's system time.

<u>Command</u>	<u>Opcode/data</u>
FTM_RAW_DATA	0x4B 0B0B 0000 0000 0000 0000 0002 AA BB CCCC DD EE FF

Where:

Byte 13 **AA** = Month
 Byte 14 **BB** = Day (must add 5 days to request)
 Bytes 15, 16 **CCCC** = Year: 2 bytes **little-endian**; Example: 0xE7 0x07 -> 0x07E7 = 2023 (must add 10 years)
 Byte 17 **DD** = Hour (24hr format)
 Byte 18 **EE** = Minute
 Byte 19 **FF** = Second

Ensure a successful response.

Example of sent data: 0x4B0B0B000000000000000000000002 **0A 06 E707 00 00 00**

Where:

0x0A = Month of October
 0x06 = Day 1st (0x06 – 5 days = 1st day)
 0xE707 = Year 2013 (2023 – 10 years = 2013)
 0x00 = Hour 0
 0x00 = Minute 0
 0x00 = Second 0

This example results in the RTC being set to 1st Oct 2013 midnight (10/01/2013 00:00:00).

**11.46 SET_PHONE_SYSTEM_TIME**

This procedure is to program the radio's system time.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Use the RTCC command to program the radio's system time.

<u>Command</u>	<u>Opcode/data</u>
RTCC	0X0029 02 + XXXX YYYYYYYY

Where; XXXX is the number of days since Jan.1, 2000. (2 bytes)

YYYYYYYY is the number of seconds since midnight (4 bytes)

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11.47 suspend

This procedure will place the radio in Suspend mode. Once in this mode, internal tasks will be suspended, call processing commands will not be available.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Once in P2K command mode, use the SUSPEND command to place the UE into test mode.

Command	Opcode/data	
SUSPEND	0x0036 00	(Enter test mode)

Read the P2K test command response from the SUSPEND command.

Verify that the SUSPEND response data = 0x00.

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11.48 Switch_To_Fastboot_Flash_Mode

This procedure puts the phone into fastboot flash mode.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Test Procedures:

Issue the FLASH command to switch the radio into fastboot flash mode.

<u>Command</u>	<u>Opcode/data</u>
FLASH	0x000D 01

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11.49 Unlock_CID_Data_Block_IMEI

This method will program CID 0 template data into the radio CID Data Block using IMEI.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Data:

DB_TYPE - Refer to the Test Command Document to get valid Data Block Types.

Test Procedures:

Issue DATABLK_SIGN (0x0C31) command with data 0x03 + DB_TYPE + CID0_DATA to Configure Data Block data.

Command	Opcode/Action
DATABLK_SIGN	0x0C31 03 00F0 + CID0_DATA

Where 0x03 represents the "configure" data block provision type.

0x00F0 represents "CID (Customer Identifier)" data block type

Verify that the returned data is 0x00.

Issue the DATABLK_SIGN (0x0C31) command with data 0x00 + 0x00F0 to request Data Block Data.

Command	Opcode/Action
DATABLK_SIGN	0x0C31 00 00F0

Where 0x00 represents the "request" data block provision type.

0x00F0 represents "CID (Customer Identifier)" data block type

Verify that the first byte of the returned data is 0x00, and then use the remaining bytes of returned data (DB_DATA) to send to the PKI server to be signed. NextTest will require the IMEI in advance before programming.

Note: IMEI/MEID and ULMA must be updated in the datablock before getting it signed. For IMEI, the rule is imei[0]A and remaining byte swapped with luhn digit being 0. For MEID, make sure it is byte swapped. For ULMA, if there is no ULMA (bluetooth address) then use all 0s as the value.

Issue DATABLK_SIGN (0x0C31) command with data 0x01 + 0x00F0 + IMEI + DB_DATA to store the CID data block using IMEI.

Command	Opcode/Action
DATABLK_SIGN	0x0C31 01 00F0 + IMEI + DB_DATA

Where 0x01 represents the "store" data block provision type.

0x00F0 represents "CID (Customer Identifier)" data block type

Verify that the returned data is 0x00.



11.50 Unlock_CID_Data_Block_MEID

This method will program CID 0 template data into the radio CID Data Block using MEID.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Data:

DB_TYPE - Refer to the Test Command Document to get valid Data Block Types.

Test Procedures:

Issue DATABLK_SIGN (0x0C31) command with data 0x03 + DB_TYPE + CID0_DATA to Configure Data Block data.

Command	Opcode/Action
DATABLK_SIGN	0x0C31 03 00F0 + CID0_DATA

Where 0x03 represents the "configure" data block provision type.
0x00F0 represents "CID (Customer Identifier)" data block type

Verify that the returned data is 0x00.

Issue the DATABLK_SIGN (0x0C31) command with data 0x00 + 0x00F0 to request Data Block Data.

Command	Opcode/Action
DATABLK_SIGN	0x0C31 00 00F0

Where 0x00 represents the "request" data block provision type.
0x00F0 represents "CID (Customer Identifier)" data block type

Verify that the first byte of the returned data is 0x00, and then use the remaining bytes of returned data (DB_DATA) to send to the PKI server to be signed. NextTest will require the MEID in advance before programming.

Note: IMEI/MEID and ULMA must be updated in the datablock before getting it signed. For IMEI, the rule is imei[0]A and remaining byte swapped with luhn digit being 0. For MEID, make sure it is byte swapped. For ULMA, if there is no ULMA (bluetooth address) then use all 0s as the value.

Issue DATABLK_SIGN (0x0C31) command with data 0x01 + 0x00F0 + MEID + DB_DATA to store the CID data block using MEID.

Command	Opcode/Action
DATABLK_SIGN	0x0C31 01 00F0 + MEID + DB_DATA

Where 0x01 represents the "store" data block provision type.
0x00F0 represents "CID (Customer Identifier)" data block type

Verify that the returned data is 0x00.

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11.51 Unlock_Subsidy_Lock

This procedure will unlock the NWSCP Lock, if locked.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Data:

NWSCP = dddddd (8 digit decimal number)

KEY_DATA = NWSCP (Encoded KEY_DATA is the 8 byte ASCII to Hex value of the NWSCP Number)

Example:

Unlock code: 12345678

ASCII to HEX: 12345678 => 3132333435363738

[8Bytes unlock code] is 3132333435363738

Note: non-production (test, development, and prototype) units should always be programmed with a default password of "12345678"

Test Procedures:

Send the following FTM RAW DATA Command to unlock unit.

Command	Opcode/Data
FTM_RAW_DATA	0x8021 16FA 00000000 + Four byte Session ID LOW + Four byte Perso feature + four byte Key length + n bytes Key data

Where:

SESSION_ID_LOW = ID from 11.35 PROGRAM_SUBSIDY_LOCK

PERSO_FEATURE = 0x00000000 (network feature)

KEY_LENGTH = Key length string

KEY_DATA = Unlock code data (typically 8 bytes)

Example of sent data: 0x8021 16FA 00000000 **993C1C16** 00000000 08000000 **3132333435363738**

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 16, 20, and 24 from the second response.

Ensure bytes 16, 20, and 24 of the returned data match the following.

Where:

Byte 16 = 0x01 (Deactivates personalization)

Byte 20 = 0x00 (Network feature)

Byte 24 = Number of unlock attempts allowed (example 0x0A)

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11.52 Unlock_CDMA_Master_Lock

This procedure will unlock the Master Subsidy Lock, if locked. The radio will again be Master Subsidy Locked following a power cycle. To disable the lock, the lockcode must be cleared, write 000000 to the passcode seem element, while the unit is in an unlocked state.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Data:

SBSDY_LCK_STRING definition:

Example:

If SBSY_LCK = 'abcdef'

Then SBSY_LCK_STRING = "00AA00BB00CC00DD00EE00FF0000"

Where: AA = ASCII value in hex of 'a'

BB = ASCII value in hex of 'b'

CC = ASCII value in hex of 'c'

DD = ASCII value in hex of 'd'

EE = ASCII value in hex of 'e'

FF = ASCII value in hex of 'f'

And: 'a, b, c, d, e, f' are numeric values.

Test Procedures:

Use the FTM_Raw_Data command to unlock the CDMA Master Subsidy lock.

Command	Opcode/data
FTM_Raw_Data	0x41 + <SBSY_LCK_STRING>

Where SBSY_LCK_STRING is the hex converted value of the lock code.

For example, if the lock code value is 000000, then its corresponding hex value, 303030303030 will be the SBSY_LCK_STRING value to be used in the command.

Compare return data to find state of lock

0x00 – invalid code, unit locked

0x01 – valid code, unit now unlocked

0x02 – number of retries exceeded allowable amount

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11.53 UTAG_Clear_Datecode

This procedure clears the date code value in UTAG memory partition.

Note: The boot loader INFOdate default value is 01-01-1970.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to clear the date code value.

Command	Opcode/Data	
UTAG	0x0108 01 000C 0001 00	(Clear date code value)

Check that the response of the test command is a success.

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11.54 UTAG_Clear_ICCID

This procedure clears the SIM ID (Integrated Circuit Card ID [ICCID]) value in UTAG memory partition.

Note: The boot loader INFOiccid default value is 0 bytes.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to clear the ICCID value.

Command	Opcode/Data	
UTAG	0x0108 01 000D 0001 00	(Clear ICCID value)

Check that the response of the test command is a success.

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11.55 UTAG_Clear_IMEI

This procedure clears the IMEI value in UTAG memory partition.

Note: The boot loader INFOimei default value is 15 digits of 0s.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to clear the IMEI value.

Command	Opcode/Data	
UTAG	0x0108 01 000A 0001 00	(Clear IMEI value)

Check that the response of the test command is a success.

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11.56 UTAG_Clear_MD5

This procedure clears the MD5 customization data checksum value in UTAG memory partition.

Note: The boot loader INFOcust_md5 default value is 0 bytes.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to clear the MD5 value.

Command	Opcode/Data	
UTAG	0x0108 01 000E 0001 00	(Clear MD5 value)

Check that the response of the test command is a success.

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11.57 UTAG_Clear_MEID

This procedure clears the MEID value in UTAG memory partition.

Note: The boot loader INFOmeid default value is 0 bytes.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to clear the MEID value.

Command	Opcode/Data	
UTAG	0x0108 01 000A 0001 00	(Clear MEID value)

Check that the response of the test command is a success.

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11.58 UTAG_Clear_SKU

This procedure clears the SKU value in UTAG memory partition.

Note: The boot loader INFOsku default value is 15 digits of 0s.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to clear the SKU value.

Command	Opcode/Data	
UTAG	0x0108 01 000B 0001 00	(Clear SKU value)

Check that the response of the test command is a success.

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11.59 UTAG_Clear_Track_ID

This procedure clears the Track ID value in UTAG memory partition.

Note: The boot loader INFOserialno default value is 0 bytes.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to clear the Track ID value.

Command	Opcode/Data	
UTAG	0x0108 01 0001 0001 00	(Clear Track ID value)

Check that the response of the test command is a success.

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11.60 UTAG_Verify_Battery_ID

This procedure reads the SKU from UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to read the SKU.

Command	Opcode/Data	
UTAG	0x0108 00 0012	(Read Battery ID)

Convert the data return from hex values to ASCII characters.

Where:

The response format is two bytes indicating the number of bytes followed by the actual battery ID.

Example:

Return data = 0008 534E4E3539353741

Battery ID = 534E4E3539353741 (hex values)

Battery ID = **SNN5957A** (ASCII characters)

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11.61 UTAG_Verify_Bluetooth

This procedure reads the Bluetooth MAC Address from UTAG memory partition.

Test Dependencies:

[3.6.5 P2K Command Mode](#)

Test Procedures:

Issue the UTAG command to read the Bluetooth MAC Address.

Command	Opcode/Data	
UTAG	0x0108 00 000F	(Read BT MAC Address)

Convert the data return from hex values to ASCII characters.

Where:

The response format is two bytes indicating the number of bytes in the MAC Address followed by the actual MAC Address value.

Example:

Return data = 0011 30303A31313A32323A33333A34343A3535

MAC Address = 30303A31313A32323A33333A34343A3535 (hex values)

MAC Address = 00:11:22:33:44:55 (ASCII characters)

**11.62 UTAG_Verify_Bootmode**

This procedure read the radio's boot mode status.

Test Dependencies:

[3.6.5 P2K Command Mode](#)

Test Procedures:

Issue the UTAG command to read the radio's boot mode status.

<u>Command</u>	<u>Opcode/Data</u>	
UTAG	0x0108 00 0009	(Read Boot Mode status)

Where Bootmode Status:

0x0000 = No data (Expected if fastboot oem config unset bootmode issued)

0x0006286E756C6C29 = (null) (Expected if UTAG 0x0108 01 0009 0000 issued)

0x0007666163746F7279 = factory

0x000862702D746F6F6C73 = bp-tools

0x000471636F6D = qcom

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11.63 UTAG_Verify_Datecode

This procedure reads the date code from UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to read the date code.

Command	Opcode/Data	
UTAG	0x0108 00 000C	(Read date code)

Convert the data return from hex values to ASCII characters.

Where:

The response format is two bytes indicating the number of bytes in the date code followed by the actual date code.

Example:

Return data = 000A 30312D30312D31393730

Date code = 30312D30312D31393730 (hex values)

Date code = 01-01-1970 (ASCII characters)



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11.64 UTAG_Verify_ICCID

This procedure reads the SIM ID (Integrated Circuit Card ID [ICCID]) from UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to read the ICCID.

Command	Opcode/Data	
UTAG	0x0108 00 000D	(Read ICCID)

Convert the data return from hex values to ASCII characters.

Where:

The response format is two bytes indicating the number of bytes in the ICCID followed by the actual ICCID.

Example:

Return data = 0014 3839303135383038303030303030313232323839

ICCID = 3839303135383038303030303030313232323839 (hex values)

ICCID = 89015808000000122289 (ASCII characters)

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11.65 UTAG_Verify_IMEI

This procedure reads the IMEI from UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Data:

The IMEI data is a 15-digit number that includes LUHN checksum digit.

Test Procedures:

Issue the UTAG command to read the IMEI.

Command	Opcode/Data	
UTAG	0x0108 00 000A	(Read IMEI)

Convert the data return from hex values to ASCII characters.

Where:

The response format is two bytes indicating the number of bytes in the IMEI followed by the actual IMEI value.

Example:

Return data = 000F 333533323136303530303134373136

IMEI = 333533323136303530303134373136 (hex values)

IMEI = 353216050014716 (converted to ASCII characters)

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11.66 UTAG_Verify_MD5

This procedure reads the MD5 customization data checksum from UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to read the MD5.

Command	Opcode/Data	
UTAG	0x0108 00 000E	(Read MD5)

Convert the data return from hex values to ASCII characters.

Where:

The response format is two bytes indicating the number of bytes in MD5 followed by the actual MD5.

Example:

Return data = 0020 4434314438434439384630304232303445393830303939384543463834323745

MD5 = 4434314438434439384630304232303445393830303939384543463834323745 (hex values)

MD5 = D41D8CD98F00B204E9800998ECF8427E (ASCII characters)

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11.67 UTAG_Verify_MEID

This procedure reads the MEID from UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Data:

The MEID data is a 14-digit number and does not include a checksum digit.

Test Procedures:

Issue the UTAG command to read the MEID.

Command	Opcode/Data	
UTAG	0x0108 00 000A	(Read MEID)

Convert the data return from hex values to ASCII characters.

Where:

The response format is two bytes indicating the number of bytes in the MEID followed by the actual MEID value.

Example:

Return data = 000E 393930303030313139353939373235

MEID = 393930303030313139353939373235 (hex values)

MEID = 99000119599725 (converted to ASCII characters)

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11.68 UTAG_Verify_Multiple_WLAN

This procedure reads the WLAN MAC Address hex representation of ASCII string from UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to read the WLAN MAC Address

Command	Opcode/Data	
UTAG	0x0108 00 0010	(Read WLAN MAC Address)

Convert the data return from hex values to ASCII characters.

Where:

The response format is two bytes indicating the number of bytes in the string followed by the WLAN MAC Address.

Example:

Return data = 0011 30303A31313A32323A33333A34343A3535

MAC Address = 30303A31313A32323A33333A34343A3535 (hex values)

MAC Address = 00:11:22:33:44:55 (ASCII characters)

If more than one mac address returned it will be separated with a comma “,”

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11.69 UTAG_Verify_MultiSIM

This procedure reads the number of supported SIMs from UTAG memory partition.

Test Dependencies:

[3.6.5 P2K Command Mode](#)

Test Procedures:

Issue the UTAG command to read the Multi SIM UTAG value.

Command	Opcode/Data	
UTAG	0x0108 00 0014	(Read Multi SIM UTAG)

Convert the data return from hex values to ASCII characters.

Where:

The response format is two bytes indicating the number of bytes in the Multi SIM UTAG followed by the actual value.

Example:

Return data = 0001 31

Multi-SIM value = 31 (hex values)

Multi-SIM value = 1 (ASCII characters)

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11.70 UTAG_Verify_SKU

This procedure reads the SKU from UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to read the SKU.

Command	Opcode/Data	
UTAG	0x0108 00 000B	(Read SKU)

Convert the data return from hex values to ASCII characters.

Where:

The response format is two bytes indicating the number of bytes in the SKU followed by the actual SKU.

Example:

Return data = 0008 58464F4E20415454

SKU = 58464F4E20415454 (hex values)

SKU = XFON ATT (ASCII characters)

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11.71 UTAG_Verify_TRACK_ID

This procedure reads the Track ID from UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to read the Track ID.

Command	Opcode/Data	
UTAG	0x0108 00 0001	(Read Track ID)

Convert the data return from hex values to ASCII characters.

Where:

The response format is two bytes indicating the number of bytes in the Track ID followed by the actual Track ID value.

Example:

Return data = 000A 4C554D45314530303435

Track ID = 4C554D45314530303435 (hex values)

Track ID = LUME1E0045 (ASCII characters)



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11.72 UTAG_Write_Battery_ID

This procedure programs the battery ID into UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to program battery ID.

Command	Opcode/Data	
UTAG	0x0108 01 0012 + length + Battery ID	(Program battery ID)

Where:

Battery ID = variable data

Encoded battery ID is the variable byte ASCII to Hex value of the data that needs to be written.

Example:

Battery ID = SNN5957A

Encoded battery ID = 534E4E3539353741

If SNN5957A needs to be written, send 0x0108 01 0012 0008 534E4E3539353741

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11.73 UTAG_Write_Bluetooth

This procedure writes the BT MAC Address hex representation of ASCII string to UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to write the BT MAC Address

Command	Opcode/Data	
UTAG	0x0108 01 000F 0011 + BT_MAC_ADDRESS	(Write BT MAC Address)

Where:

0011 = Number of bytes to be written

BT_MAC_ADDRESS = Alphanumeric BT MAC ADDRESS converted to hexadecimal

Note: For BT_MAC_ADDRESS include the colon separators.

Example:

If 00:11:22:33:44:55 needs to be written, send 0x0108 01 000F 0011 30303A31313A32323A33333A34343A3535

**11.74 UTAG_Write_Bootmode_Normal**

This procedure sets radio's boot mode to out of factory status.

Test Dependencies:

[3.6.5 P2K Command Mode](#)

Test Procedures:

Issue the UTAG command to program boot mode to out of factory.

<u>Command</u>	<u>Opcode/Data</u>	
UTAG	0x0108 01 0009 0000	(Program Boot Mode)

Where:

0x0000 = Write zero bytes of data (NULL)

For alternate Fastboot method refer to Section 11.16

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11.75 UTAG_Write_Datecode

This procedure programs the date code into UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to program date code.

Command	Opcode/Data	
UTAG	0x0108 01 000C + length (2byte) + DATECODE	(Program date code)

Where:

DATECODE = dddddddddddddddddd (10 byte data)

Encoded DATECODE is the 10 byte ASCII to Hex value of the DATECODE data that needs to be written.

Example:

DATECODE = 01-01-1970

Encoded DATECODE = 30312D30312D31393730

If 01-01-1970 needs to be written, send 0x0108 01 000C 000A 30312D30312D31393730

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11.76 UTAG_Write_ICCID

This procedure programs the SIM ID (Integrated Circuit Card ID [ICCID]) into UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to program ICCID.

Command	Opcode/Data	
UTAG	0x0108 01 000D + length (2byte) + ICCID	(Program ICCID)

Where:

ICCID = dddddd (20 byte data)

Encoded ICCID is the 20 byte ASCII to Hex value of the ICCID data that needs to be written.

Example:

ICCID = 89015808000000122289

Encoded ICCID = 38393031353830383030303030313232323839

If 89015808000000122289 needs to be written;

send 0x0108 01 000D 0014 38393031353830383030303030313232323839

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11.77 UTAG_Write_IMEI

This procedure programs IMEI into UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to program IMEI.

Command	Opcode/Action	
UTAG	0x0108 01 000A 000F + IMEI	(Program IMEI)

Where:

IMEI = dddddddddddddd (15-digit number)

Encoded_IMEI is the 15 byte ASCII to Hex value of the IMEI Number

Example:

IMEI = 353216050014716

Encoded_IMEI = 333533323136303530303134373136

If **353216050014716** needs to be written, send 0x0108 01 000A 000F 333533323136303530303134373136

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11.78 UTAG_Write_MD5

This procedure programs the MD5 customization data checksum into UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to program MD5.

Command	Opcode/Data	
UTAG	0x0108 01 000E + length (2byte) + MD5	(Program MD5)

Where:

MD5 = ddddddddddddddddddddddddddddddddddd (16 byte data)

Encoded MD5 is the 32 byte ASCII to Hex value of the MD5 data that needs to be written.

Example:

MD5 = D41D8CD98F00B204E9800998ECF8427E

Encoded MD5 = 4434314438434439384630304232303445393830303939384543463834323745

If D41D8CD98F00B204E9800998ECF8427E needs to be written;

send 0x0108 01 000E 0020 4434314438434439384630304232303445393830303939384543463834323745

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11.79 UTAG_Write_MEID

This procedure programs MEID into UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to program MEID.

Command	Opcode/Action	
UTAG	0x0108 01 000A 000E + MEID	(Program MEID)

Where:

MEID = dddddddddddddd (14-digit hexadecimal number)

Encoded_MEID is the 14 byte ASCII to Hex value of the MEID Number

Example:

MEID = 99000119599725

Encoded_MEID = 3939303030313139353939373235

If 99000119599725 needs to be written, send 0x0108 01 000A 000F 3939303030313139353939373235

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11.80 UTAG_Write_Mutiple_WLAN

This procedure writes the WLAN MAC Address hex representation of ASCII string to UTAG memory partition

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to write the WLAN MAC Address

Command	Opcode/Data	
UTAG	0x0108 01 0010 00xx + WLAN_MAC_ADDRESS	(Write WLAN MAC Address)

Where:

00xx = Number of bytes to be written

WLAN_MAC_ADDRESS = Alphanumeric WLAN MAC ADDRESS converted to hexadecimal

Note: For multiple WLAN_MAC_ADDRESS include the comma "," separator.

Programming 2 WLAN MAC addresses

WLAN MAC1 address – STA (station) Mode

WLAN MAC2 address – WiFi Direct Mode

Example:

If 10:11:22:33:44:55,20:11:22:33:44:55 needs to be written, send 0x0108 01 0010 0023

30303A31313A32323A33333A34343A35352C32303A31313A32323A33333A34343A3535

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11.81 UTAG_Write_MultiSIM

This procedure writes the number of supported SIMs from UTAG memory partition.

Test Dependencies:

[3.6.5 P2K Command Mode](#)

Test Procedures:

Issue the UTAG command to write the Multi-SIM UTAG Value

Command	Opcode/Data
UTAG	0x0108 01 0014 0001 + Multi-SIM UTAG Value

Where:

0001 = Number of bytes to be written

Multi-SIM UTAG Value = Alphanumeric Multi-SIM UTAG Value converted to hexadecimal

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11.82 UTAG_Write_SKU

This procedure programs the SKU into UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to program SKU.

Command	Opcode/Data	
UTAG	0x0108 01 000B + length (2byte) + SKU	(Program SKU)

Where:

SKU = dddddddddddddd (variable data)

Encoded SKU is the variable byte ASCII to Hex value of the SKU data that needs to be written.

Example:

SKU = XFON ATT

Encoded SKU = 58464F4E20415454

If XFON ATT needs to be written, send 0x0108 01 000B 0008 58464F4E20415454

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11.83 UTAG_Write_Track_ID

This procedure programs the Track ID into UTAG memory partition.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Issue the UTAG command to program Track ID.

Command	Opcode/Action
UTAG	0x0108 01 0001 000A + TRACK_ID (Program Track ID)

Where:

TRACK_ID = 10 digit alphanumeric converted to hexadecimal

Example:

If **LUME1E0045** needs to be written, send 0x0108 01 0001 000A 4C554D45314530303435

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11.84 Validate_HOB

This procedure will check that the RF data in the NV Table exactly matches the data in the HOB.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Test Procedures:

Issue the following HOB command to validate HOB

<u>Command</u>	<u>Opcode/data</u>	
HOB	0x0D07 04	(Verify HOB Restore)

Verify that returned data is 0x0000.

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11.85 Verify_Battery_Gauge

This procedure verifies the battery capacity.

Test Dependencies:

- Radio must be in P2K Mode. (Sec 3.6.5)
- Radio must be in Suspend mode. (Sec 3.6.6)

Test Procedures:

Send the following test command to read gas gauge and check battery capacity.

Command	Opcode/Action	
BATT	0x0085 040000	(Read Capacity in percentage)

Ensure a successful response

4 bytes of data are returned as a percentage value.

Convert the 4 bytes of response data to decimal to get percentage value.

Ensure battery voltage level is between 40% and 100% as specified in section [2.1 Test Limits](#).

Example:

Below is an example of the response data.

Response data in hex: 8223008500060000000000040000002F

The last four bytes here are 0000002F, whose decimal value is 47 which is the battery charge voltage value in percentage format.

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11.86 VERIFY_BP_RTC_FTM

This procedure reads the radio's system time.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

SET_BP_RTC_FTM (Sec 11.45)

Test Procedures:

Send the following FTM RAW DATA command to read the radio's system time.

Command	Opcode/data
FTM_RAW_DATA*	0x4B 0B0B 0000 0000 0000 0000 0102

Ensure a successful response.

Read and parse the response data. Compare the MONTH, YEAR, and HOUR to the expected value.

Where response:

Byte 14 **aa** = Month

Byte 15 **bb** = Day (must subtract 5 days from **response**)

Bytes 16, 17 **cccc** = Year: 2 bytes **little-endian**; Example: 0xE7 0x07 -> 0x07E7 = 2023 (must subtract 10 years from response)

Byte 18 **dd** = Hour (24hr format)

Byte 19 **ee** = Minute

Byte 20 **ff** = Second

**Note: The RTC time is offset by 10 years and 5 days from the radio's system time. So when reading it back out subtract 10yrs and 5 days to get the radio's system time.*

Example of received data: 0x4B0B0B000010000000000000000002 0A06 E707 00 00 03 0000723B7E

Where:

0x0A = Month of October

0x06 = Day 1st (0x06 – 5 days = 1st day)

0xE707 = Year 2013 (2023 – 10 years = 2013)

0x00 = Hour 0

0x00 = Minute 0

0x03 = Second 3

This example results in the RTC being read as 1st Oct 2013 midnight (10/01/2013 00:00:03).

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11.87 Verify_DHOB

This procedure will verify the DHOB.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Test Procedures:

Issue the following FTM raw command to verify DHOB

Command	Opcode/data	
FTM_RAW_DATA	0x4B 0B00 8040 4005	(Verify DHOB)

Ensure the response matches the expected string.

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11.88 Verify_FSG_Version

This Procedure reads the FSG version out of the Radio and compares with the expected version.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Test Procedures:

Use the RDELEM command to retrieve the FSG data from SEEM.

Command	Opcode/data	
RDELEM	0x0020 1F6D 0001 0000 0050	(Read FSG Version)

Convert the data return from hex values to ASCII characters.

Compare the character string to the expected version string of the FSG.

11.89 Verify_IMEI_FTM

This procedure read the IMEI from the radio and compares it to the expected value.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Data:

The IMEI data bytes are determined as follows:

Byte1: 0x08 → Upper nibble = '0', Lower nibble = '8'
 Byte2: Upper nibble = TAC 1, Lower nibble = 'A'
 Byte3: Upper nibble = TAC 3, Lower nibble = TAC 2
 Byte4: Upper nibble = TAC 5, Lower nibble = TAC 4
 Byte5: Upper nibble = FAC 1, Lower nibble = TAC 6
 Byte6: Upper nibble = Serial 1, Lower nibble = FAC 2
 Byte7: Upper nibble = Serial 3, Lower nibble = Serial 2
 Byte8: Upper nibble = Serial 5, Lower nibble = Serial 4
 Byte9: Upper nibble = '0' (LUHN Digit), Lower nibble = Serial 6

Definitions:

FAC – Factory Code

LUHN – Check digit for the IMEI

TAC – Type Approval Code

Test Procedures:

Issue the NV Read (0X26) command with data 0x2602 + 128 bytes of 0x00.

Command	Opcode/data
---------	-------------

[illegible]

The response will be of the form 0x262602 + 128 bytes of data + 2 bytes of status.

If the 2 bytes of status are 0x0500,

The IMEI has not yet been programmed and the data is not valid.

If the 2 bytes of status are 0x0000,

The IMEI is the first 9 bytes of the data.

Byte swap the returned data to get the correct IMEI.



11.90 Verify_MBM_Version

This Procedure reads the version of the MBM in the radio and compares with the expected version.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Test Procedures:

Issue the VERSION (0x039) command with data 0x1200.

Compare the character string to the expected version string of the MBM.

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11.95 Verify_Phone_Clean

This procedure verifies that the phone has no personalized content.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Send the following test command to verify that there is no customized data.

Command	Opcode/Action
CUSTOMIZE	0x 010C 03

(Verify Clean)

Ensure the response is 0x00, otherwise fail unit.

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11.96 Verify_Phone_System_Time_APK

This procedure reads the radio's system time.

Test Dependencies:

Radio must be turned on in factory mode

Test Procedures:

Use the System Time Activity command to read the radio's system time.

<u>Command</u>	<u>Activity</u>	
START	SystemTime	(Start the SystemTime test activity)

<u>Command</u>	<u>Activity</u>	<u>Data</u>	
TELL	SystemTime	GET_SYSTEMTIME	(Read radio's system time)

Ensure a successful response

Read the returned response and compare the YEAR, MONTH, and HOUR to the expected value.

Where response:

YEAR=<year>

MONTH=<month>

DAY=<day>

HOUR=<hour>

MINUTE=<minute>

SECOND=<second>

AM_PM=<am or pm>

Send the following command to stop the System Time Activity.

<u>Command</u>	<u>Activity</u>	
STOP	SystemTime	(Stop the system time activity)

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11.98 Verify_Ready_To_Ship_Status

This procedure verifies that the eFuse contents on the radio are programmed.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Test Procedures:

Use the EFUSE command to verify ready to ship Efuse status.

Command	Opcode/Data
EFUSE	0x0C2E 01

Verify that the return data is 0x01.

Possible Return data

0x00 – NOT Ready to Ship

0x01 – Is Ready to Ship

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11.99 Verify_SBK

This procedure will verify the SBK status of the UUT and validate the key pair.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Test Procedures:

Use the PKI_DATA_PROVISION command to request the status of the UUT and validate key pair.

Command Opcode/data

PKI_DATA_PROVISION 0x006A 0201 00EB (PKI Verify Key Pair)

Verify Key Pair Status:

Byte [0-15]: 1 byte status per PKI Data Type

0x00 = Valid Key Installed

0x01 = Invalid Key Pair

0x02 = Unsupported PKI type

0xFF = Unused PKI types

Ensure that byte 0 of the response is 0x00 representing success.

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11.100 Verify_Setup_Wizard_APK

This test verifies that Setup Wizard has been completed at CFC.

Test Dependencies:

11.10 FASTBOOT_FLASH

11.18 FULL_POWER_ON_CHECK (add 25s delay if APK with GET_SETUPWIZARD_STATUS is missing)

Radio must be turned on in factory mode

UUT must have the CommServer running and communication establish over BLAN Port 2631.

Test Procedures:

Send the following command to run the Setup Wizard Status test.

Command	Activity	data
START	SetupWizard	
TELL	SetupWizard	GET_SETUPWIZARD_STATUS

The response back is Setup Wizard status. Record the value and compare to the passing string: COMPLETED.

Send the following command to stop the Setup Wizard Activity.

Command	Activity	data
STOP	SetupWizard	

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11.101 Verify_Subsidy_Lock_Active

This procedure verifies that the subsidy lock is active.

Test Dependencies:

Radio must be in FTM Mode. (Sec 3.6.1)

Test Procedures:

Send the following FTM RAW DATA command to register a client ID.

<u>Command</u>	<u>Opcode/Data</u>	
FTM_RAW_DATA	0x8021 00FA 00000000	(Register a client ID)

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 16,17,18,19 from the second response. Make this the 4 byte **Client ID HIGH**.

Send the following FTM RAW DATA Command to use to open a session.

<u>Command</u>	<u>Opcode/Data</u>
FTM_RAW_DATA	0x8021 02FA00000000 + Four byte Client ID HIGH + 00000000 + 12bytes of 0x00

Example of sent data: 0x802102FA 00000000 **748A3629** 00000000 00000000 00000000 00000000

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Extract bytes 20,21,22,23 from the second response. Make this the 4 byte **Session ID LOW**.

Send the following FTM RAW DATA Command to check if subsidy lock is active.

<u>Command</u>	<u>Opcode/Data</u>
FTM_RAW_DATA	0x8021 17FA 00000000 + Four byte Session ID LOW + 00000000

Example of sent data: 0x8021 17FA00000000 **993C1C16** 00000000

Read the first FTM command response.

Poll the response buffer and wait up to 10 seconds for the second response. Check byte 16 from the second response. This is the radio's Network Indicator Status.

Ensure the response is 0x01, otherwise fail unit.

Where NW_IND_STATUS:

0x00 = inactive

0x01 = active

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11.102 Verify_SW_Version

This Procedure reads the software version out of the Unit under Test and compares with the expected version.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Test Procedures:

Use the VERSION command to read the software version from the UE.

<u>Command</u>	<u>Opcode/data</u>	
VERSION	0x0039 FFFF	(Read release label)

Convert the data return from hex values to ASCII characters and remove any non-alphanumeric characters.
 Compare the character string to the expected version string of the flex.

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11.103 Verify_Track_ID

This procedure verifies the factory tracking number stored in the radio with the expected value from business system.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Test Procedures:

Use the RDELEM command to read the unit Track ID.

Command	Opcode/data
RDELEM	0X0020 2726000100000080

Convert the data read from the radio from hex values to ASCII characters
Compare the number string to the expected number string of the Track ID.

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11.104 Verify_Values_Through_APK

This procedure verifies the BP version and IMEI or MEID values (as applicable) programmed into radio and compares them with the expected values from business system.

Test Dependencies:

Radio must be turned on in factory mode

Test Procedures:

Use the VERSION Activity command to read the software version from the UE.

<u>Command</u>	<u>Activity</u>	
START	VERSION	(start the Version test activity)

<u>Command</u>	<u>Activity</u>	<u>Data</u>	<u>Data</u>	
TELL	VERSION	GET_VERSION	BP_VERSION DEVICE_ID	(Read the Version info)

Read the VERSION responses from the UE.

<u>Command</u>	<u>Activity</u>	
STOP	VERSION	(stop Version test activity)

Verify that the reported BP version and IMEI or MEID (as applicable) values are correct for the current model and build profile.

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11.105 Verify_XCVR

This procedure reads the model from the factory information element and compares it to the expected Model.Care should be taken to determine the data format of the information in this element, so that the correct information is compared.

Test Dependencies:

Radio must be in P2K Mode. (Sec 3.6.5)

Radio must be in Suspend mode. (Sec 3.6.6)

Test Procedures:

Use the RDELEM command to retrieve the factory information.

Command	Opcode/data
RDELEM	0X0020 2726000100000080

From the 2 bytes of data starting at byte 125, determine if the format is any of the defined versions.
If the versions are defined,

Read the data in the first 10 bytes as the ASCII representation of the radio model

Else

Read the data in the 12 bytes starting at the 9th byte in the string as the ASCII representation of the radio model
Compare the model string to the expected model string.

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12.0 Appendices

12.1 Appendix A: NV Item Description

Due to changing NV item descriptions and not wanting to duplicate documentation, please see Select RadioOne RF NV Items document 80-V3722-1 for a complete description of all NV items.

For a copy of the latest Select RadioOne RF NV Items document 80-V3722, please visit Qualcomm site at the following location:

URL: <https://downloads.cdmatech.com>



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12.2 Appendix B: GSM Band Divisions

The table below defines how the operating channels are divided for phasing compensation across the frequency bands.

Band Divisions			
Band	GSM900	DCS1800	PCS1900
Division 1	975-1009	512-586	512-571
Division 2	1010-1023, 0-20	587-661	572-631
Division 3	21-55	662-736	632-691
Division 4	56-90	737-811	692-751
Division 5	91-124	812-885	752-810

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12.3 Appendix C: Formulas

Fractional number to 7.8 format

This is a description of the conversion of a fractional (floating-point) number to signed 7.8 format hexadecimal number (2's complement with 2^{-8} precision).

Pseudocode:

$S7.8 = \text{INT16} (n \times 256)$ where: n is a floating-point number ($256=2^8$)

Calculator conversion method:

$S7.8 = \text{HEX16} (\text{ROUND} ((n \times 256) - 65536))$ where: n is a floating-point number

Description:

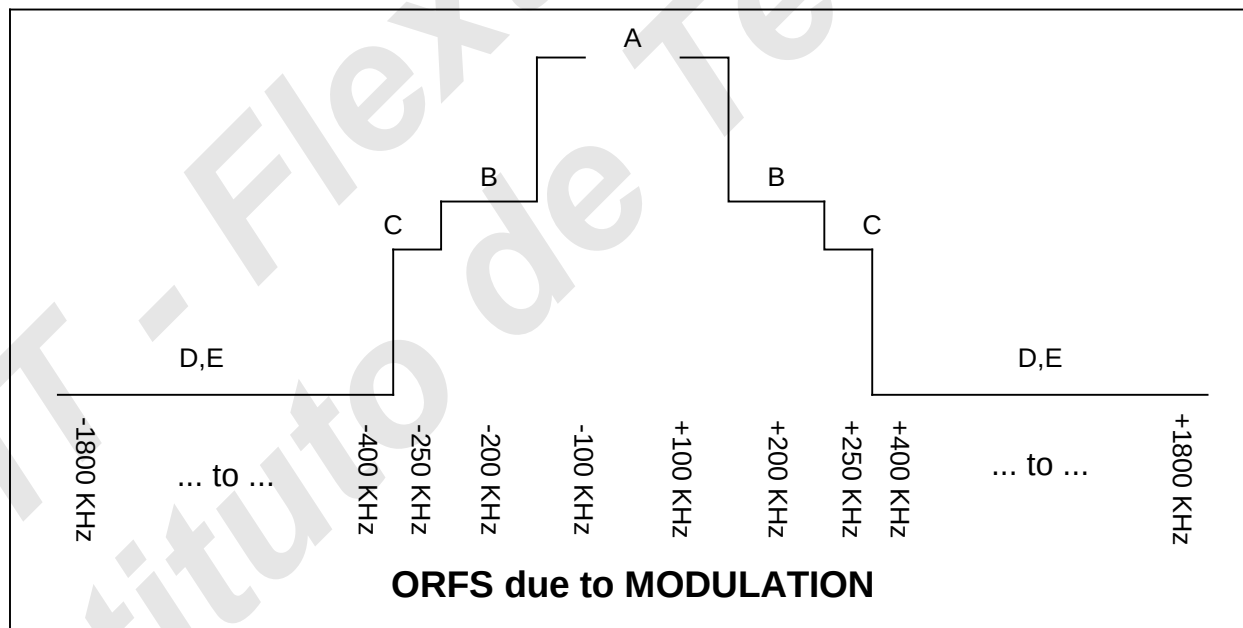
- 1.) Multiply the number by 256.
- 2.) Subtract 65536.
- 3.) Round to the nearest integer and convert to hexadecimal.
- 4.) Truncate to 16 bits.

12.4 Appendix D: GSM Output RF Spectrum Requirements

OUTPUT RF Spectrum due to Modulation Limits						
Band		A	B	C	D	E
		+/- 100 kHz	+/- 200 KHz	+/- 250 kHz	+/- 400 kHz	+/- 600 – 1800 kHz
GSM 400, 700, 850, 900	Modulation Spec	+5 dBc	-30 dBc	-33 dBc	-60 dBc	-60 dBc
	Max Level Spec	-36 dBm	-36 dBm	-36 dBm	-36 dBm	-51 dBm*
DCS 1800 / PCS1900	Modulation Spec	+5 dBc	-30 dBc	-33 dBc	-60 dBc	-60 dBc
	Max Level Spec	-36 dBm	-36 dBm	-36 dBm	-36 dBm	-56 dBm*

*Exceptions: Up to 3 measurements in the frequency range ≥ 600 kHz above and below the carrier may measure up to, but may not exceed, -36 dBm.

Note: The on-channel power measurement must be done with RBW = 30 kHz.



Appendix D: GSM Output RF Spectrum Requirements (Continued)

GSM 900 RF Spectrum due to Switching Transients				
Power Level	Maximum Level for Various Offsets from Carrier Frequency			
dBm	400 KHz	600KHz	1200 KHz	1800 KHz
33	-19 dBm	-21 dBm	-21 dBm	-24 dBm
31	-21 dBm	-23 dBm	-23 dBm	-26 dBm
29	-23 dBm	-25 dBm	-25 dBm	-28 dBm
27	-23 dBm	-26 dBm	-27 dBm	-30 dBm
25	-23 dBm	-26 dBm	-29 dBm	-32 dBm
23	-23 dBm	-26 dBm	-31 dBm	-34 dBm
<=21	-23 dBm	-26 dBm	-32 dBm	-36 dBm

DCS 1800, PCS 1900 RF Spectrum due to Switching Transients				
Power Level	Maximum Level for Various Offsets from Carrier Frequency			
dBm	400 KHz	600KHz	1200 KHz	1800 KHz
30	-22 dBm	-24 dBm	-24 dBm	-27 dBm
28	-23 dBm	-25 dBm	-26 dBm	-29 dBm
26	-23 dBm	-26 dBm	-28 dBm	-31 dBm
24	-23 dBm	-26 dBm	-30 dBm	-33 dBm
22	-23 dBm	-26 dBm	-31 dBm	-35 dBm
<=20	-23 dBm	-26 dBm	-32 dBm	-36 dBm

Note: All measurements must be done using a gated measurement technique such that the spectrum is measured only while the uplink carrier is on and only between bits 87 to 132 of the burst.

The detailed requirements of the ORFS measurement may be found in 3GPP TS 51.010-1 V5.0.0.

(Continued on next page)

Appendix D: GSM Output RF Spectrum Requirements (Continued)
Fail Code Format

The following table defines the format of the reported value for the ORFS measurement. Map each of the failed frequency offsets to a bit in the 4-byte failure code as shown in the table below. Set bit 22 if the measurements exceed the requirements for ORFS due to modulation.

Bit	Freq Offset (kHz)	Pass/Fail	Modulation / Transient Measurement
30 (msb)	- 1800	0/1	T
29	- 1200	0/1	T
28	- 600	0/1	T
27	- 400	0/1	T
26	+ 400	0/1	T
25	+ 600	0/1	T
24	+ 1200	0/1	T
23	+ 1800	0/1	T
22		PASS/FAIL Flag for Modulation	M
21	-1800	0/1	M
20	-1600	0/1	M
19	-1400	0/1	M
18	-1200	0/1	M
17	-1000	0/1	M
16	- 800	0/1	M
15	- 600	0/1	M
14	- 400	0/1	M
13	- 250	0/1	M
12	- 200	0/1	M
11	- 100	0/1	M
10	+ 100	0/1	M
9	+ 200	0/1	M
8	+ 250	0/1	M
7	+ 400	0/1	M
6	+ 600	0/1	M
5	+ 800	0/1	M
4	+ 1000	0/1	M
3	+ 1200	0/1	M
2	+ 1400	0/1	M
1	+ 1600	0/1	M
0 (lsb)	+1800	0/1	M

**12.5 Appendix E: Time Mask Parameters**

Time Mask Measurement points:

<u>Time (uS)</u>	<u>Lower Limit</u>	<u>Upper Limit</u>
-28	-999	-59***
-18	-999	-30**
-16	-999	-6*
-14	-999	-6*
-12	-999	-6*
-10	-999	-6*
-8	-999	+4
-6	-999	+4
-2	-999	+4
0	-1	+1
542.8	-1	+1
552.8	-999	-6*
560.8	-999	-30**
570.8	-999	-59***

*For GSM900 MS:

- 4 for power control level 16
- 2 for power control level 17
- 1 for power control levels 18 & 19

*For GSM1800/1800 MS:

- 4 for power control level 11
- 2 for power control level 12
- 1 for power control levels 13, 14 & 15

*For GSM1900/1900 MS:

- 4 for power control level 11
- 2 for power control level 12
- 1 for power control levels 13, 14 & 15

**For GSM850 and GSM900 MS:

- 30dBc or -17dBm, whichever is the higher

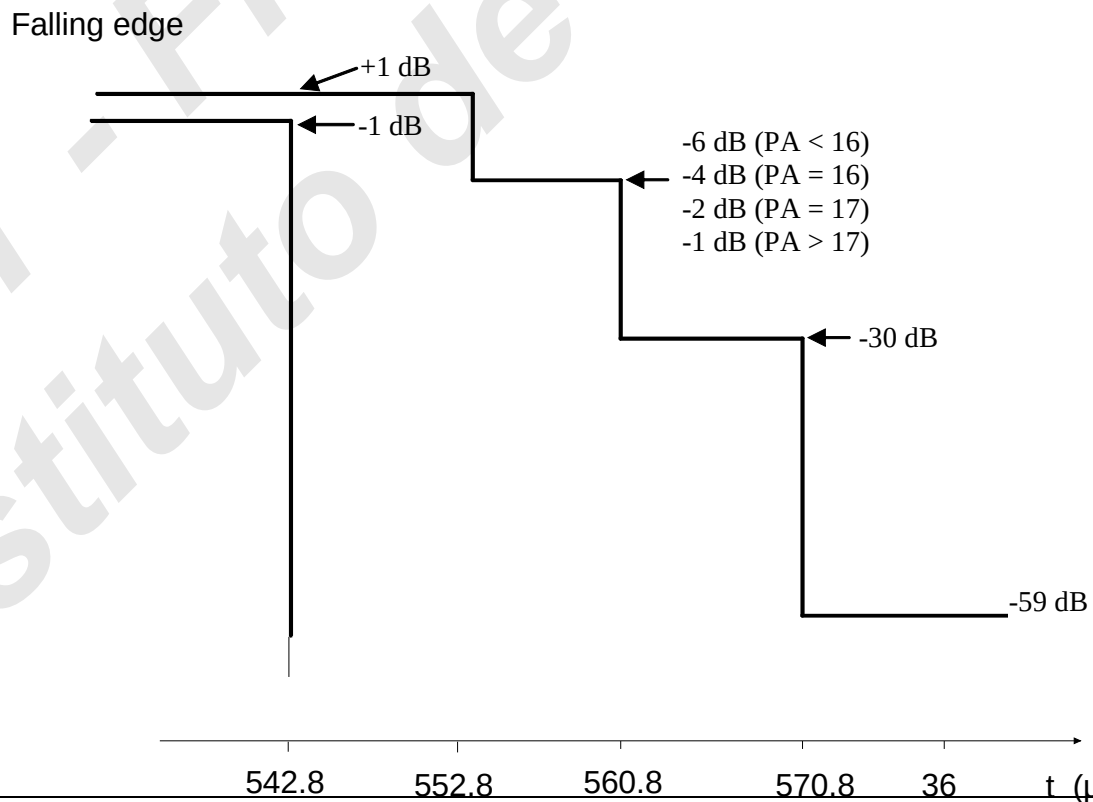
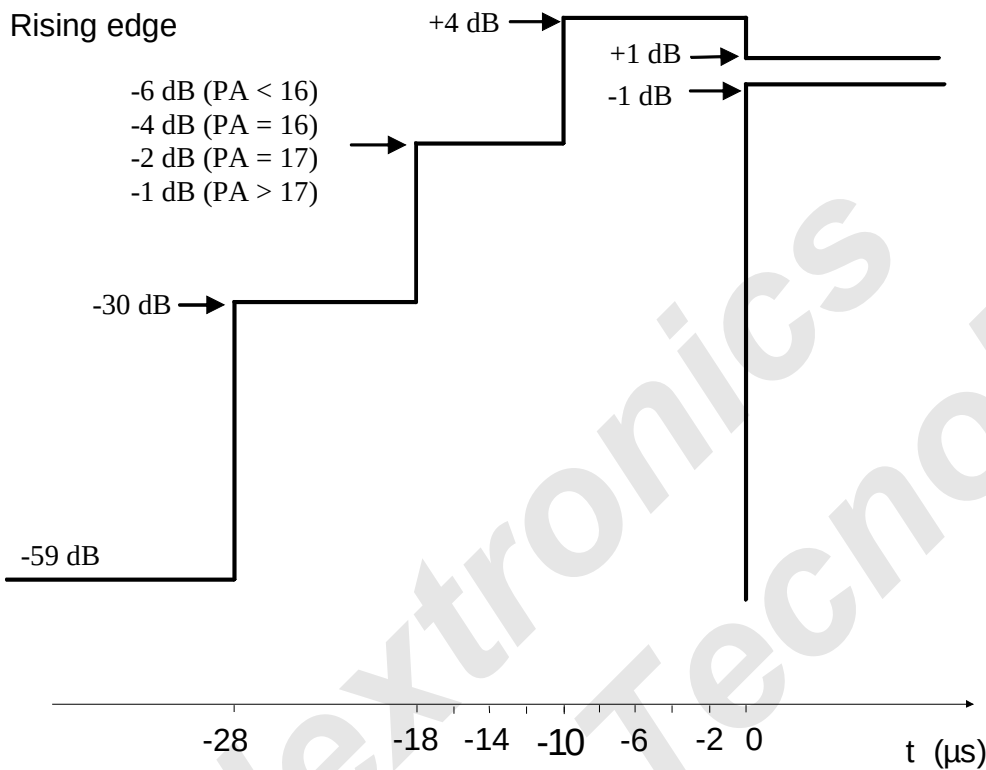
** For GSM1800 and GSM1900 1800 MS:

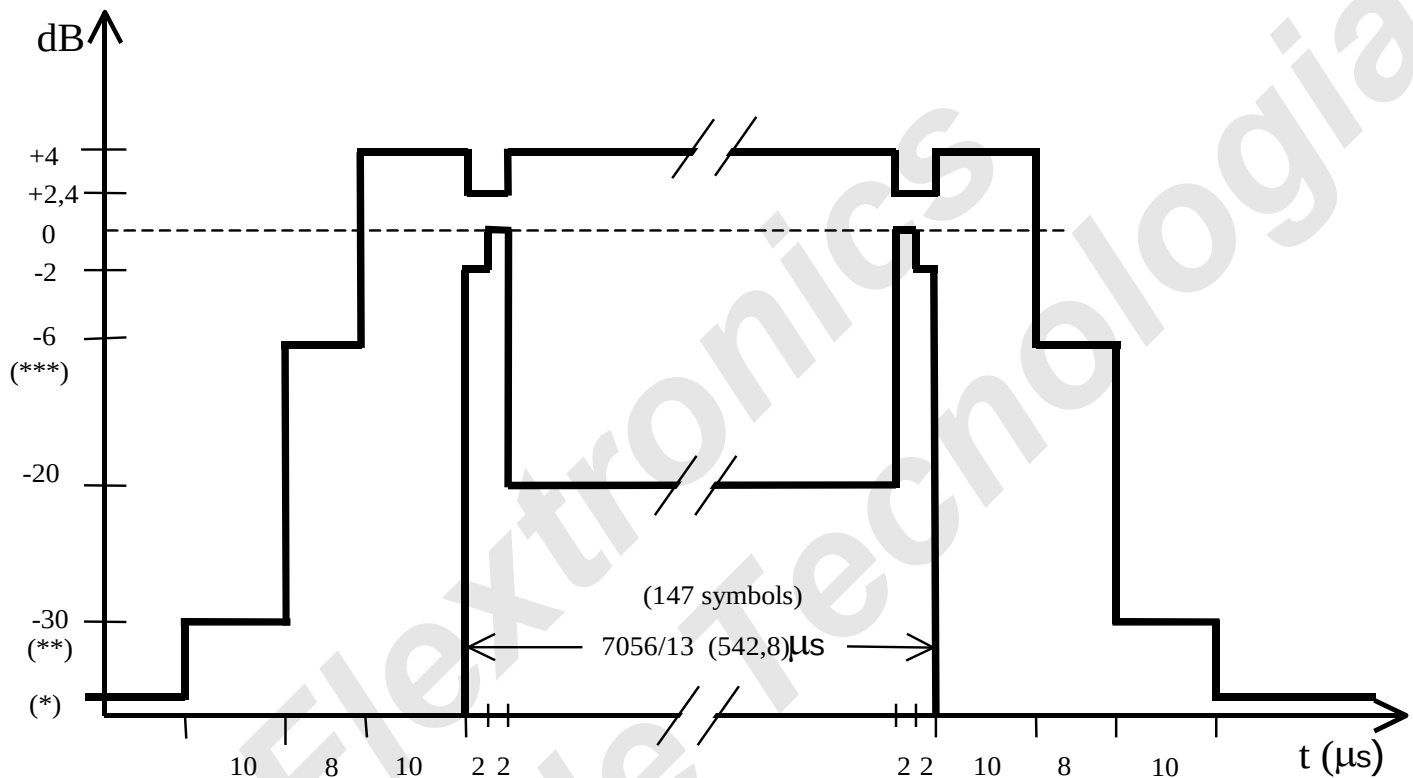
- 30dBc or -20dBm, whichever is the higher

	Lowest Limits
GSM850 and GSM900	-59dBc or -54dBm, whichever is the highest, except for the Timeslot preceding the active slot, for which the allowed level is -59dBc or -36dBm, whichever is the highest. Note: These limits are required for conducted testing only.
GSM1800 and GSM1900	-48dBc or -48dBm, whichever is the highest. Note: These limits are required for conducted testing only.

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8-PSK Modulation


(*) For GSM 400, GSM 850 and GSM 900 MS:

For GSM1800 and GSM1900 MS:

(**) For GSM 400, GSM 900 and GSM 850 MS:

For GSM1800 MS:

 (***) For GSM400, GSM850 and GSM900 MS:
 For GSM1800 MS

-59 dBc or -54 dBm, whichever is higher

NOTE: This is required for conducted testing only

-48 dBc or -48 dBm, whichever is higher.

NOTE: This is required for conducted testing only

-4 dBc for power control level 16;

-2 dBc for power level 17;

-1 dBc for power level controls levels 18 and 19.

-4dBc for power control level 11,

-2dBc for power level 12,

-1dBc for power control levels 13,14 and 15

-30 dBc or -17 dBm, whichever is higher.

-30dBc or -20dBm, whichever is higher.

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12.6 Appendix F: Camera Test Conditions

Ensure that the light level illuminating the Focus test chart is between 300 and 700 lux (target 500 lux) using light source of 3400°K to 7000°K. The light level is measured by having the back of the meter against the test chart. The Focus test chart is evenly lit with the maximum delta allowed is 200 lux from any spot on the chart to another spot on the chart with black and white blocks being evenly lit with less than 50 lux across each individual blocks.

For Blemish and Relative Uniformity Test testing, the diffused light source is the test target. The light source is 4100°K to 5500°K. The diffuser is an Opal Diffuser. The light level is between 400 lux to 1800 lux with light meter sensor touching opal diffuser. The light level illumination across the Opal Diffuser does not change more than 8% from maximum point to minimum point on the diffuser using a Digital Still Camera as a reference.

Materials needed:

Light box to provide controlled lighting for focus test chart
3400°K to 7000°K light source or equivalent for Focus test

Minolta CL-200 or equivalent precision light meter

Slanted SFR Focus test chart (Rotation, Tilt X, Tilt Y, FOV and Focus Tests)

Opal Diffuser Test Fixture (Blemish Test, Relative Illumination, Relative Uniformity, and Illumination Check)

Link to camera Charts: <https://drive.google.com/a/motorola.com/#folders/0B7eiu1wvlowwd1dzMVp6UmJ3QzQ>

Test Chart:

Image test	Test chart(s)	Chart distance	Replacement Frequency
Focus @ 10cm	SFR Black and White Chart	10cm	Every 3 mos.
Focus @ 35cm		35cm	
Focus @ 53cm		53cm	
Focus @ 60cm		60cm	
Blemish Relative Uniformity	Opal Diffuser Light Source	For opal diffuser 50mmx50mm <2.4cm For opal diffuser 100mm x 100mm or larger 2.4cm to 7.0cm	Verify Light Level Monthly

35cm and 60cm Focus SFR Test Chart Specs:

Black Ink Density 1.3 to 1.5

Dark Gray Ink Density 0.38 to 0.46

Light Gray Ink Density 0.16 to 0.24

Paper Brightness 85% to 91% using uncoated paper

10cm Focus SFR Test Chart Specs:

Black Ink Density 1.3 to 1.5
 Dark Gray Ink Density 0.68 to 0.80
 Light Gray Ink Density 0.46 to 0.58
 Paper Brightness 79% to 89%

Still Image Test Conditions:

Test	Light Source*	Light Intensity (Lux)	Test chart	Chart distance
Focus @ 10cm	3400°K to 7000°K	300-700 (Target=500)	SFR Black and White Chart	10cm
Focus @ 35cm				35cm
Focus @ 53cm				53cm
Focus @ 60cm				60cm
Blemish Relative Uniformity	4100°K to 5500°K Opal diffuser light	400 lux to 1800 lux ²	Opal Diffuser Light	For opal diffuser 50mmx50mm <2.4cm For opal diffuser 100mm x 100mm or larger 2.4cm to 7.0cm

Note 1: Focus light is measured by having the back of the meter against the test chart

Note 2: Blemish light is measured by having light meter sensor touching opal diffuser

Note 3: Both light intensity and light source color temperature needs to be validated.

The Region of Interest for the Focus Test is at 60% from the center of the Chart with 0 degrees of Tilt X and Tilt Y and 0 degrees of rotation. With Tilt X and Tilt Y not at 0 degrees and/or rotation not at 0 degrees, the center and corners will not be in the optimal locations.

The focus chart is mounted so that the longer edge of the chart matches the same orientation to the sensor array longer edge in the phone.

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Focus Test Black and White Chart

Chart Size: 14cm x 11cm placed at a distance of 10cm from the image sensor.

Chart Size: 50cm x 38cm placed at a distance of 35cm from image sensor.

Chart Size: 81cm x 64cm placed at a distance of 53cm or 60cm from image sensor.

Chart Picture: (for reference only)

Focus Chart @ 10cm





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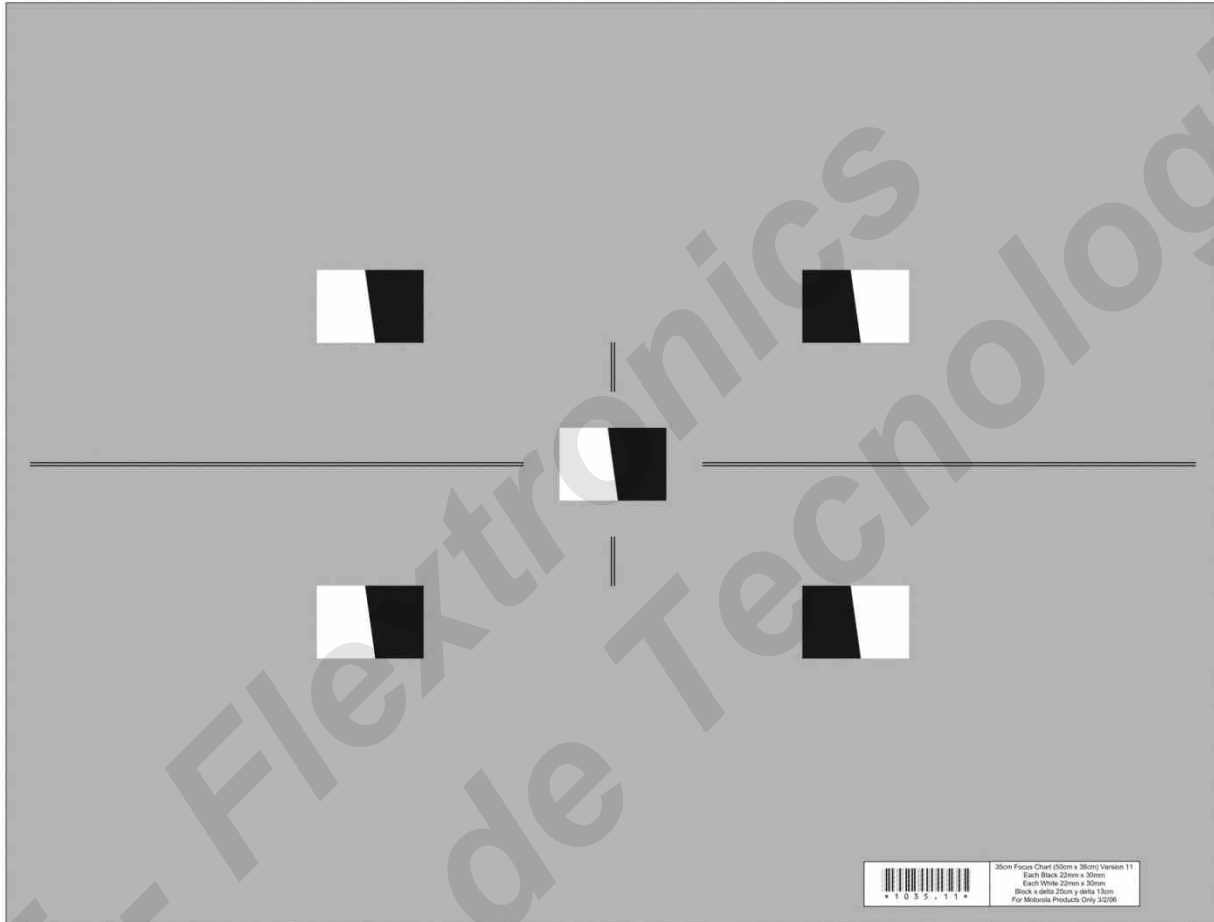
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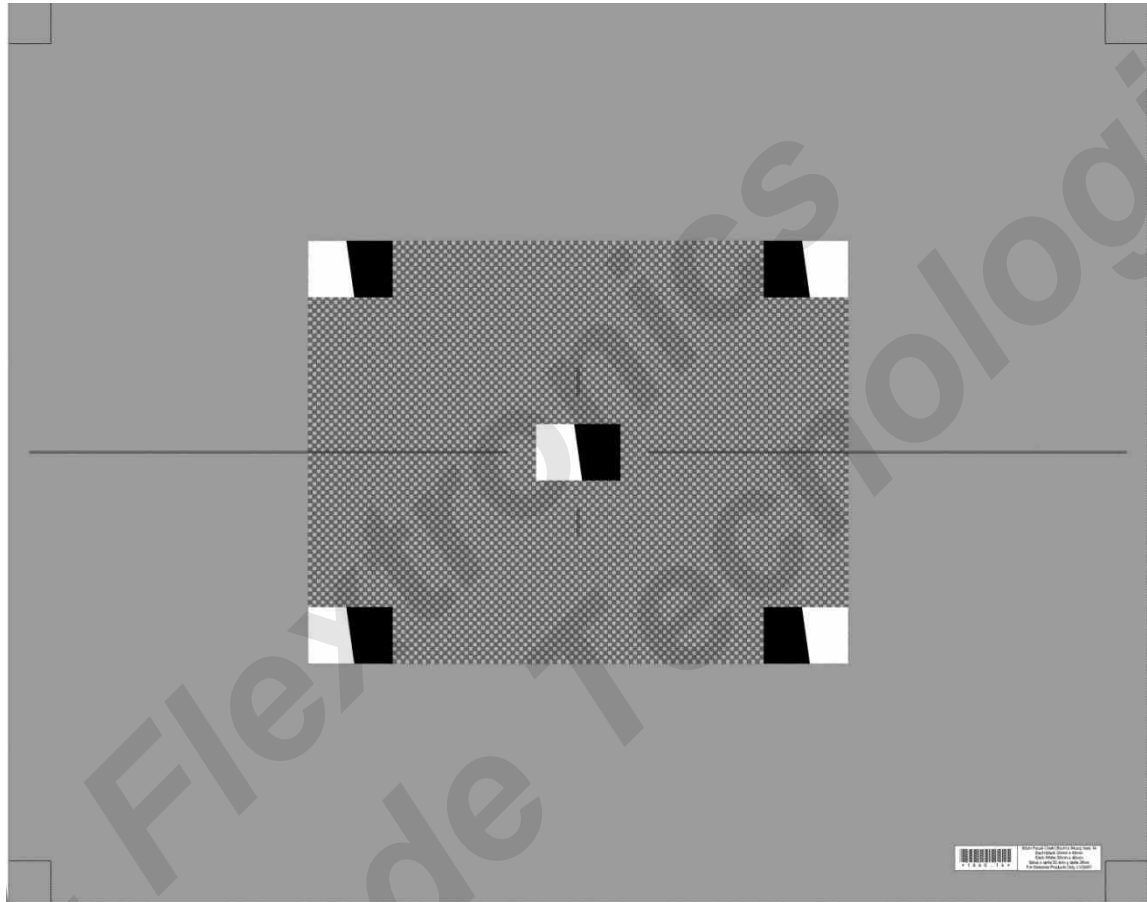
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Focus Chart @ 35cm



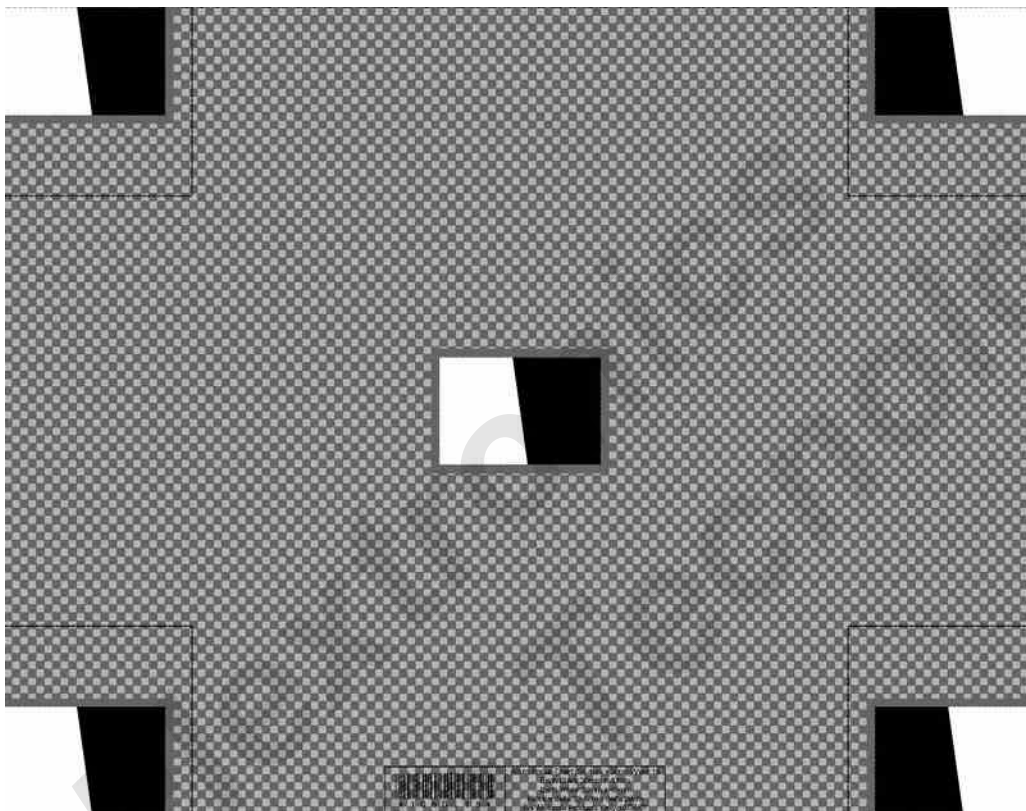
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Focus Chart @ 53cm and 60cm (Size 81cm x 64cm)



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Focus Chart for Flip Test 60cm (Size 38.4cm x 30cm)



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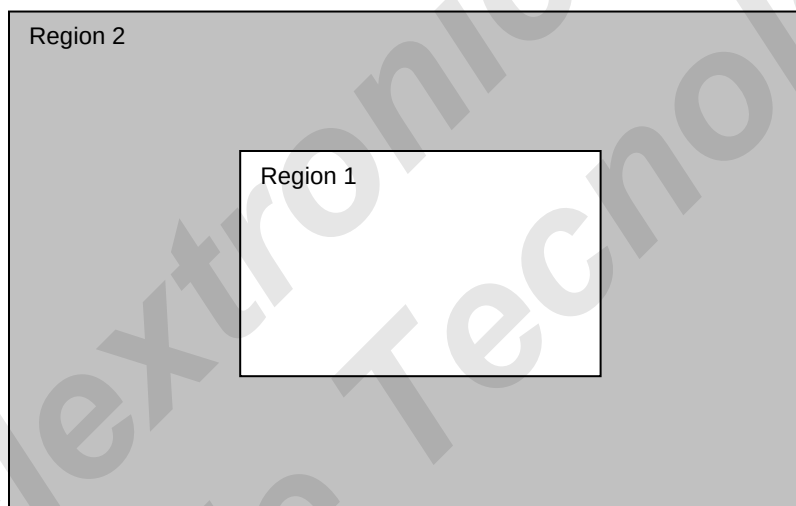
Blemish Test Conditions:

Position the camera with the lens facing directly towards the chart.

The following method will hold true for all display resolutions and orientations:

The Region 1 areas will always be a rectangle centered in the image and that rectangle will be half the dimensions of the image.

The Region 2 areas will consist of everything outside the Region 1 areas.
(See Drawing Below)



Example:

For a VGA landscape (640 wide x 480 high) Region 1 size will be 320 wide x 240 high, and centered in the image. Using the image coordinate system, where (0,0) is upper left and (639,479) is lower right for VGA landscape, Region 1 is defined as:

REGION 1 MAX X	479	Blemish Test - Maximum x coordinate of image in Region 1
REGION 1 MAX Y	359	Blemish Test - Maximum y coordinate of image in Region 1
REGION 1 MIN X	160	Blemish Test - Minimum x coordinate of image in Region 1
REGION 1 MIN Y	120	Blemish Test - Minimum y coordinate of image in Region 1

Region 1 Min X = Max Resolution in X / 4

Region 1 Max X= Max X value – (Max Resolution in X / 4)

Region 1 Min Y = Max Resolution in Y / 4

Region 1 Max Y = Max Y value – (Max Resolution in Y / 4)

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Image Analysis Tool Control Limits

Image analysis tool has output parameters in the MTF.txt file that must be evaluated (pass/fail) to determine if the capture image is valid and will produce accurate readings.

Issues that are monitored by these control limits are:

- 1) Captured the image is correct.
- 2) Lighting is even.
- 3) Black and / or white blocks are not cut off.
- 4) No error when analyzing the image.

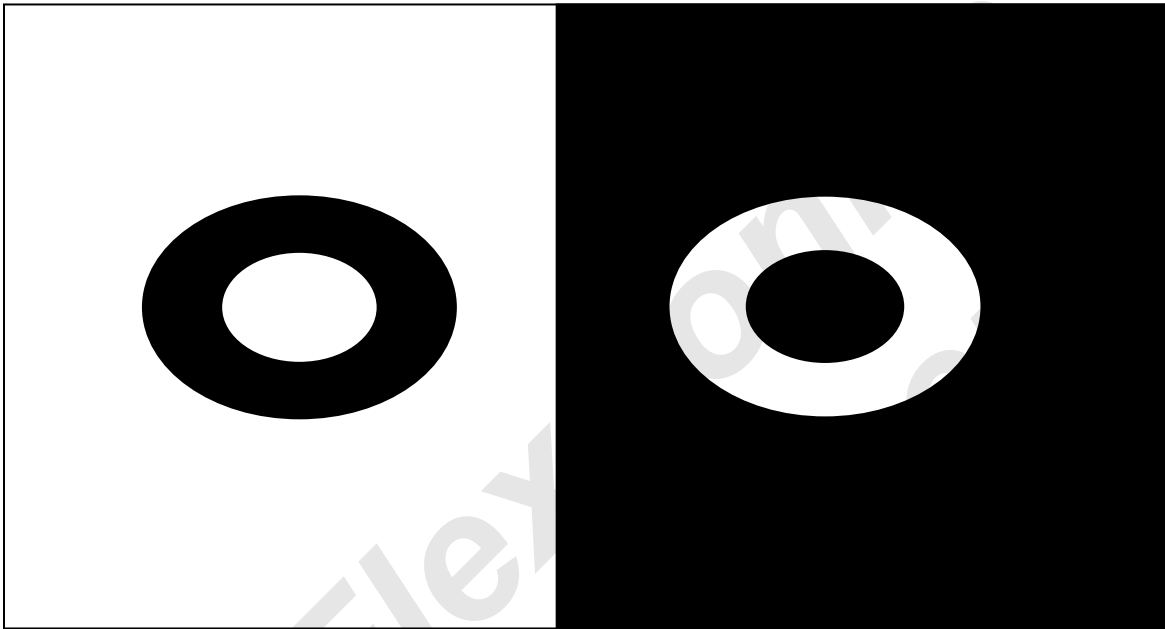
Below is the table of items to monitor.

<u>Master Test Name</u>	<u>Low Spec</u>	<u>Hi Spec</u>	<u>Comment</u>
Average Black Value Upper Left	0	100	If measurement fails this limit, then suspect that image is wrong.
Average Black Value Upper Right	0	100	If measurement fails this limit, then suspect that image is wrong.
Average Black Value Lower Left	0	100	If measurement fails this limit, then suspect that image is wrong.
Average Black Value Lower Right	0	100	If measurement fails this limit, then suspect that image is wrong.
Average Black Value Center	0	100	If measurement fails this limit, then suspect that image is wrong.
Average White Value Upper Left	100	250	If measurement fails this limit, then suspect that image is wrong.
Average White Value Upper Right	100	250	If measurement fails this limit, then suspect that image is wrong.
Average White Value Lower Left	100	250	If measurement fails this limit, then suspect that image is wrong.
Average White Value Lower Right	100	250	If measurement fails this limit, then suspect that image is wrong.
Average White Value Center	100	250	If measurement fails this limit, then suspect that image is wrong.
Black STDEV Upper Left	0	10	If measurement fails this limit, then suspect that analysis missed the region of interest or did not find the region of interest or lighting is not even.
Black STDEV Upper Right	0	10	If measurement fails this limit, then suspect that analysis missed the region of interest or did not find the region of interest or lighting is not even.
Black STDEV Lower Left	0	10	If measurement fails this limit, then suspect that analysis missed the region of interest or did not find the region of interest or lighting is not even.
Black STDEV Lower Right	0	10	If measurement fails this limit, then suspect that analysis missed the region of interest or did not find the region of interest or lighting is not even.
Black STDEV Center	0	10	If measurement fails this limit, then suspect that analysis missed the region of interest or did not find the region of interest or lighting is not even.
Error Code Upper Left	0	0	Analysis tool failure flag. If not zero, review Mot_image user guide.
Error Code Upper Right	0	0	Analysis tool failure flag. If not zero, review Mot_image user guide.
Error Code Lower Left	0	0	Analysis tool failure flag. If not zero, review Mot_image user guide.
Error Code Lower Right	0	0	Analysis tool failure flag. If not zero, review Mot_image user guide.
Error Code Center	0	0	Analysis tool failure flag. If not zero, review Mot_image user guide.
White STDEV Upper Left	0	10	If measurement fails this limit, then suspect that analysis missed the region of interest or did not find the region of interest or lighting is not even.
White STDEV Upper Right	0	10	If measurement fails this limit, then suspect that analysis missed the region of interest or did not find the region of interest or lighting is not even.
White STDEV Lower Left	0	10	If measurement fails this limit, then suspect that analysis missed the region of interest or did not find the region of interest or lighting is not even.
White STDEV Lower Right	0	10	If measurement fails this limit, then suspect that analysis missed the region of interest or did not find the region of interest or lighting is not even.

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White STDEV Center	0	10	If measurement fails this limit, then suspect that analysis missed the region of interest or did not find the region of interest or lighting is not even.
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Black and White Circle Chart – Used for operator viewfinder testing.





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12.7 Appendix G: Focus Image Analysis Table

Image Sensor Resolution	Diagonal Field Of View	Test Chart	PIXEL_SIZE	CORNER_X_DELTA	CORNER_Y_DELTA	MTF_SCALE	TARGET_DISTANCE	X_SEARCH	Y_SEARCH
VGA	66	1060.22	See Dev Eng.	25	25	20	60	19	19
1.3MP	60	1060.22		25	25	25	60	45	45
1.3MP	66	1060.22		25	25	25	60	40	40
2MP	60	1060.22		25	25	25	60	56	56
2MP	66	1060.22		25	25	25	60	49	49
3MP	60	1060.22		25	25	25	60	72	72
3MP	66	1060.22		25	25	25	60	63	63
5MP	60	1060.22		25	25	25	60	90	90
5MP	66	1060.22		25	25	25	60	81	81
8MP	60	1060.22		25	25	25	60	114	114
8MP	66	1060.22		25	25	25	60	102	102

Note: FOCAL_LENGTH is no longer used for image analysis with mtfanalysis.dll version 2.0 and higher. Default it to 5.0.

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12.8 AppendixH: Audio Conversion Equations.

Convert from mV to dB SPL

$$P = \text{AudioReading (mV)} / \text{MicCalFactor} \left(\frac{mV}{Pa} \right)$$

$$\text{dB SPL} = 20 \log \left(\frac{P}{P_{ref}} \right)$$

Where

Pref = 0.00002 Pa (The threshold of human hearing)

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12.9 Appendix I: Defaults, Batch Files and Audio Masks

The link below contains all product SPI Defaults, PA Defaults, or other files required for manufacturing.

<https://drive.google.com/a/motorola.com/#folders/0BxpO5vaJFvVSaFdKQVcyMml3WXc>

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12.10 Appendix J: Backup Phasing Templates

The link below contains backup phase data templates required for manufacturing.

<https://drive.google.com/a/motorola.com/#folders/0BxpO5vaJFvVSekF1cDhwTGRKbmM>



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12.11 Appendix K: PA Gain State Naming Convention

Products which use a two (2) gain state PA:

Mode of operation	PA_ON	PA_R1	PA_R0	RadioComm	Factory Designation	NV Designation
Power down / PA Off	Low	X	X	X	X	X
High Power Mode	High	Low	High	PA_R1	R1	MASTER1
Low Power Mode	High	High	Low	PA_R2	R2	MASTER0

Products which use a three (3) gain state PA:

Mode of operation	PA_ON	PA_R1	PA_R0	RadioComm	Factory Designation	NV Designation
Power down / PA Off	Low	X	X	X	X	X
High Power Mode	High	Low	High	PA_R1	R1	MASTER2
Medium Power Mode	High	High	High	PA_R3	R3	MASTER1
Low Power Mode	High	High	Low	PA_R2	R2	MASTER0

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12.12 Appendix L: WLAN Time Mask Parameters

The transmit power-on ramp for 10% to 90% of maximum power shall be no greater than 2 μ s. The transmit power-on ramp is shown in Figure 12.12.1.

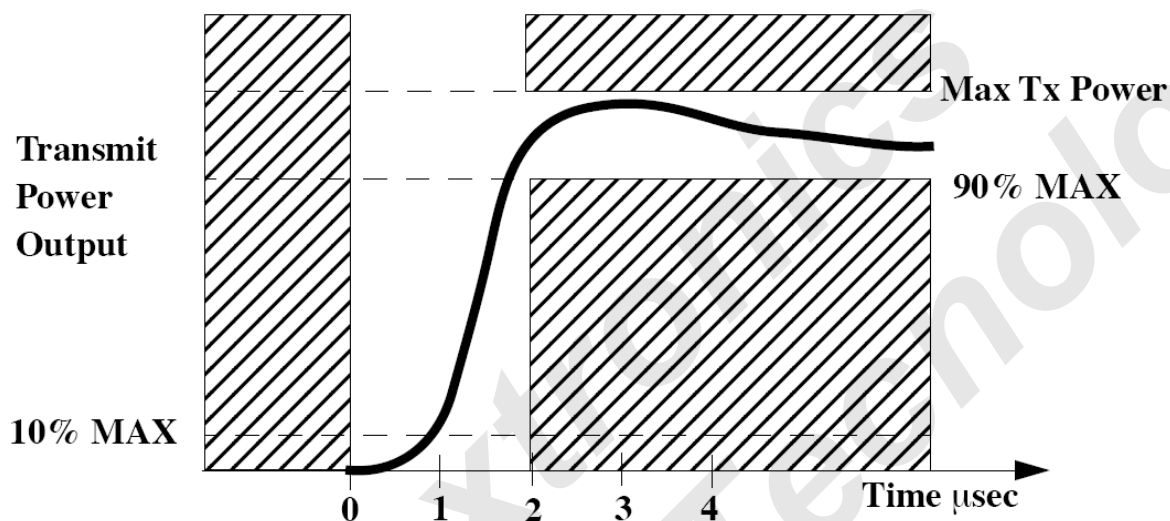


Figure 12.12.1 Transmit power-on ramp

The transmit power-down ramp for 90% to 10% maximum power shall be no greater than 2 μ s. The transmit power-down ramp is shown in Figure 12.12.2.

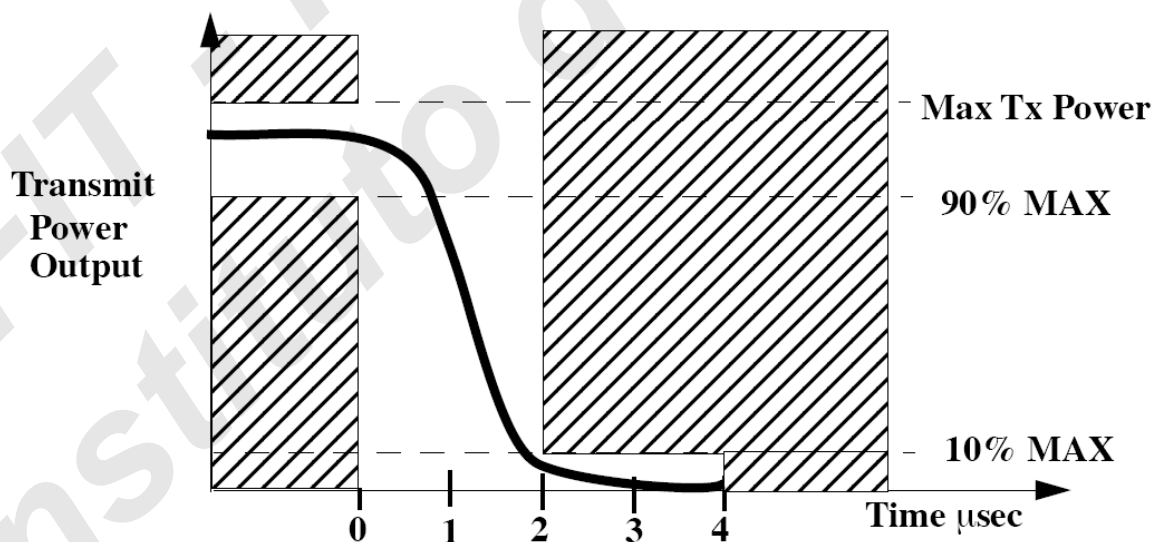


Figure 12.12.2 Transmit power-on ramp

12.13 Appendix M: WLAN Spectral Mask Parameters

The average energy of the constellations in each of the spectral lines $-16..-1$ and $+1..+16$ will deviate no more than ± 2 dB from their average energy. The average energy of the constellations in each of the spectral lines $-26..-17$ and $+17..+26$ will deviate no more than $+2/-4$ dB from the average energy of spectral lines $-16..-1$ and $+1..+16$. The data for this test shall be derived from the channel estimation step.

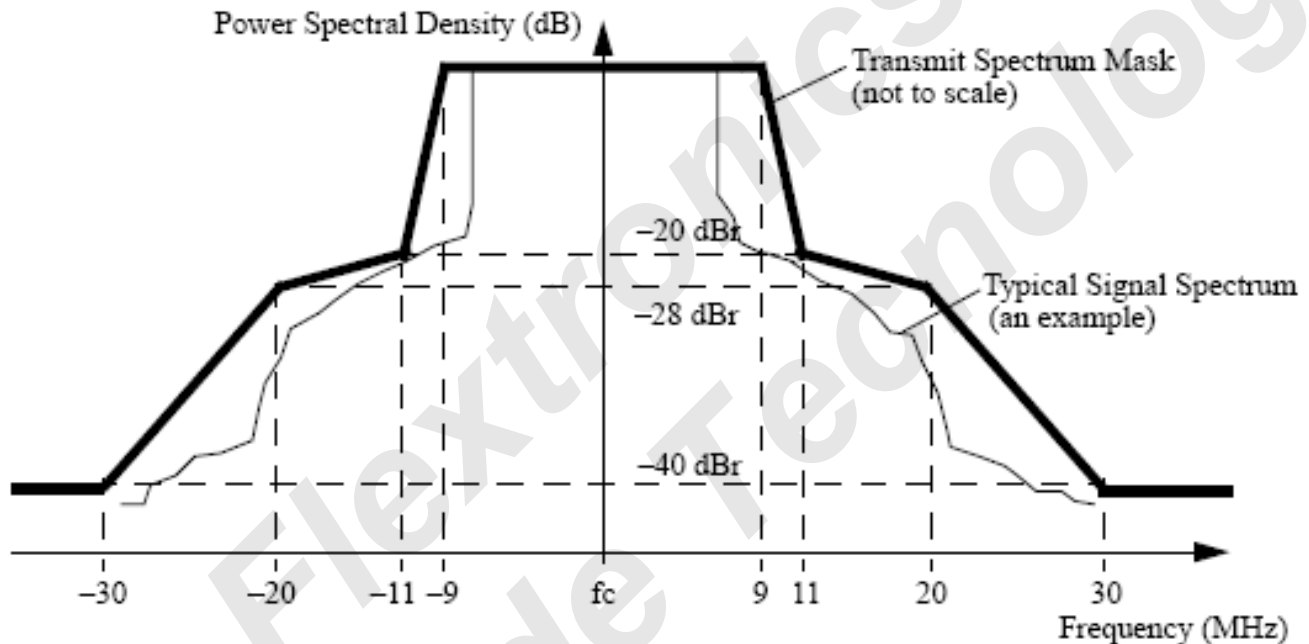


Figure 12.13.1 Transmit spectrum mask (802.11a/g) for data rates 6, 9, 12, 18, 24, 36, 48, 54 Mbit/s.

Data Rate (Mbit/s)	Transmission Type	Modulation
6	OFDM	BPSK
9	OFDM	BPSK
12	OFDM	QPSK
18	OFDM	QPSK
24	OFDM	16-QAM
36	OFDM	16-QAM
48	OFDM	64-QAM
54	OFDM	64-QAM

Table 12.13.1 Modulation scheme (802.11a/g)

Data Rate (Mbit/s)	EVM (dB)	EVM % ($\text{EVM}(\%) = 10^{\text{EVM}(\text{dB})/20}$)
6	-5	56.2
9	-8	39.8
12	-10	31.6
18	-13	22.4
24	-16	15.8
36	-19	11.2
48	-22	7.9
54	-25	5.9

Table 12.13.2 Maximum RMS EVM (802.11a/g)

Data Rate (Mbit/s)	Minimum Sensitivity (dBm)	Adjacent Channel Rejection, 25 MHz (dB)
6	-82	16
9	-81	15
12	-79	13
18	-77	11
24	-74	8
36	-70	4
48	-66	0
54	-65	-1

Table 12.13.3 Receiver Performance Requirements (802.11a/g)

The transmitted spectral products shall be less than -30 dBr (dB relative to the SINx/x peak) for

$f_c - 22 \text{ MHz} < f < f_c - 11 \text{ MHz}$; and
 $f_c + 11 \text{ MHz} < f < f_c + 22 \text{ MHz}$;

and shall be less than -50 dBr for

$f < f_c - 22 \text{ MHz}$; and
 $f > f_c + 22 \text{ MHz}$.

where

f_c is the channel center frequency.

The transmit spectral mask is shown in Figure 145. The measurements shall be made using a 100 kHz resolution bandwidth and a 100 kHz video bandwidth.

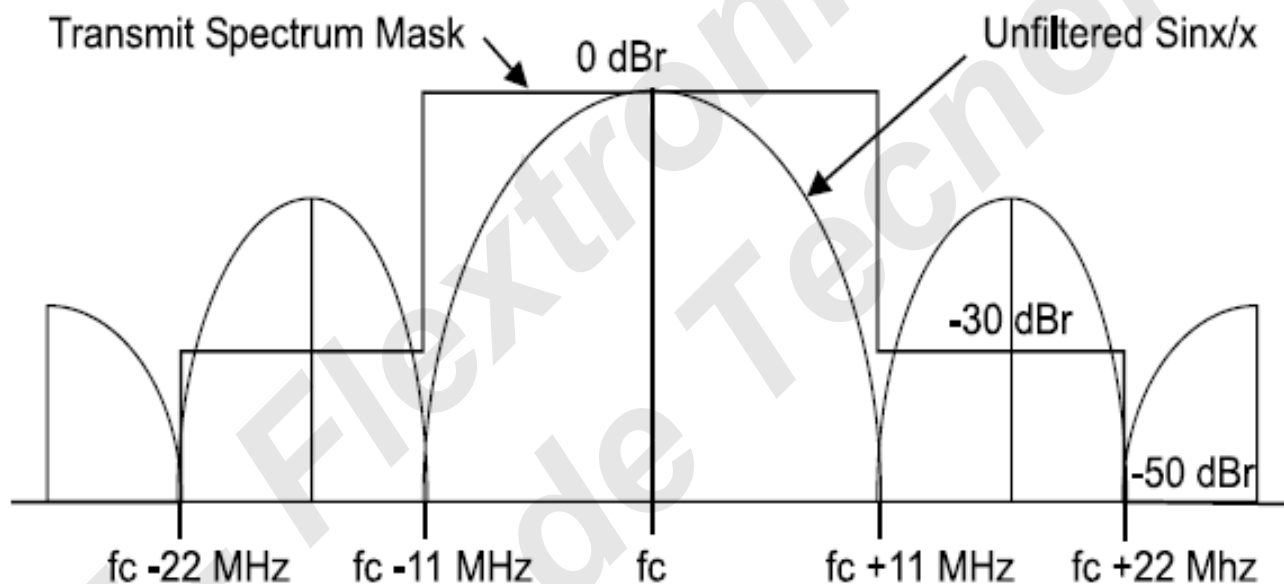


Figure 12.13.2 Transmit spectrum mask (802.11b) for data rates 1, 2, 5.5, 11, 22, and 33 Mbit/s.

Data Rate (Mbit/s)	Transmission Type	Modulation
1	DSSS	DBPSK
2	DSSS	DQPSK
5.5	DSSS	CCK/DQPSK
11	DSSS	CCK/DQPSK
22	DSSS	PBCC/QPSK
33	DSSS	PBCC/QPSK

Table 12.13.4 Modulation scheme used in 802.11b

CCK = Complementary Code Keying

PBCC = Packet Binary Convolutional Code

Data Rate (Mbit/s)	EVM (dB)	EVM % ($EVM(\%) = 10^{EVM(dB)/20}$)
1	-9	35
2	-9	35
5.5	-9	35
11	-9	35
22	-9	35
33	-9	35

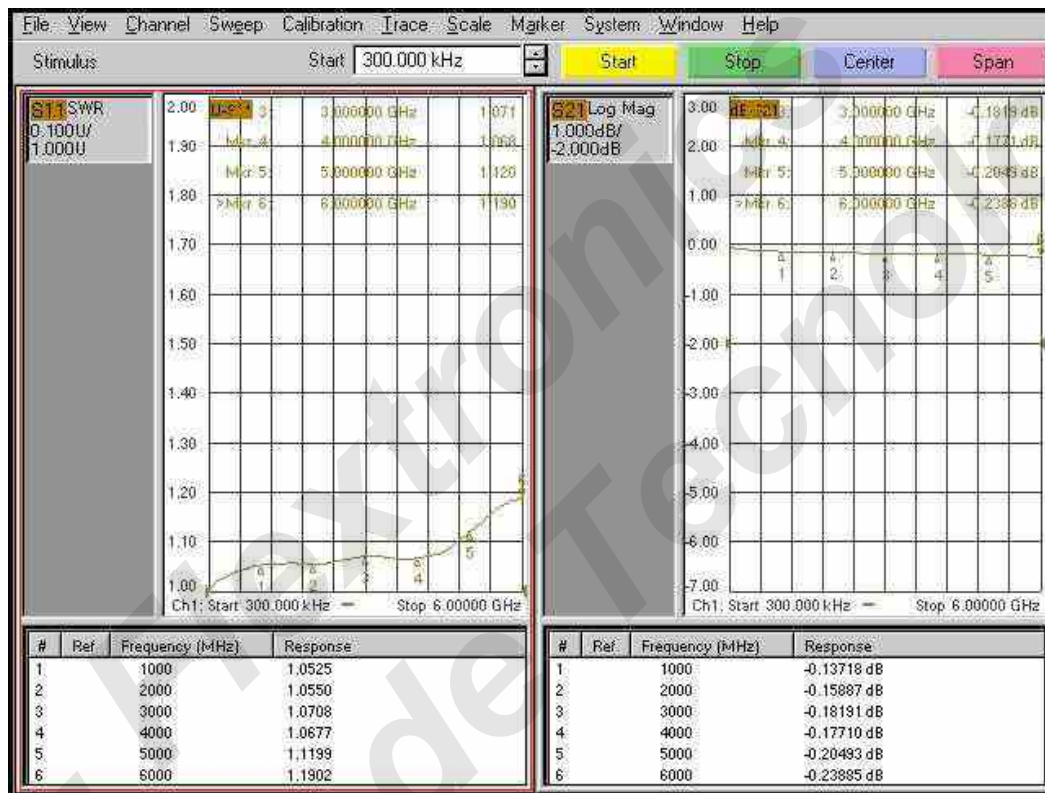
Table 12.13.5 Maximum RMS EVM (802.11b)

Data Rate (Mbit/s)	Minimum Sensitivity (dBm)	Adjacent Channel Rejection, 25 MHz (dB)
1	-76	35
2	-76	35
5.5	-76	35
11	-76	35
22	-76	35
33	-74	35

Table 12.13.6 Receiver Performance Requirements (802.11b)

12.14 Appendix N: Hirose Connector Loss

The data below is for the Hirose connector 2 part combination (Hirose part number HRMJ-U.FLP-ST2 and part number HMR-PA-PJ(F)-1(40)) The left side is the VSWR and the right side is the insertion loss for the 2 part combination.



12.15 Appendix O: LTE Test Frequencies

The test frequencies are based on the E-UTRA frequency bands. Refer to [7] 3GPP TS 36.508; Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Common test environments for User Equipment (UE) conformance testing.

12.15.1 Test frequencies for operating band 1

Table 7.1.1: Test frequencies for E-UTRA channel bandwidth for operating band 1

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	5	18025	1922.5	25	2112.5
	10	18050	1925	50	2115
	15	18075	1927.5	75	2117.5
	20	18100	1930	100	2120
Mid Range	5/10/15/20	18300	1950	300	2140
High Range	5	18575	1977.5	575	2167.5
	10	18550	1975	550	2165
	15	18525	1972.5	525	2162.5
	20	18500	1970	500	2160

12.15.2 Test frequencies for operating band 4

Table 7.1.2: Test frequencies for E-UTRA channel bandwidth for operating band 4

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	1.4	19957	1710.7	1957	2110.7
	3	19965	1711.5	1965	2111.5
	5	19975	1712.5	1975	2112.5
	10	20000	1715	2000	2115
	15	20025	1717.5	2025	2117.5
	20	20050	1720	2050	2120
Mid Range	1.4/3/5/10/15/20	20175	1732.5	2175	2132.5
High Range	1.4	20393	1754.3	2393	2154.3
	3	20385	1753.5	2385	2153.5
	5	20375	1752.5	2375	2152.5
	10	20350	1750	2350	2150
	15	20325	1747.5	2325	2147.5
	20	20300	1745	2300	2145

12.15.3 Test frequencies for operating band 7

Table 7.1.3: Test frequencies for E-UTRA channel bandwidth for operating band 7

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	5	20775	2502.5	2775	2622.5
	10	20800	2505	2800	2625
	15	20825	2507.5	2825	2627.5
	20 ^[1]	20850	2510	2850	2630
Mid Range	5/10/15 20 ^[1]	21100	2535	3100	2655
High Range	5	21425	2567.5	3425	2687.5
	10	21400	2565	3400	2685
	15	21375	2562.5	3375	2682.5
	20 ^[1]	21350	2560	3350	2680

12.15.4 Test frequencies for operating band 13

Table 7.1.4: Test frequencies for E-UTRA channel bandwidth for operating band 13

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	5 ^[1]	23205	779.5	5205	748.5
	10 ^[1]	23230	782	5230	751
Mid Range	5 ^[1] /10 ^[1]	23230	782	5230	751
High Range	5 ^[1]	23255	784.5	5255	753.5
	10 ^[1]	23230	782	5230	751

12.15.5 Test frequencies for operating band 17

Table 7.1.5: Test frequencies for E-UTRA channel bandwidth for operating band 17

Test Frequency ID	Bandwidth [MHz]	N _{UL}	Frequency of Uplink [MHz]	N _{DL}	Frequency of Downlink [MHz]
Low Range	5 ^[1]	23755	706.5	5755	736.5
	10 ^[1]	23780	709	5780	739
Mid Range	5 ^[1] /10 ^[1]	23790	710	5790	740
High Range	5 ^[1]	23825	713.5	5825	743.5
	10 ^[1]	23800	711	5800	741

12.16 Appendix P: Main Tx/Rx & Diversity Rx Support

Main Antenna TX/RX and Diversity Antenna Rx product HW configuration to indicate how the factory should route the signals from the call box to the phone, to test the intended path.

Product	UMTS Band	Phone's DL Primary HW Path Configuration			Phone's UL Primary HW Path Configuration	
		Main	Diversity	Dedicated_RX_2100	Main	Diversity
	Band 1	X	X	N/A	X	N/A
	Band 2	X	X	N/A	X	N/A
	Band 5	X	X	N/A	X	N/A
	Band 8	X		N/A	X	N/A

This table shows the HW configuration on the phone to indicate how the factory should route the signals from the call box/phone to test the intended path.

Table 12.16.1

Product	GSM Band	Phone's DL Primary HW Path Configuration		Phone's UL Primary HW Path Configuration	
		Main	Diversity	Main	Diversity
	850	X	N/A	X	N/A
	900	X	N/A	X	N/A
	1800	X	N/A	X	N/A
	1900	X	N/A	X	N/A

This table shows the HW configuration on the phone to indicate how the factory should route the signals from the call box/phone to test the intended path.

Table 12.16.1

Product	CDMA 1X Band	Phone's DL Primary HW Path Configuration		Phone's UL Primary HW Path Configuration	
		Main	Diversity	Main	Diversity
	BC0	X	N/A	X	N/A
	BC1	X	N/A	X	N/A

This table shows the HW configuration on the phone to indicate how the factory should route the signals from the call box/phone to test the intended path.

Table 12.16.1

Product	CDMA EVDO Band	Phone's DL Primary HW Path Configuration		Phone's UL Primary HW Path Configuration	
		Main	Diversity	Main	Diversity
	BC0	X	X	X	N/A
	BC1	X	X	X	N/A

This table shows the HW configuration on the phone to indicate how the factory should route the signals from the call box/phone to test the intended path.

Table 12.16.1



motorola
a lenovo company

Material or Methods Specification

No. 12014512001

Title: **Factory Test Specification for LTE / 3GSM / CDMA / TD-SCDMA Handsets**

Revision Date: **22-June-2016**

MEMO:

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Product	LTE Band	Phone's DL Primary HW Path Configuration		Phone's UL Primary HW Path Configuration	
		Main	Diversity	Main	Diversity
	B13	X	X	X	N/A
<i>This table shows the HW configuration on the phone to indicate how the factory should route the signals from the call box/phone to test the intended path.</i>					

Table 12.16.1

12.17 Appendix Q: Murata Factory Antenna and RF Probe Insertion Loss and Isolation

The following data specifies typical insertion loss factors for the Murata RF Probe (MM206615).

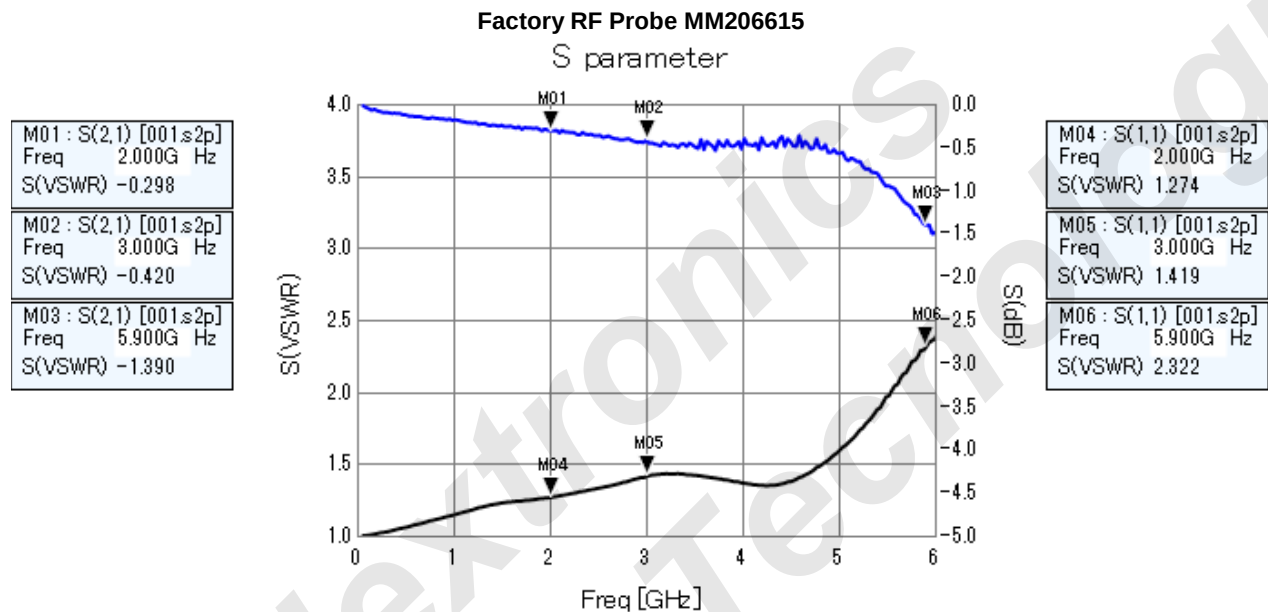


Figure 12.17.1 RF Data

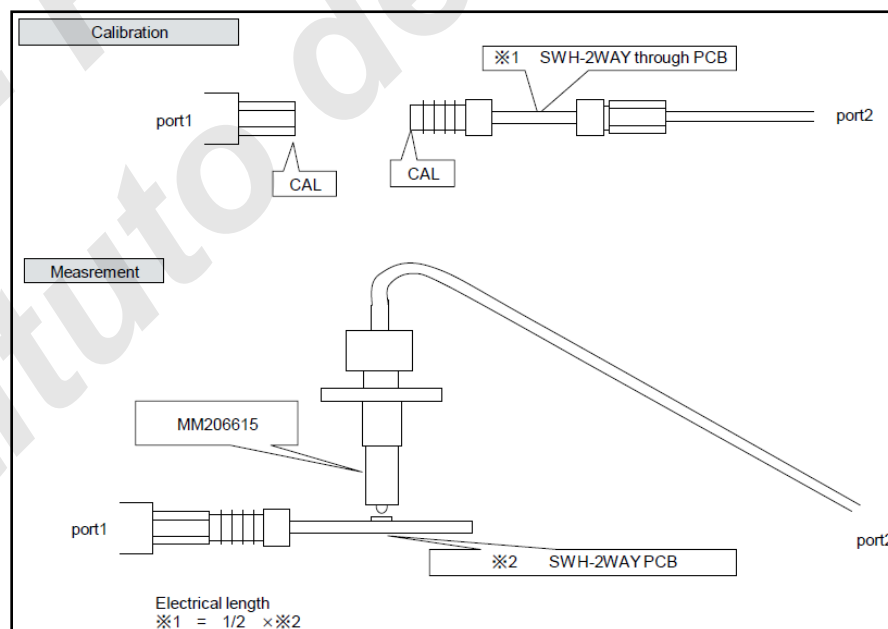


Figure 12.17.2 Calibration Method

 motorola a lenovo company	Material or Methods Specification	No. 12014512001
Title: Factory Test Specification for LTE / 3GSM / CDMA / TD-SCDMA Handsets		
Revision Date: 22-June-2016	MEMO:	Issue: 03
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12.18 Appendix R: Installing APK Files to a Phone

Phone software typically comes pre-loaded with the necessary apk (Android application package) files needed to launch the touch screen or display apps. If there is a need to install these applications, the following command can be sent from the command line.

Note: In the below example the default location of the apk files are located in the c:\prod\config directory.

Send the following ADB command to install APK file
adb install -r c:\prod\config\CQATest.apk

 motorola a lenovo company	Material or Methods Specification	No. 12014512001
Title: Factory Test Specification for LTE / 3GSM / CDMA / TD-SCDMA Handsets		
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12.19 Appendix S: Enable/Disable ADB

Enable ADB

Send the following command to enable the ADB USB Interface

Command	Opcode/data
CTRL_ADB	0x0097 0001

Disable ADB

Send the following command to disable the ADB USB Interface

Command	Opcode/data
CTRL_ADB	0x0097 0000

When enabling or disabling the ADB state the native ADB USB interface will enumerate or de-enumerate. However, when setting either ADB state, this command will also cause the test command BLAN USB interface to de-enumerate and then re-enumerate.

 motorola a lenovo company	Material or Methods Specification	No. 12014512001
Title: Factory Test Specification for LTE / 3GSM / CDMA / TD-SCDMA Handsets		
Revision Date: 22-June-2016	MEMO:	Issue: 03 Page 444 of 444

12.20 Appendix T: Total Percent Sampler Details

High volume manufacturing reduces testing to historically worse case channel to reduce cost of test. Due to concerns that post-SA shifts in parts performance or tolerances may cause undetected failures, a form of test sampling can be enabled. This method of sampling is named TotalPercent. The idea is that we can define blocks of tests and have the sampler alternate which block gets tested for each phone tested. For example, we define a block for 800 band Low channel, Mid Channel, and High channel tests. Then 1 phone would test only the low channel. The next phone would test only mid channel. Next phone tests High channel. This pattern then repeats.

One major concern raised about this sampler was that if a phone failed one channel, then when it was retested it may be tested on different channel and pass. This could possibly let a failure escape. This concern is addressed thru the disabling of the sampler if a phone is determined to be a retest that has previously failed a sampled test. We are using the flow control bytes stored in the phone's factory info block to determine if a phone is a retest. The flow control bytes store not only if the phone has passed a certain station but also how many times it has been tested at that station (maxes out at 15) and if it failed a sampled test. If we read that the phone has been tested at the current station and previously failed a sampled test, then a flag is set in the sampler to not sample (i.e. not to skip) any of the tests.

Also, the sampler can be configured to define how many units must pass all tests before sampling is enabled. This means if it is configured to 1, then using the example above, the first phone tested must pass low, mid, and high channel tests. The next phone would start sampling cycle. Also configurable is the number of failures to allow before sampling is disabled. If it is disabled, the phones will test all blocks. Lastly, there is another configurable parameter to control how many units must pass all blocks before sampling is once again enabled. All this tracking of how many units passed, failed, what is next block in sampler to be tested, etc. is tracked in a database stored locally on each test bench. This means that if one bench is bad (say it has a bad RF cable) then all benches will not have sampling disabled, only the bad one would. There is the capability to configure a line monitor to monitor passes and failures across an entire line of test benches. However, this was determined to not be value added so was not implemented. At this time the line monitor is not approved for use by any product.

Below are the control tables for each sampled testing block. The only tests that are approved for sampling must have a S in the required column in Section 2.0 for the product being tested. Make sure that the token called: ENABLE_SAMPLING_FOR_SECOND_RUN is NOT in the general setup of the sampler. Sampling for second run units is NOT approved.

All Bands (S1)

Percent	33.33333
InitialTestLim	1
LocalFailLim	3
LocalPassLim	3
LineFailLim	0
LinePassLim	0

All Bands (S2)

Percent	10
InitialTestLim	1
LocalFailLim	1
LocalPassLim	10
LineFailLim	0
LinePassLim	0

All Bands (S3)

Percent	1
InitialTestLim	5
Local FailLim	1
LocalPassLim	10
LineFailLim	0
LinePassLim	0